

FIELD MANUAL

**HIGH FREQUENCY
RADIO COMMUNICATIONS**

HEADQUARTERS, DEPARTMENT OF THE ARMY

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HIGH FREQUENCY RADIO COMMUNICATIONS

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PART ONE

GENERAL

CHAPTER 1

INTRODUCTION

1-1. Purpose

The purpose of this manual is to aid personnel in using high-frequency (hf) radio systems to achieve successful communications. The manual is intended to provide theoretical and practical information for use in setting up and operating hf radio systems.

1-2. Scope

a. Part One of the manual provides fundamental principles and setup and operating guidance for all hf radio operators and other personnel responsible for hf radio communications. The information in Part One is primarily for use by tactical operators as well as personnel using non-tactical transportable systems. Information in Part One includes hf radio-wave propagation fundamentals, transmission line fundamentals, and hf antenna principles.

b. Part Two of the manual presents more complicated techniques for the most efficient selection, siting, and orientation of antennas, and for frequency selection. Part Two describes the radio propagation services provided by the US Army Communications-Electronics Engineering Installation Agency (USACEEIA). Also included is information on handling problems peculiar to short skywave paths; information on problems in communicating in mountains, desert, jungle, arctic terrain, and equatorial areas; and information on hf net communications.

c. A glossary of terms commonly used in radio propagation is found in the rear of the manual, along with appendixes covering:

- (1) The usage of decibels.
- (2) Required signal-to-noise ratios for hf systems.
- (3) A description of USACEEIA hf propagation services.
- (4) The use of ionospheric sounders.
- (5) Fresnel zone measurements and effects.
- (6) Solar effects and ionospheric disturbances with their cause and corrective techniques

for radio-wave propagation.

- (7) HF exercise operational checklist.

1-3. Introduction to Hf Radio

a. For successful communications, radio-frequency performance depends upon the type of emission, the amount of transmitter power output, the characteristics of the transmitter antenna, the amount of propagation path loss, the characteristics of the receiver-antenna, the amount of received noise, and the sensitivity and selectivity of the receiver. A signal-to-noise ratio must be present at the input of the receiver which is sufficient for satisfactory receiver performance.

b. The challenge facing hf radio personnel is tremendous. They are expected to use their hf radio systems to complete important missions. In combat or in peacetime emergencies or disasters, these missions are usually critical. The challenge of making hf radio systems work can be illustrated by contrasting them with the commonly used line-of-sight (LOS) radio systems. A well designed, properly maintained LOS system will operate hour after hour, day after day, and year after year with insignificant outages. On the other hand, even if the hf system is initially well designed, the hf radio operator must continually adjust that system to compensate for an ever-changing ionosphere and an ever-changing terrestrial environment (interference from other stations, atmospheric interference, manmade noise, etc.).

c. Despite the additional human effort required to make them work, hf radio systems still are vitally necessary because they do some jobs that other systems cannot do, or at least cannot do economically. It is practically impossible at the present time to match the combination of simplicity, economy, transportability, and versatility of hf systems. Using readily available equipment, these systems can be used over almost any distance, air-dropped, installed quickly, and dismantled quickly. Under ideal conditions 15 watts of transmitter power may be used to successfully

communicate over thousands of miles. Knowledgeable operators, backed up by antenna design and propagation predictions from a highly developed propagation engineering service, are the key to successful hf radio-frequency system performance.

1-4. User's Comments

You are encouraged to submit comments and recommended changes to this manual to improve its usefulness. Key each comment to the specific page, paragraph, and line of text where the change

is recommended. Provide a reason for each comment so that the evaluator understands clearly the reason for the recommended change. Submit your comments on DA Form 2028 (Recommended Changes to Publications and Blank Forms) and mail them to Commander, US Army Communications Command, ATTN: CC-OPS-SM, Fort Huachuca, Arizona 85613. Mail an information copy to Commander, US Army Signal Center and Fort Gordon, ATTN: ATZHTD-P-L, Fort Gordon, Georgia 30905. If DA Form 2028 is not available, a letter is acceptable.

CHAPTER 2

RADIO WAVE PROPAGATION FUNDAMENTALS

2-1. The Radio Spectrum

The radio spectrum is generally considered to be that portion of the electromagnetic wave spectrum extending from about 30 Hz to about 3×10^{12} Hz (3000 GHz). The lower limit is completely arbitrary; the upper limit approximates the lower limit of the infrared portion of the spectrum. The various portions of the radio spectrum have been

given the names listed in table 2-1. The arbitrary divisions listed in the table give the lower limit of the high frequency band as 3 MHz. In this manual it will be considered as extending to 2 MHz since there is no essential difference in the manner of wave propagation between a frequency of 3 MHz and one of 2 MHz. The upper limit of 30 MHz corresponds to a marked break in the manner of wave propagation.

Table 2-1. Divisions of the Radio Spectrum

Frequency range	Metric subdivision	Designation
30 to 300 hertz	Megametric waves	elf Extremely low frequency
300 to 3000 hertz		vf Voice frequency
3 to 30 kilohertz	Myriametric waves	vlf Very-low frequency
30 to 300 kilohertz	Kilometric waves	lf Low frequency
300 to 3000 kilohertz	Hectometric waves	mf Medium frequency
3 to 30 megahertz	Decametric waves	hf High frequency
30 to 300 megahertz	Metric waves	vhf Very-high frequency
300 to 3000 megahertz	Decimetric waves	uhf Ultra-high frequency
3 to 30 gigahertz	Centimetric waves	shf Super-high frequency
30 to 300 gigahertz	Millimetric waves	ehf Extremely high frequency
300 to 3000 gigahertz	Decimillimetric waves	-----

2-2. The Atmosphere

Wave propagation deals with the properties and the nature of the atmosphere through which radio waves must travel from the transmitting antenna to the receiving antenna. The atmosphere is not uniform, but varies with the altitude, geographic location, time of day or night, season, and year. A knowledge of the composition and properties of the atmosphere aids in the solution of problems that arise in planning radio communication paths and in predicting the reliability of communications.

a. Troposphere. The troposphere is that portion of the earth's atmosphere extending from the surface of the earth to heights of approximately 10 km (6¼ mi). Within the troposphere, the bending of radio waves by refraction causes the distance to the radio horizon to exceed the distance to the optical horizon. Tropospheric refraction (bending caused by sudden changes in the characteristics of air in a lower atmosphere) affect the received signal at distances beyond the radio horizon.

b. Stratosphere. The stratosphere is that por-

tion of the earth's atmosphere lying between the troposphere and ionosphere, about 10 to 60 km (6¼ to 37 mi) above the earth. The temperature in this region is more nearly constant than in the troposphere.

c. Ionosphere. The ionosphere is that portion of the earth's atmosphere above the lowest level at which ionization (splitting of molecules into positive and negative charges, or ions) of low-pressure gases affects the transmission of radio waves. It extends from about 60 to 1,000 km (37 to 600 mi) above the earth. The ionosphere is composed of several distinct layers in which ionization occurs at different levels and intensities.

d. Propagation in the Atmosphere. There are two principal paths by which radio waves can travel from a transmitter to a receiver. One is by groundwaves, which travel near the ground from the transmitter to the receiver. The other is by skywaves, which travel up to the ionosphere and are bent downward, or refracted, back to the earth (fig. 2-1). Long distance hf radio transmission is achieved by the use of skywaves. Short-distance transmission is by means of ground-

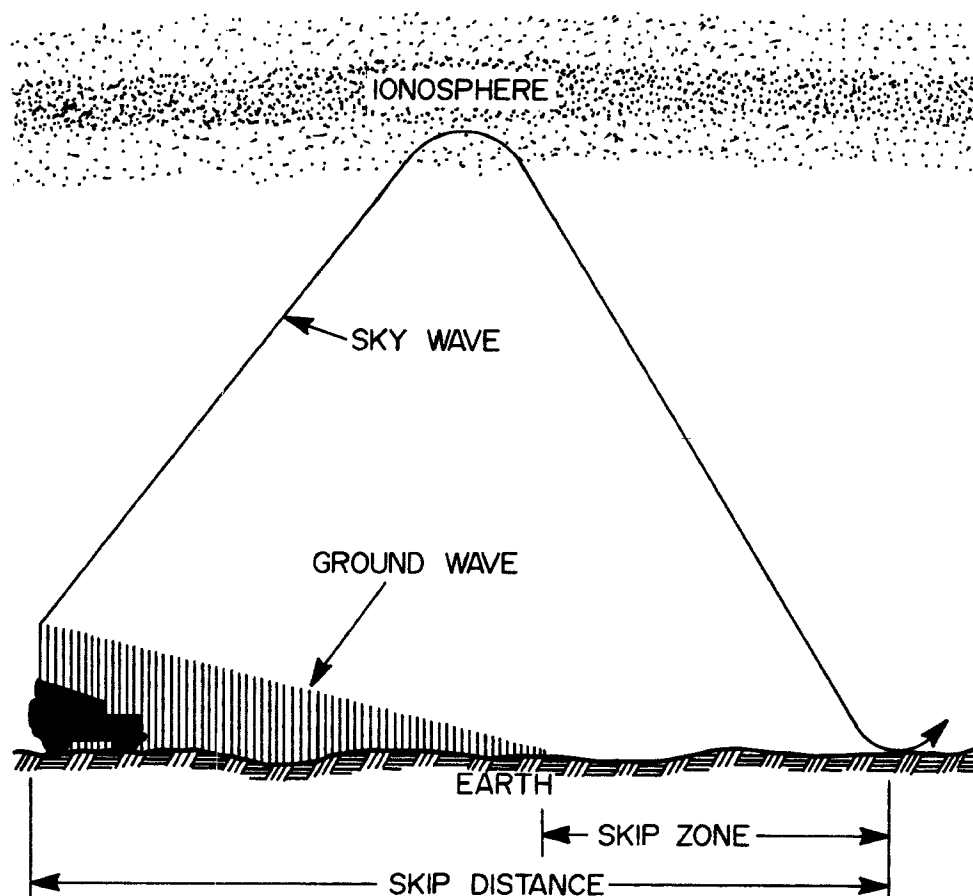


Figure 2-1. Hf radio wave components.

waves. Groundwave propagation is affected by the electrical characteristics of the earth and by the amount of diffraction, or bending, of the wave around the curvature of the earth. These characteristics vary in different localities, but they are relatively constant with respect to time and the season. Skywave propagation is variable, since the constantly changing state of the ionosphere has a definite effect on the refraction of the waves.

2-3. Groundwave Propagation

a. General. The groundwave is considered to be made up of three components: the direct wave, the ground-reflected wave, and the surface wave, as shown in figure 2-2. The field intensity of groundwaves depends upon the radio frequency, transmitter power, transmitting antenna characteristics, and electrical characteristics (conductivity and dielectric constant) of the terrain. Low and very low frequencies are propagated much better by surface wave than are high frequencies. When high-powered transmitters and efficient antennas are used, the surface wave has a maximum range of about 500 km (300 mi) at 2 MHz and the range decreases with increasing frequency. About 80 km (50 mi) represents the usual maximum

range. The range of the groundwave from commonly used transportable equipment is less.

b. Direct Component.

(1) The direct wave is that component of the entire wave front which travels directly from the transmitting antenna to the receiving antenna. In terrestrial communications, this component of the groundwave is limited by the distance to the horizon from the transmitter, plus the small distance added by the atmospheric refraction of the wave around the curvature of the earth. This is essentially the line-of-sight distance. This distance can be extended by increasing the height of either the transmitting or the receiving antenna, or both. The direct wave is also useful for extraterrestrial communications.

(2) The direct wave is useful in air-ground communications, but most short-distance air-ground services are now on very high frequencies (vhf) or ultra high frequencies (uhf).

c. Ground-Reflected Component. The ground-reflected component, as its name indicates, is the portion of the radiated wave that reaches the receiving antenna after being reflected from the ground or from the sea. Upon reflection from the earth's surface, a near-grazing wave undergoes a

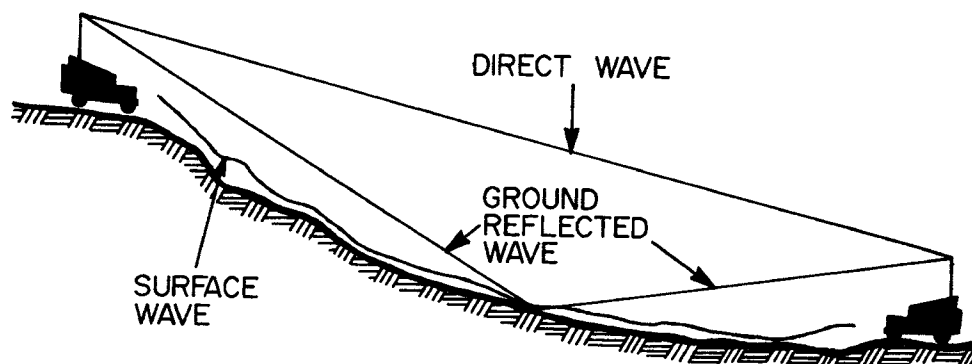


Figure 2-2. Possible routes for groundwaves.

phase shift of 180° . Since the reflected component travels a longer time in reaching its destination, a phase displacement somewhat greater than the 180° shift caused by reflection results. The net result near the ground is a weakening of the direct wave, by an amount approximately equal to the strength of the reflected wave.

d. Surface Wave Component.

(1) The surface wave is that component of the groundwave that is affected primarily by the conductivity and the dielectric constant of the earth. When both transmitting and receiving antennas are close to the ground, the direct and ground-reflected components of the wave tend to cancel out, and the resulting field intensity is principally that of the surface wave. The surface-wave component is not confined to the earth's surface, however, but extends up to considerable heights, diminishing in field strength with increased height. Its intensity becomes negligible at a height about one wavelength over ground and five to ten wavelengths over sea water.

(2) Absorption of part of the surface wave's energy is by the ground, which attenuates the electric intensity of the surface wave. This attenuation depends on the relative conductivity of the surface over which the wave travels. Table 2-2 shows the relative conductivity for various types of terrain. The best type of surface for surface-wave transmission is sea water. The electrical properties of the terrain that determine the attenuation of the surface wave field intensity vary little, and therefore this type of transmission has relatively stable characteristics.

(3) The earth has a short-circuiting effect, which results in severe attenuation, on the electric field of a horizontally polarized wave. However, the earth has much less effect on vertically polarized waves. For this reason, a vertically polarized wave is usually transmitted when the surface-wave component is used for communications. Radio waves travel more slowly in the earth than in the air, resulting in a forward tilt of the wavefront. This forward tilt means that

Table 2-2. Propagation Characteristics of Various Types of Terrain

Type of surface	Relative conductivity	Dielectric constant
Sea water	Good	80
Large bodies of fresh water	Fair	80
Wet soil	Fair	30
Flat, loamy soil	Fair	15
Dry, rocky terrain	Poor	7
Desert	Poor	4

the wave is being directed toward the surface. This explains the satisfactory performance of antennas only slightly above the ground. Poor conducting surfaces cause high loss and greater tilt.

2-4. Polarization

a. Radio waves consist of an electric (E) field

and a magnetic (H) field. These fields are at right angles to each other and also at right angles to the direction of propagation. A horizontally polarized wave has an electric field parallel to the earth's surface. A vertically polarized wave has an electric field perpendicular to the earth's surface. Figure 2-3 shows horizontally and vertically polarized waves.

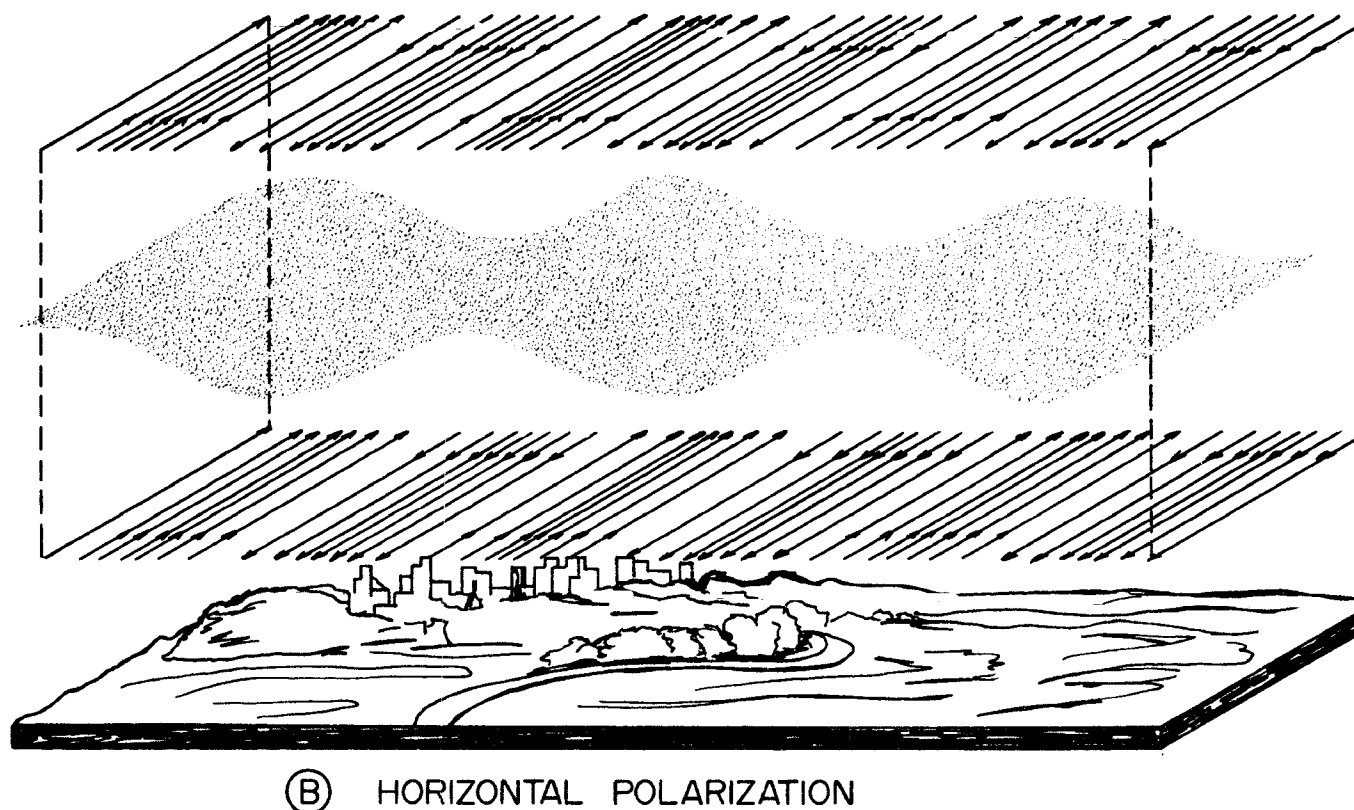
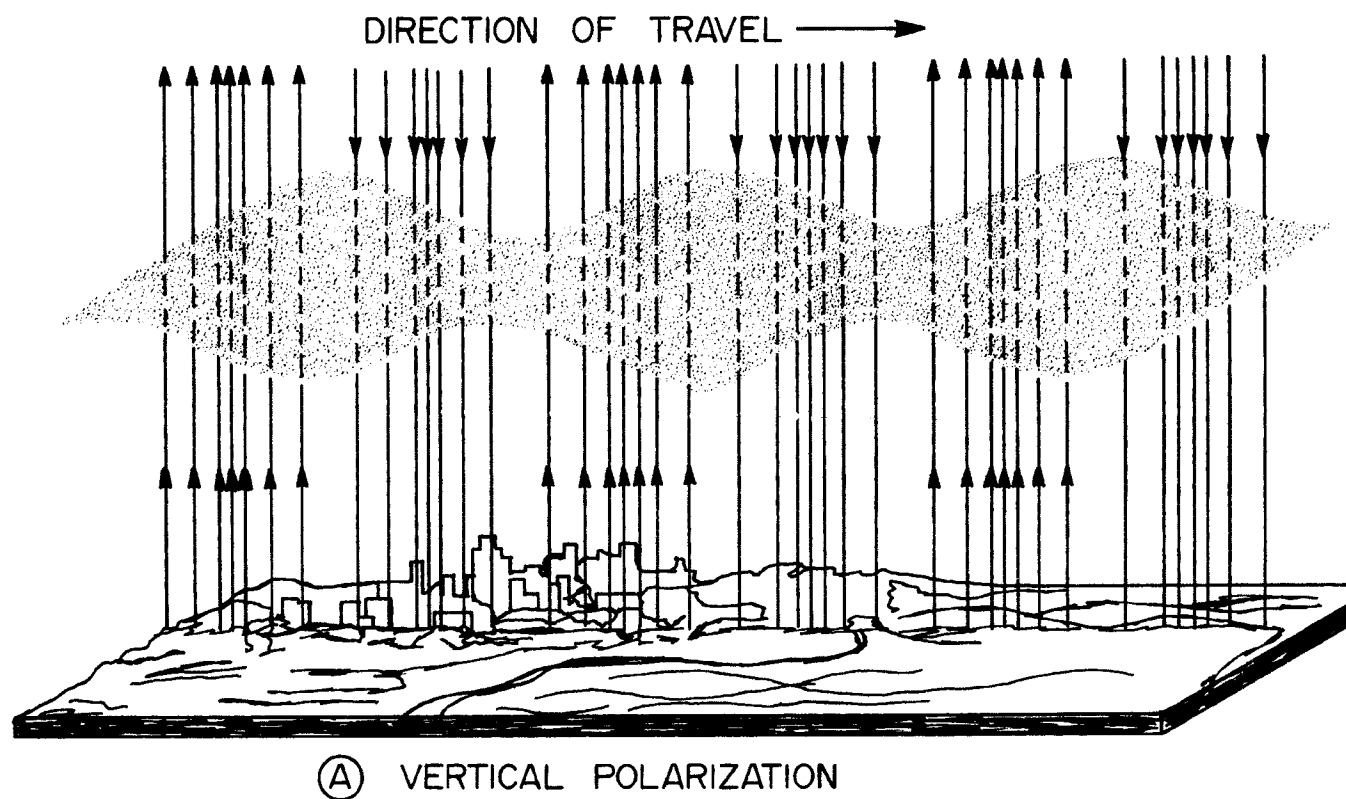


Figure 2-3. Vertical and horizontal polarization.

b. The polarization of the propagated wave is determined initially by the type and arrangement of the transmitting antenna. As a rule, a vertical conductor radiates a vertically polarized wave, and a horizontal conductor radiates a horizontally polarized wave. More complex forms, such as circular and elliptical polarization, in which the direction of maximum voltage rotates in space at the frequency of transmission, are also possible. These complex waves are generated by special antennas, or may be developed accidentally when linearly polarized waves pass through nonuniform media such as the ionosphere. The performance of a receiving antenna is improved if it can be oriented to take advantage of the polarization of the incident wave. However, as a consequence of random changing of the polarization of high-frequency waves as they travel through the ionosphere, matching the polarization of transmitting and remote receiving antennas is not essential. If possible, both vertically and horizontally polarized antennas should be tried. Where circuit requirements dictate surface-wave propagation, it is important that the antennas at both ends of the path have the same polarization. However, vertically polarized antennas provide the most effective surface-wave coverage.

2-5. The Ionosphere

a. *General.* Long-distance, high frequency communication is made possible by reflections of radio waves from the ionized layers in the earth's upper atmosphere. Under favorable conditions attenuation of the skywave is relatively low, and communication over a path thousands of miles in length is possible with very low transmitter power. However, the ionosphere is in a constant state of flux, and communication by skywave is much less stable than by groundwave. The communicator in the field other than the forward area tactical communicator will usually have available long-term propagation predictions, as well as current information and short-term forecasts of solar activity and ionospheric disturbances. (Appendixes D and G describe the services available.) This discussion on skywave propagation is intended to aid the communicator in understanding this material available to him.

b. Formation of the Ionosphere.

(1) The ionosphere is that portion of the earth's atmosphere which is sufficiently ionized that the presence of free electrons significantly affects the propagation of radio waves. It extends from approximately 60 km (37 mi) to about 1,000 km (600 mi) above the earth's surface. The funda-

mental cause of the ionized region is ultraviolet radiation from the sun. Radiation and particles from outside the solar system also contribute. A highly variable source of ionizing agents is the *solar wind*. The solar wind consists of ions thrown out from the surface of the sun as a result of turbulent processes there. As particles travel out through space, some come close enough to the earth to be trapped by the earth's magnetic fields.

(2) At great heights the ionizing agents are very intense, but the atmosphere is rare and there are few molecules present to be ionized. Therefore in this region the ionization density (number of ions or electrons per unit volume) is very low. As height is decreased, the atmospheric pressure and ionization density increase until a height is reached where the ionization density is a maximum. Below this height the atmospheric pressure continues to increase, but the ionization density decreases because a large portion of the ionizing energy has been absorbed or used up in the process of ionization.

c. Structure of the Ionosphere.

(1) *Layers in the ionosphere.* Figure 2-4 shows typical day and night profiles of electron density in the ionosphere. The figure shows several regions in which the electron density increases with height. These regions are called the D, E, E_s, F, F₁, and F₂ layers, as shown in the figure. The existence of layers is accounted for by the fact

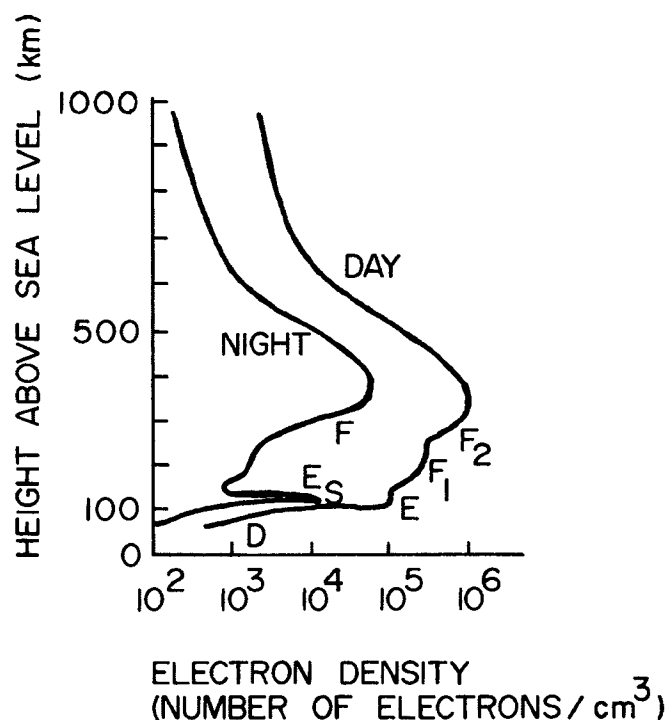


Figure 2-4. Typical day and night electron-density profiles in the ionosphere.

that the atmosphere is a mixture of gases which differ in their susceptibility to ionizing radiations, and which thus produce maximum ionization at different altitudes. The degree of ionization and height of these layers is affected by time of day, season of year, and variations in solar activity. Some characteristics of the ionosphere layers are summarized in table 2-3. Figure 2-5 is a pictorial representation of the ionospheric layers (the E_s (sporadic E) layer is omitted because of its irregular occurrence and limited geographic extent).

(2) *D layer*. The D layer extends from about 60 to 90 km (37 to 55 mi) above the earth. Normal D-layer ionization is produced by solar ultraviolet light and X-rays during daylight hours. The normal D-layer ionization decreases soon after sunset and increases at dawn. Additional D-layer ionization may be produced at any time of day

or night, especially at high latitudes, by high-energy electrons and protons originating from the sun. This type of increase in D-layer ionization is likely to be associated with geomagnetic disturbances. High frequency radio waves are not reflected by the D layer. The layer is important because of its adverse effect in absorbing energy from waves traversing it. The absorption is small at night and greatest about midday. It is quite variable, in general increasing with sunspot number.

(3) *E layer*. The E layer extends from about 100 to 140 km (60 to 85 mi) above the earth. The ionization is produced by solar ultraviolet and X-ray radiation, hence the ionization drops to relatively low values at night. During the day, reflections from the E layer are useful for communicating at distances up to about 2,000 km

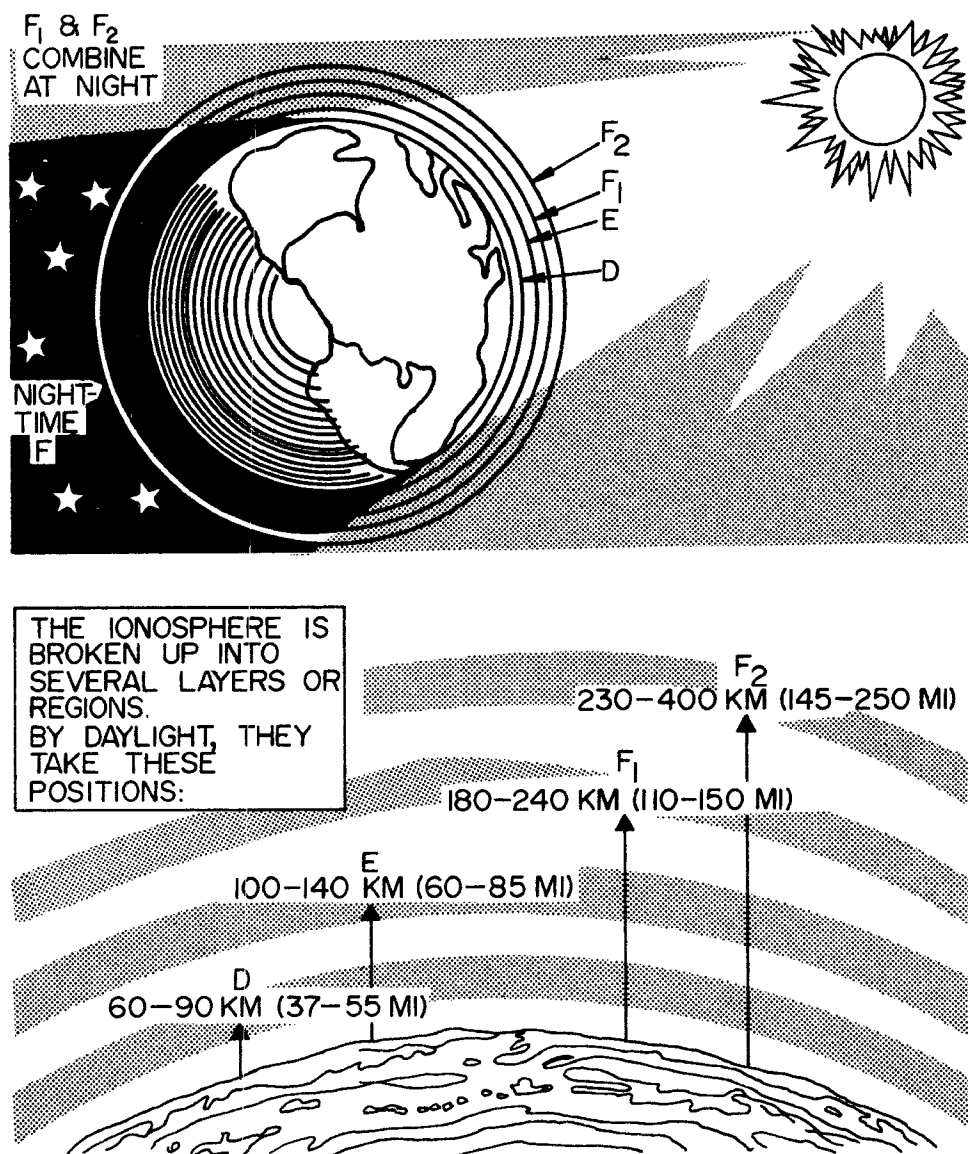


Figure 2-5. Average layer distribution of the ionosphere.

Table 2-3. Characteristics of Ionospheric Regions

Characteristics	D region	E region	F1 layer	F2 layer
Likely origin	(a) Ionization of nitrous oxide with Lyman-alpha radiation (b) Ionization of all gases by soft X-rays	Ionization of all gases by soft X-rays	Ionization of oxygen with fast decrease of recombination coefficient with height	Ionization of oxygen by ultraviolet, X-ray and probably corpuscular radiation
Height, km	60-90 by day; weakens at night	100-140	180-240 by day; disappears by night	230-400
Electron or ion density, per cubic centimeter.	10^2 - 10^3 for electrons; 10^6 - 10^8 for ions	Up to 4.5×10^3 by day; fixed at about 5×10^3 to 10^4 at night	2×10^5 - 4.5×10^5	Max. 2×10^6 by day in winter; max. 2×10^5 by day in summer, 3×10^5 at night in winter

(1,250 mi.). The E layer is not useful for communications at night.

(4) *Sporadic E layer*. Within and somewhat above the region normally occupied by the E layer, a thin, patchy layer of high electron density sometimes occurs. Such a layer is known as a *sporadic E* (E_s) layer. E_s is spotty in geographical extent and in time. The cause of the E_s layer is not definitely known. In the arctic regions, it is frequently associated with auroral activity. The existence of E_s may be detrimental or beneficial to communications. The E_s layer may offer a useful reflection medium, or it may interfere with the use of higher ionospheric layers.

(5) *F region*. As a factor in high frequency communications, the F region is the most important portion of the ionosphere. Most skywave transmissions involve one or more reflections from the F region. During the day there are two separate layers, F1 and F2, in the F region. The

F1 layer extends from about 180 to 240 km (110 to 150 mi) above the earth. It is useful for communication during the daylight hours. It disappears at night. The F2 layer has a range of from about 230 to 400 km (145 to 250 mi) above the earth. The height and electron density of the F2 layer vary with geographical location, solar activity, and local time. It is most useful for communications at night.

2-6. Skywave Propagation

a. Refraction in the Ionosphere.

(1) *Refraction and reflection*. As an oblique radio wave enters a region of increasing electron density, its phase velocity increases in proportion to the density. This increase in phase velocity results in a *refraction* of the wave away from the direction of increasing electron density, as shown in figure 2-6. A ray starting at an oblique angle from A is bent toward the horizontal and, finally,

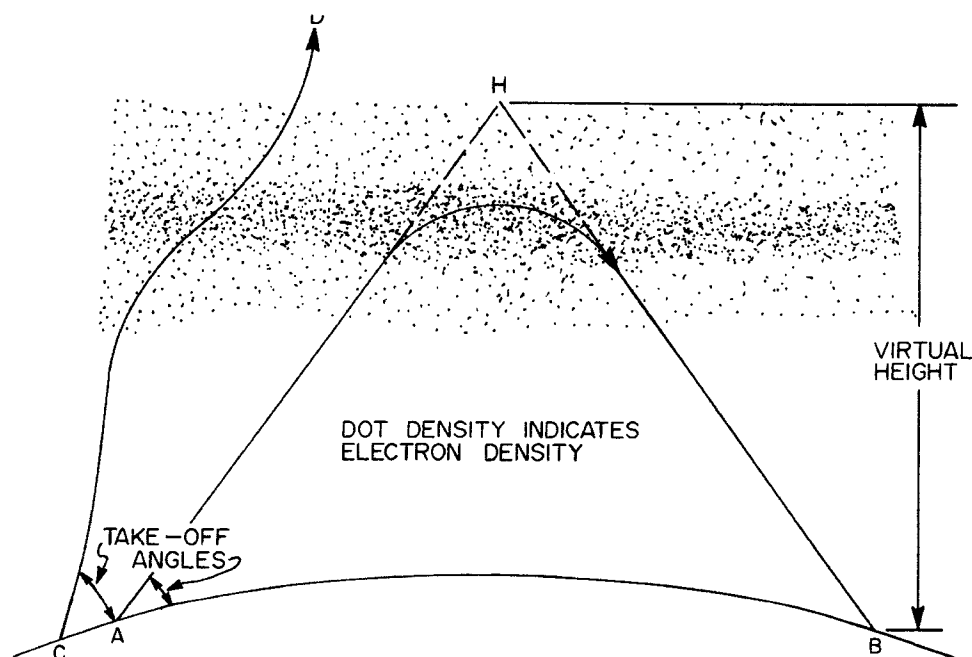


Figure 2-6. Refraction of radio waves by variation in electron density.

downward, ending at B. In many ways, the behavior of the ray is as though it had been *reflected* from the point H. It is commonly said that radio waves are reflected from the ionized layers, when in fact they are refracted by the layers. This manual follows the usual practice of referring to reflected waves when no ambiguity in meaning results.

(2) *Virtual height.* The height of the point H (fig 2-6), from which the wave appears to have been reflected, is referred to as *virtual height* of the layer.

(3) *Critical angle.* Figure 2-6 shows the ray CD which is not bent back to earth because its take-off angle is too large. The largest take-off angle which allows the ray to return to earth is called the *critical angle*. Because the amount of refraction depends upon frequency, the critical angle is frequency dependent and the frequency should always be stated when the value of the critical angle is given.

(4) *Critical frequency.* As the frequency of the transmitted signal at vertical incidence is increased, the wave is returned to earth from successively high layers. As the increase in frequency continues, a frequency will be reached which will penetrate the F2 layer and will not be returned to earth. The highest frequency at which a vertical signal will be returned to earth is known as the *critical frequency*.

(5) *Maximum usable frequency.* Although the critical frequency for any layer represents the highest frequency that will be reflected back from that layer at vertical incidence, it is not the highest frequency that can be reflected from the layer. The highest frequency that can be reflected depends upon the angle of incidence, and hence, for a given layer height, upon the horizontal length of the hop. The maximum frequency that can be reflected back for a given transmission path is called the *maximum usable frequency* (MUF) for that path. The MUF is closely related to the critical frequency and, like the critical frequency, it changes with the time of day, season, solar activity, and location.

(6) *Frequency of optimum operation.* Because of the strong increase of absorption with decreasing frequency it is desirable to use as high a frequency as possible. From a purely physical point of view, a frequency very close to the MUF would be most suitable. This, however, is impracticable since the MUF changes considerably from day to day and, for operational reasons, the working frequency cannot be adapted to these changes. Even the monthly median of the MUF is unsuit-

able since it is reached only during 50 percent of the days. In practice a frequency is chosen which corresponds to a 90 percent probability of reflection. It is considered to be approximately 0.85 times the monthly median value of the MUF. The frequency thus chosen is called the *frequency of optimum operation* (FOT). (The abbreviation FOT is derived from the international term "fréquence optimum de travail.") It is also called *optimum working frequency* and *optimum traffic frequency*.

b. *Ordinary and Extraordinary Waves.* In general a wave propagating perpendicular to the earth's magnetic field will have components of the electric field both parallel and perpendicular to the magnetic field. Such a wave in the ionosphere will split into two refracted waves which travel different paths with different time delays. The process is termed *magnetoionic splitting*. The waves are called the *ordinary* and the *extraordinary wave*. The ordinary and extraordinary waves are shown in figure 2-7. The extraordinary wave suffers greater absorption at high frequencies and has a slightly higher critical frequency than the ordinary wave.

c. *Measurement of Layer Heights and Critical Frequency.* The simplest method of measuring heights of ionospheric layers is by means of a vertical-incidence sounding. A vertical-incidence sounding station is a combination of a transmitter and a receiver placed side-by-side, often using the same antenna. The transmitter sends out pulses of electromagnetic energy, which, during a period of about a minute, sweep in frequency through the range from 0.75 to 25 MHz. The receiver is synchronized with the frequency of the transmitter. The output of the receiver is coupled to a display (usually, a cathode-ray tube) which indicates the time required for the transmitted pulses to be returned as a function of the swept frequency of the transmitter. A permanent record can be obtained by photographing the display. The travel time is used to determine the virtual height of the ionized layers. A plot of the measured height versus frequency is called an *ionogram*. A simplified ionogram is shown in figure 2-8. The figure shows that at the lowest frequency of the sweep, the signal is reflected from the E layer. As frequency is increased, the height of reflection increases, until, at the frequency designated "f_o E" on the figure, the E layer is penetrated (f_o E can be thought of as the critical frequency for the E layer). Similarly, the F1 layer is picked up and then penetrated. Above the critical frequency, no return signal is received. Appendix E

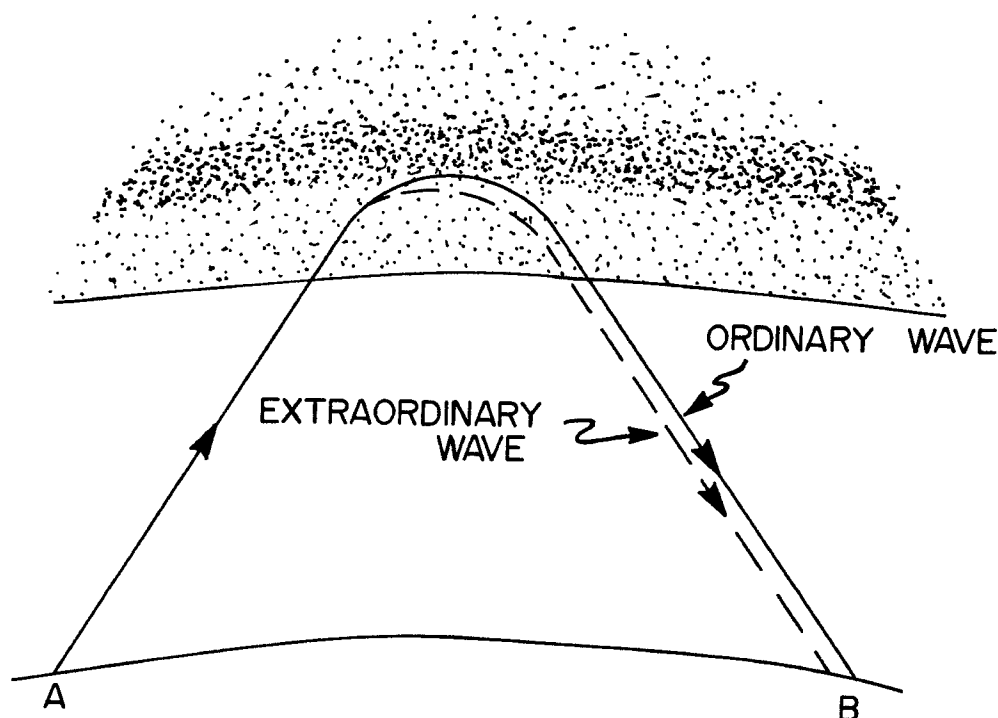


Figure 2-7. Ordinary and extraordinary waves.

provides more complete information about ionospheric sounders and interpretation of ionograms. Both vertical incidence and oblique sounders are discussed in the appendix.

d. Multihop Paths. The ray paths in figures 2-6 and 2-7 have shown a single reflection from the ionosphere. Actually, the radio waves are also reflected by the earth's surface and may be again reflected by an ionospheric layer. Several such reflections can take place. Communication paths involving multiple reflections are termed *multihop*

paths. Figure 2-9 shows the common reflection modes. The designation under each diagram is descriptive of the mode. For example, "3E" indicates three hops with all ionospheric reflections from the E layer. The designations "N" and "M" are used because their diagrams suggest those letters. The three diagrams on the lowest row represent complex paths. Long circuits require the use of multihop paths. However, each reflection increases the signal loss and increases the possibility of multipath rays. Where possible, use a

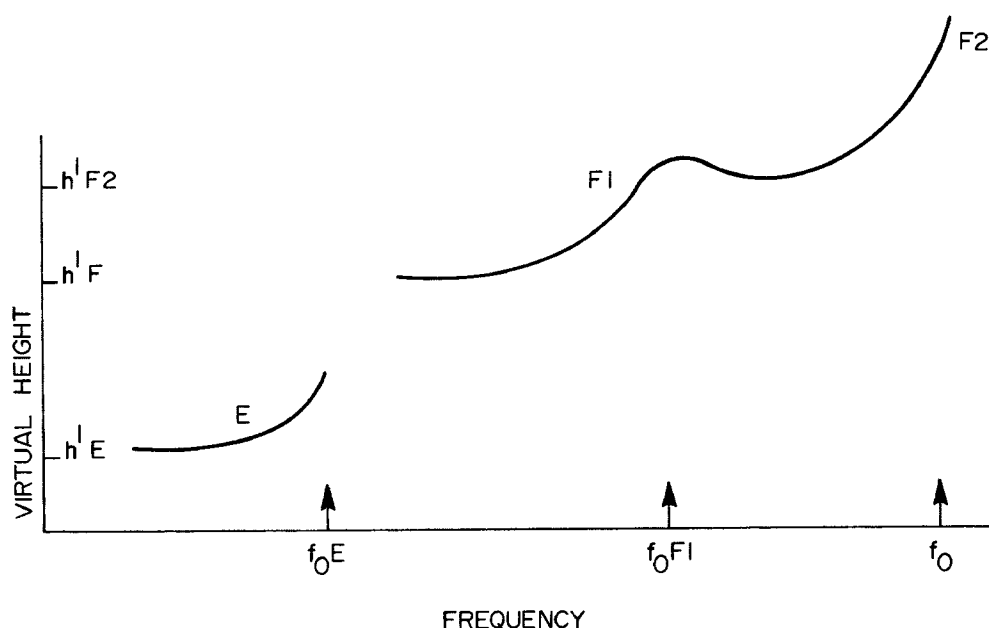


Figure 2-8. Simplified vertical incidence ionogram.

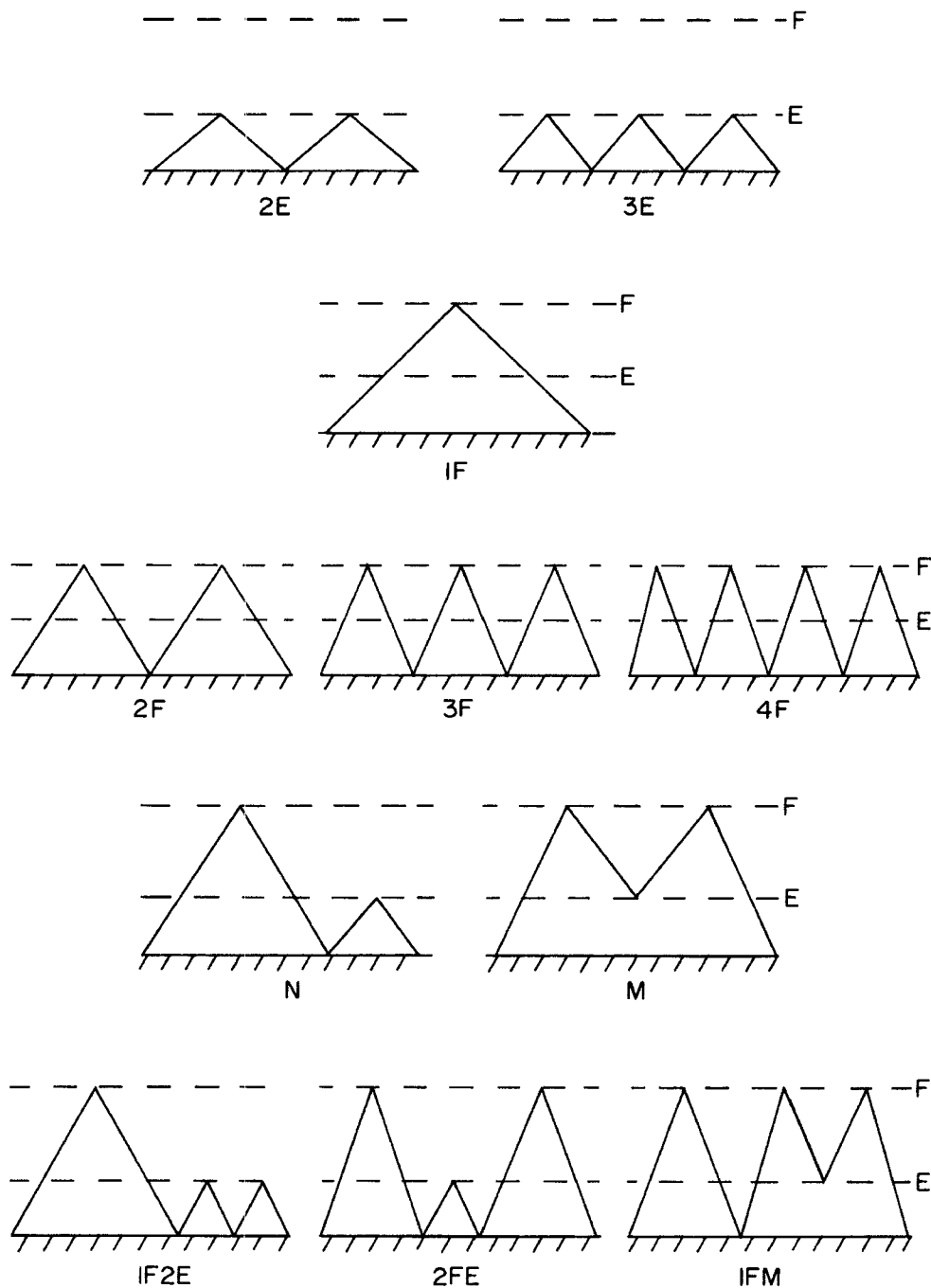


Figure 2-9. Common ionospheric propagation modes.

higher frequency for communications, and thus avoid a large number of hops.

2-7. Multipath and Differential Delay

Sometimes a signal can follow more than one path between transmitter and receiver. Because the paths are usually not the same length, the transit times over the paths are not identical. A sample ionogram in appendix E shows a multipath skywave situation. When the groundwave is being used as the principal communications mode, under some conditions the skywave will also be

received. Differential delay is most troublesome when the amplitudes of signals arriving by differing paths are almost equal. The differential delays on practical circuits do not affect voice transmissions appreciably but they are a limiting factor on speed of digital traffic. Because of this limitation, it is common to multiplex numerous low-speed teletypewriter channels, with or without voice channels, on a single high frequency circuit. It is possible to discriminate against undesired multipath by proper frequency and antenna selection. In general, a frequency near the MUF and

antennas favoring low elevation angles should be used.

2-8. Fading

a. *General.* Because of fluctuations in ionospheric conditions, the received intensity of the skywave is not constant, but varies with time. The term *fading* is applied to these variations, some of which occur during a space of seconds, or even fractions of a second. In general, fading is more sudden on high than on low frequencies. A type of fading known as *selective fading* also can cause distortion in radio-telephone signals. In such cases, the fading affects certain frequencies more than others and, therefore, may affect the side bands and the carrier wave differently. Selective fading has less effect on single sideband operation than on double sideband. Fading falls into four principal classes: interference fading, polarization fading, absorption fading, and skip fading. Most of the rapid fading in the input to a receiver is a combination of the first two types; the other two are responsible for slower changes.

b. Rapid Fading.

(1) *Interference fading.* Interference fading is caused by phase interference of waves arriving at the receiver over slightly different paths. The relative phases of the waves arriving over the different paths vary with time as a result of ionospheric changes, causing alternate reinforcement and cancellation of the field intensity. If the waves are of almost equal amplitude, severe fading can result. Figure 2-10 represents a case of severe, rapid fading at 18 MHz. The vertical scale, receiver automatic gain control voltage, is a function of signal strength but without a calibration curve cannot be equated to it.

(2) *Polarization fading.* Variation in the

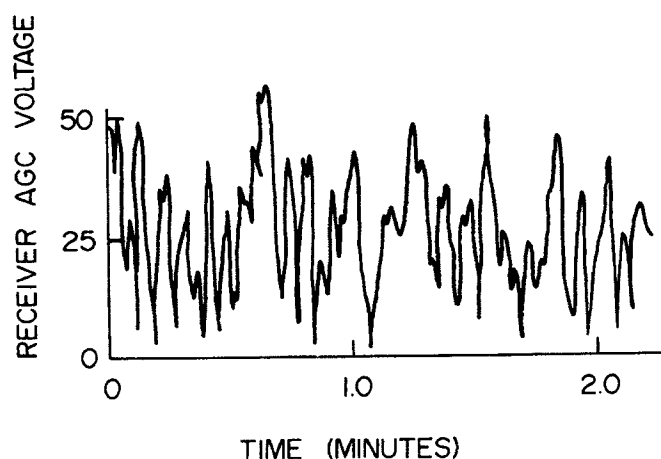


Figure 2-10. Example of rapid fading at 18 MHz.

signal intensity at the receiving antenna can result from changes in the state of polarization of the incoming wave relative to the orientation of the antenna. This variation is called *polarization fading*. In general, the incoming skywave's state of polarization is changing constantly, mainly because its two oppositely polarized components, the ordinary and the extraordinary wave, each have varying amplitudes and phases. As a result, values change randomly and constantly in the amplitude and orientation of the electric field with respect to the receiving antenna. The state of polarization of skywaves varies more rapidly the higher the frequency, which accounts in part for the rapid fading on the higher frequencies.

(3) *Terrain screening.* Apparent fading, especially that associated with mobile tactical operations, may be due to groundwaves or sky-wave screening. Examples of the natural form of radio path screening are dense brush and trees with heavy, leafy foliage which absorbs radio waves. Mobile tactical operation lends itself to natural screening when the unit enters a valley. Mobile units are also subject to screening by such manmade obstructions as tunnels, underpasses, or steel bridges. Pole-mounted wire lines, such as telephone and power, absorb power from radiating antennas located in their vicinity. Buildings located between stations, particularly steel and reinforced concrete structures, hinder transmission and reception.

c. Long Term Fading.

(1) *Absorption fading.* Absorption fading is caused by variations in the amount of energy lost from the wave because of absorption in the ionosphere. In general, the ionospheric absorption changes slowly.

(2) *Skip fading.* Skip fading is observed at places near the limit of the skip distance, and is caused by changes in refraction. Near sunrise and sunset, when the ionization density of the ionosphere is changing, it may happen that the MUF for a given transmission path fluctuates about the actual operating frequency. When the skip distance moves past the receiving station, the received intensity abruptly drops by a factor of 100 or more, and just as abruptly increases when the skip distance moves in again. This may take place many times before steady conditions for transmission are established.

d. Counteracting Fading by System Design.

(1) *General.* A very elementary high frequency radio system can be made up of a low-powered transmitter, a simple receiver, and half-wave horizontal antennas. Such a system will

function very well under favorable propagation conditions. The principal purpose of system design is to increase the percent of time that satisfactory communications prevail. Only the principles of combating fading by system design are discussed in this paragraph. Application details are discussed in chapter 5.

(2) *Diversity*. *Diversity* means receiving the signal by two or more methods and either combining the received signals or choosing the best one. The use of diversity is covered in paragraph 5-8b.

(3) *Antenna selection*. During system design the optimum elevation angles of departure and arrival for various seasons and times of day will be determined. These angles will be selected for the minimum number of hops and will be lower than for many of the multipath interfering signals. Selecting an antenna which favors the desired angle will tend to minimize fading caused by interference.

e. Fading Control by Operator.

(1) The adverse effects of ionospheric disturbances can often be minimized by a change in operating frequency. Appendix G describes services available to the operator from the Air Force Global Weather Central and from the Space Environment Services Center, and also provides guidance to the operator regarding effective use of these services. On the basis of these services the operator may be able to inform the responsible person that a frequency change is required.

(2) If the operator has oblique ionospheric soundings available, frequencies having good signal strength with a minimum of multipath can be chosen. The operator may, on the basis of the ionogram, select an operating frequency much higher than the forecast FOT. The use of ionospheric sounding in choosing an operating frequency is discussed in appendix E.

2-9. Ionospheric Variables Related to Communications

a. Long Term Variables.

(1) *Sunspot number*. Since radiation from the sun is the principal cause of ionization in the earth's atmosphere, a correlation between solar activity and conditions in the ionosphere would be expected. Sunspots are disturbances in the extremely hot and highly ionized atmosphere of the sun. An index of sunspot activity, called the *sunspot number*, has been developed by astronomers. Records of sunspot numbers extend back about 200 years. Sunspot numbers have a well documented 11-year cycle which is also found in ionospheric conditions. Figure 2-11 shows how noon and midnight critical frequencies vary in phase with variations in the sunspot number. The curves in the figure are based on smoothed data. Knowledge of phase of the sunspot cycle is useful for predicting average sunspot numbers a few months in the future. Attempts at forecasting years in advance have not been successful because of variations in the 11-year cycle.

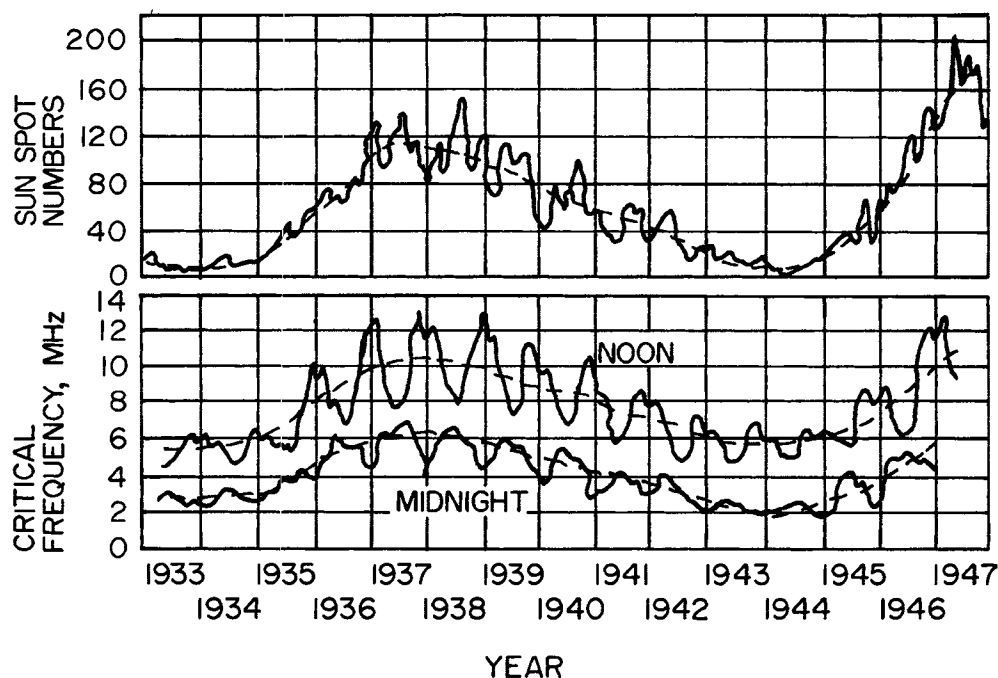


Figure 2-11. Variations in sunspot number and in noon and midnight critical frequencies (smoothed data).

(2) *Ottawa 10.7-cm solar flux.* Another basic index of solar activity is the 10.7-cm solar flux. This flux is routinely measured by the Algonquin Radio Observatory, Ottawa, Canada. Monthly mean values of the 10.7-cm flux and the E-layer critical frequency correlate well. The 10.7-cm flux is gaining favor over sunspot number because it

is a measured parameter whereas the sunspot number is determined subjectively. Figure 2-12 is a plot of the 10.7-cm flux which shows a 27-day cycle which persisted through two cycles then decayed. (Twenty-seven days is the approximate period of rotation of the sun at its midlatitudes.)

(3) *Sunspot number of solar flux conver-*

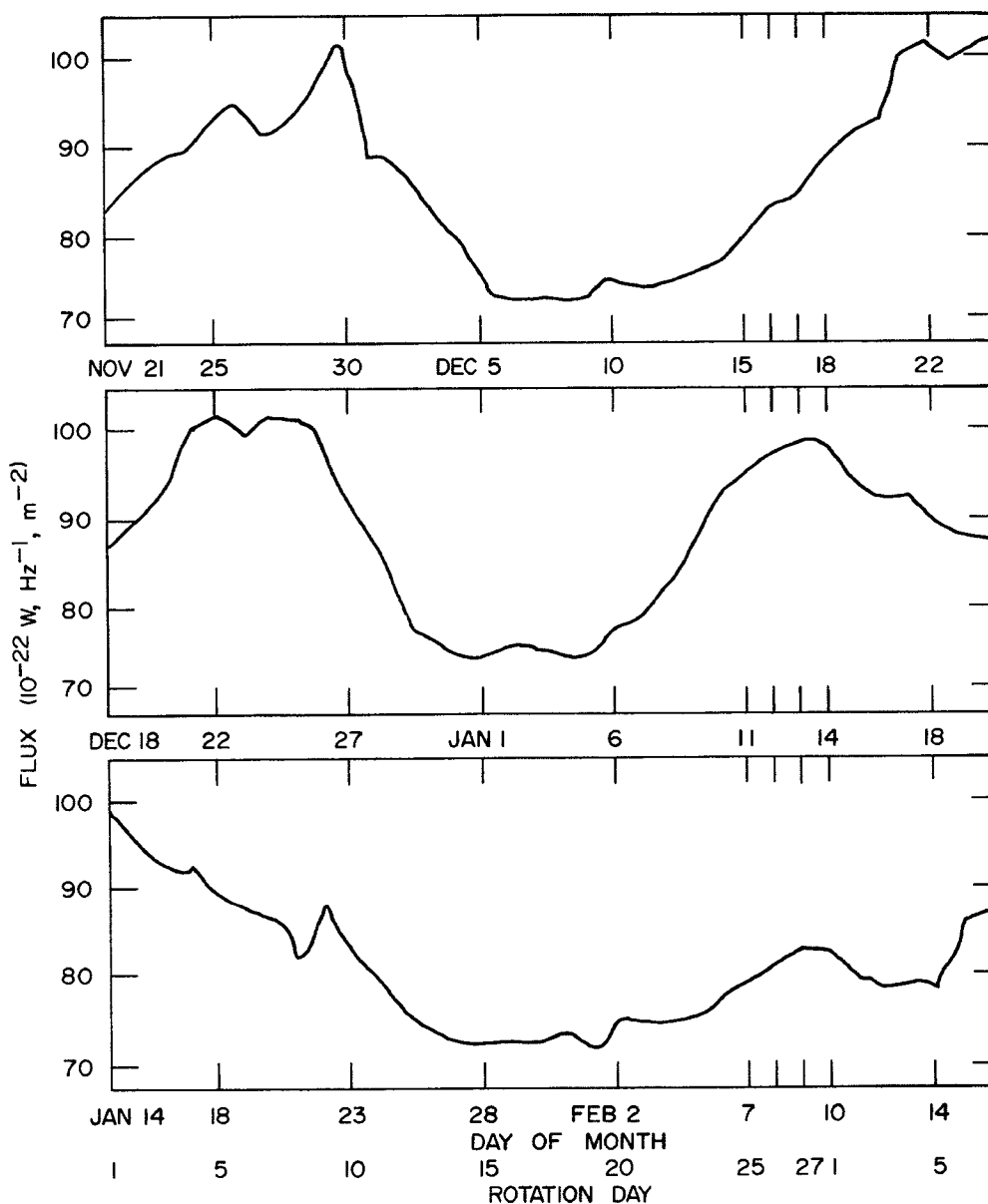


Figure 2-12. Solar 10.7-cm flux over three solar rotations.

sion. There is good but not exact correlation between the 10.7-cm solar flux and sunspot number. A best fit showing this correlation is presented in figure 2-13.

b. Short Term Variables.

(1) *Daily variation.* One of the outstanding characteristics of the ionosphere is the change in structure which it undergoes daily. The D layer

is ionized during the day but weakens and is unimportant at night. The E layer is useful for short range communications during the day, but not at night. The single F layer which exists at night becomes the F1 and F2 layers during the day. These regular variations dictate the need for at least two frequencies for all around-the-clock sky-wave links—a minimum of a *day frequency* and a *night frequency*.

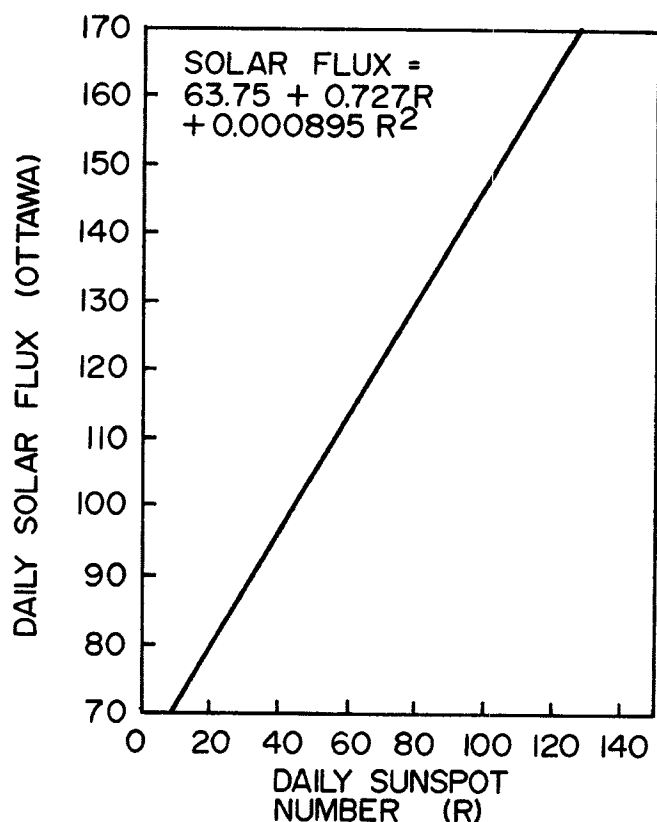


Figure 2-13. Relationship between 10.7-cm solar flux and sunspot number.

(2) *Irregular variations.* In addition to the regular changes in the ionosphere, there are ir-

regular variations related to disturbances on the sun. The effect of these disturbances on communications can range from a decrease in circuit quality on a few circuits for a few hours to an almost complete blackout of all hf skywave circuits for a day or two. In general, high latitudes are affected more than low latitudes. Ionospheric disturbances and their effect on hf communications are discussed in appendix G.

2-10. Noise

a. Potential Noise Sources. The desired signal is always in conflict with both internal noise in the receiver and external radio noise from various sources. Thermal agitation of electrons creates noise in all components. At the very minimum, the noise at the input of a receiver is the thermal noise generated in the antenna. Thermal noise is given by the expression—

$N = kTB$ watts, where

k is Boltzman's constant

T is the absolute temperature (frequently assumed as about 290 K for antennas)

B is the bandwidth of the receiver in hertz.

Manmade, atmospheric, and galactic noise are added to this thermal noise. Because the thermal noise is always present, noise levels are frequently expressed as dB above kTB . Figure 2-14 indicates the spectrum of potential sources of noise external

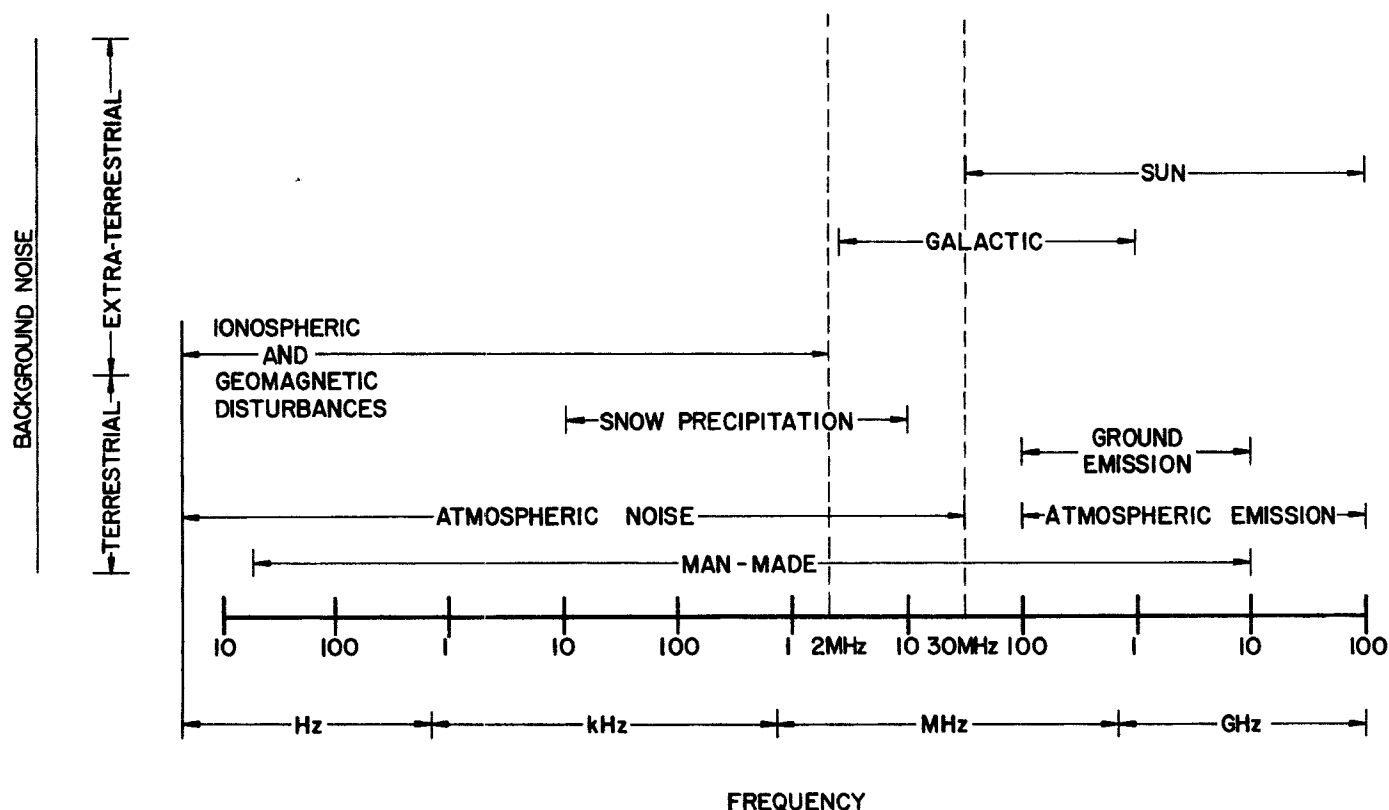


Figure 2-14. Potential noise sources.

to the receiver. Blowing snow and sand result in charged particles striking antennas. This can be an occasional source of noise. Other noise sources are relatively unimportant. The desired signal can also be in conflict with other types of interference resulting from emanations from other C-E systems in the area. See paragraph 5-9 for a discussion of electromagnetic compatibility (EMC) and potential sources of interference and noise.

b. Relation of Antenna to Received Noise. Antennas are responsive principally to waves having the same polarization as the antenna. For maximum noise reduction, the antenna should have a polarization 90° from that of the noise. For other reasons, this may not always be practical. Better site selection may aid in minimizing noise.

c. Atmospheric Noise. Atmospheric noise originates in lightning. A lightning flash starts with a predischARGE, an initial stroke during which a pilot streamer slowly moves from the cloud toward the ground, followed by a leader with a current of about 200 amperes, moving in a series of steps of 25 to 30 microseconds duration. The main discharge is the return stroke with a peak current of from 10 to 100 kiloamperes, progressing from the earth to the cloud through the path blazed by the stepped leader. When atmospheric noise is received from a distance, its spectrum at the receiving site is influenced by the propagation path. The spectrum of median values of distant atmospheric noise at Washington, D.C., for summer and winter is shown in figure 2-15. Noise levels in figure 2-15 are in dB's above kTB .

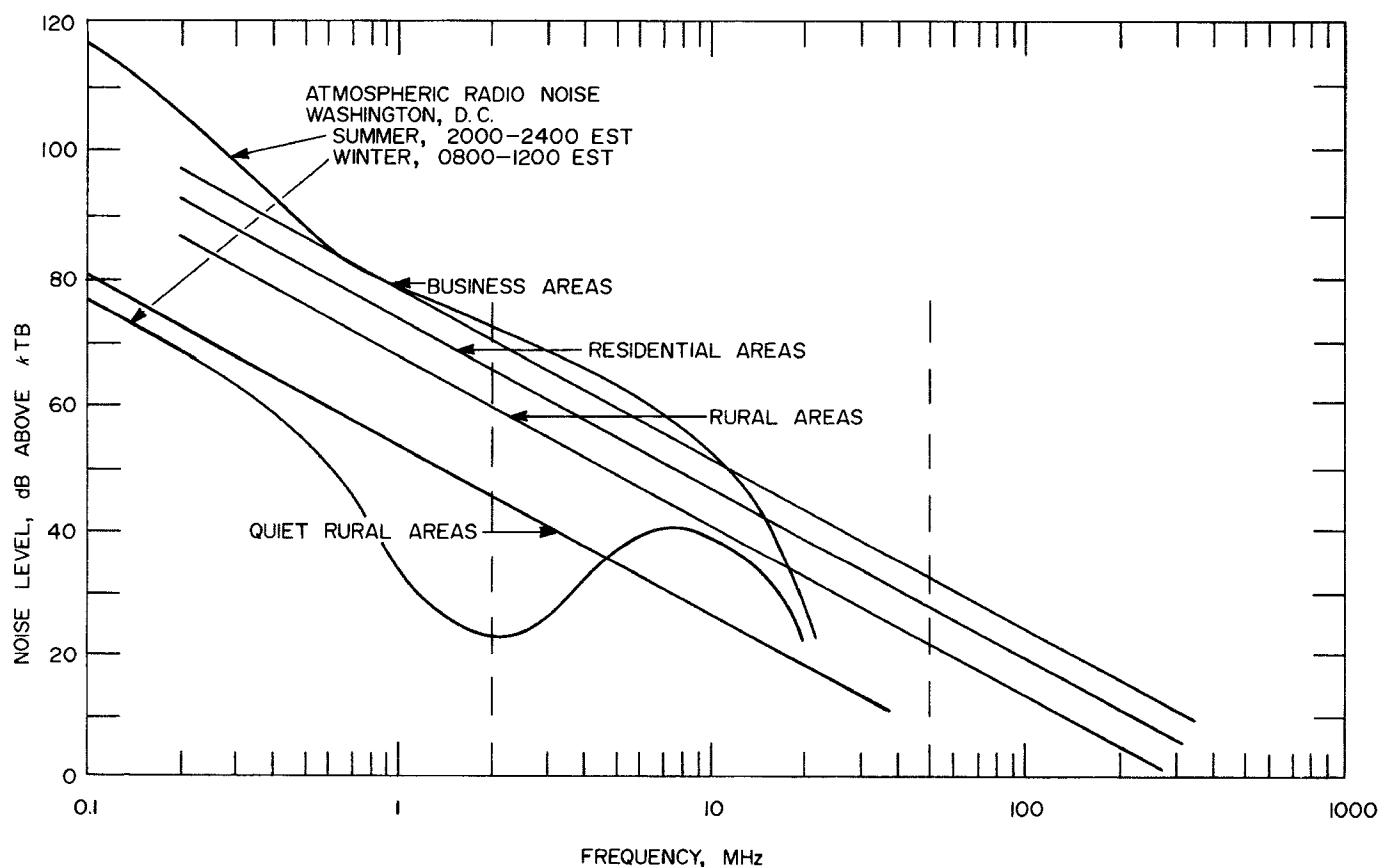


Figure 2-15. Comparison of atmospheric and manmade noise.

d. Manmade Noise.

(1) A spectrum of estimated median values of manmade noise expected at typical rural locations is included in figure 2-15. The amplitudes are different, but the spectrum is similar for the different areas—business, residential, and rural. Figure 2-16 indicates the power spectrum and amplitude of ac powerline noise from measurements taken while moving parallel to a 115-kV

line in rural Wyoming, both under the line and $\frac{1}{4}$ mile from the line. At a distance of $\frac{1}{4}$ mile, the powerline noise is of the same order of magnitude as the usual manmade background noise. Manmade radio noise can be generated by switching transients in relay operation, vibrating contacts, ignition transients, commutator noise on rotating equipment, pulsed currents in radars, X-ray and diathermy equipment, and corona and

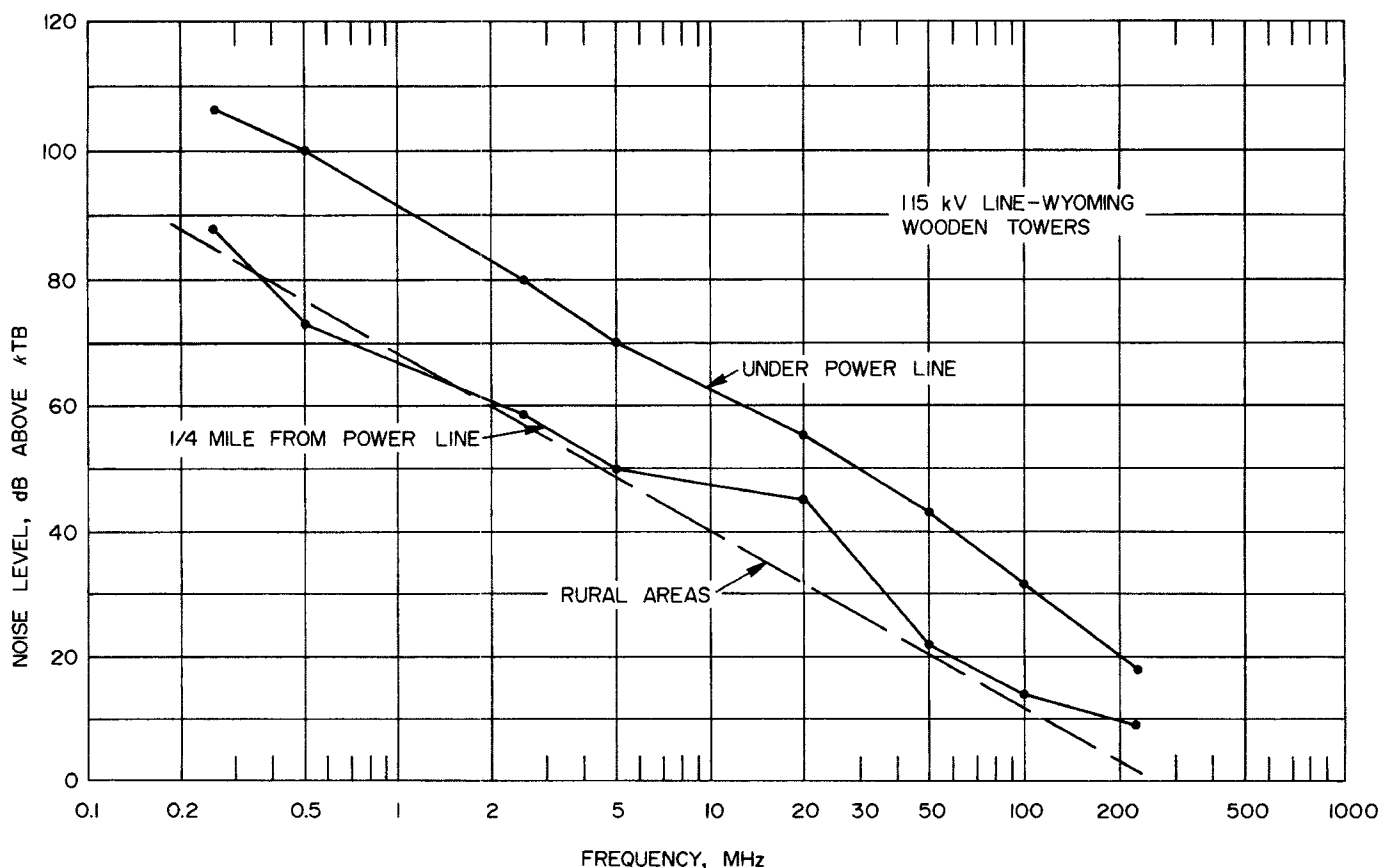


Figure 2-16. Comparison of powerline and normal rural area noise.

other electrostatic discharges. Much manmade noise is periodic.

(2) A simple technique for locating noise sources is to carry a portable radio around the area surrounding the receiving site. A receiver which tunes over the frequency range of interest is best, but a set tuned to the upper end of the broadcast band is acceptable. The receiver should be tuned to a frequency at which no radio signal is received and the volume should be set so that the noise is audible. The antennas on most portable radios have at least some directivity. By changing the azimuth of the set as it is being moved, moving in the direction in which the signal becomes stronger, and tuning higher in frequency as the noise becomes stronger, it is frequently possible to locate noise sources. Mobile sources and sources of intermittent interference may be difficult to find.

(3) The operator at a site develops a "feel" for the usual noise reading on the "S" meter for a particular frequency band, antenna, and receiver combination. An increase in that reading, except from thunderstorm, precipitation, or blowing-sand static, is probably caused by a new noise source. The new noise source should be located at once.

(4) The cure for interfering noise is sometimes simple. Generator brushes may need replacing. Noise from an ignition system may be eliminated by replacing the plugs with the resistive type and installing resistive ignition wire. The distributor cap may require replacement. For more difficult problems, the procedure is to filter and shield the offending item. Dimmer controls on lights and speed controls on motors generate noise spikes by their switching action. A capacitor across the line at the control may be sufficient, but an L-C filter is more effective. If filtering does not solve the problem in a particular case of noisy equipment, enclose the noise source in a ferrous shield, if possible, and ground the shield. The shield need not be solid—at hf, hardware cloth or similar material is effective.

(5) Excessive noise from a powerline caused by leaky insulators, transformers, or other hardware can be corrected by proper maintenance. Proper maintenance will also minimize corona discharge from a powerline. Suspected powerline deficiencies should be reported to the cognizant agency—the commercial power company or the base civil engineering organization.

(6) A decrease of 3 dB in overall noise is an increase of 3 dB in the signal-to-noise ratio. This

has the same effect as doubling the power of the transmitter. Hence, it is obvious that noise reduction yields worthwhile benefits. However, noise reduction can be effective only at a naturally noise-free site. Little can be done about the noise from multiple sources in an urban area.

e. Galactic Noise. The galactic noise spectrum is shown in figure 2-14. Above 20 MHz its amplitude may be greater than atmospheric noise or quiet, rural-area manmade noise. In most rural areas and in residential areas, galactic noise is, in general, less than manmade noise.

f. Solar Noise. Solar noise is usually insignifi-

cant as an hf noise source. This is because the hf region is far from the peak of solar radiation and because much of the hf radiation is reflected by the ionosphere.

g. Atmospheric Emission Noise. Atmospheric noise does not become significant until the operating frequency is about 3 GHz or higher, and is not severe below about 10 GHz. This source can be completely neglected in the hf band.

h. Ground Emission Noise. Ground emission noise is relatively constant, but it has a very low value. It is not important in the hf band.

CHAPTER 3

TRANSMISSION LINE FUNDAMENTALS

3-1. General

It is sometimes desirable or necessary to situate a radio transmitter or receiver some distance away from the associated antenna. When this is done, a transmission line is used as the connecting link. The purpose of a transmission line is to transfer radio frequency energy with minimum loss. Also, an ideal transmission line should neither radiate to nor receive energy from surrounding space.

3-2. Types of Transmission Lines

a. Two Parallel Conductors. Two parallel conductors are frequently used as a transmission line (fig. 3-1). The type of line shown at the left of figure 3-1, known as *twin-lead*, is commonly used for home television and frequency-modulated radio lead-in. It is usually about half an inch wide. The *open wire* line shown at the right of figure 3-1 may be constructed in almost any desired size, but usually widths are 6 inches or less. Solid, cellular, or air dielectric can be used between conductors. Air has the least loss. A transmission line made up of two parallel conductors is symmetrical and is called a *balanced* line.

b. Coaxial. Most transmission lines currently in use are of the coaxial type. A coaxial transmission line consists of an inner conductor, a dielectric, and a concentric outer conductor (fig 3-2). Air or other gas dielectric, foam dielectric, or solid dielectric may be used. A coaxial line may be rigid, semiflexible, or flexible. Most coaxial lines are less than one-half inch in diameter. However, much larger coaxial lines are occasionally used. Coaxial transmission lines are usually operated with the outer conductor at ground potential. Because of the lack of symmetry, coaxial transmission lines are called *unbalanced*.

3-3. Properties of Transmission Lines

a. Characteristic Impedance. A parallel-conductor line has distributed series resistance and inductance, and also capacitance and conductance across the line. Figure 3-3, showing a series of closely spaced resistors, inductances, and capacitors, approximates a real transmission line. When an alternating voltage is placed upon such a line, each resistance and inductance limits the rate at which the following capacitor is charged. The effect of the chain is to establish a definite relationship between the voltage applied to the line and the current. The relationship is known as

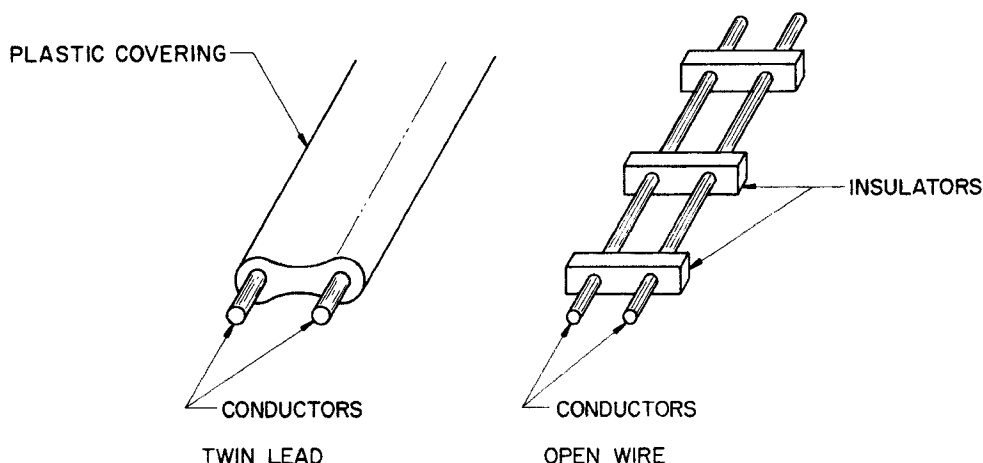


Figure 3-1. Parallel-conductor transmission lines.

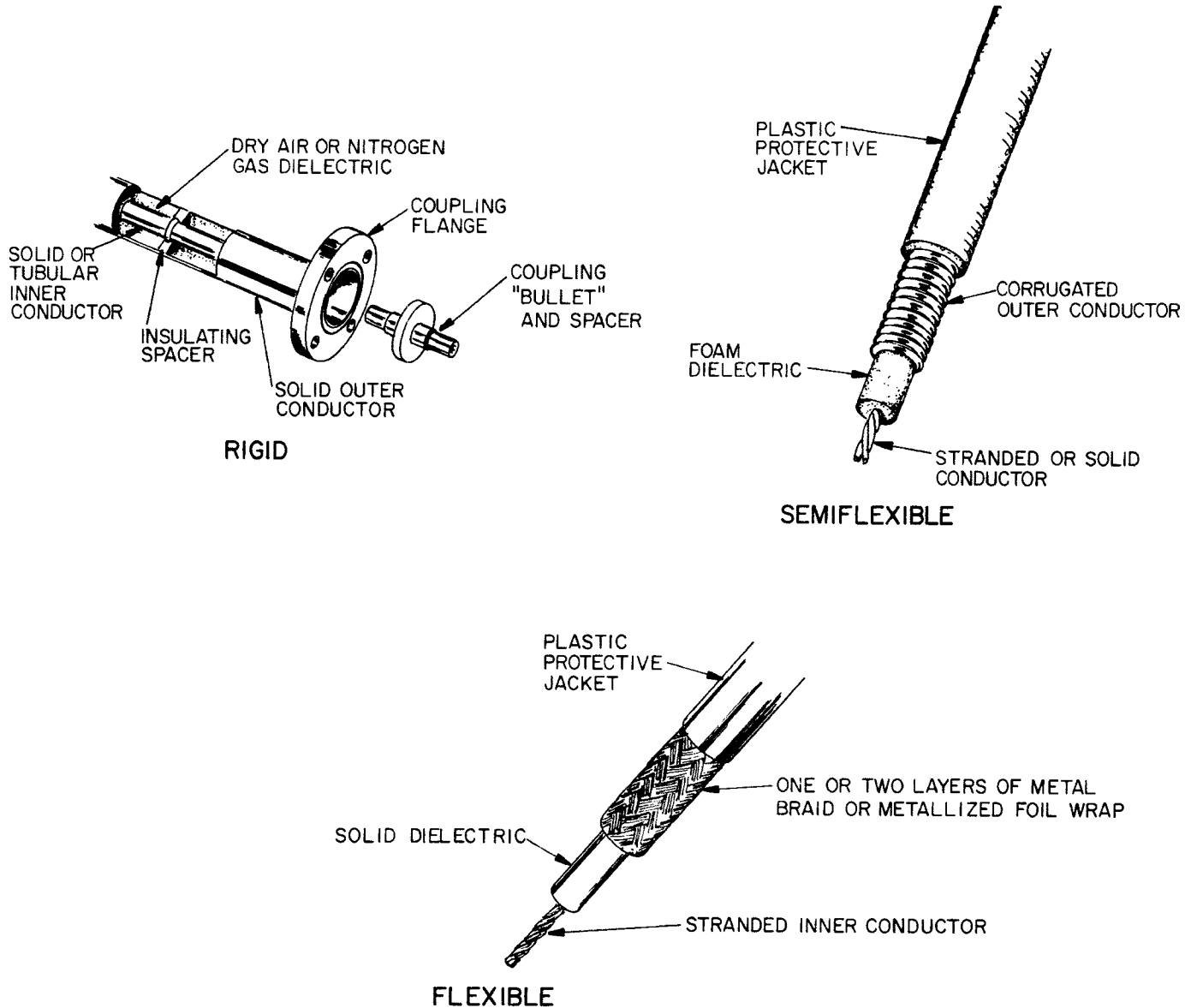


Figure 3-2. Coaxial transmission lines.

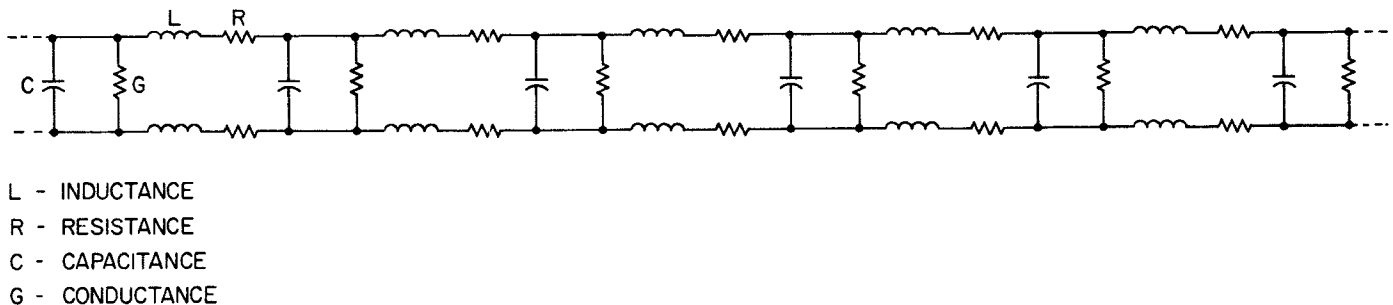


Figure 3-3. Simplified concept of a transmission line.

characteristic impedance, Z_0 . For a given line, this value is a constant for a line of infinite length. It is also this same constant for a line terminated in its characteristic impedance, regardless of line length. The values of distributed resistance, inductance, and capacitance, and hence the imped-

ance, depend upon such physical factors as conductor size, conductor spacing, and the dielectric constant. Parallel lines with large, closely spaced conductors have a relatively low characteristic impedance. Lines with thin, widely spaced conductors have a relatively high impedance. The

practical range for parallel-conductor lines is 200 to 800 ohms. For coaxial lines, the important dimension is the ratio of the inner diameter of the outer conductor to the outer diameter of the inner conductor. The larger this ratio, the higher the impedance. Coaxial lines usually range from 50 to 150 ohms. Figures 3-4 and 3-5 provide graphs and formulas for determining characteristic impedances from physical factors.

b. Attenuation. Attenuation is the loss which occurs in the signal as it progresses along a transmission line. Since low loss is always desirable, the attenuation characteristic is usually the most important factor to consider in choosing a transmission line. The loss per unit distance is called the attenuation constant, α . In the United States, manufacturers usually state the attenuation as loss in decibels per 100 feet. (See appendix B for an explanation of decibels.) For any given transmission line, the attenuation constant increases with frequency. The total attenuation of a line at any frequency is the attenuation constant at that

frequency multiplied by the length of the line. The length of the line must be expressed in the same unit of length as that used in the attenuation constant. Figure 3-6 shows the attenuation of some commonly used coaxial transmission lines. In general, the larger the diameter of coaxial lines, the less the attenuation. Although not included in the figure, the loss of coaxial lines having air dielectric is very low. Likewise, the loss of parallel conductor lines using air dielectric and high quality insulators is very low.

c. Velocity of Propagation. In some applications it is necessary to know the wavelength of a signal in a transmission line. Wavelength depends upon frequency and velocity of propagation. The ratio of the velocity of propagation of a given transmission line to that of free space is the *velocity of propagation factor*. It is always less than one. Typical values range from about 0.7 for solid dielectric to above 0.9 for very low-loss lines. The velocity of propagation factor is equal to $1/\sqrt{\epsilon}$, where ϵ is the dielectric constant. However, the

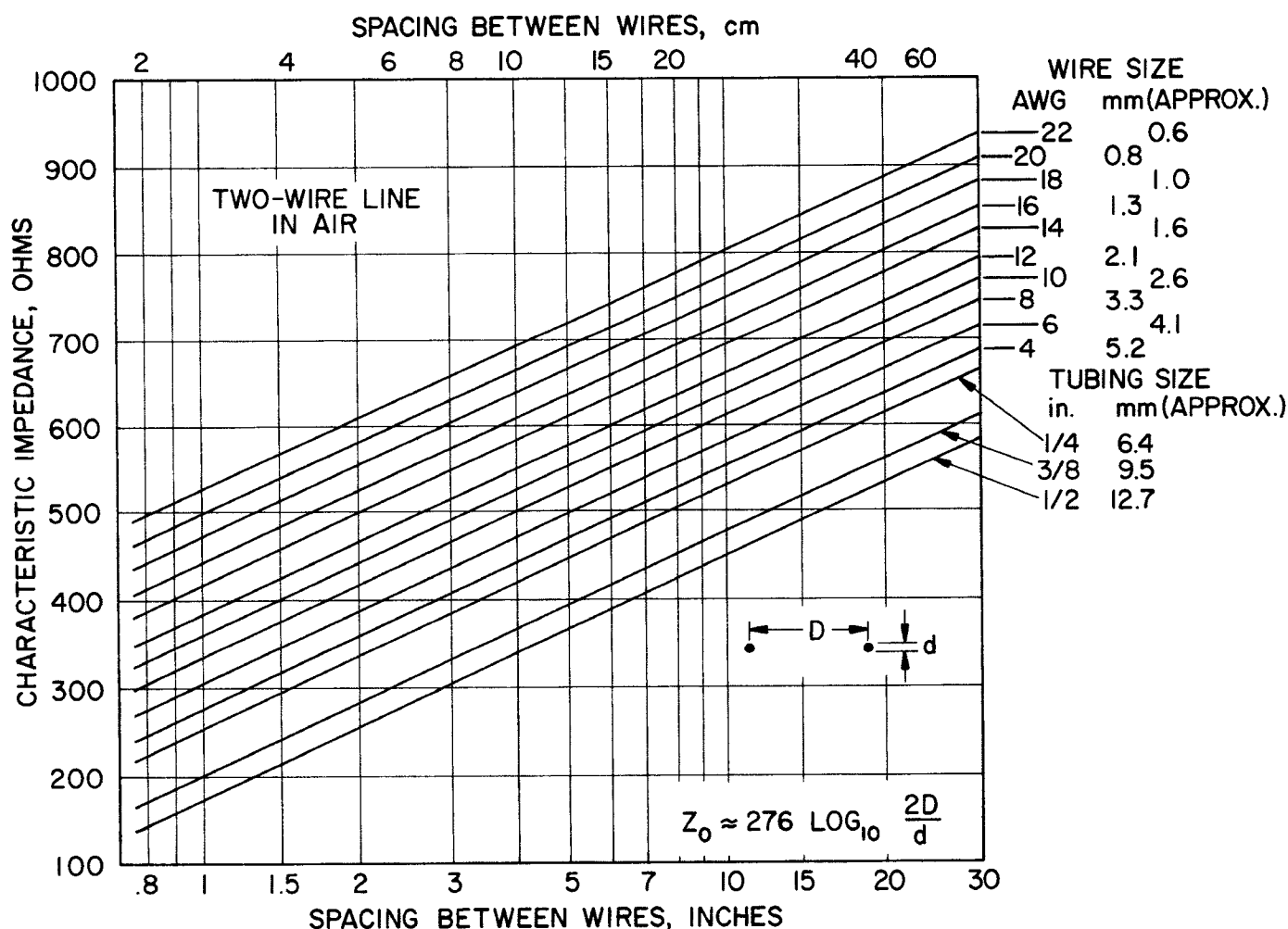


Figure 3-4. Characteristic impedance of parallel-conductor transmission line.

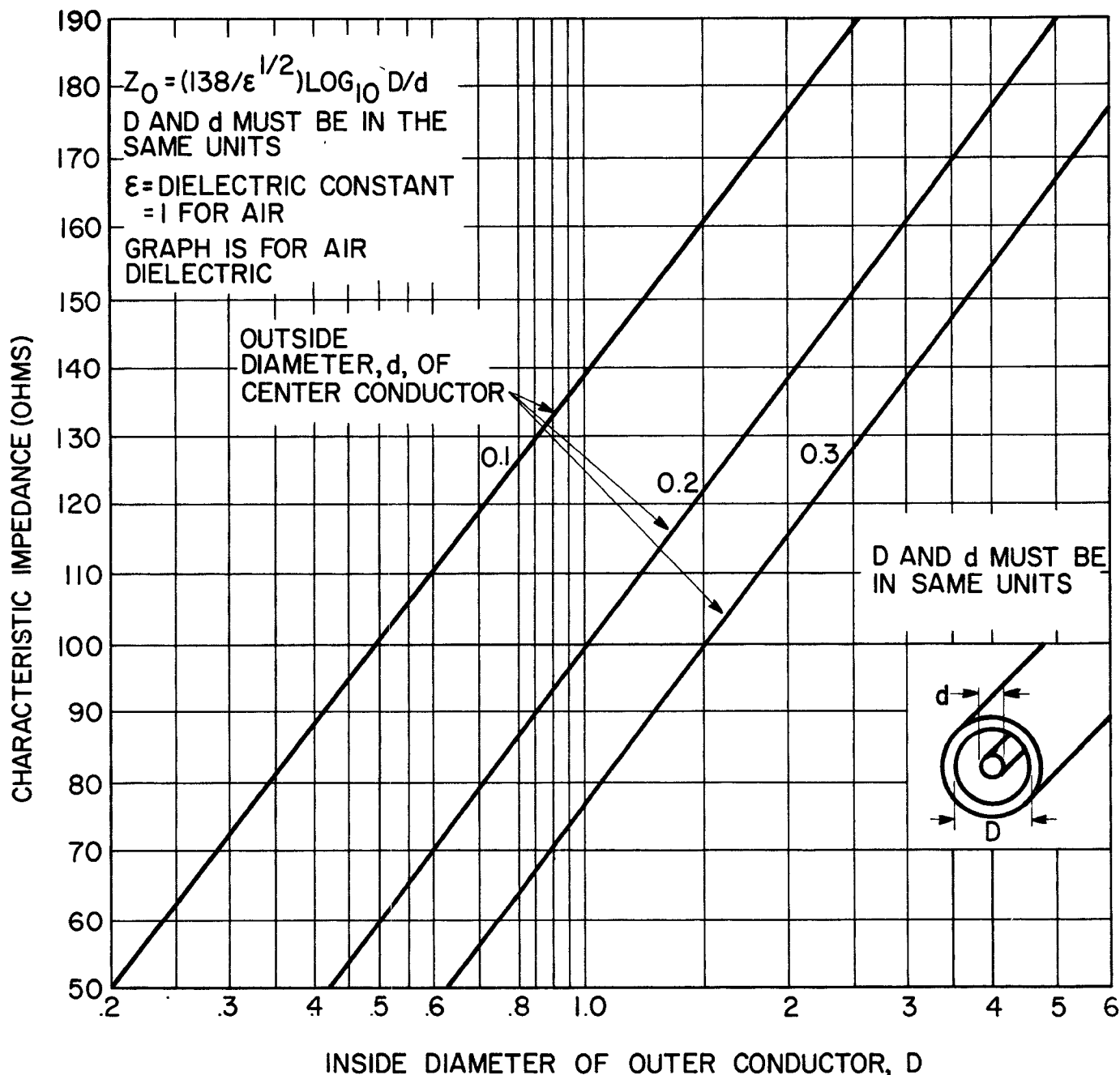


Figure 3-5. Characteristic impedance of coaxial transmission line.

factor is usually given directly in a manufacturer's descriptive information. The wavelength of a signal in a transmission line is its wavelength in free space multiplied by the velocity of propagation factor.

3-4. Resonant and Nonresonant Transmission Lines

a. *Voltage Standing Wave Ratio.* The load on a transmission line may be either the transmitting antenna or the receiver. If the load presents a

proper impedance match to the transmission line, all the energy flowing in the line is absorbed and the voltage-to-current ratio at any point on the line is equal to the characteristic impedance, Z_0 . In general, the load does not exactly match the characteristic impedance of the line and only a portion of the power flowing in the line is accepted by the load. The remainder is reflected into the line, resulting in waves traveling in both directions. The voltage at any point along the line is the sum of the incident voltage and the reflected voltage, and standing waves exist along the line.

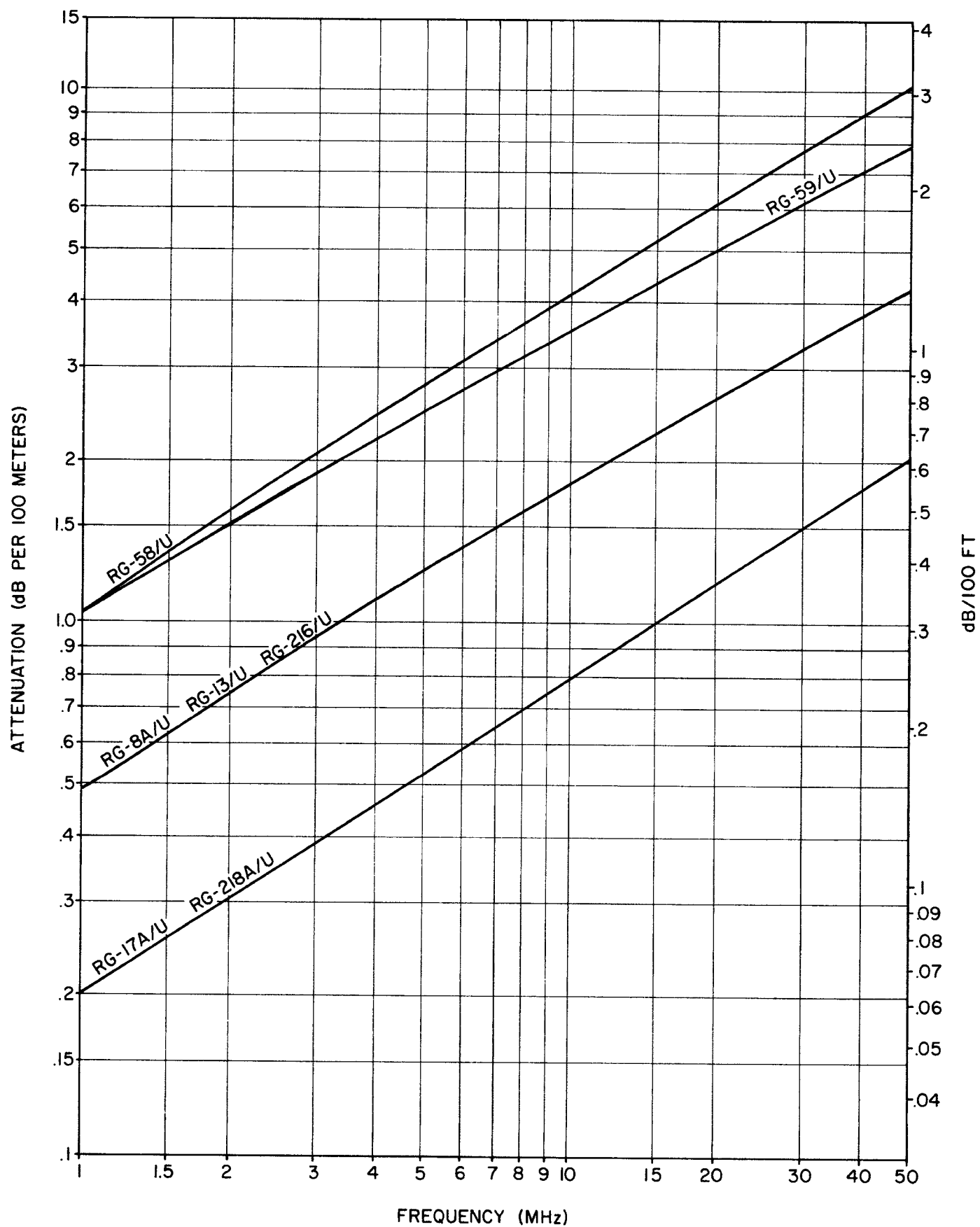


Figure 3-6. Attenuation of selected coaxial transmission lines.

Figure 3-7 is a graph of the voltage along a transmission line, showing the standing waves. The ratio of a maximum voltage ($E_{\max}$) to an adjacent minimum voltage ($E_{\min}$) is the *voltage standing wave ratio* (VSWR). It is sometimes called merely the *standing wave ratio* (SWR). The points of maximum voltage are called *voltage loops*; the points of minimum voltage are called *voltage nodes*. Standing waves of current accompany the standing waves of voltage. Points of current maximums (current loops) occur at voltage nodes and points of current minimums (current nodes) occur at voltage loops.

b. Converting Incident and Reflected Power into VSWR. VSWR may be measured directly. However, it may also be computed from other parameters. Many transmitters have a built-in meter which reads incident and reflected power. Figure 3-8 provides a convenient nomograph for converting these readings into VSWR. If a calculator with a square-root function is available, VSWR can be calculated directly from the formula:

$$\text{VSWR} = \frac{1 + \sqrt{P_r/P_i}}{1 - \sqrt{P_r/P_i}}$$

P_i is incident power and P_r is reflected power. Figure 3-8 illustrates the example of 180 W incident power and 2.7 W reflected power. The VSWR can be read as approximately 1.27, which

can be rounded off to 1.3. The same conditions are used as an example of applying the formula.

$$\begin{aligned} \text{VSWR} &= \frac{1 + \sqrt{2.7/180}}{1 - \sqrt{2.7/180}} \\ &= \frac{1 + \sqrt{0.015}}{1 - \sqrt{0.015}} \\ &= \frac{1 + 0.1225}{1 - 0.1225} \\ &= 1.279 \approx 1.3 \end{aligned}$$

c. Nonresonant Lines. Exact matching of the impedance of the load to the line is a sufficient condition to ensure a VSWR of 1:1. A line with this exact match is a nonresonant line. Although the ideal cannot be realized in practice, the VSWR should be as low as practical. A VSWR of 1.5:1 is usually considered good and up to 2:1, acceptable. A line with a VSWR of 2:1 or less can be considered a nonresonant line for practical purposes. The length of a nonresonant line is not critical. Any VSWR above 1:1 introduces additional transmission line loss, because of the increased I^2R loss at current nodes and increased loss in the dielectric at voltage loops. However, as shown in figure 3-19, the additional loss is not great for a VSWR of less than 2:1.

d. Resonant Lines. The input impedance of a transmission line with a high VSWR depends upon the length of the line and is resistive only when the line is an integral number of one-quarter

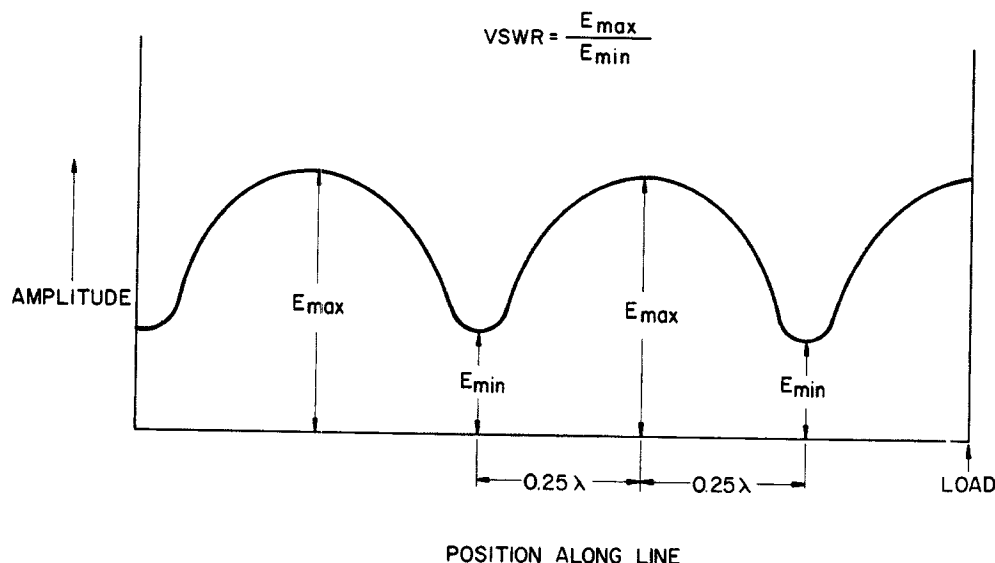


Figure 3-7. Graph of voltage standing wave ratio.

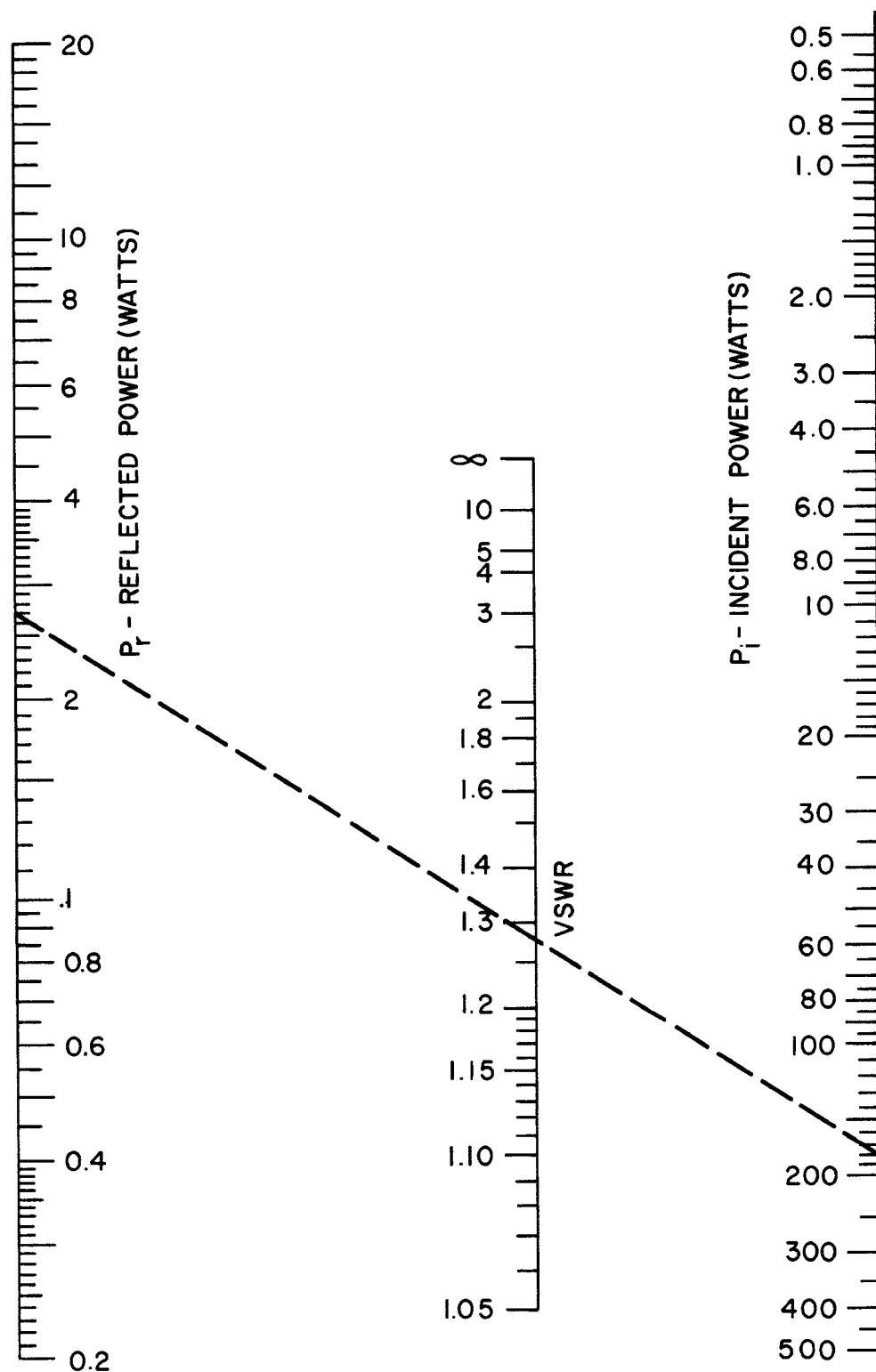


Figure 3-8. VSWR nomograph.

wavelengths. Lines cut to such a length and operated with a VSWR greater than about 3:1 are called *resonant lines*. A resonant line is used when it is not practical to match the line to the load. For example, if a resonant antenna is operated on several harmonically related frequencies, it is

impossible to obtain a match that will hold for all frequencies.

e. Radiation from Transmission Lines. Coaxial lines are essentially nonradiating because the outer conductor is at ground potential. When

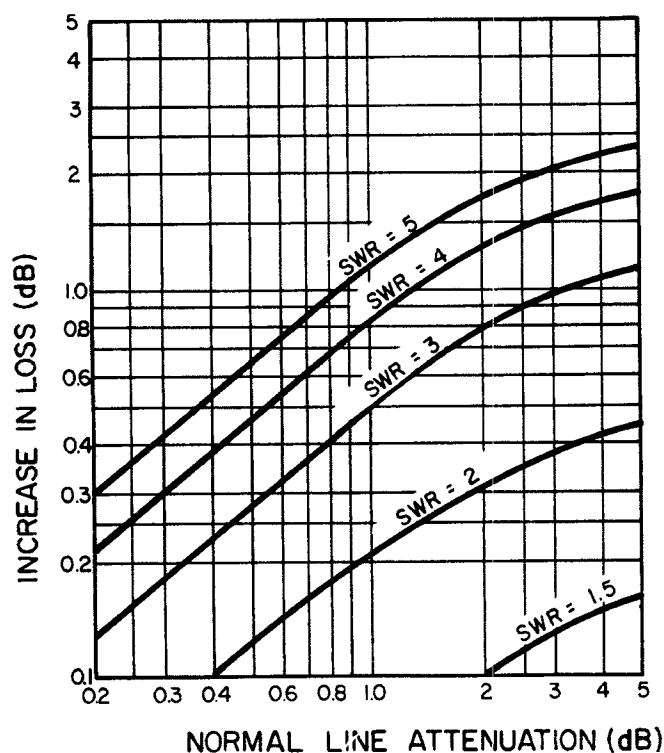


Figure 3-9. Additional transmission line loss because of standing waves.

the conductors of parallel-wire transmission lines are equally spaced throughout the length of the line and equally distant from other objects, the line is symmetrical. The currents in the two conductors of a symmetrical line are equal but opposite in direction. Cancellation of the fields about the conductors results. Any departure from symmetry results in some radiation, the radiation being least if there are no standing waves. However, if spacing between the conductors is 0.01 wavelength or less, radiation is negligible even with a high VSWR, provided the currents in the two conductors are equal.

3-5. Power Ratings of Coaxial Transmission Lines

The power-handling capability of coaxial transmission lines is related to the distance between the inner and outer conductors and the ability of the dielectric to withstand the heat generated in the inner conductor. The greater the distance between inner and outer conductors, the higher the power that can be safely handled. Polyethylene dielectric can be damaged by temperatures in excess of 80° C, while high-temperature dielectrics can withstand temperatures on the order of 250° C. Table 3-1 gives maximum power ratings at 30 MHz for the most commonly used military cable

Table 3-1. Power Ratings of Coaxial Transmission Lines

Cable type (RG/U)	O.D. (cm)	Maximum power rating (kilowatts) at 30 MHz
13	1.1	1.35
17A	2.2	7.4
58	0.5	.55
59	0.6	.7
213	1.0	1.7
216	1.1	1.35
218	2.2	7.4

types. Higher powers can be applied at frequencies below 30 MHz, with caution to prevent damaging overheating. Since ambient temperature is a factor in the temperature of the cable, high ambient temperatures will result in lower power handling capabilities than those given in the table. The same is true of high VSWR. Since standing waves cause "hot spots" along the cable, the power handling capability falls off as VSWR increases.

3-6. Optimum Transfer of Signal Energy

The importance of matching the load to the transmission line was discussed in the preceding paragraph. Provided this has been done, the greatest transfer of energy from the signal source to the line occurs when the source impedance matches the transmission line. Hence, the receiving antenna and the receiver should both be matched to their connecting transmission line. Likewise, the transmitter and the transmitting antenna should both be matched to their connecting transmission line. The two transmission lines, however, may be of different impedances.

3-7. Matching Between Transmitter and Line

Transmitters are designed to present a characteristic impedance close to that of some standard transmission line: 50 and 70 ohms for coaxial line, and 300 and 600 ohms for open-wire line. Hence, the obvious means of impedance matching is selection of a transmission line with impedance corresponding to that of the transmitter. Multicouplers are used when it is necessary to connect two or more transmitters to a single antenna. Multicouplers built to DCA standards are equipped with meters to measure VSWR on each line while tuning the multicoupler. Where connections between balanced equipment and unbalanced lines are required, baluns are used (para 3-8b). Baluns also accomplish impedance transformations. If it is necessary to connect the unbalanced, low-impedance output of a transmitter

to a higher impedance, balanced line, one of the methods shown in figure 3-10 can be used. If it is necessary to match the coaxial line from a transmitter to a coaxial transmission line having a high VSWR, one of the circuits shown in figure 3-11 can be used. The circuits can be adjusted to give a VSWR close to 1:1 on the line to the transmitter.

3-8. Matching to the Antenna

a. Balanced Line. Even though coaxial transmission lines are usually used, occasionally there is an application for a balanced line. The two types described in this paragraph are simple and may be used under emergency conditions.

(1) *Twisted-pair feed.* An emergency method for center-feeding a half-wave antenna uses a twisted-pair line, the characteristic impedance of which is approximately 70 to 80 ohms (fig 3-12). The twisted pair will usually give a fair match to the center of a half-wave antenna. This method normally is not used because of the high line losses associated with the twisted pair. In emergencies, however, a twisted pair formed from a lamp cord or field telephone wire can be used as a transmission line.

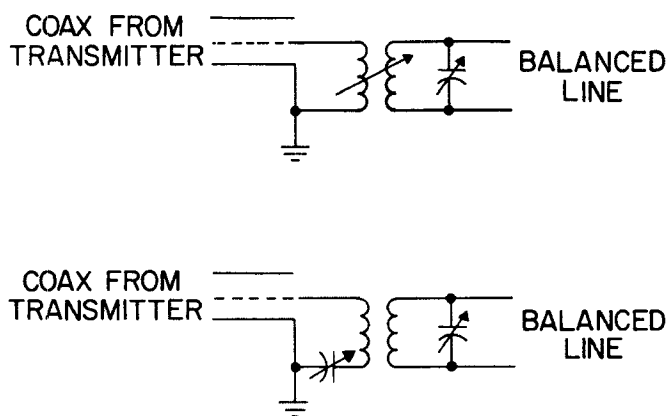


Figure 3-10. Matching coaxial line to balanced line.

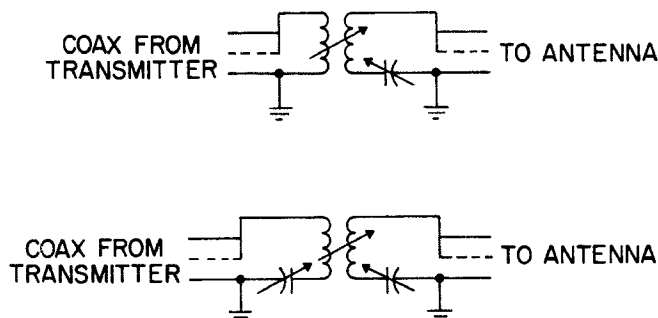


Figure 3-11. Feeding coaxial line with high VSWR.

(2) *Two-wire feed using delta match.* Because of constructional difficulties, the open two-wire line cannot always have a characteristic impedance. A practical line of this type has a Z_0 of 300 to 600 ohms. If this line is used to center-feed a half-wave antenna having a 73-ohm input impedance, some type of impedance transformation is necessary. In the delta match this transformation is obtained by spreading the transmission line as it approaches the antenna (fig 3-13). Electrically speaking, the delta section is actually part of the antenna. The delta, therefore, introduces a radiation loss, which is one of its disadvantages. Another disadvantage is that although formulas

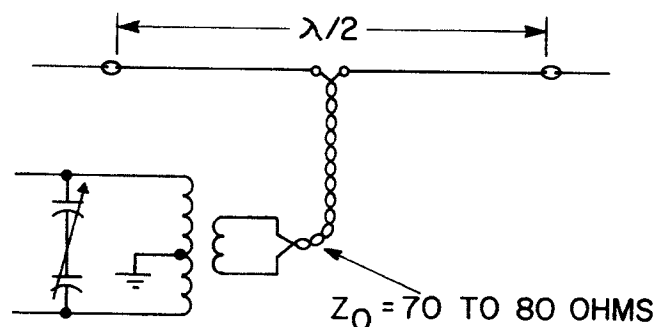


Figure 3-12. Twisted pair transmission line.

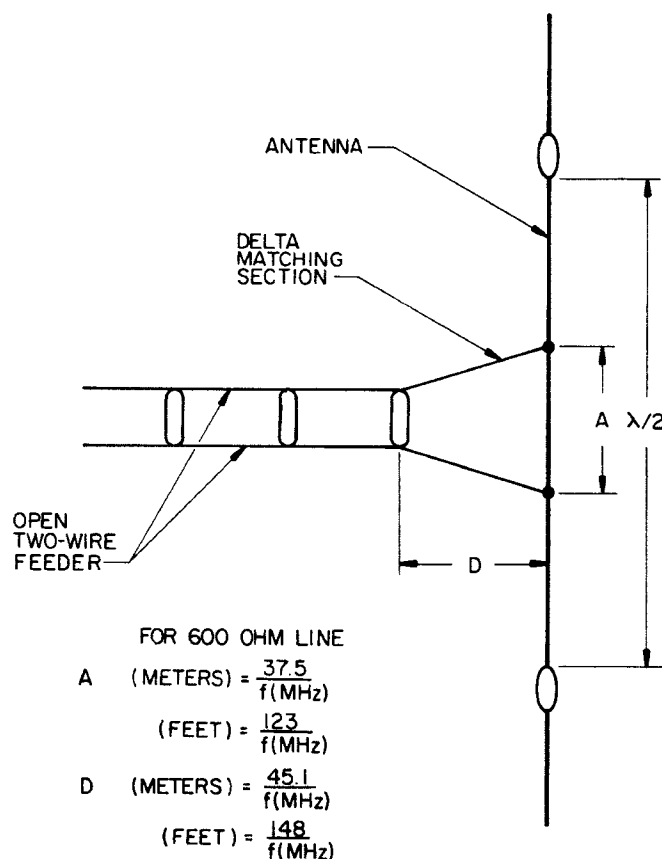


Figure 3-13. Delta match to antenna.

can be used for the first trial dimensions, cut-and-try methods must be used to determine the optimum dimensions of the delta section. Since two dimensions can be varied, adjustment is laborious. To avoid equipment damage, follow the dimensions given in figure 3-13 closely for the first trial.

b. Unbalanced Lines. Coaxial transmission lines are available which provide a good match to the center of a half-wave antenna. If a balun is not available, the inner conductor of the cable can be connected to one section of the antenna and the outer conductor to the other. This configuration, however, gives a skew to the radiation pattern, and the use of a balun, as shown in figure 3-14 or

3-15, is preferred. Figure 3-15 shows a balun which gives either a 1:1 or 1:4 transformation ratio. A balun transformer is very often used to couple the 50- or 75-ohm output of a transmitter to a balanced transmission line leading to the antenna. The various types of rf transformers and baluns exhibit different frequency and power-handling capabilities. The specifications for power handling capability, operating frequency range, and maximum permissible VSWR for the various types of RF transformers and baluns should be carefully observed. Balun transformers make possible the ease of operation, safety, and simplicity of switching coaxial lines inside the station while permitting the use of economical and efficient open

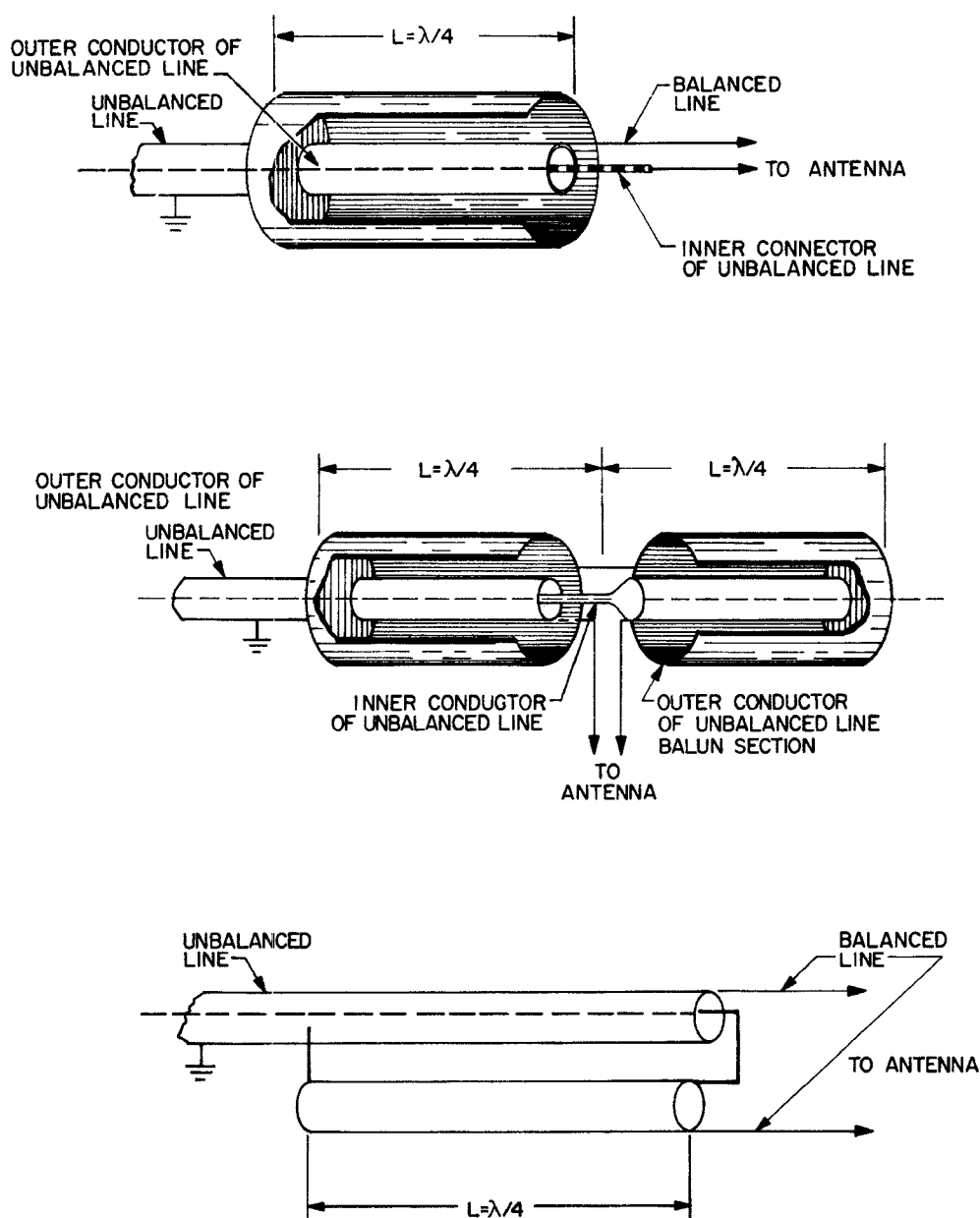
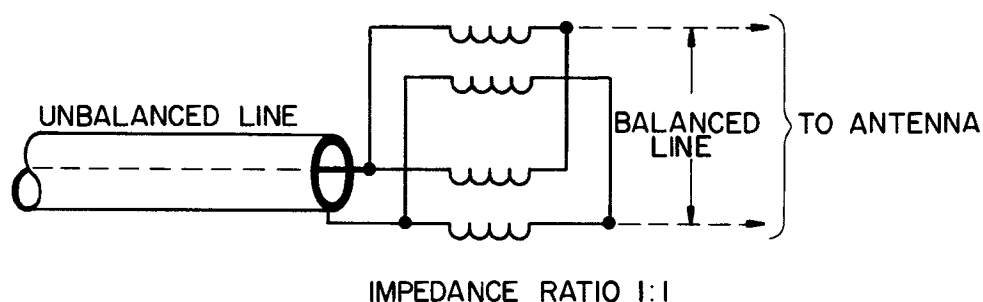


Figure 3-14. Baluns with 1:1 transformation ratio.

wire lines outside. A well-built balun is practically maintenance-free, dependable, and, when properly enclosed, suitable for long-term outdoor service in severe weather. The only maintenance required

is occasional cleaning and external inspection. Baluns are usually equipped with coaxial connectors for the unbalanced side and standoff insulator connections for the balanced side.

CHARACTERISTIC COIL IMPEDANCE EQUAL
TO 2X BALANCED INPUT IMPEDANCE



CHARACTERISTIC COIL IMPEDANCE EQUAL
TO 1/2X BALANCED INPUT IMPEDANCE

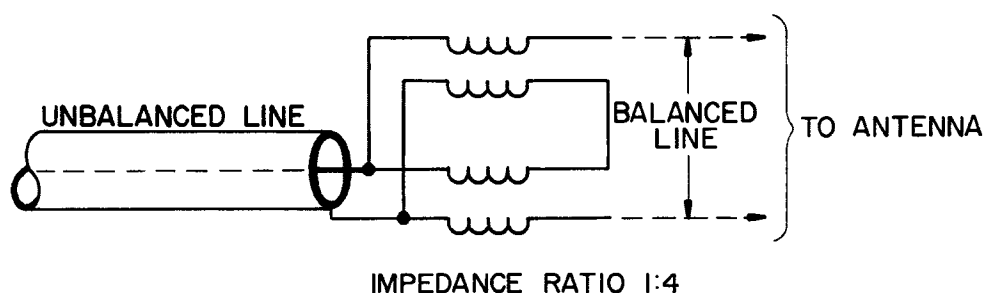


Figure 3-15. Balun with 1:1 or 1:4 transformation ratio.

CHAPTER 4

HIGH FREQUENCY ANTENNA PRINCIPLES

4-1. General

It is necessary to couple the output power of the transmitter to some type of structure capable of radiating electromagnetic waves into space and, likewise, to connect the input of the receiver to a structure capable of receiving such waves from space. An antenna is such a structure. Antennas are available with a wide variety of electrical characteristics. They vary in horizontal directivity, vertical angle characteristics, polarization, bandwidth capability, and method of feed. Power handling capability must also be considered for transmitting antennas. Various physical forms have been designed to attain the wide variety of desired electrical characteristics. Proper antenna selection is one of the most important parts of hf radio system design.

4-2. Isotropic Antenna

a. To understand actual radio transmitters and the associated wave propagation, it is helpful to examine an idealized model. Such a model is an isotropic radiator in free space. An isotropic antenna is one that radiates equally well in all directions. Free space means that the earth, its atmosphere, its ionosphere, and other objects are far enough away that they have no influence. Radio waves are transmitted through free space without loss. The power flow from an isotropic radiator in free space is represented in figure 4-1. Power originates from a point at the center of the imaginary sphere and radiates equally well in all directions. The amount of energy flowing through the area *abcd* on the surface of the sphere is exactly the same, no matter where on the surface it is placed.

b. A useful concept is implied in figure 4-1. The area of a sphere is $4\pi r^2$ where r is the radius. Consider a sphere having a 1-foot radius with a 1-W transmitter at its center. One watt of radiation is spread over a surface of 4π ft², or 12.57 ft². The intensity of radiation (power density) is 1 W/12.57 ft², or 0.0796 W/ft². Consider a second sphere, with a radius of 2 feet, with the same 1-W

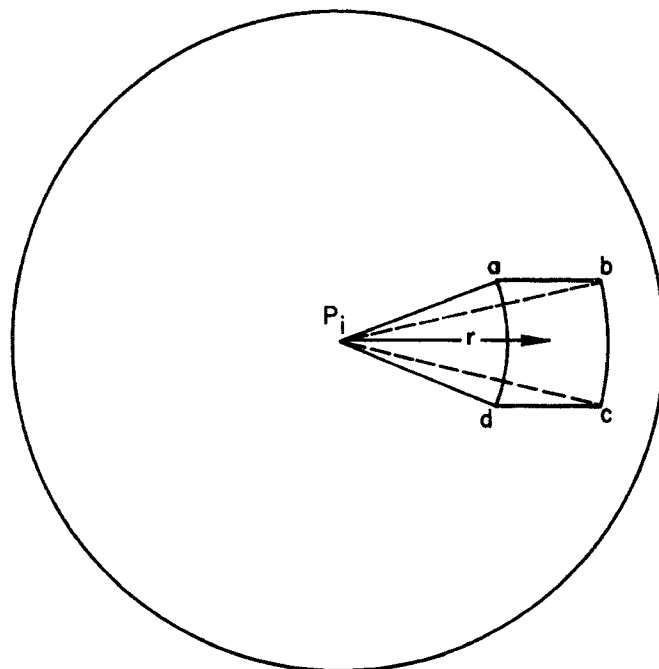


Figure 4-1. Power flow from an isotropic radiator in free space.

transmitter. The area of the second sphere is $4\pi \times 4$ ft², or 50.27 ft². The power density is 1 W/50.27 ft², or 0.0199 W/ft². This is one-fourth of the previous density. Computations for a sphere of 3-ft radius will yield an intensity or power density of one-ninth the original. We can conclude that radiation intensity in free space varies inversely as the square of the distance. This relationship is known as the *inverse square law*.

4-3. Radiation from a Wire

An electric current flowing in a wire is accompanied by a magnetic field around it. When the magnetic field is changing, as it does with alternating current a changing electric field is also present. When an rf current is fed into a short, isolated, straight piece of wire, radiation of a portion of the magnetic and electric fields results. By definition, the direction of polarization of the radiated wave is the direction of the lines of flux of the electric field. If the wire is in a horizontal position, the radiation is horizontally po-

larized. In free space, the radiated wave travels at the speed of light (3×10^8 m/s or 186,000 mi/s). The power density of the wave varies inversely with the square of the distance. At 200 miles the power density is one-fourth that at 100 miles.

4-4. Half-Wave Dipole Antenna

a. A *dipole* is a straight antenna, usually fed at the center, and producing a maximum of radiation in the plane normal to its axis. A very common and quite useful antenna is a center-fed dipole which is one-half wavelength long. This antenna is called a half-wave dipole. The antenna may be horizontal, sloping or vertical, but at high frequencies it is usually horizontal. Figure 4-2 shows a sketch of a center-fed, half-wave dipole. In the discussion of transmission lines, chapter 3, it was pointed out that proper termination of a transmission line prevented, or at least held to a minimum, standing waves. Although standing waves are undesirable on transmission lines, they are an inherent characteristic of a resonant antenna. Figure 4-2 shows the standing waves of voltage and current and the variation in impedance along the antenna. The minimum impedance, about 73 ohms, represents the impedance encountered by the radiating electromagnetic waves, and is called *radiation resistance*.

b. Unlike the isotropic antenna, the half-wave dipole antenna does not radiate equally in all directions, but is directional. Figure 4-3 shows the electromagnetic field created by a half-wave dipole

in free space. This pattern is discussed further in paragraph 4-11.

4-5. Quarter-Wave Vertical Antenna

a. The ground is a fairly good conductor for medium and low frequencies and acts as a large mirror for the radiated energy. This results in the ground reflecting a large amount of energy that is radiated downward from an antenna mounted over it. It is just as though a mirror image of the antenna is produced, the image being located the same distance below the surface of the ground as the actual antenna is located above it. Even in the high frequency range, very significant ground reflections occur, especially if the antenna is erected over highly conducting earth, salt water, or a ground screen.

b. Utilizing this characteristic of ground, a quarter-wave vertical antenna operating against a ground can be considered as being the upper half of a vertical dipole (fig 4-4).

c. Because the quarter-wave antenna physically consists of half a dipole, it is frequently called a *monopole*. The term monopole is applied generally to vertical antennas which appear to have an image antenna.

d. There are two modes of operating the quarter-wave vertical antenna. The base of the antenna may be insulated from the ground and the antenna fed at the base. Or, the base may be grounded and the feed connected a short distance

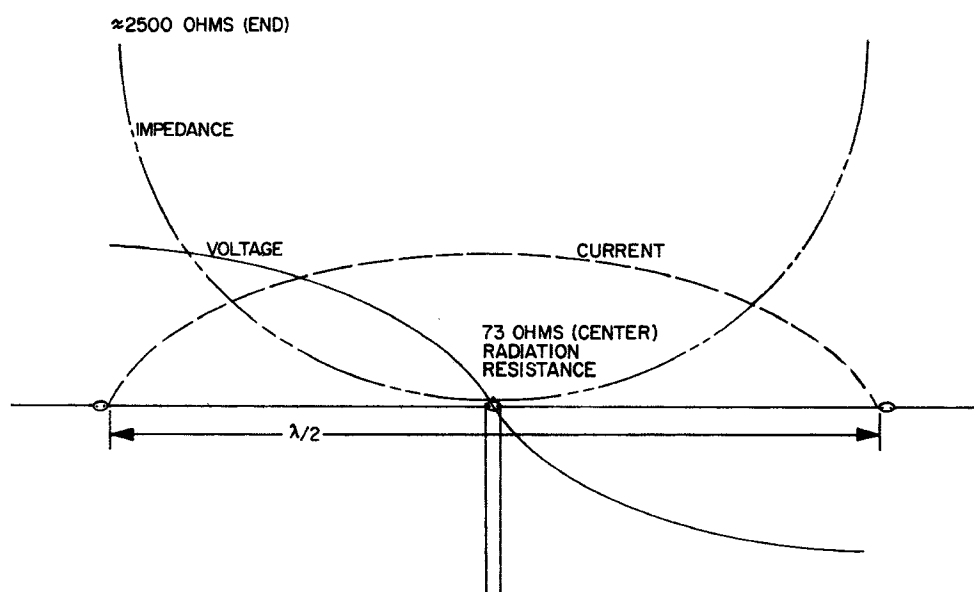


Figure 4-2. Voltage, current, and impedance characteristics of a half-wave dipole antenna.

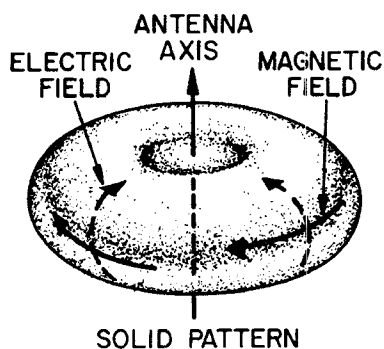
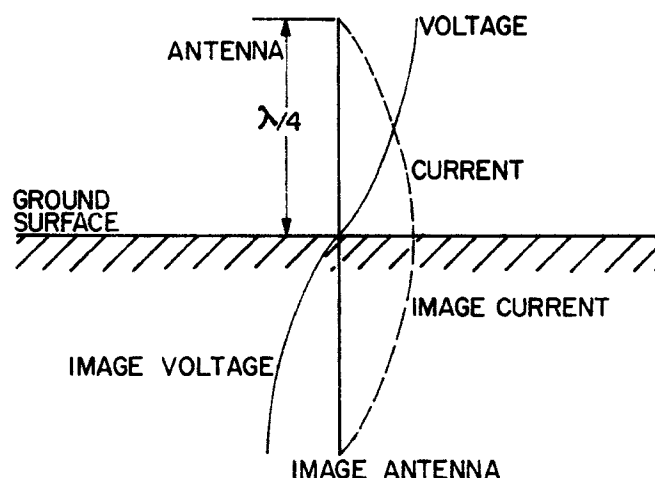
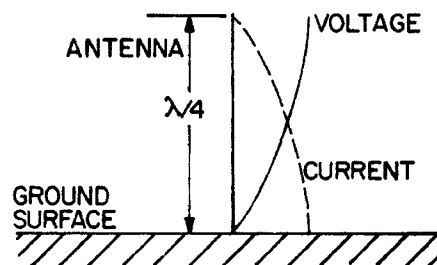
above the base, at a point where the antenna's input impedance matches that of the transmission line. In either case, coaxial transmission line with grounded outer conductor is used.

e. The radiation pattern of the quarter-wave antenna is shown in figure 4-5. It resembles half of the pattern of a half-wave antenna in free space.

4-6. Power Gain and Directivity

a. It can now be seen that different types of antennas result in different patterns of propagation. Given the same location point with reference to the antennas, different values of power density will be measured for different types of antennas. This is true even though the input

power is the same for both types. Power gain and directivity are concepts which are useful in describing antenna patterns.



NOTE: THE ARROWS INDICATE INSTANTANEOUS DIRECTIONS OF THE FIELDS. THEY REVERSE AT THE FREQUENCY OF THE ENERGIZING SIGNAL.

Figure 4-3. Solid radiation pattern of half-wave dipole in free space.

Figure 4-4. Voltage and current distribution in quarter-wave vertical antenna.

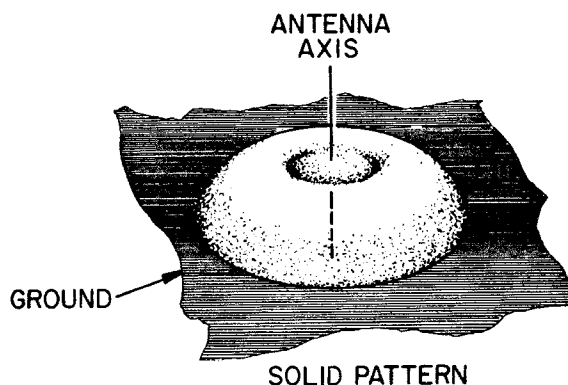
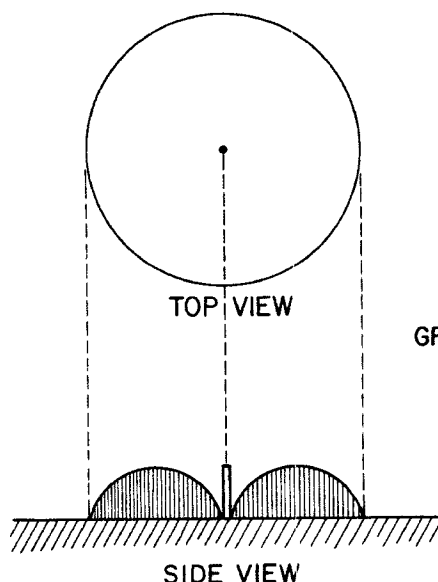


Figure 4-5. Radiation pattern of quarter-wave vertical antenna.

b. The power gain of an antenna is the ratio of the maximum power density radiated by the antenna to the maximum power density radiated by the reference antenna (e.g., the isotropic) when both antennas have equal input powers.

c. The directivity of the antenna, which is sometimes confused with antenna power gain, is the ratio of the maximum power density radiated by the antenna to the average power density radiated by the antenna.

d. Antenna power gain takes antenna power losses into account, whereas directivity does not. Since all real antennas have some power losses, the directivity of the antenna will exceed the antenna power gain.

e. Gain and directivity are discussed further in paragraph 4-10c.

4-7. Antenna Resistance and Impedance

a. *Resistance.* A current flowing in an antenna must contend with three types of resistance. The power expended in radiation can be thought of as an I^2R_r loss, where R_r is the *radiation resistance*. The current flowing in the antenna itself has an I^2R_o loss, where R_o is the ohmic resistance of the antenna. There is an additional I^2R loss because of leakage resistance. In most practical antennas of good design, the radiation resistance is large compared with the other resistances and is the only resistance which needs consideration. The radiation resistance of a half-wave dipole in free space is about 73 ohms.

b. *Impedance.* An antenna has standing waves of current and voltage. The impedance of the antenna itself at any point can be thought of as the value of the voltage divided by the value of the current at that point. For a half-wave antenna, the theoretical impedance would range from infinite at the ends to zero at the center. Because of some capacitive effects, the impedance at the ends is not infinite, but several thousand ohms. If the antenna is cut to resonance, the impedance at the center, is 73 ohms. Figure 4-2 shows the variation in impedance of a half-wave dipole in free space.

4-8. Antenna Bandwidth

The frequency range over which an antenna can be used is an important consideration. Both the impedance and the directivity of a given antenna change with frequency. For transmitters, the bandwidth is determined by the frequency range over which the VSWR can be held to 2:1 or less.

The criterion for receivers, which are more tolerant of VSWR than are transmitters, is the frequency range over which acceptable directional gain can be obtained. The half-wave dipole, discussed in paragraph 4-4, belongs to the class of antennas called *resonant*. A simple resonant antenna characteristically has a bandwidth of only a few percent of its resonant frequency. Another class of antennas, called *nonresonant*, consists of antennas which are deliberately designed for great bandwidth. Nonresonant antennas characteristically operate over bandwidths of 2:1 up to 4:1, but a range of 10:1 is possible.

4-9. Velocity of Propagation and Antenna Length

a. In free space an electrical wavelength in free space can be computed from the formula

$$\lambda f = c$$

where

λ = wavelength

f = frequency in hertz

and

c = the velocity of propagation in the same units used for λ (3×10^8 m/s or 9.843×10^8 ft/s)

The equation can be transposed into

$$\lambda = c/f$$

The half-wavelength, which is more frequently of interest, is

$$\lambda/2 = c/2f$$

In the metric system,

$$\lambda/2 \text{ (meters)} = 150/f \text{ (MHz)}$$

In the English system,

$$\lambda/2 \text{ (feet)} = 492/f \text{ (MHz)}$$

Two factors cause the wavelength in an antenna to be slightly less than that in free space.

(1) The velocity of propagation in a finite conductor is slightly less than in free space.

(2) Stray capacitance increases the dielectric constant and lowers the wave velocity. This capacitance can be caused by the transmission line, the insulators supporting the antenna, or nearby objects. This second factor, termed the *end effect*, is the more important of the two.

b. In the hf band, a practical half-wavelength antenna is about 5 percent shorter than a half-wavelength in free space. The previous formulas become—

$$\lambda/2 \text{ (meters)} = 143/f \text{ (MHz)}$$

$$\lambda/2 \text{ (feet)} = 468/f \text{ (MHz)}$$

4-10. Harmonic Operation of Linear Antennas

a. Length. A harmonic antenna is a multiple of half-wavelengths in electrical length. The end effect discussed in paragraph 4-9 occurs only twice, regardless of the antenna length. Hence, an antenna one wavelength long electrically will not physically be twice as long as an electrical half-wavelength antenna. The formulas for physical length of harmonic antennas are—

$$l \text{ (meters)} = 150(L-0.05)/f \text{ (MHz)}$$

$$l \text{ (feet)} = 492(L-0.05)/f \text{ (MHz)}$$

where L is the number of half-wavelengths for which the antenna is being designed.

b. Standing Waves. As the length of the antenna is increased so that it operates on higher harmonic frequencies, or as the frequency applied to an antenna of fixed length is raised, a greater number of half-waves of voltage and current occur on the antenna. This is shown in figure 4-6. The standing waves of voltage and current are 90° out of phase.

c. Gain and Directivity.

(1) Very little gain occurs when the antenna is only a few wavelengths long. When the length becomes appreciable, however, considerable power gain results, and increased power gain is accompanied by greater directivity. This is true since an increase of power in certain directions is attained by reduction of power in other directions. The approximate power gain of harmonic

antennas of various lengths in free space is shown in the tabulation below.

Antenna length (wavelengths)	Power gain (over half-wave antenna)
1	1.2
2	1.4
4	2.1
6	3.1
8	4.3
10	5.6
12	7.2

(2) Figure 4-7 shows the radiation patterns of harmonic antennas up to three wavelengths. The pattern produced by the half-wave antenna is shown for comparison. Note that as the antenna length is increased, more lobes are produced. The $1\frac{1}{2}$ -wavelength antenna, which operates on the third harmonic, has three lobes—two major lobes and one minor lobe, the latter lying at right angles to the antenna. The 3-wavelength antenna, which operates on the sixth harmonic, has six lobes—two major lobes and four minor lobes.

d. Feeding Harmonic Antennas. Since a point on the antenna which is a current loop on one harmonic does not remain a current loop on the next higher harmonic, an antenna which is current-fed at a single fixed point can operate only at odd harmonics of the original frequency. To operate on all harmonics, end-feeding is required.

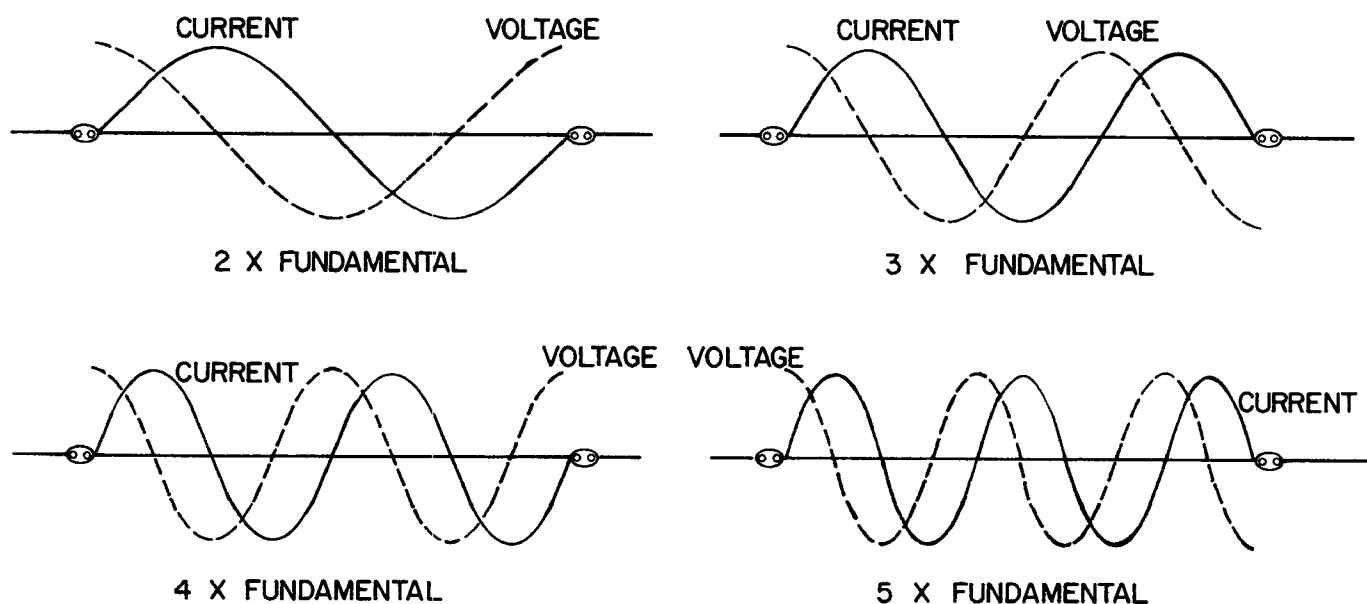


Figure 4-6. Standing waves on harmonic antennas.

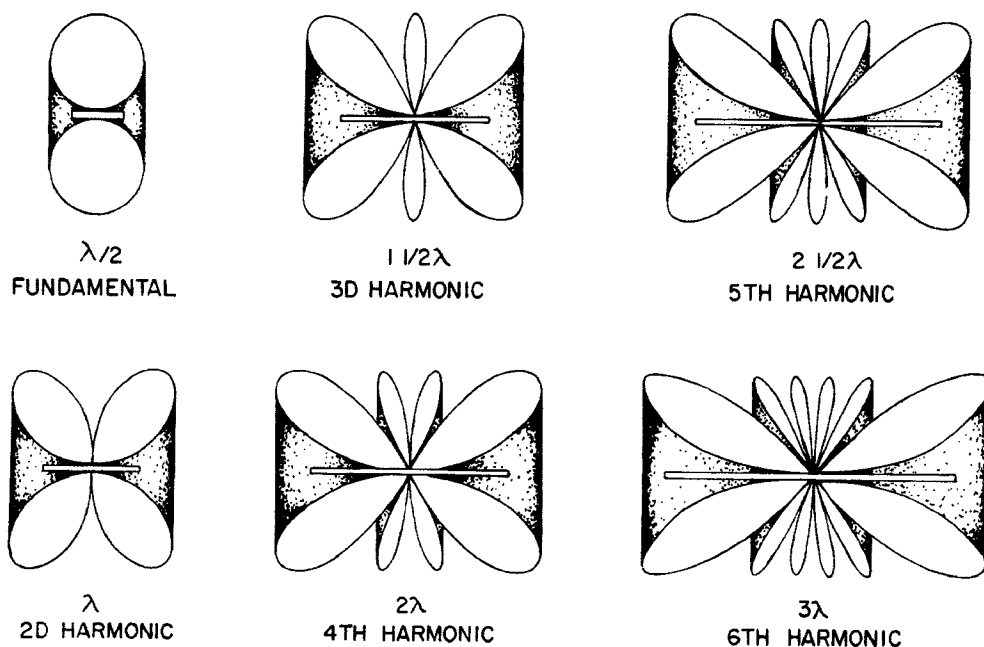


Figure 4-7. Solid radiation pattern of harmonic antennas.

4-11. Radiation Patterns

a. Figure 4-8 shows how the solid radiation pattern of figure 4-3 can be transformed into horizontal and vertical plane patterns. The figure shows patterns for the antenna axis vertical and for the axis horizontal. In a plane through the axis, the pattern is highly directional. In the plane at right angles to the axis, the pattern is symmetrical. The free space patterns of the half-wave dipole are greatly modified when the antenna is near the ground, but the symmetry in a plane at right angles to the axis is retained for a vertical antenna. The patterns of the half-wave dipole are important because the patterns of many practical antennas can be derived from them.

b. Figure 4-8 is merely showing a concept, so no scale is shown. Patterns representing practical antennas will have a scale showing relative power. A plot of power may be in decibels (dB) relating the antenna performance to a standard antenna. Comparison to the standard antenna, the isotropic, is indicated by dBi. A half-wave dipole in free space is also used as a reference antenna. The various methods of plotting antenna patterns will be shown by examples in this manual as specified antennas are discussed.

c. Figure 4-9 shows a horizontal radiation pattern. The figure states that gain is with respect to an isotropic radiator. The plotted curve shows that the forward gain is 12 dB above an isotropic radiator. At azimuths of 60° and 300° the radi-

ated power is equal to that of an isotropic radiator. The rear lobe is down 15 dB compared with isotropic.

d. Some patterns may show power in dB compared to the maximum intensity of the main lobe. Some examples given later in this chapter show field strength in millivolts per meter at one mile for one watt input at antenna. Other patterns may show the relative power factor, with zero at the center and circles labeled 0.5, 1.0, 2.0, 3.0, etc., up to 4.0 or higher. In many cases the pattern gives qualitative information only and the units are not stated. For engineering purposes, precise information such as shown in figure 4-9 is required.

e. Vertical radiation patterns also exist in various forms. Again, for engineering purposes, a pattern with a clearly defined scale is required.

f. For convenience in indicating the degree of directivity, the parameter known as *beamwidth* has been developed. Beamwidth is the angle bounded by the halfpower points on the pattern. To locate the half-power points on a pattern representing relative power, draw an arc whose radius is one-half the distance from the center to the maximum of the lobe. The intersections of this arc with the pattern are the half-power points. If the pattern is plotted in terms of voltage, the boundary lines are the points where the voltage is 0.707 times the maximum. (Power is

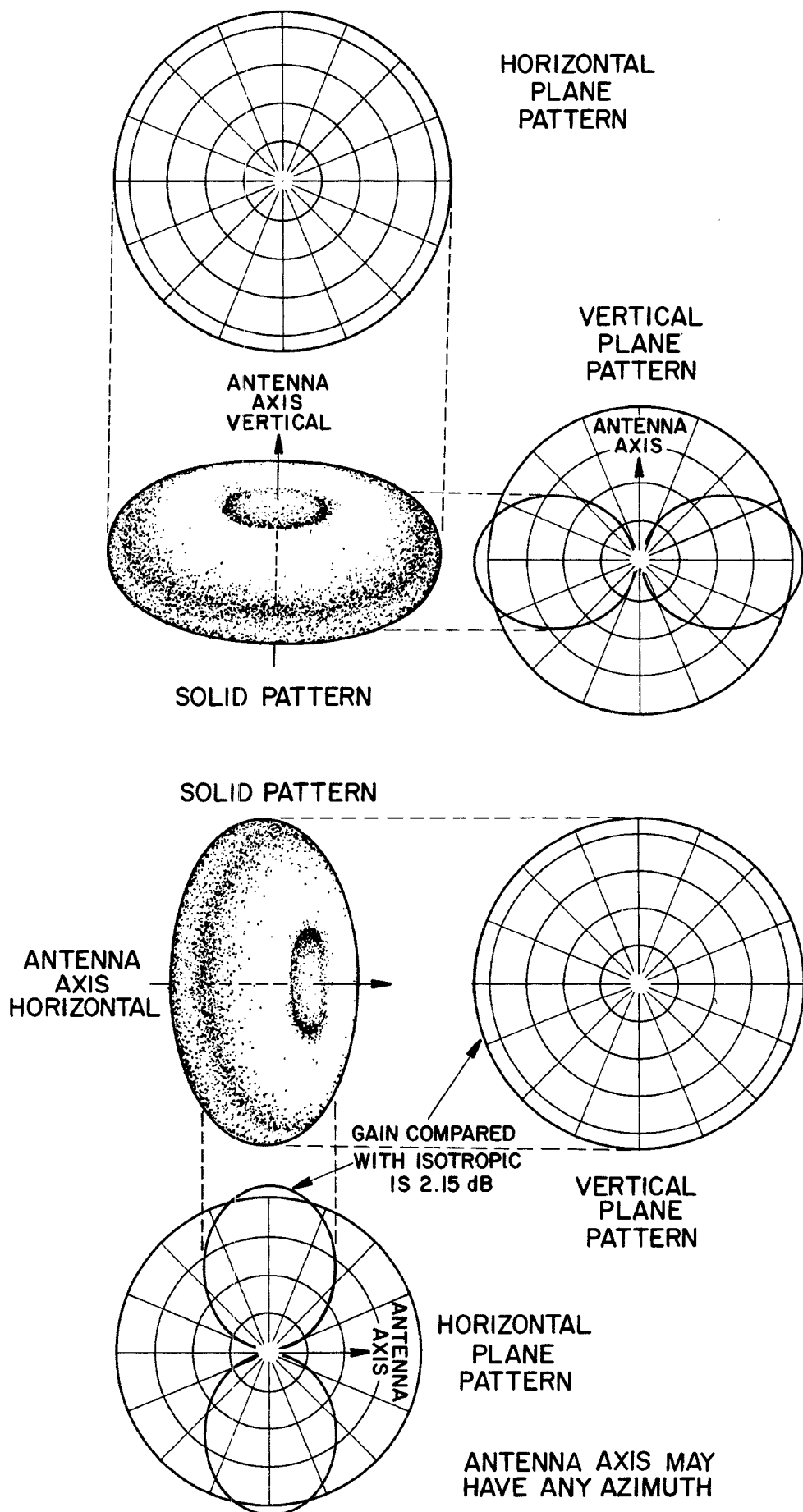


Figure 4-8. Horizontal and vertical patterns of half-wave dipole in free space.

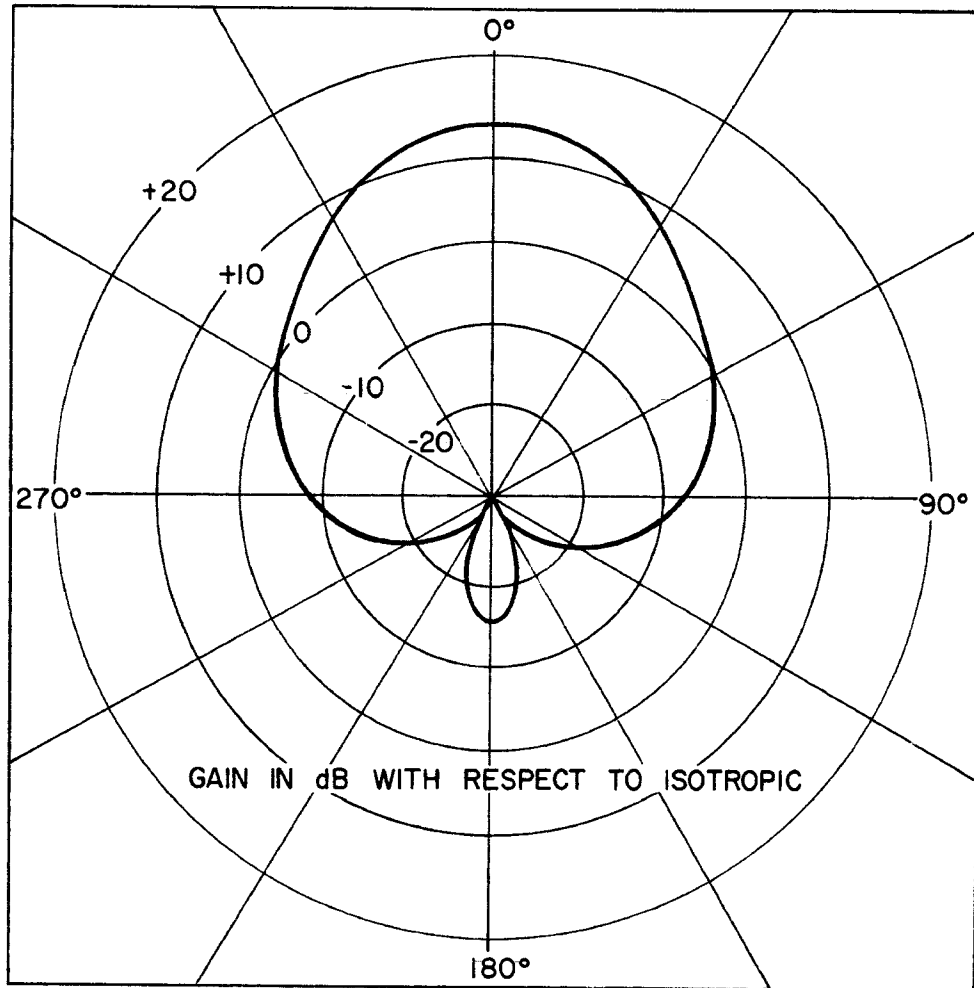


Figure 4-9. Typical horizontal radiation pattern.

proportional to the square of the voltage $(0.707)^2 \approx 0.5$.) Figure 4-10 shows the radiation pattern for an antenna plotted in both relative field strength and relative power. In either case, the beamwidth is 90° . If the antenna plot is in decibels (dB), the half-power point is 3 dB below the lobe maximum.

4-12. Radiation and Induction Fields

a. When rf power is fed to an antenna, two fields are set up. One is the radiation field (the far field), which transmits the desired signal. The other is the induction field (the near field), which is associated with the stored energy in the vicinity of the antenna. Very near the antenna, the induction field is stronger than the radiation field. However, its power density falls off in proportion to the fourth power of the distance. At a distance of one and one-half wavelengths the induction field of a small antenna can usually be disregarded.

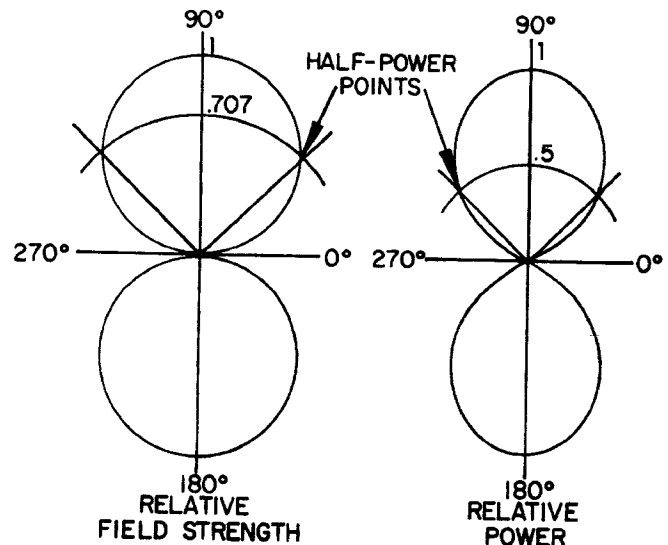


Figure 4-10. Beamwidth measured on relative field strength and relative power patterns.

b. The induction field may cause interference problems by overloading receivers or creating intermodulation. With powerful transmitters, personnel safety is a consideration. The pattern of the induction field is not the same as that of the radiated field. Hence, it is not safe to assume that induction field problems will be confined to the main lobe of the transmitting antenna.

4-13. Reciprocity

a. The half-wave antenna has the property of reciprocity since, when it is used for transmitting, it has the same directivity characteristics as when used for receiving. A half-wave antenna used for transmitting radiates its maximum energy at right angles to the antenna itself. When such an antenna is used for reception, it receives maximum energy at right angles to itself. A vertically mounted half-wave antenna radiates energy equally in all horizontal directions. A similar antenna used for reception receives energy equally in all horizontal directions. Consequently, a pattern which shows the radiation of an antenna can be used also to show the reception pattern of that antenna.

b. In general, this reciprocity of radiation and reception applies not only to the half-wave antenna but also to more complicated antennas. If an antenna is highly directive as a transmitting antenna, it will have similar directivity as a receiving antenna.

c. Even though antenna reciprocity holds, path reciprocity may not always exist. For example, assume that two identical antennas, A and B, are used for communication by way of the ionosphere. When antenna A is transmitting to antenna B, the radiated energy follows a certain path up toward the ionosphere and then down toward antenna B. When antenna B is transmitting to antenna A, it is possible that a slightly different path is followed by the radiated energy in traveling toward antenna A. Under these conditions, the received wave at A will arrive at a slightly different vertical angle, and a different part of the directive pattern will be used. For complete path reciprocity to exist, it is also necessary that antennas be terminated similarly when transmitting and receiving.

4-14. Effect of Perfect Ground on Radiation Pattern

a. *Reflection Factor.* The reflection factor is a

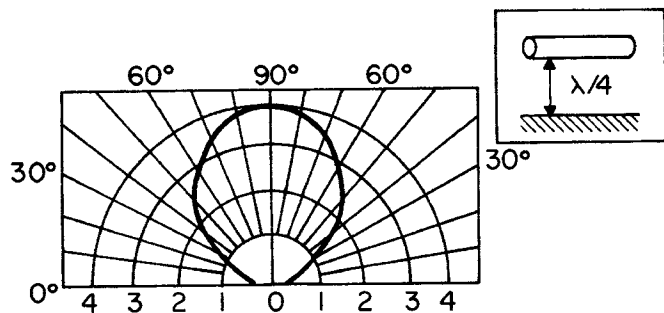
term by which the free-space radiation pattern of an antenna must be multiplied in order to determine the radiated field strength of a practical antenna at a given vertical angle. The maximum value of the reflection factor is two. At vertical angles having a factor of two, the direct and reflected waves are in phase, and twice the free-space signal strength occurs. The minimum value of the reflection factor is zero. At vertical angles having a factor of zero, the direct and reflected waves are of opposite phase, and complete cancellation occurs. Two and zero are limiting values which occur only with a perfectly reflecting ground. Actual values will be greater than zero and less than two. The value of the reflection factor depends on the height of the antenna above the ground plane as well as the vertical angle.

b. Effect of Ground on Horizontal Half-Wave Antenna.

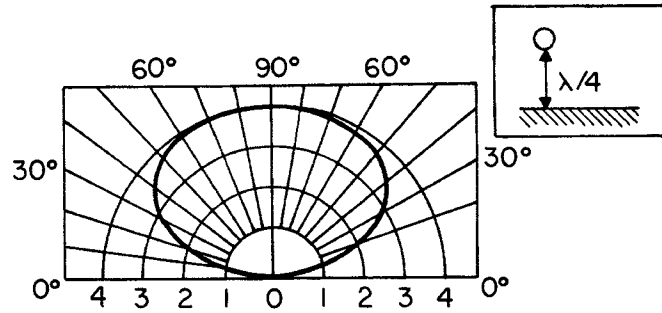
(1) When reflection factors are applied to the free-space radiation pattern of a horizontal half-wave antenna, the patterns shown in figure 4-11 are produced. Patterns A, C, E, and G are the vertical radiation patterns in the plane of the antenna itself. B, D, F, and H are the vertical radiation patterns in the plane which is at right angles to the antenna. Patterns A and B are for antenna heights of a quarter-wavelength; C and D are for antenna heights of a half-wavelength; E and F are for heights of three quarter-wavelengths; G and H are for heights of one wavelength.

(2) Although vertical patterns are shown here for only four specific heights above ground, it is not difficult to predict the patterns produced at intermediate heights. This is because the patterns do not change abruptly as the height of the antenna is increased gradually. There is a smooth transition from the pattern shown for a height of a quarter-wavelength to the pattern shown for a height of a half-wavelength.

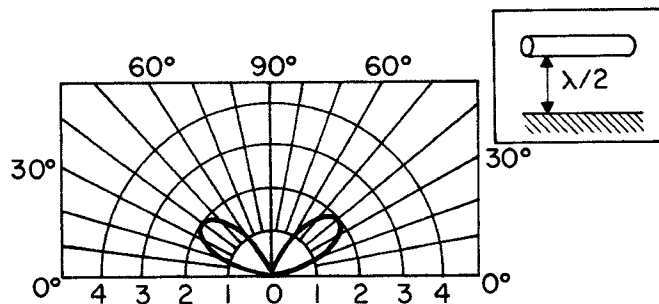
(3) At heights less than a quarter-wavelength above ground, the vertical patterns produced by a horizontal half-wave antenna are almost circular. As the antenna is raised, the vertical pattern is flattened somewhat at its top, at a vertical angle of 90° (B, fig. 4-11). As the height is increased above a quarter-wavelength, a depression begins to appear at the top of the pattern, and the pattern width increases. The depression grows deeper and deeper as the antenna height approaches a half-wavelength. Finally, at a height of a half-wavelength, the pattern splits into two separate lobes. The radia-



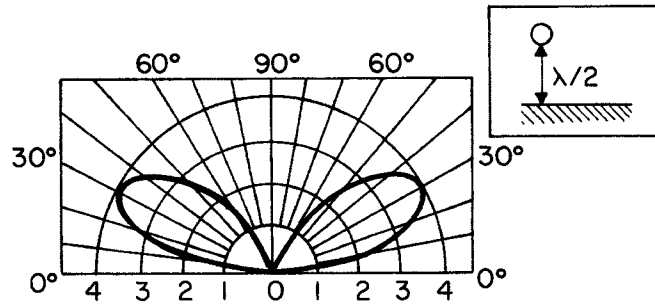
A



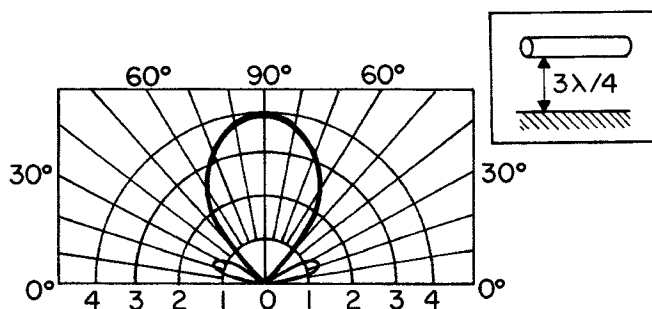
B



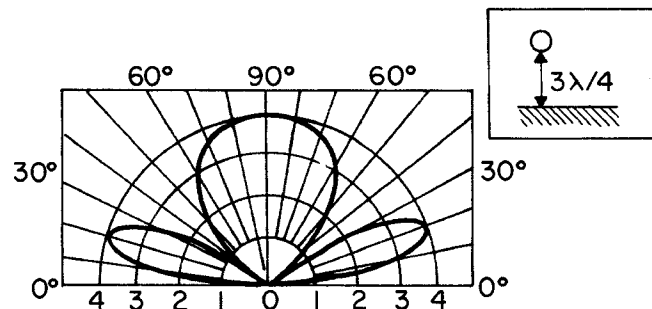
C



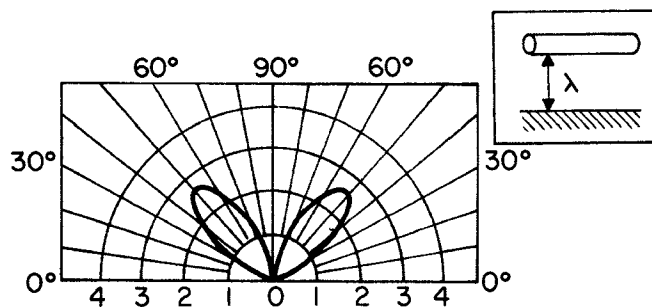
D



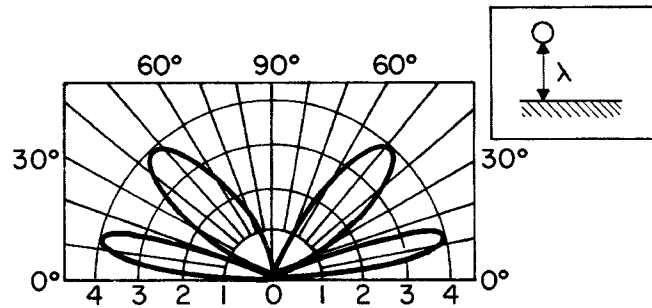
E



F



G



H

SCALES ARE MILLIVOLTS PER METER AT 1 MILE
FOR 1 WATT INPUT TO ANTENNA

Figure 4-11. Vertical-plane radiation patterns of horizontal half-wave antenna above perfectly conducting ground.

tion at a vertical angle of 90° (straight up) is zero at this height, as in D of figure 4-11. As the antenna height increases still more, a lobe of radiation begins to grow out of the center of the pattern at a vertical angle of 90° . As this lobe increases in amplitude with increasing antenna height, the two side lobes spread farther apart so that their maximums occur at lower vertical angles. This vertical lobe has its maximum amplitude and begins to flatten somewhat (F, fig 4-11) at an antenna height of three quarter-wavelengths. As the antenna height increases still more, the vertical lobe develops a depression that grows deeper as the height increases. Finally, at a height of one wavelength, the center lobe splits into two separate lobes and the radiation at a vertical angle of 90° is again zero. Now four distinct lobes exist (H, fig 4-11).

(4) The patterns that are produced at antenna heights in excess of one wavelength also can be estimated by studying figure 4-11. When the height of the horizontal antenna is an odd number of quarter-wavelengths above ground, a lobe of maximum radiation is produced at a vertical angle of 90° . When the height of the antenna is an even number of quarter-wavelengths above ground, a null (zero radiated energy) occurs at the 90° vertical angle.

(5) At a height of one quarter-wavelength above ground, the radiation pattern is seen to consist of one lobe only. At a height of two quarter-wavelengths ($\lambda/2$) above ground, the radiation pattern consists of two lobes. At a height of three quarter-wavelengths, the pattern consists of three lobes. At a height of four quarter-wavelengths (λ) above ground, the radiation pattern consists of four lobes. Consequently, the number of vertical lobes produced is numerically equal to the height of the antenna above ground in quarter-wavelengths and continues for any antenna height. It is possible to get a fairly good idea of the vertical radiation pattern of a horizontal half-wave antenna at any height above ground. For example, if the antenna is located at a height of two wavelengths above ground, which is an even number of quarter-waves, a null is produced at a vertical angle of 90° . Then, since two wavelengths represent eight quarter-wavelengths, the radiation pattern consists of eight lobes.

(6) Antenna heights which are an even number of quarter-wavelengths should not be used for short-distance links. For very long distances, which require a low elevation angle, considerable height is desirable. In both cases, the axis of the

major lobe should be in line with the direction of communication.

c. *Vertical Half-Wave Antenna.* When reflection factors are applied to the free-space radiation pattern of a vertical half-wave antenna, the patterns shown in figure 4-12 are produced. Only a single-plane view need be shown here because the vertical half-wave antenna is nondirectional in the horizontal plane. Its free-space horizontal radiation pattern is a circle. Therefore, the effect of the reflection factor is the same in all horizontal directions. At all antenna heights, the vertical half-wave antenna produces a null at 90° . As the antenna is raised above ground, a greater number of lobes appear (A, fig 4-12). When the antenna is raised to a half-wavelength, four lobes appear, as in B. The amplitude of the upper lobes is much smaller than that of the lobes which lie along the ground. At a height of three quarter-wavelengths, there are still four lobes, but the amplitude of the upper lobes has increased, as shown in C. When the antenna is raised to a height of a full wavelength, as in D, six lobes appear. A vertical antenna is useful where low elevation angles are required. However, the lack of directivity in the horizontal plane must also be considered. This lack of directivity can be desirable or undesirable, depending upon the communication requirements.

d. *Harmonic Antenna.* Some horizontal radiation patterns at various elevation angles for harmonic antennas parallel to the ground are shown in figure 4-13. The general shapes of the patterns are not altered by the earth. The effect of the earth is to change the relative amplitude of a pattern, as can be seen by comparing the pattern for a 10° elevation angle with the pattern for a 15° elevation angle on A, B, or C.

4-15. Effect of Imperfect Ground on Antenna Height and Radiation Pattern

a. *Effect on Antenna Height.* The depth to which a radio wave penetrates the surface of the earth varies with frequency and type of soil. The tabulation below gives some representative values of penetration depths. The values given in the tabulation are the depths at which the signal is reduced to about 37% ($1/e$, where e is 2.718 . . . , the base of natural logarithms).

Frequency	Depth (m)		
	Sea water	Good soil	Mountains
3 MHz	0.14	5	17
10 MHz	0.08	2	9

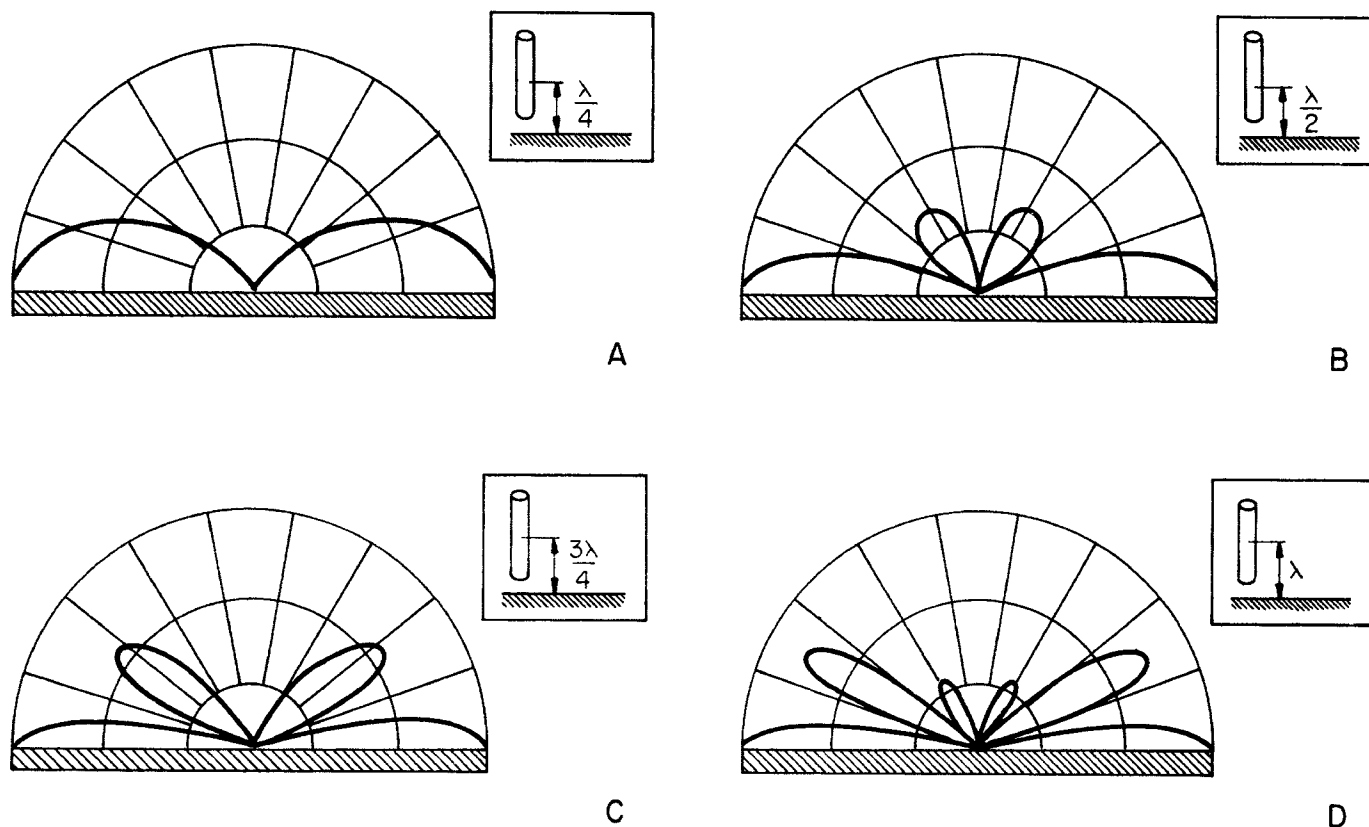


Figure 4-12. Vertical-plane radiation patterns produced by vertical half-wave antenna above perfectly conducting ground.

The penetration of the earth by the signal results in an increase in effective antenna height. However, this effect is important only for the upper part of the hf band over poor ground.

b. Effect on Radiation Pattern.

(1) *Horizontal antennas.* When a radio wave strikes an imperfect ground, there is no sharp plane of reflection. This means that the sharp reinforcements and cancellations of a perfect ground are tempered.

(2) *Vertical antennas.* In general, vertical antennas yield stronger signal strengths over good ground than poor ground, but the effect is not pronounced. When the performance of a vertical antenna depends upon a good ground, a counterpoise or other ground system should be provided, as discussed in paragraph 4-19.

4-16. Effect of Ground on Radiation Resistance

a. The radio-frequency current flowing in an antenna is made up of two components. One component is determined by the free-space radiation resistance and the power supplied by the transmitter. The second component is produced by radio waves which, after being directed earthward from the antenna, are reflected upward

from the earth. Upon passing back across the antenna, these reflected waves induce a current in the antenna. The phase and strength of this induced current depends on the height of the antenna above ground or reflecting surface. Although the induced current is small compared to that supplied by the transmitter, it cannot be neglected. At some antenna heights, the two currents are in phase and they combine to produce a resultant current which is greater than that expected from the free-space radiation resistance. At other heights, the two currents are out of phase so that the resultant current is reduced. Resistance equals power divided by the square of the current,

$$R = P/I^2.$$

Therefore, if the antenna current is changed by changing the antenna height while holding the power input constant, it is equivalent to changing the radiation resistance of the antenna.

b. Figure 4-14 shows the variation in radiation resistance for half-wave antennas at various heights above ground. The curves in this figure have been plotted for antennas erected above a highly conducting ground. If an imperfectly con-

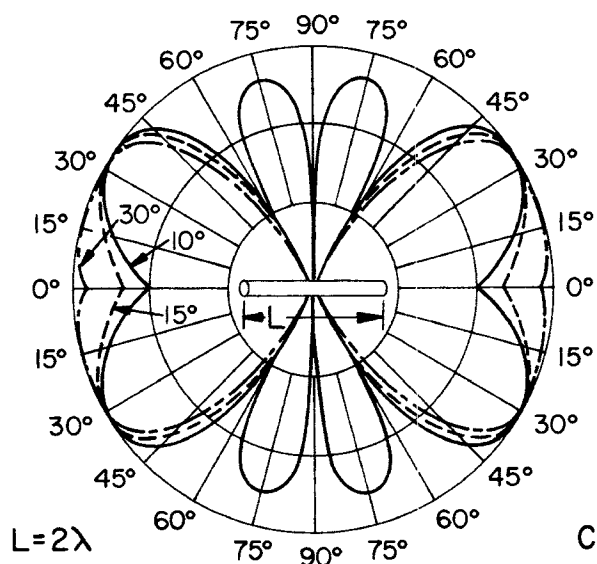
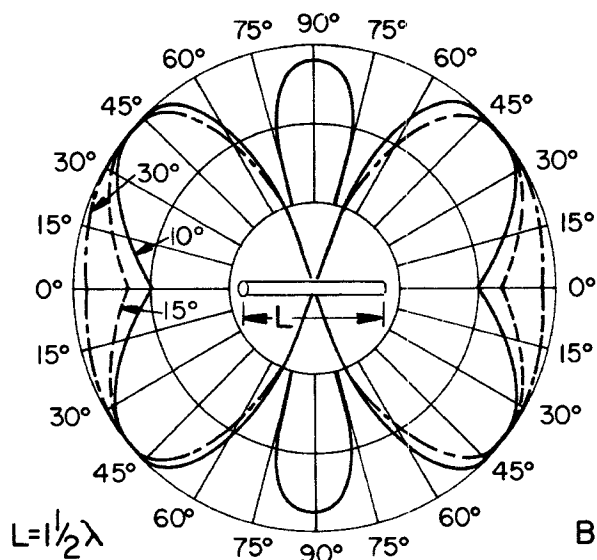
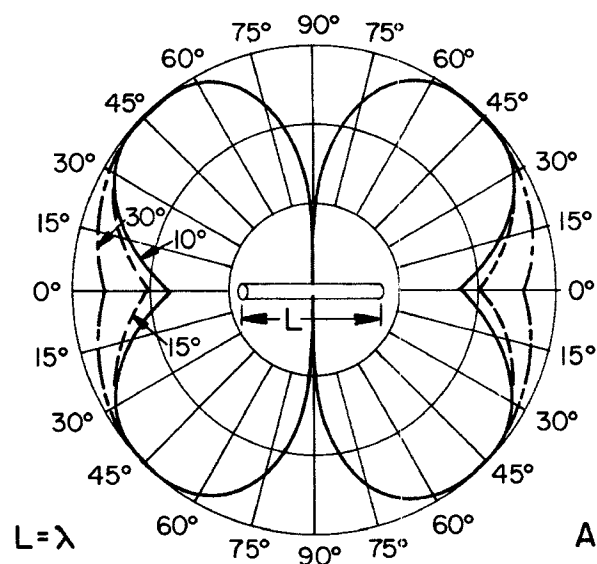


Figure 4-13. Horizontal patterns of harmonic antenna at various vertical angles.

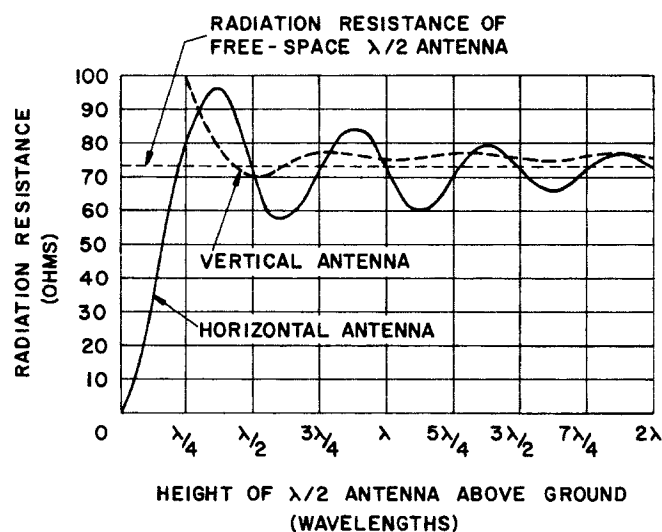


Figure 4-14. Effect of ground on radiation resistance of half-wave antenna.

ducting ground is used, the curves are modified. In general, the use of an imperfectly conducting ground shifts the curves shown slightly toward the left. In addition, the curves do not rise to as high values nor do they fall to as low values as when a highly conducting ground is used. The effect is to smooth out the curves and reduce the amount of change in radiation resistance as the antenna height is increased above ground. If appropriate, allowance should be made for the increase in effective height caused by penetration of the radio waves into the earth (para 4-15a).

4-17. Antenna Tuners

a. An antenna tuner is very useful in meeting many antenna feed requirements which occur under field conditions. By using an antenna tuner, it is possible to match any transmitter output or any transmission line to any desired antenna feed point. As examples, a half-wave conductor can be fed from the end rather than from the middle, or a random length of wire which does not resonate at the operating frequency can be matched to the transmitter. Many of the transportable and field-expedient antennas described subsequently in this chapter require an antenna tuner or other impedance-matching device.

b. An antenna tuner consists of variable capacitances and inductances which can be arranged in a network and tuned to provide the desired impedance match. The antenna tuner may be part of the transmitter or may be a separate device mounted on or near the transmitter cabinet. Tuning, which is accomplished at

very low power levels, lowers the VSWR to near 1:1. Tuners are available which automatically adjust the VSWR to an acceptably low level.

4-18. Commonly Used HF Antennas

a. General. A variety of practical antennas have been developed to meet various performance requirements and to suit various operating conditions. The more elaborate, higher performance antennas are usually used only at fixed permanent stations. Transportable stations usually use simple antennas and sometimes may use expedient versions having reduced effectiveness.

b. Horizontal Half-Wave Dipole.

(1) The horizontal half-wave dipole, which formed the basis for much of the theoretical discussion in the early portion of this chapter, is a very serviceable operational antenna. Its limited bandwidth (about 5 percent) is a handicap for general application. However, it is useful for communicating by skywave over short and moderate distances if only a single frequency or a narrow band of frequencies is involved. Even though several antennas may be needed because of frequency requirements, its use may be economical. The radiation pattern is bidirectional, with maximum lobes at right angles to the antenna axis. At heights less than a quarter-wavelength the antenna radiates and receives well at the steep angles involved in skywave transmission over

distances of 400 km (250 mi) or less. At these heights the orientation of the antenna is not important; it has little azimuthal directivity. At heights approaching a half-wavelength, the antenna is suitable for greater distances (up to 2,000 km, or 1,200 mi), but it then exhibits a bidirectional radiation pattern. The antenna is not ideal for extremely great distances. Horizontal half-wave antennas that are fed in the center are called *doublets* (fig 4-15).

(2) Because of its simplicity, the doublet is ideal for transportable applications. Its radiation pattern is well suited to short and medium range skywave communication. The antenna can be supported from two poles, or from trees or buildings. In a form specially designed for field use, the antenna consists of two reels of phosphor-bronze wire rope stored in a cylindrical housing. For operation at a particular frequency, each wire is pulled out of the housing to the correct length as indicated by a dial calibrated in frequency. The wires are secured at the desired length with wire-lock knobs mounted in the housing. The end of each wire is attached through a swivel to a length of nylon rope. The antenna is then raised to the desired height by passing each rope over a tree limb, tower, or other vertical structure. A ferrite balun contained in the housing transforms the balanced dipole circuit to a 50-ohm unbalanced line. Connection to the radio equipment is made with standard 50-ohm coaxial cable. The antenna

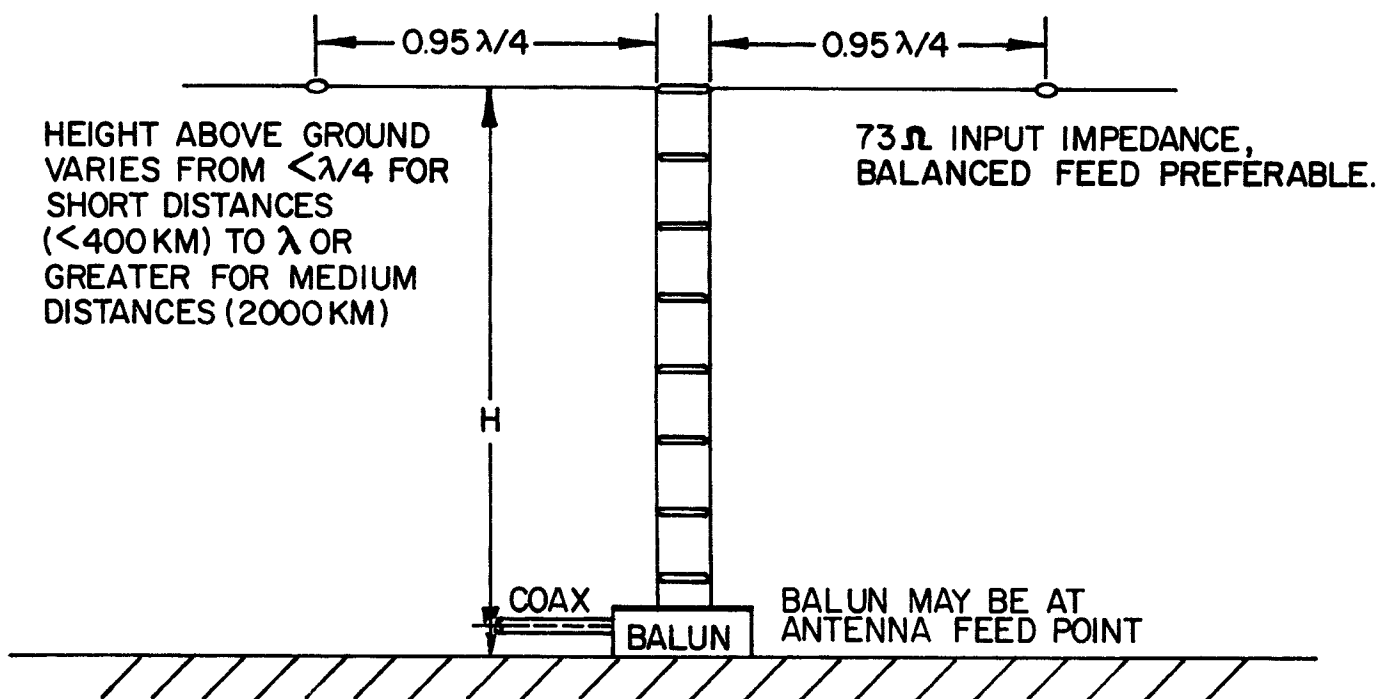


Figure 4-15. Doublet antenna for field use.

weighs less than 5 pounds and can be deployed by one man in 5 to 20 minutes.

(3) The transportable doublet can be furnished in the form of sufficient length of radiating wire and many spare insulators, making it easy to devise an antenna whose resonant frequency can be changed easily. The concept of such an antenna is shown in figure 4-16. A glass or ceramic insulator is used at the feed point. However, in an emergency any nonconducting material such as rope, wood, plastic, or leather may be used as long as it is moisture free. In wet weather, only good quality insulators can be used. To facilitate the changes in length required to accommodate frequency changes, insulators are placed in the antenna at points corresponding to the desired resonant frequencies. Jumpers are placed across insulators in order to yield the operational length required. When it is necessary to change frequency, the antenna is lowered and the jumper arrangement is changed. The configuration shown in figure 4-16 corresponds to the length designated as L_3 . The formula for length on figure 4-16 can also be used for a dipole for a single frequency.

(4) Antenna support can be furnished by means of ropes tied to either end of the antenna and to supporting structures such as buildings, trees, poles, etc. For best results, keep the antenna arms well away from foliage, parallel wires, or any other types of conductors.

(5) Guidance about antenna height is fre-

quently expressed in terms of wavelength or fraction of a wavelength. Figure 4-17 may be used to convert operating frequencies into wavelengths or fractions of wavelengths. As an approximation, a height of 10 meters (33 ft) will work reasonably well for most of the hf band.

c. Horizontal Half-Wave Dipole Variations.

(1) Sometimes it is necessary to depart from the standard doublet configuration because of lack of space or lack of supporting structures. Some degradation in antenna performance results from such variations. To keep the degradation to a minimum, the central portion of the antenna should be the proper height above the ground. Most radiation takes place from the central portion, where the current is high.

(2) Loading can be very effectively used on a doublet. By careful design, the size of the antenna can be reduced by one-half while the efficiency is lowered by only about 2 dB. When the need for such an antenna is anticipated, the antenna design services of USACEEIA should be requested, following the instructions provided in appendix D. The details of design are quite important in order to maintain a reasonably good efficiency with the reduced size.

(3) Another method of reducing the space requirement for a half-wave antenna is shown in figure 4-18. The figure illustrates a situation in which there is not enough space between two trees for a standard half-wave dipole. The an-

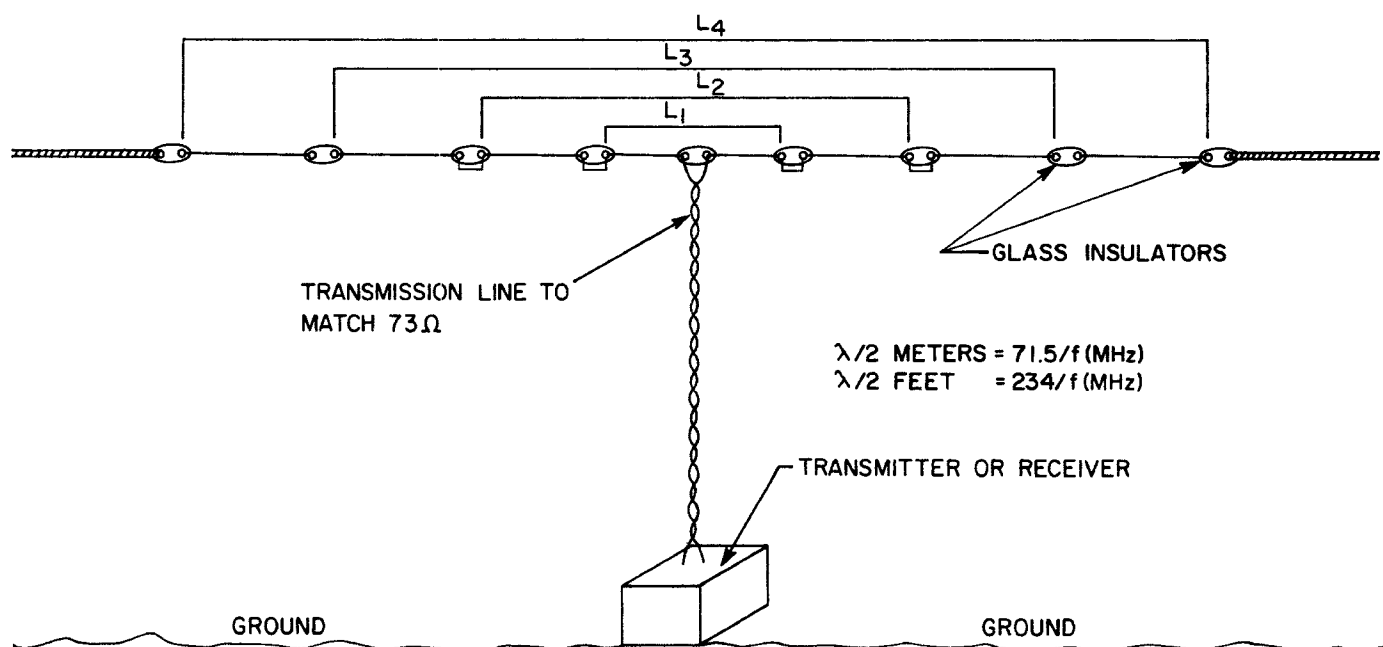


Figure 4-16. Segmented wire doublet.

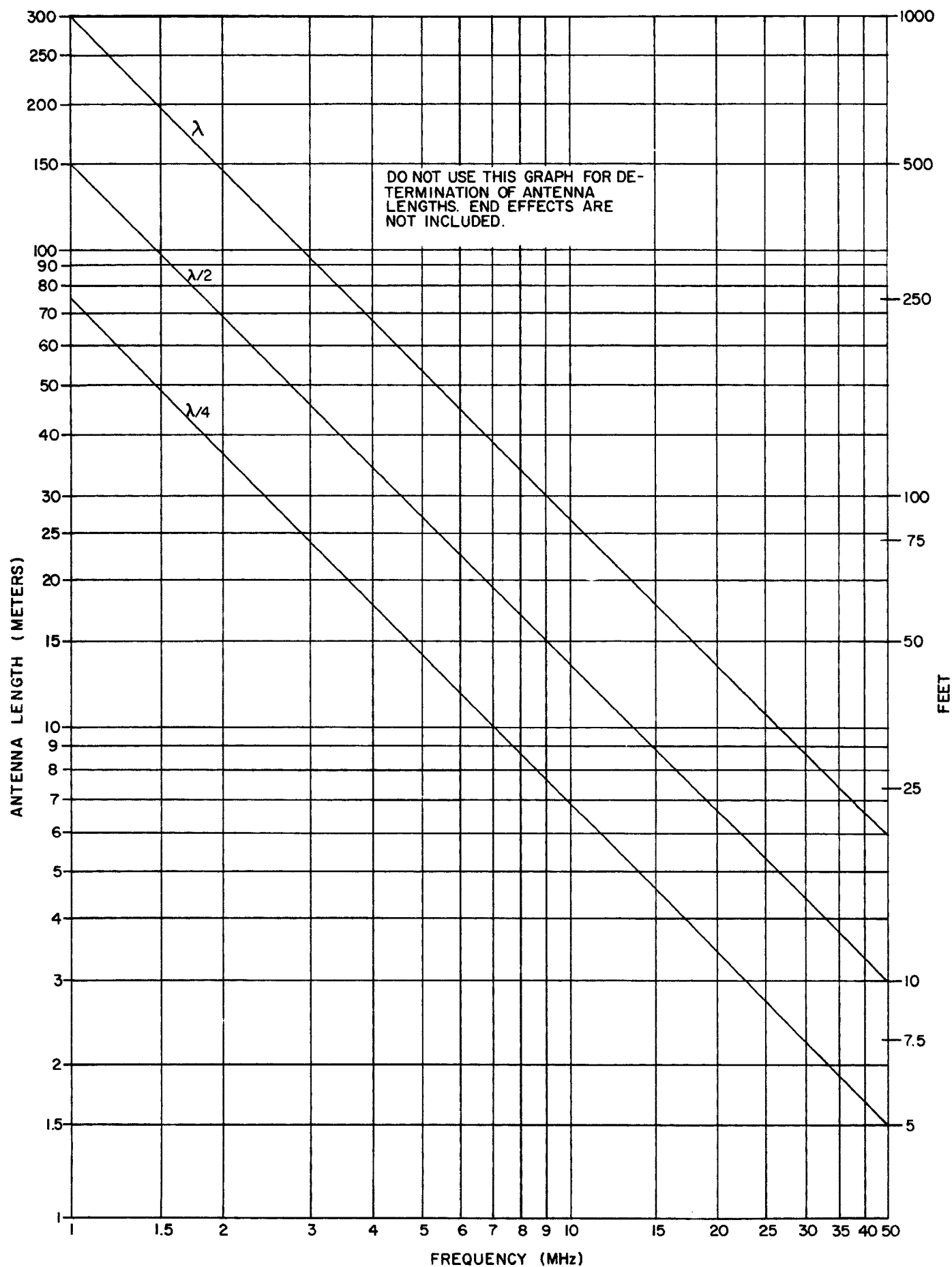


Figure 4-17. Conversion of radio frequency into wavelengths and fractions.

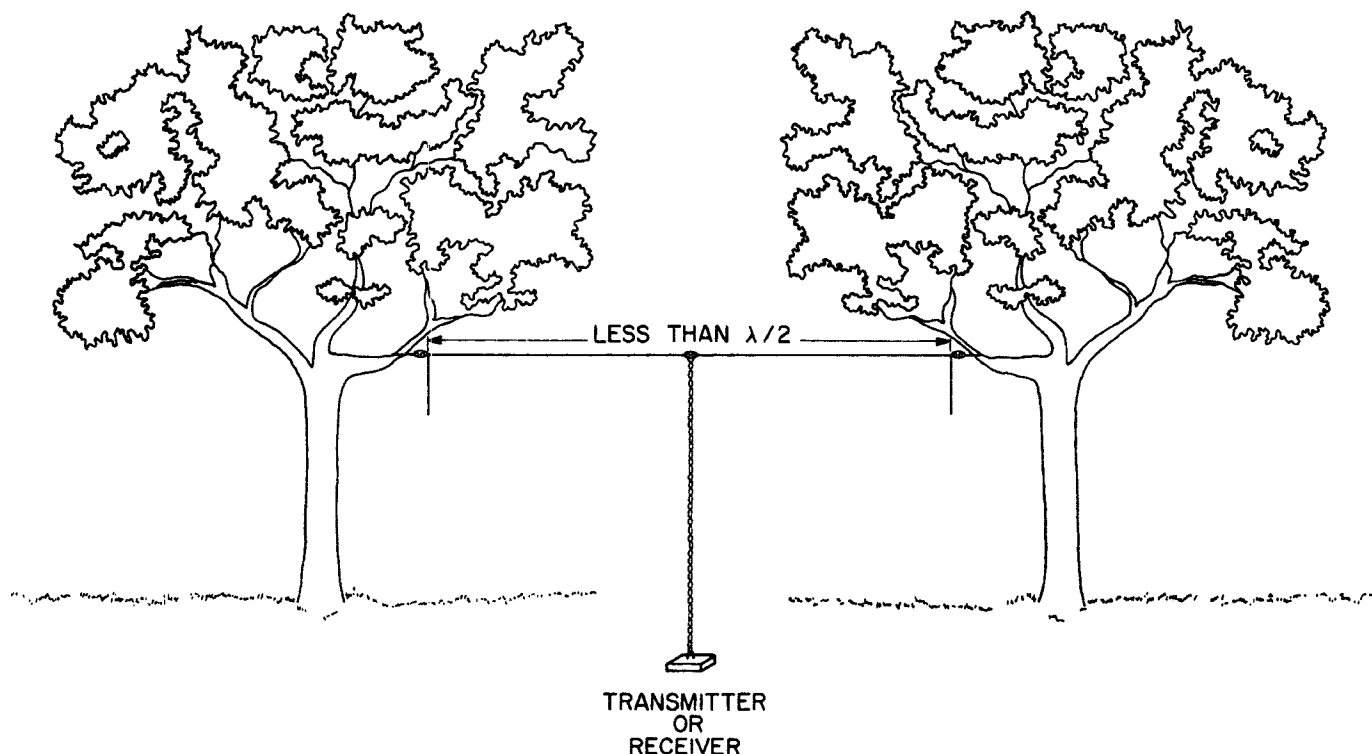


Figure 4-18. Modified half-wave dipole in restricted space.

tenna is cut to the standard length, but the ends are allowed to droop.

(4) If an antenna is to be erected outdoors where only a single tree or other natural supporting structure is present, a sloping wire antenna or the inverted V antenna can be considered. Such a situation is shown in figure 4-19. The use of an inverted V antenna places more of the important radiating portion of the antenna at the highest point. But if the tree has much foliage, a sloping wire antenna may be better if the central portion can be kept higher and in the clear.

d. Vertical Monopole.

(1) Vertical quarter-wave antennas are useful for groundwave service because they radiate vertically polarized waves at low elevation angles. Simple quarter-wave vertical antennas are resonant, the length of the grounded antenna being given by the equations—

$$H \text{ (meters)} = 71.5/f \text{ (MHz)} \text{ and}$$

$$H \text{ (feet)} = 234/f \text{ (MHz)}$$

The height is measured from the tip of the antenna to the point of connection to ground or to a counterpoise. The antenna may be a self-supporting conductor such as a metal rod, a structure supported by insulated guys, or a wire supported by rope. The antenna may be taped to a wall as an expedient measure. Installation need

not be stripped from the wire except for connection to the transmitter, receiver, or transmission line. The antenna will perform just as well with the insulation in place.

(2) The monopole may be insulated from the ground and fed at the base, or it may be connected to the ground and fed a short distance above the base. A good ground is essential for effective operation of the monopole. Unbalanced feed is required.

(3) At the upper end of the hf band (30 MHz), a resonant monopole is 2.4 m (7.9 ft) long. Hence, self-resonant monopoles are practical at the upper end of the hf band. For lower frequencies, loading is usually utilized in mobile operations and at transportable stations. For other than mobile operation, a sloping antenna cut to resonance according to one of the formulas in (1), above, can be used. A sloping antenna is useful in the vicinity of 2 MHz where groundwaves are very effective, but a quarter-wave-length antenna is about 35 m (115 ft) long.

e. Top Loaded Antennas.

(1) Forming the radiating conductor in the shape of a physical inverted L, as shown in figure 4-20, is a method of reducing the height of a vertical quarter-wave antenna. The top portion of the antenna can be bent over in a horizontal direction without severely altering its resonant

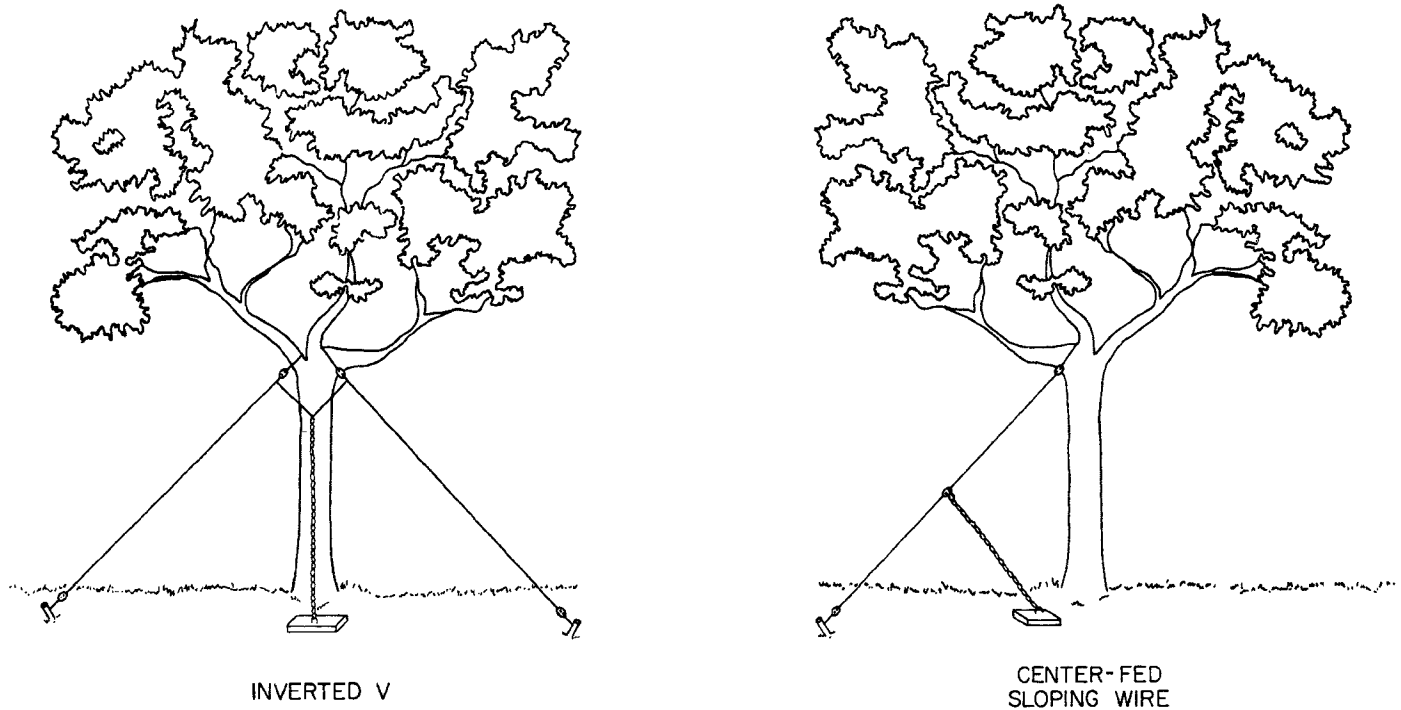


Figure 4-19. Use of a tree for antenna support.

frequency. The portion of the antenna which is bent into a horizontal direction is removed as a contributor to the desired radiation pattern. Because the current is smallest at the end of the wire away from the transmitter, bending this portion away from the vertical will have the smallest degrading effect on performance. No more of the wire than is absolutely necessary should be bent over. Often one or more additional wires are connected at the point where the bend in wire occurs in order to add more capacitance to the vertical portion of the wire. Loading the antenna in this fashion allows it to be tuned to resonance at the operating frequency.

(2) Various other examples of top-loading are also shown in figure 4-20. The schemes shown are arranged in the order of preference, with the best arrangement on the left. The "cross" is preferred because it permits the greatest amount of top-loading in the smallest space, but does require room to extend wire in all directions from the antenna. There is no limitation on the number of wires which can be used to top-load. Each additional wire has less effect than the preceding one, however. The "Tee" is simply a two-wire version of the cross. The "crow's foot" is a compromise version of the cross used when it is impossible to deploy the wires in every direction about the azimuth plane. Spreading the wires of the crow's foot apart as far as possible tends to increase the top-loading effect. The inverted L

represents the simplest method of top-loading. The length of the vertical and horizontal wires together should total a quarter-wave. The last two examples shown on the figure should be avoided, if possible, because the downward flow of current in the top-loading wires tends to partially cancel the radiation current in the monopole.

f. Whip Antennas.

(1) Whip antennas are the most commonly used antennas for field radio communication over relatively short distances with portable and mobile radio equipment. Most whip antennas are constructed of sections of metal tubing that can be taken apart or telescoped when not in use. In this way, the antenna has a minimum length during transit and portability is increased. In some lightweight portable equipment, the antenna can be collapsed completely into the equipment itself so that none of it is exposed. Although there is no definite upper limit on the physical length of mobile whips, few are longer than 18 feet. Whip antennas in the hf band are usually less than a quarter-wavelength long, and are loaded to obtain resonance.

(2) When used with a suitable ground as an antenna for a transportable station, the whip's characteristics are similar to those of other vertical antennas. When a whip is mounted on a vehicle, the metal of the vehicle affects the opera-

tion of the antenna. As a result, the direction in which the vehicle is facing may also affect transmission and reception, particularly when distant stations or other weak signals are involved. A vehicle with a whip antenna mounted on its left rear side will transmit its strongest signal in a line running from the antenna through the right front side of the vehicle. Similarly, an antenna mounted on the right rear side of the vehicle radiates its strongest signal in a direction toward the left front side (fig 4-21). The best reception is obtained from signals traveling in the directions shown by the broken lines. In some cases, the

best direction can be determined by driving the vehicle in a small circle until the best position is located. Normally, the best direction for receiving from a distant station is also the best direction for transmitting to that station.

(3) A vertical monopole is not ordinarily used for short-distance skywave service because the radiation pattern favors low angles. However, if a whip antenna is bent as shown in figure 4-22, its ability to receive and transmit high-angle skywaves is enhanced.

g. *Inverted L Antenna.* One form of inverted L antenna, in which the horizontal section of the

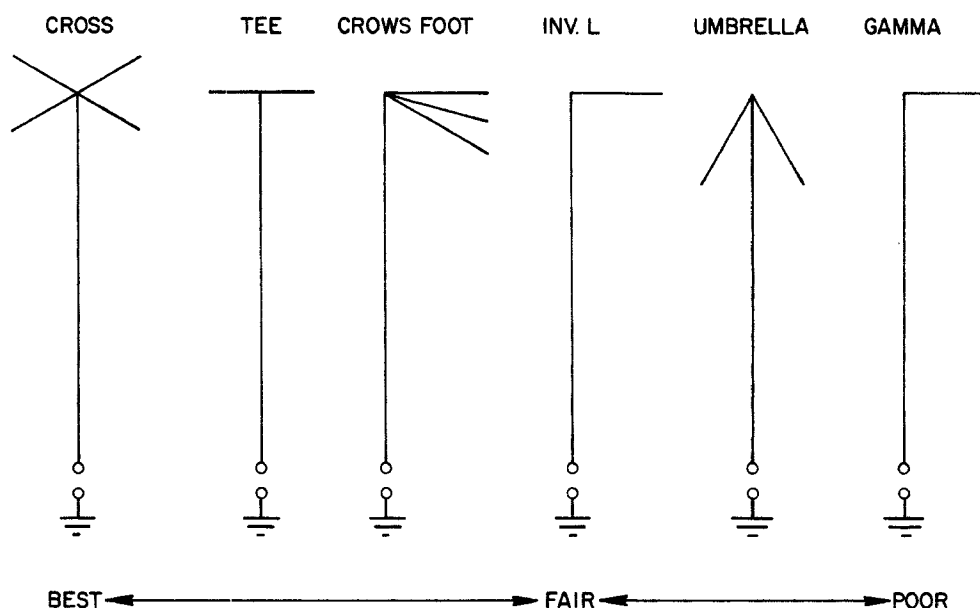


Figure 4-20. Top-loading of vertical antenna.

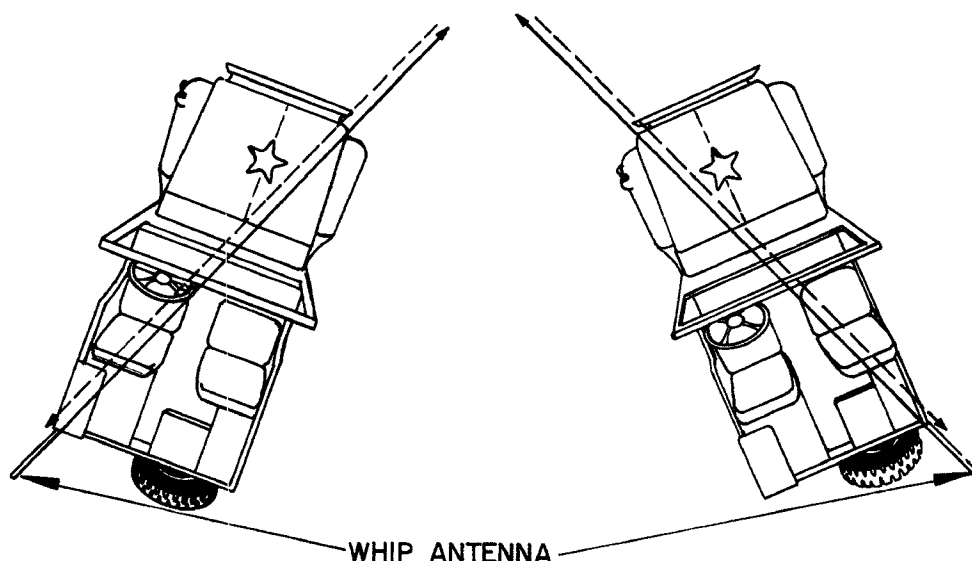


Figure 4-21. Directional effects of vehicle on whip antenna.

antenna is used primarily as a means of top loading the vertical section, was discussed in *e* above. In another common form of inverted L antenna, the horizontal section is made long enough to provide, in addition to top loading, a useful amount of horizontally polarized radiation for use on short and medium distance sky-wave paths.

h. Horizontal V Antenna. The horizontal V antenna consists of two wires arranged in the form of a V as shown in figure 4-23. It is a low-angle, medium-gain, highly directive, broadband antenna of simple, inexpensive construction. It can be designed to handle high power. The two wires are fed at the apex of the V by means of a balanced line or balun, so that equal currents of opposite phase flow in corresponding parts of the two wires. The apex angle is so chosen that the main lobes of the wires reinforce each other along the bisector of the apex angle. The lobes

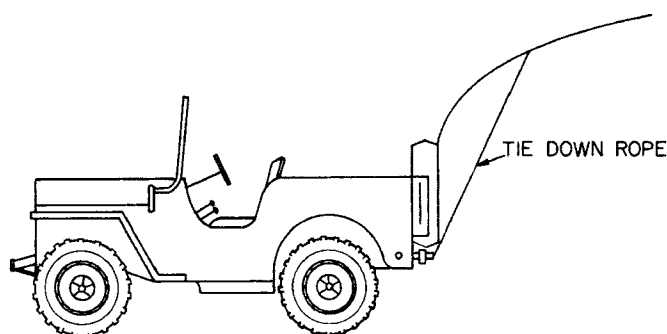


Figure 4-22. Whip antenna tied down for high-angle skywave service.

in other directions tend to cancel. The V antenna may be made unidirectional by terminating the far end of each leg to ground with a noninductive resistor. Each resistor should have a value of approximately 500 ohms and should be capable of dissipating at least a third of the power supplied to the antenna. An excellent ground should be used on each terminator (para 4-19d).

i. Sloping V Antenna. The sloping V antenna (fig 4-24) is a very useful field antenna. It has the advantage of requiring only a single support, which should have a height of about 23 meters (75 ft) and not less than 15 meters (49 ft). The best practice is to elevate the terminated ends about 8 meters (26 ft) and not less than 4.5 meters (15 ft). Each terminating resistor should be noninductive, have a resistance of 360 to 400 ohms, and be capable of dissipating one half of the power supplied to the antenna. An excellent ground should be used on each terminator (para 4-19d).

j. Rhombic Antenna. Where space is available, the rhombic antenna is often the best choice for long distance links. The rhombic antenna consists of four spans of wire of equal length, arranged in the form of a diamond, or rhombus, supported in a horizontal plane. A rhombic antenna is shown in figure 4-25. The important rhombic design parameters are the length of each span, L , the tilt angle, ϕ , and the height above ground, H . Varying these parameters changes the radiation pattern of the antenna. The tilt angle, ϕ , is the angle each span makes with the minor axis, as shown in the figure; it

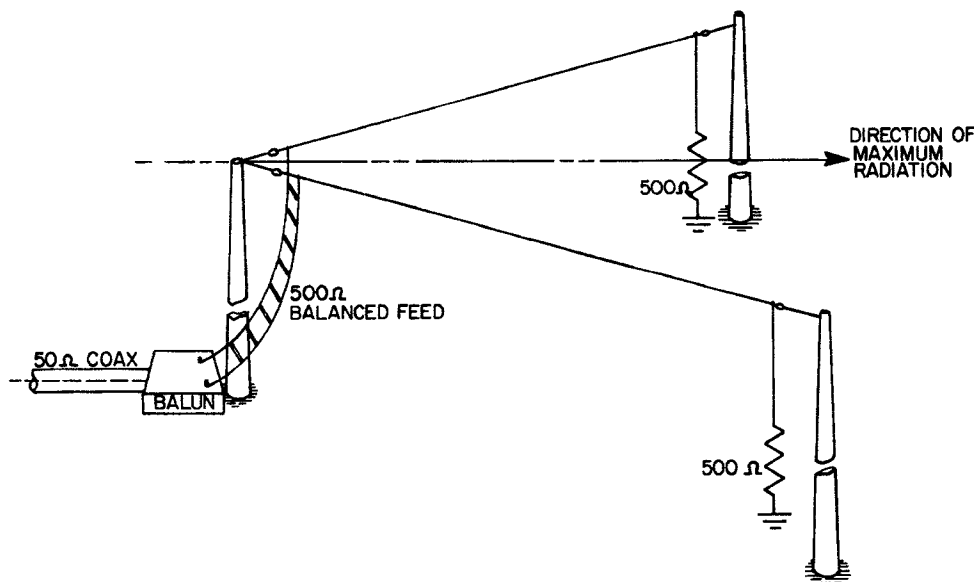


Figure 4-23. Horizontal V antenna.

RANGE (MILES)	FREQUENCY BAND	APEX ANGLE (α)
500-1500	4-15 MHz	40 DEGREES
1500-2500	4-20 MHz	35 DEGREES

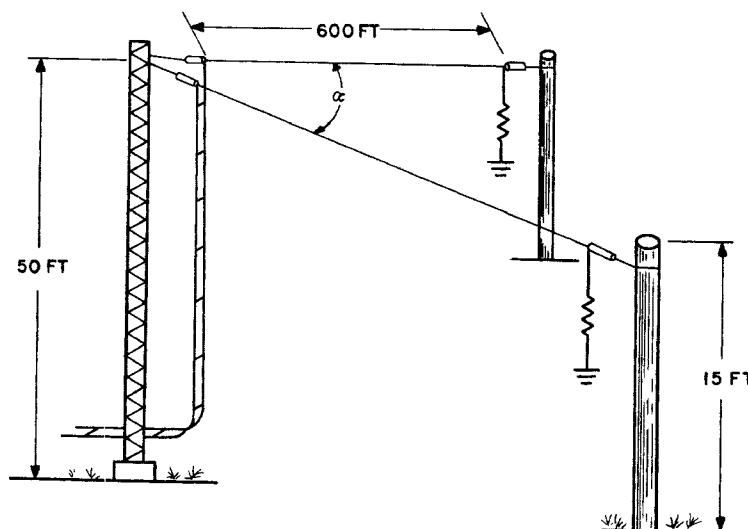


Figure 4-24. Sloping V antenna.

is not the angle of maximum radiation. The antenna is fed at the acute angles at one end and usually terminated at the opposite end by a resistor or dissipation line which matches the characteristic impedance of the antenna (about 500 ohms). This termination should be capable of dissipating one-third of the input power to the antenna. This termination makes the rhombic antenna unidirectional, with direction of maximum propagation towards the terminated end. If bidirectional characteristics are desired, the termination is omitted. Each leg of the terminated rhombic antenna has its own radiation pattern as shown in figure 4-26. The radiation patterns of all four legs combine with each other, and with the reflection from the ground, to produce one overall main pattern for the antenna as shown in figure 4-27. Some advantages of the rhombic antenna in hf applications are: the radiated energy can be concentrated in one major lobe to provide a high order of gain in the direction of transmission or reception; the major lobe can be directed at the low vertical radiation angles necessary for long distance coverage; the characteristics

of the radiation pattern remain satisfactory over a wide frequency range, typically four to one. The chief disadvantage of the rhombic antenna is that it requires a large area. Where space is limited, the rhombic performance can be approached by an appropriate log-periodic antenna.

k. Vertical Half-Rhombic Antenna. A vertical half-rhombic, shown in figure 4-28, has the advantage that only a single support is required. When terminated, the half rhombic is unidirectional and broadband. Due to the low efficiency of the antenna at low frequencies, the terminating resistor should be capable of dissipating all the transmitter output power.

l. Log-Periodic Antenna. The log-periodic antenna approaches the performance of a rhombic and has the advantage of requiring less space. Bandwidths over a frequency range of ten to one can be achieved with a practical log-periodic antenna. A wide variety of log-periodic antennas are available.

(1) The horizontal log-periodic dipole antenna consists of parallel and coplanar elements

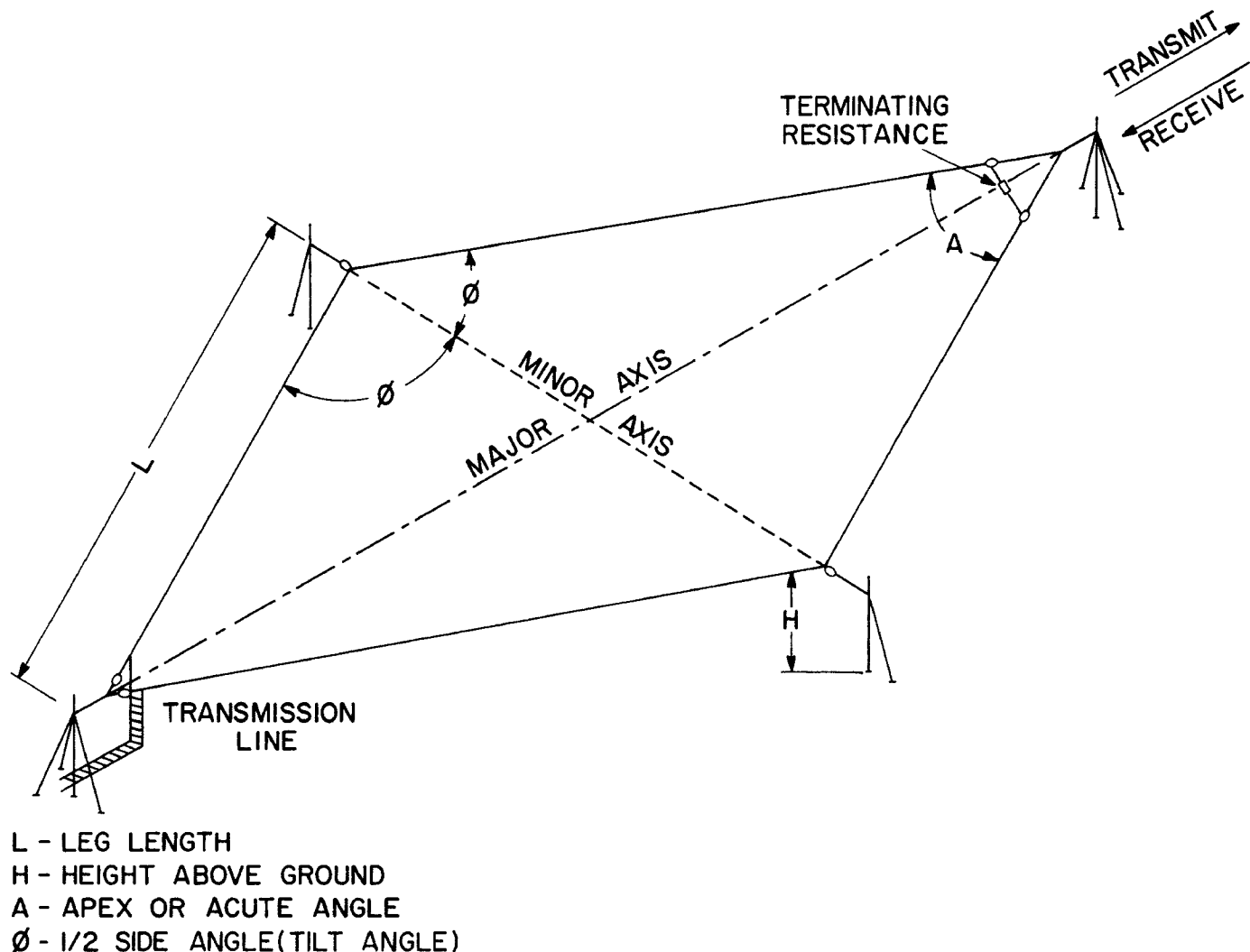


Figure 4-25. Rhombic antenna.

in which the element length and spacing are logarithmically periodic, proceeding from the small apex out to the longest elements (fig 4-29). The low-frequency cutoff occurs when the longest dipole is approximately one-half wavelength long. The high-frequency cutoff occurs when the shortest dipole is about a quarter-wavelength long. At any frequency between the cutoff frequencies, the antenna currents are largest in the several elements that are about a half-wave long. These elements contribute most of the radiation and form the active region of the antenna. This antenna is fed with a balanced two-wire line entering at the apex and running through the center of the structure, transposed between adjacent elements so that all adjacent elements are fed 180° out of phase. The height, H , of the shortest element is about 3 meters. A typical array slope angle is approximately 30° above the horizontal.

(2) A vertical log-periodic antenna may be of either the dipole or monopole type. The dipole

type is very similar to the horizontal log-periodic except for the method of suspension. A common type of vertical log-periodic antenna arrangement is composed of monopoles (one-half of the dipole system). This requires an extensive ground system to provide the image equivalent of the other half-dipoles. A vertical log-periodic requires only one tall tower and is easier to install than the horizontal (fig 4-30). A log-periodic monopole array cannot be considered as simply one-half of a dipole array fed against ground. This is because the transmission line cannot be transposed from one element to the next as is done in the dipole array. The solution is to provide the proper phase shift between elements by other means, involving lumped constant T-sections that use series inductance and shunt capacitance between elements. These sections also act as filters near cut-off to confine the currents in the array to the elements that are nearly a quarter-wave long. An unbalanced feed is required.

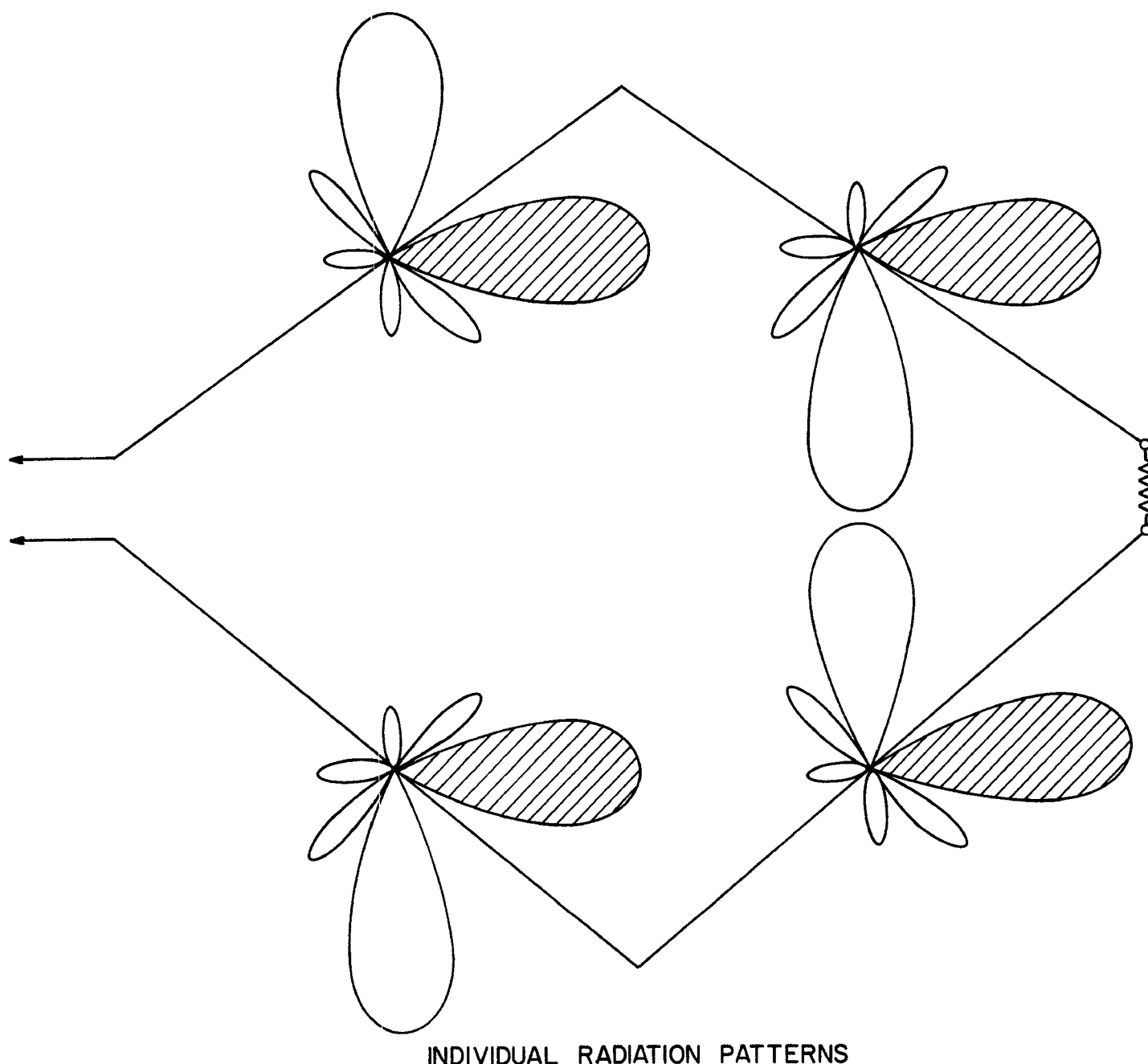


Figure 4-26. Radiation patterns of legs of rhombic antenna.

(3) Rotatable log-periodic antennas are available in transportable configurations. Rotatable log-periodics are of the horizontal type, as shown in figure 4-31.

m. Long-Wire Antenna. A low wire many wavelengths long makes an effective antenna which is highly directional in the general direction of the wire. Gain and directivity increase with length. Long-wire antennas are characteristically about 300 m (1,000 ft) long. The wire may be 1 to 6 m (3 to 20 ft) high, the lower wire

having the advantage of low visibility. Figure 4-32 shows a schematic of a long-wire antenna. When terminated, the antenna is nonresonant and unidirectional. The input impedance of the antenna is about 500 ohms which is about the characteristic impedance of a single-wire transmission line with ground return. A long-wire antenna is also called a Beverage wave antenna. Due to the low efficiency of the antenna, the terminating resistor should be capable of dissipating all of the output power of the transmitter.

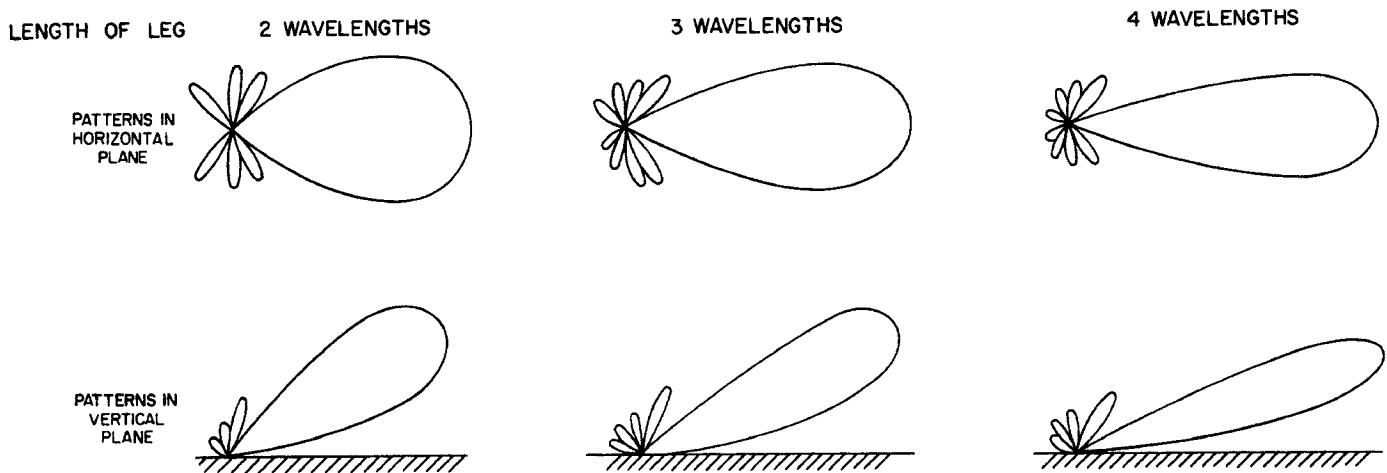


Figure 4-27. Radiation patterns of rhombic antenna.

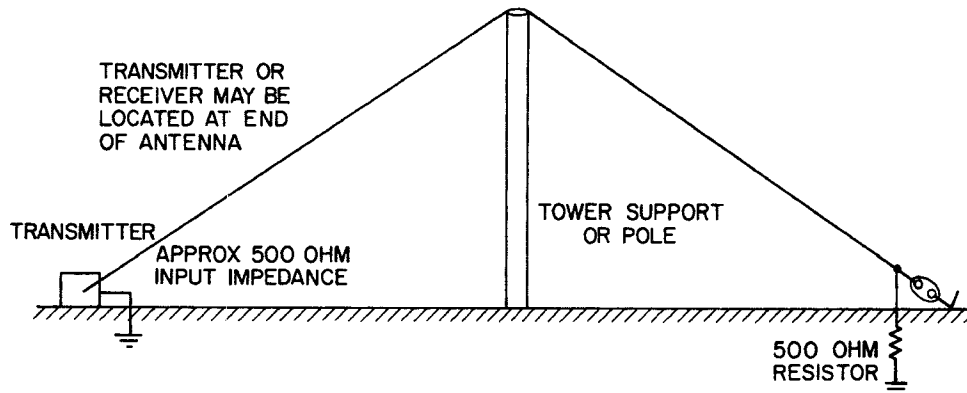


Figure 4-28. Vertical half-rhombic antenna.

n. Random Length of Wire As an Antenna. In an emergency, any random piece of wire may be used as an antenna. To the extent possible, the random wire should be in the clear, and be configured so as to optimize the polarization desired. The performance of a random piece of wire will usually not approach that of a well designed antenna, but may be adequate for temporary requirements. The use of the makeshift antenna may occur merely during the period while a more suitable antenna is being constructed. When the antenna is used for transmitting, an antenna tuner must be placed between the antenna and transmitter to provide proper impedance match.

o. Summary of Antenna Characteristics. Table 4-1 summarizes the important characteristics of the practical antennas discussed in this chapter.

4-19. Antenna Grounding

a. General. It is important in the design of an antenna to determine the extent of its ground system. In general, a ground system will be re-

quired for a ground-mounted antenna if the antenna is fed in a manner which makes the ground the return path for the current flowing in the antenna. A ground system thus provides a low-loss path for this return current and acts to improve the radiation efficiency of the antenna. Lightning arresters may be required if occasional lightning storms prevail in the area where the antenna is located. Ground systems for improvement of antenna efficiency may also serve as a portion of the system required for lightning protection, although the converse is not necessarily true.

b. Radial Ground Systems

(1) Experience has shown that the most practical configuration of wires in an antenna ground system is a radial distribution with its center at the base of the vertical portion of the antenna. This arrangement is dictated by the radial nature of the paths followed by returning ground currents. If site factors require crossing of radials, they should be insulated at the cross-

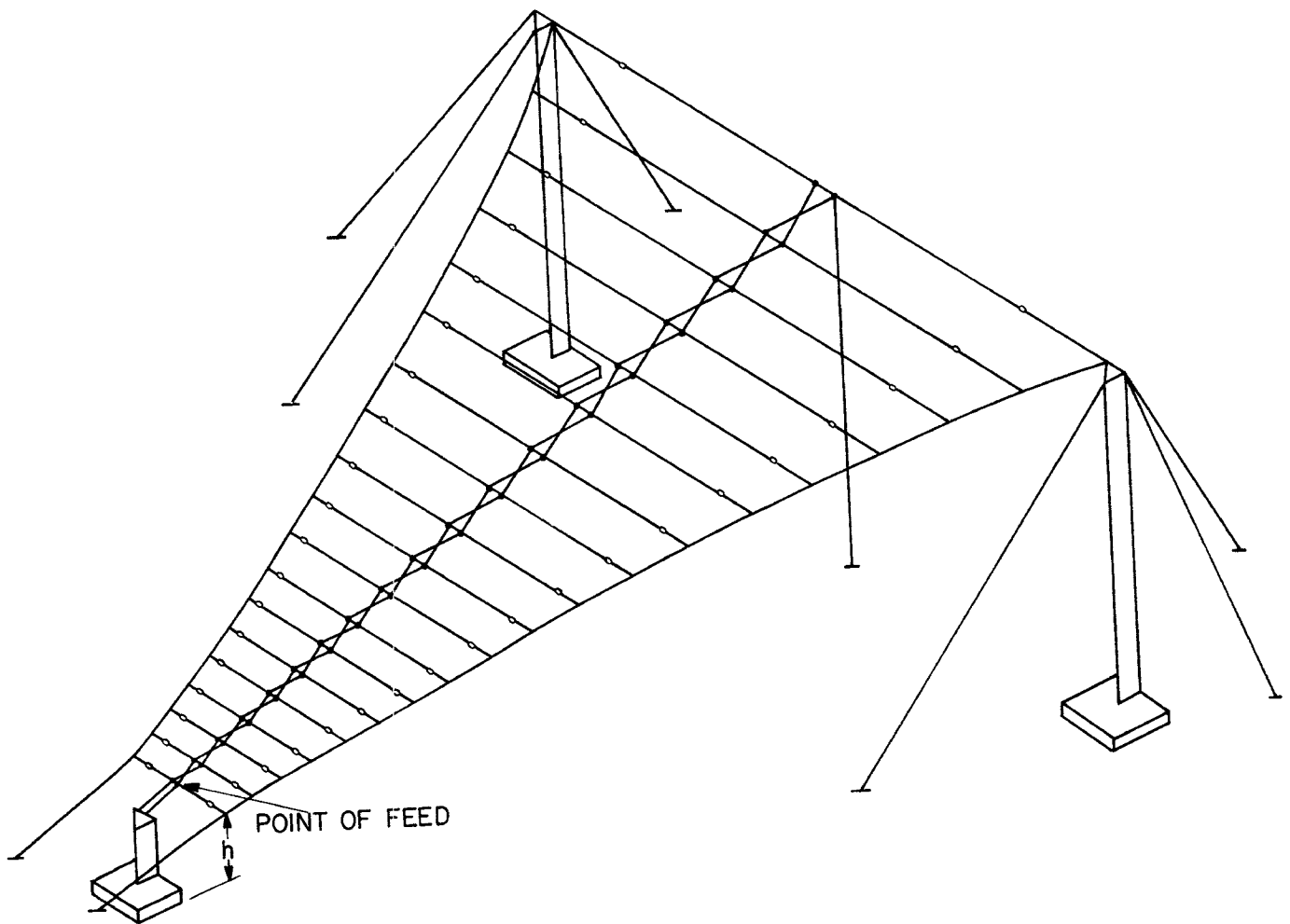


Figure 4-29. Horizontal log-periodic antenna.

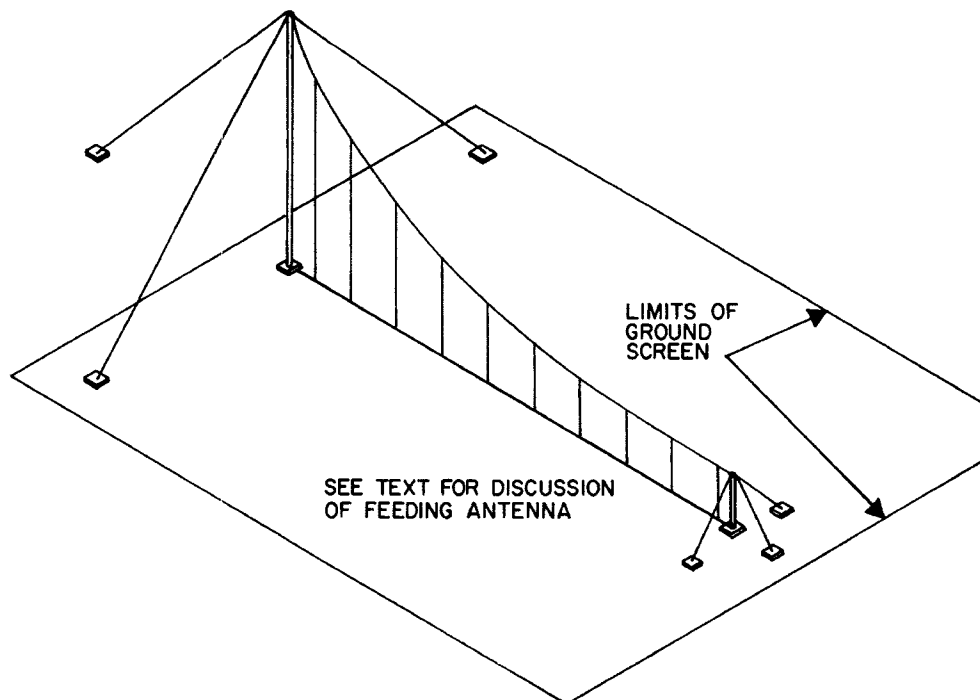


Figure 4-30. Vertical log-periodic monopole antenna.

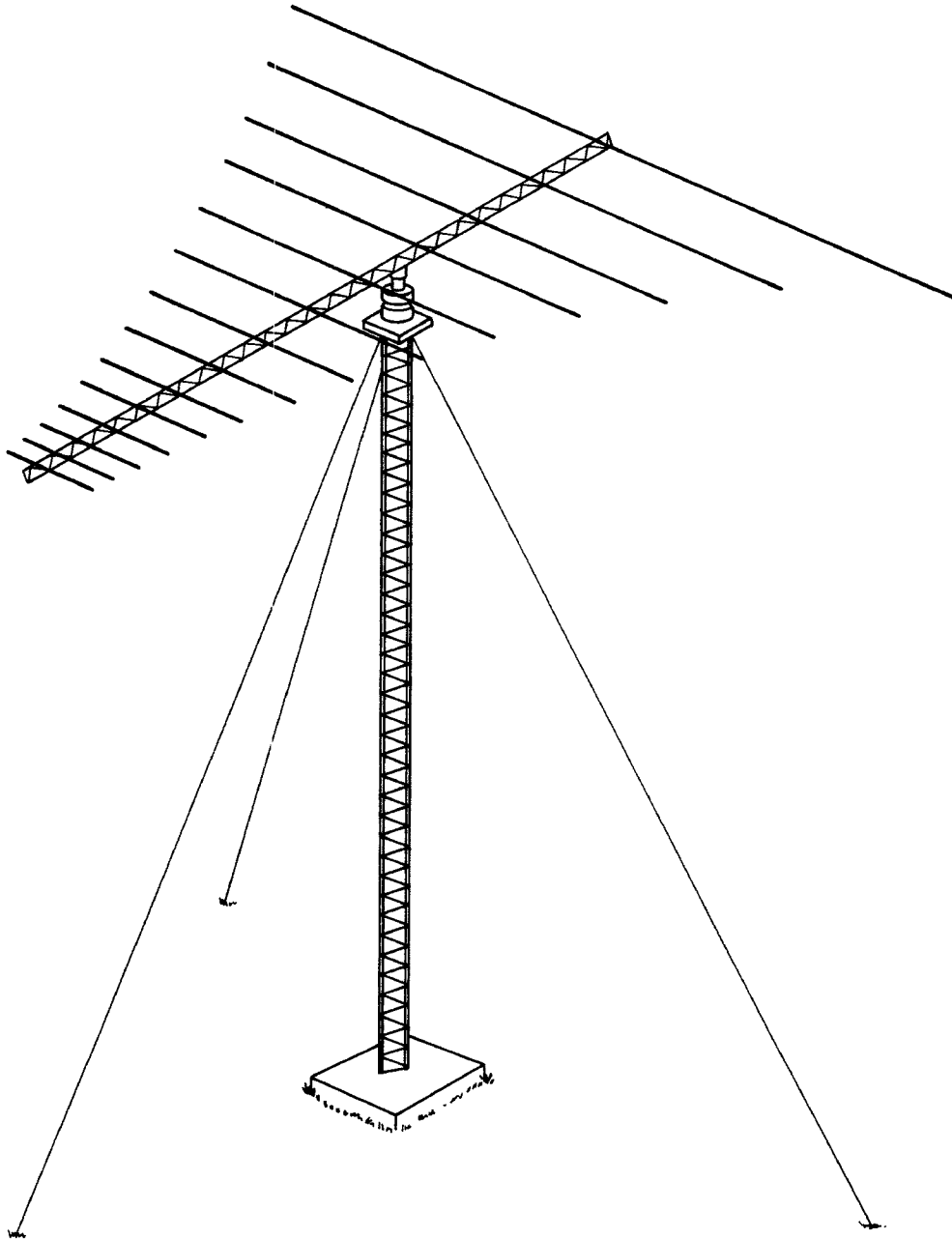


Figure 4-31. Rotatable log-periodic antenna.

over points. The radials should lead out from the antenna base in a straight line.

(2) Vertical antennas, which are often located with the antenna feed point at the ground surface, require a system of radial ground wires extending from the antenna to a sufficient distance to provide a low-resistance return path for the ground currents produced by the induction fields. For short vertical antennas, the radial length should be approximately equal to the wavelength divided by 2π ; for longer antennas the length is approximately equal to the antenna height. In addition, near the antenna base, where

the fields are intense, a grid-type ground mat is frequently used.

(3) It is recommended that No. 8 or No. 10 AWG annealed copper or copper-clad steel wire be used for the radials. If more copper is desired in the ground system, it is best to lengthen the radial wires rather than to increase the cross-sectional area, unless the antenna current has a large value, causing high current densities in the individual wires. In general, the wire of the radial conductors has little effect on the performance of the ground system. When the above conditions are met, the gage need only be large enough to with-

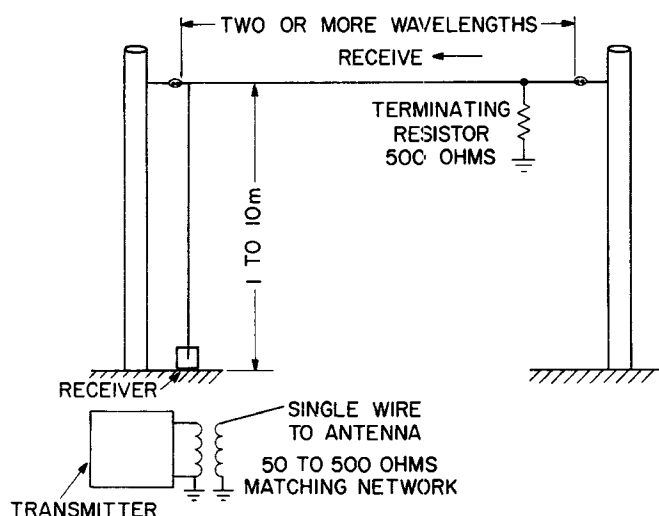


Figure 4-32. Long-wire antenna.

stand the mechanical stresses of the burying process.

(4) If the plot of ground that is to be used is not large enough for the circular layout, then compromises must be made. Under certain boundary conditions the radial wires may not be the required optimum length and ground rods will have to be used. These ground rods should be driven into the earth at the end of the shorter radials to a depth equal to the skin depth to minimize ground loss. For the range of frequencies which are of interest, computed skin depths indicate that rod depths of 30 feet would be required for normal soil conditions and depths of 45 feet or more for poor soil conditions. From a practical standpoint, however, it will usually be adequate to drive a 10-foot rod to terminate the ends of such radials. If the property is so small as to limit all of the radial wires to a length shorter than optimum length, then ground rods and peripheral bonding should be used.

(5) If the radials are optimum length, sufficient to have zero current at the end of the wires, there is no need for ground rods. The test for this is to drive one ground rod, connecting one of the radial wires to it, and measure the current distribution along the wire. If an appreciable current exists at the end of the wire, a ground rod is necessary. The same applies to the circular bonding wire around the perimeter of the ground system. It is only when the radials are too short that it is necessary to use circular bonding.

(6) The depth of the wire is immaterial, whether the soil is moist or dry. HF antennas requiring a ground system perform best with ground radials on the surface of the soil. Bury

radials only as deep as necessary for physical protection. The wires should be brought to the surface near the base of the antenna so that a good ground screen is formed above the soil where the electric field strengths are high.

(7) In some instances, the earth's surface under the antenna is largely rocky in nature so that radials cannot be buried or ground rods driven. In this event, radials should be laid on the surface and staked at frequent intervals to keep them in place. Shifting of grids on the surface of soil or rock formation during high winds is detrimental to signal-to-noise ratio.

c. Ground Screens. When high base currents are encountered, it is desirable to use a copper mesh ground screen at the base of an antenna to further insure against ground system power loss. The ground screen is normally covered only to the extent necessary to protect it against physical damage. Typical screens have dimensions of 12 by 12 feet or larger and are fabricated of expanded copper mesh or a grid of welded copper wire. Each radial wire must be carefully bonded to the screen. In all bonding operations only silver soldering, brazing or copper welding should be used.

d. Antenna Termination Grounds. In terminating non-resonant antennas, BALUN cases, and antenna masts, it is recommended that a star-ground be used. Five or six ground rods (preferably, at least eight feet long) should be driven into the earth in a circle around a central ground-rod. The radius of the circle should be equal to the length of the rod. The rods should be bonded to each other.

4-20. Tuning To Correct Mismatch to Antenna

a. Matching the antenna to the transmission line involves adjusting the point or points of feed and, in the case of a resonant antenna, adjusting the length of the antenna. Matching the transmitter to the transmission line is accomplished by adjusting a network which either is a part of the transmitter or is inserted between the transmitter and the transmission line. The accuracy of the match is indicated by the VSWR.

b. Monitoring the VSWR of a transmission line or individual antennas in an array without interrupting normal operation may be accomplished with a directional coupler. The coupler is permanently installed in each line or antenna input. A jack is provided for an external VSWR indicator which measures incident and reflected power.

Table 4-1. Comparison of Antenna Characteristics

Antenna types, listed by azimuthal patterns and polarization	Power gain (referred to isotropic) (dB) (Note 2)	Usable radiation angles (degrees) (Note 3)	Land required (acres)	Power handling capability (kW pep) (Note 4)	Appropriate material cost (thousands of dollars) (Note 5)
Unidirectional patterns					
Horizontal rhombic	8 to 23	4-35	5-15	Limited by termi- nation	5-10
Horizontal terminated V	4 to 10	5-30	4-9		3-6
Horizontal $\lambda/2$ dipole	8 to 15	5-45	2-4	40	15-25
Vertical log-periodic (dipole)	6 to 10	3-25	3-5	40	20-30
Vertical log-periodic (monopole)	4 to 8	3-25	3-5	20	20-30
Horizontal $\mu/2$ dipole	5 to 7	5-80	<1	20	1-2
Vertical half rhombic	Note 7				
Inverted L	Note 8				
Long wire (Beverage)	Note 9				
Random wire	Note 10				
Omnidirectional pattern (vertical)					
Vertical monopole	5 to 7	5-45	<1	20	

Table 4-1. Comparison of Antenna Characteristics—continued

Antenna types, listed by azimuthal patterns and polarization	Nominal bandwidth ratio (Note 6)	Nominal bandwidth limited by	Horizontal beamwidth (-3 dB) (degrees)	Horizontal beamwidth (-10 dB) (degrees)	Sidelobe suppression (dB)	Transmit maximum VSWR	Nominal input impedance (ohms)
Unidirectional patterns							
Horizontal rhombic	$\geq 2:1$	Pattern	6°-26°	11°-46°	≥ 6	2:1	500
Horizontal terminated V	$\geq 3:1$	Pattern	8°-36°	11°-48°	≥ 6	2:1	500
Horizontal log-periodic	$\geq 8:1$	VSWR	55°-75°	75°-120°	≥ 12	2:1	50/300
Vertical log-periodic (dipole)	$\geq 8:1$	VSWR	90°-140°	150°-180°	≥ 12	2:1	50
Vertical log-periodic (monopole)	$\geq 8:1$	VSWR	90°-140°	150°-180°	≥ 12	2:1	50
Horizontal $\lambda/2$ dipole	$\geq 5\%$	VSWR	80°-180°/ lobe	180°/lobe	NA	1.5:1	73
Vertical half rhombic	$\geq 2:1$	Pattern	Note 7				500
Inverted L	$\geq 5\%$	VSWR	Note 8				
Long wire (Beverage)	Note 9						500
Random wire	Note 10						
Omnidirectional pattern (vertical)							
Vertical monopole	$\geq 5\%$	VSWR	NA	NA	NA	1.5:1	Note 11

Note 1. All antennas listed can be designed for any frequency from 2 to 30 MHz.

Note 2. Typical power gains are gains of antennas over good earth for vertical polarization and poor earth for horizontal polarization.

Note 3. Usable radiation angles are typical radiation angles over good earth for vertical polarization and poor earth for horizontal polarization; lower angles may be possible for vertical antennas over better earth, e.g., salt water.

Note 4. The power limitation criteria apply to the antenna only. The use of certain baluns may result in a lower power limitation. These criteria should be used as a basis of comparison only, since any antenna can be engineered, at increased cost, to provide increased power-handling capability.

Note 5. Approximate material costs include steel towers, guys, and installation hardware. Costs are established only to provide a relative basis of comparison of one antenna against another.

Note 6. Nominal bandwidth is the ratio of the two frequencies within which the stated VSWR will not be exceeded or within which the desired pattern will not suffer more than 3 dB degradation.

Note 7. No standard design for this antenna. Performance highly variable, depending upon construction details.

Note 8. No standard design for this antenna.

Note 9. No standard design for this antenna. Beamwidth becomes narrower and gain increases with increasing length.

Note 10. No standard design for this antenna. Performance highly variable, depending upon construction details. Antenna tuner required.

Note 11. Input impedance about 50 ohms; depends upon location of feed point.

Models are available for coaxial lines as well as open-wire lines. Maximum power rating on most commercial types is limited to 1,000 watts. The directions for the individual meter must be followed in determining VSWR.

c. Emergency VSWR indicators for use with open-wire transmission lines can be fabricated from parts normally available in the field. A crude indication of the presence of standing waves on a transmission line can be obtained by coupling a neon lamp or fluorescent light to one conductor of the transmission line and observing whether the lamp brilliance changes when the lamp is moved along the transmission line. On a well-matched transmission line, the brilliance should be constant regardless of location on the line. If three identical lamps are available, they can be mounted at one-eighth wavelength intervals and antenna adjustments made until all lamps have equal brilliance, indicating that the standing waves have been eliminated.

d. If the transmitter contains a meter which reads incident power and reflected power, the method described in paragraph 3-4b can be used to obtain the VSWR. If the VSWR is 1.5 or less, no action is required.

e. If the VSWR is above 3, check for bad connections or a short circuit. If the reading is between 2.0 and 3, adjustment is required. If the reading cannot be lowered below 2.0 by adjusting the balancing network, adjustments must be made at the antenna. The length of a doublet can be

varied. If the mismatch becomes worse when the frequency is lowered a bit, the antenna requires lengthening. If it becomes better, shortening is needed. The point, or points, at which the transmission line is connected to the antenna can be adjusted. By a combination of all adjustments, obtain the lowest reading. Anything below 2.0 is generally acceptable although 1.5 is preferred.

4-21. Antenna Siting

a. In selecting an antenna site, the terrain and the height of the antenna should be chosen in such a way that the ground-reflected wave reinforces the direct wave at the desired angle. Also, a site should be selected in which there are no obstacles to interfere with the transmitted or received radio wave. For receiving sites, there are the additional considerations of interfering radio signals and manmade noise. In selecting the site for a fixed station, every step practical should be taken to ensure optimum conditions. In establishing field stations, the best site available under the circumstances should be chosen.

b. A satisfactory horizon clearance exists when no obstruction exceeds half the elevation angle of the radio beam in that direction. If the elevation angle of the transmitted or received beam in a given direction is 10° , the horizon in that direction can be as much as 5° .

c. Short-range, skywave circuits using one-hop, high-angle radiation give a great latitude in the choice of sites. For F-layer transmission to dis-

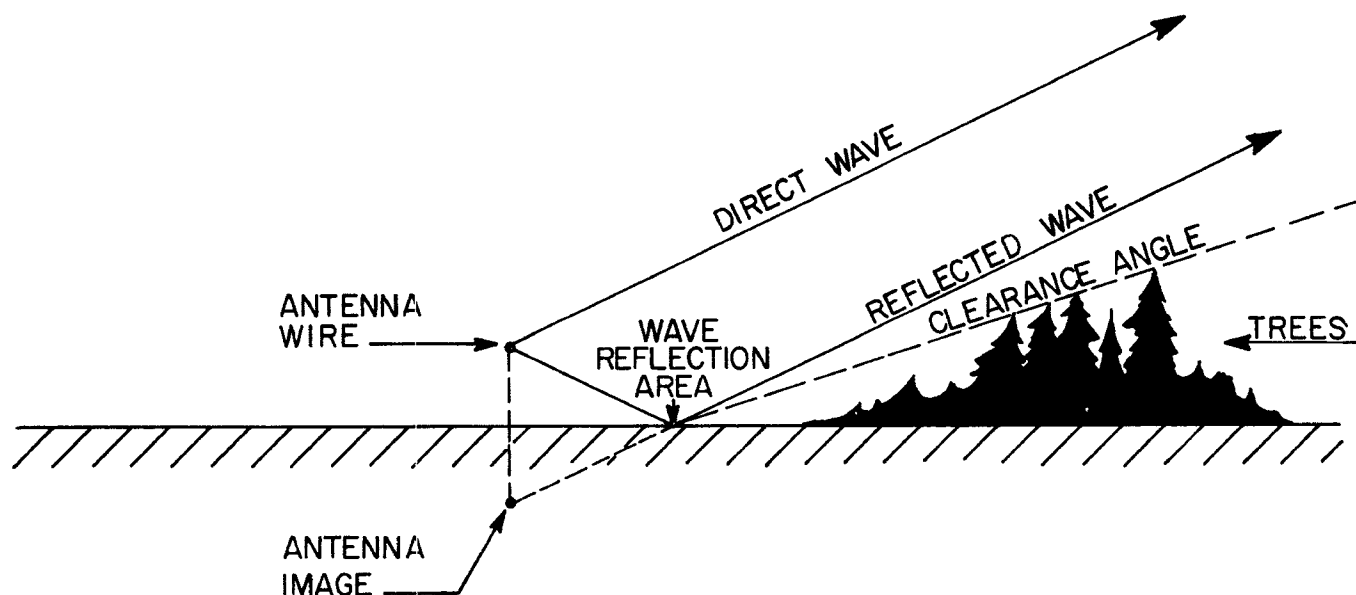


Figure 4-33. Clearance requirements for good wave reflection.

tances of 800 km (500 mi) and less, the vertical beam angles are always greater than 30°. Satisfactory sites for such transmission can often be located in rather deep valleys without compromising the circuit performance.

d. Where a choice of soil type is possible, the site with soil having the highest conductivity should be chosen. A site by a sea or lake, which can use the water as a reflecting surface, is excellent.

e. For groundwave service, there should be a minimum of intervening hills. If groundwaves, especially at the lower end of the band, are not

strictly line-of-sight but have some tendency to follow the earth's surface. However, substantial obstructions do block transmission.

f. Appendix F provides detailed instructions for calculating clearances required in the vicinity of antennas for skywave service. However, the use of figure 4-38 provides sufficient accuracy and detail for the operation of transportable equipment.

g. Receiving sites have an additional requirement that manmade noise not be excessive, preferably less than atmospheric noise at the location.

PART TWO

SYSTEM APPLICATIONS

CHAPTER 5

RADIO FREQUENCY PERFORMANCE ASPECTS OF HIGH FREQUENCY SYSTEMS

5-1. General

a. Implementation of an hf link of the type being considered in this manual involves making the best possible use of physical resources already on hand or readily obtainable by the using organization. This chapter gives general guidance to aid the field communicator in making the best choice of the options available. Assistance, in the form of propagation engineering services, is available from HQ USACEEIA, ATTN: CCC-EMEO-PED, Ft. Huachuca, AZ 85613. Appendix D describes the services available and provides instructions for requesting assistance. However, guidance is given in this manual regarding frequency choice when USACEEIA services are not available.

b. This chapter includes discussions of system parameters such as diversity, coding, and signal processing to reduce bit error rate, and electromagnetic compatibility (EMC).

c. An early decision must be made as to whether the groundwave or skywave mode of transmission is to be utilized. Eighty km (50 mi) is the approximate upper limit for the groundwave mode in military field operations. In rugged terrain, the maximum distance is less. Groundwave propagation is most effective at the lower frequencies, that is, 2 to 5 MHz. Unfortunately, noise (both atmospheric and manmade), is more detrimental at the lower frequencies, thus requiring a careful choice of frequency.

d. A compact, precise, internationally accepted method of designating the characteristics of transmitted radio signals has been developed. The complete scheme covers all types of radio communications. Table 5-1 shows the emission designations applicable to hf communications. The first character, a letter, tells whether the

carrier is amplitude (A), frequency (F), or pulse (P) modulated. The second character is a numeral designating the type of transmission, and the last character is a letter designating supplementary characteristics. By placing a numerical prefix before the emission designation, it is possible to include the bandwidth required in kilohertz. For example, the standard military independent sideband (isb) hf emission type and bandwidth are indicated as 12A9B. Modulation is discussed in TM 11-486-3. Modulation and emission terms are defined in the glossary. The signal-to-noise ratio required for satisfactory communications depends in part upon the type of emission being utilized.

5-2. Power Budget in an Hf System

a. The performance of any communications system depends upon the signal-to-noise ratio at the receiver. An adequate signal-to-noise ratio means good intelligibility for voice, a clear picture on TV, undistorted music on entertainment fm, and a low error rate on digital traffic. Appendix C presents and discusses the required signal-to-noise ratio for various types of high-frequency services.

b. Figure 5-1 is a diagram of the main components of an hf system. As shown in the figure, the signal-to-noise ratio at the input of the receiver is a function of all other components of the system.

c. In this power budget discussion, transmitter power is expressed in dBw so that received power can be computed by adding and subtracting path gains and losses. The transmission line loss is typically a few dB. Antennas are chosen and installed so as to favor the desired direction of transmission. Antenna gain will vary from a couple of dB to perhaps 15 dB.

Table 5-1. Designation of Radio Emissions

Type of modulation of main carrier	Type of transmission	Supplementary characteristics	Symbol
Amplitude modulation	With no modulation		A0
	Telegraphy without the use of a modulating audio frequency (by on-off-keying)		A1
	Telegraphy by the on-off keying of an amplitude-modulating audio frequency or audio frequencies, or by the on-off keying of the modulated emission (special case: an unkeyed emission amplitude modulated)		A2
	Telephony	Double sideband, full carrier	A3
		Single sideband, reduced carrier	A3A
		Single sideband, suppressed carrier	A3J
		Two independent sidebands, reduced carrier	A3B
	Facsimile (with modulation of main carrier either directly or by a frequency-modulated subcarrier)	Single sideband, reduced carrier	A4 A4A
	Television	Vestigial sideband	A5C
Frequency (or phase) modulation	Multichannel voice-frequency telegraphy	Single sideband, reduced carrier	A7A
	Cases not covered by the above, such as a combination of telephony and telegraphy	Two independent sidebands	A9B
	With no modulation		F0
	Telegraphy by frequency-shift keying without the use of a modulating audio frequency: one of two frequencies being emitted at any instant		F1
	Telegraphy by the on-off keying of a frequency-modulating audio frequency or by the on-off keying of a frequency-modulated emission (special case: an unkeyed emission, frequency modulated)		F2
	Telephony		F3
	Facsimile by direct frequency modulation of the carrier		F4
	Television		F5
	Four-frequency duplex telegraphy		F6
	Cases not covered by the above, in which the main carrier is frequency modulated		F9

d. The path loss is usually large and highly variable. Path losses may typically be 90 to 140 dB on a 1,200-km (745 mi) path.

e. The receiving antenna gain and transmission line loss will be comparable to those at the transmitting end. Most of the received noise is picked up by the receiving antenna. A highly directional antenna will discriminate against noise from directions other than that of the arriving signal. The receiver will usually accept only a narrow band of frequencies, the exact bandwidth depending upon the type of service. Noise outside this frequency band is of no consequence unless

it is of sufficient intensity to overload the front end of the receiver.

f. Any change in any portion of the system has an effect on the signal-to-noise ratio. Assuming a barely adequate signal-to-noise ratio, an increase of 3 dB in the received noise level requires an increase of 3 dB in transmitter power. An increase of 3 dB means doubling the power. Each segment of the system is important in determining the signal-to-noise ratio.

g. As an example, assume the following sample system:

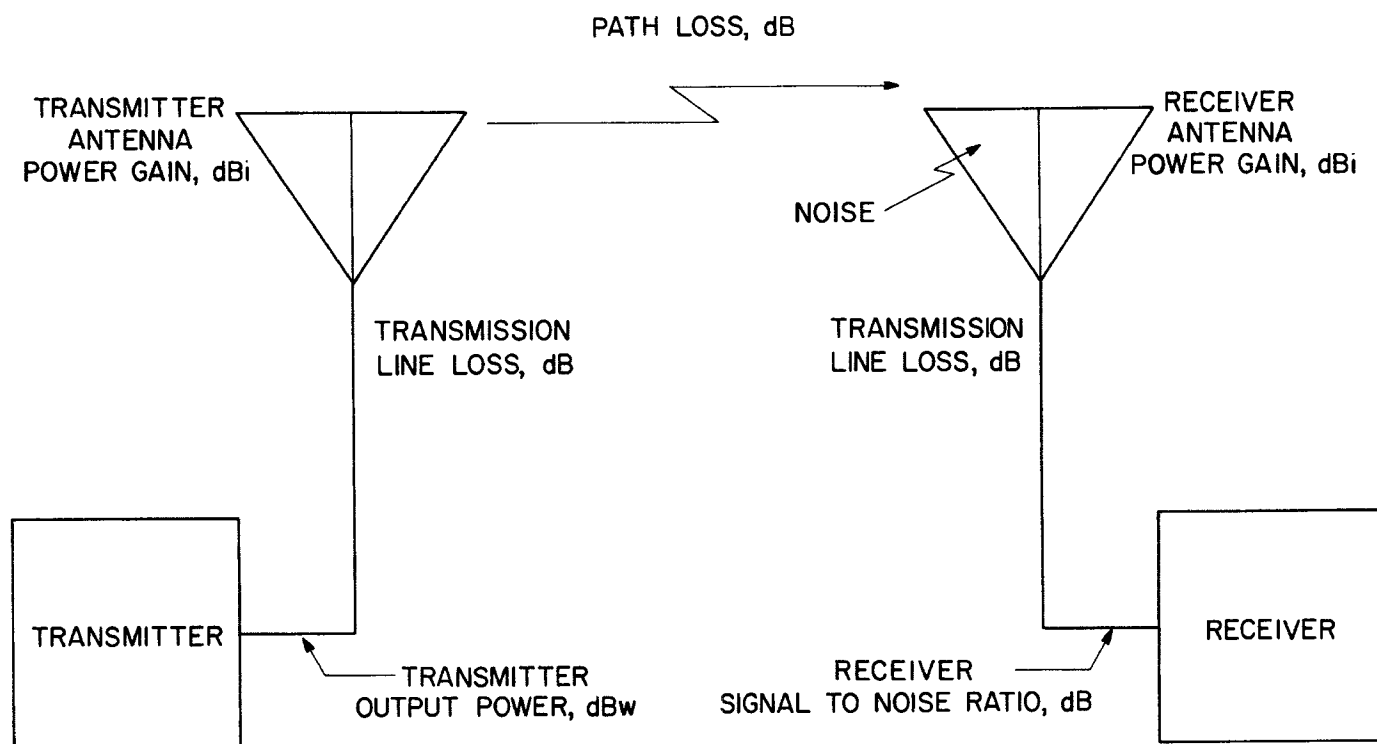


Figure 5-1. Power budget in a high frequency radio system.

- | | |
|---|----------|
| (1) Transmitter power (100 W) | 20 dBw |
| (2) Transmitter transmission line (coaxial) loss | 2 dB |
| (3) Transmitter antenna gain | 6 dBi |
| (4) Path loss | 120 dB |
| (5) Receiving antenna power gain | 8 dB |
| (6) Receiver transmission line loss | 2 dB |
| (7) Calculated received power at input to receiver | |
| $(20 \text{ dBw} - 2 \text{ dB} + 6 \text{ dB} - 120 \text{ dB} + 8 \text{ dB} - 2 \text{ dB}) = -90 \text{ dBw}$ | |
| (8) Noise power at receiver input | -150 dBw |
| (9) Signal-to-noise ratio at input to receiver | 60 dB |

5-3. Radio Frequency Performance Analysis

a. General. A detailed radio frequency performance analysis is based upon the concept of the power budget presented in paragraph 5-2. Path loss and antenna gain are examined in great detail.

b. Path Loss. Path loss is by far the most complex item in the power budget. Each reflection from an ionized layer and from ground involves a loss. Also, passage through lower ionized layers before and after reflection involves losses. These ionospheric losses vary with time of day, season,

sunspot number, and latitude. If multiple hops are involved, reflections from the earth's surface introduce losses.

c. Mathematical Simulation. The only feasible method of performing the large number of calculations involved is to use a large, high-speed electronic computer. The USACEEIA performance analysis program uses mathematical modeling in conjunction with such a computer.

d. Antenna Selection. In designing a new system, path loss calculations are carried out for isotropic antennas. Based on engineering judgment various combinations of antennas and transmitter powers are then incorporated into the system until the most suitable (usually the most economical) combination is found which will provide the required performance. For operating systems, performance calculations are made for the existing equipment.

e. Requesting Performance Analysis Service. Appendix D provides details on requesting performance analyses and other USACEEIA propagation engineering services. The appendix lists the information the user must provide and explains a sample frequency reliability table. The skywave prediction program considers the ionospheric conditions expected to occur during the

lifetime of the hf radio system. The groundwave program predicts coverage for systems using vertically polarized antennas.

5-4. Antenna Selection and Construction

a. Antenna selection is one of the most important steps in the design of an hf system. Proper antenna choice will contribute much toward achieving the most cost-effective system. The factors to be considered are—

- (1) Frequencies to be utilized.
- (2) Takeoff and arrival angles.
- (3) Gain required (or desired).

Figure 5-2 gives takeoff and arrival angles for various distances and modes of propagation.

b. Transportable equipment will often be used on relatively short paths, for which the half-wave horizontal dipole is well suited. The effectiveness of the horizontal dipole is dependent upon its being erected at the proper height. Figure 5-3 provides guidance on the optimum height. Because it may not be possible to achieve the optimum height under field conditions, the figure

also gives the height at which a 3-dB penalty is incurred.

c. If greater gain is required than that provided by a dipole, a more complex antenna such as a rhombic or log-periodic may be chosen. The sloping V is very useful under field conditions. As explained in appendix D, USACEEIA antenna advisory and engineering services are available upon request.

5-5. Antenna Siting and Orientation

a. *Siting.* Paragraph 4-21 provided guidance regarding antenna siting. For a relatively permanent site, the rules provided in appendix F regarding Fresnel zone clearance should be followed. For horizontal antennas, clearance to the first Fresnel zone is required in order to obtain maximum reinforcement of the direct ray by the ground-reflected ray. For very short skywave paths siting is not a serious problem. Siting problems are greater for groundwave service and for intermediate and long skywave paths.

b. *Orientation.*

- (1) If the azimuth of the radio path is not

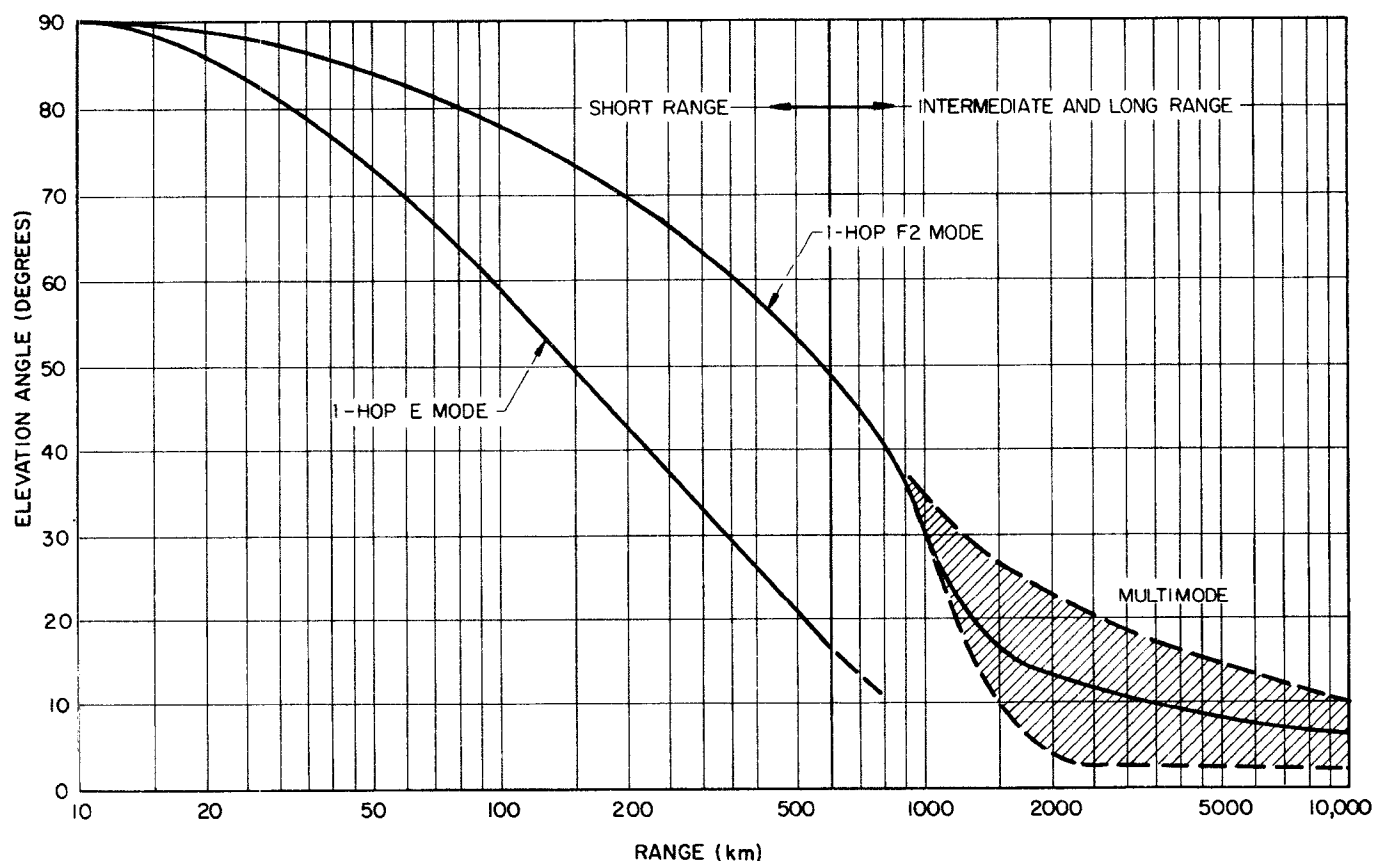


Figure 5-2. Takeoff and arrival angles for ionospheric transmissions as a function of range and mode.

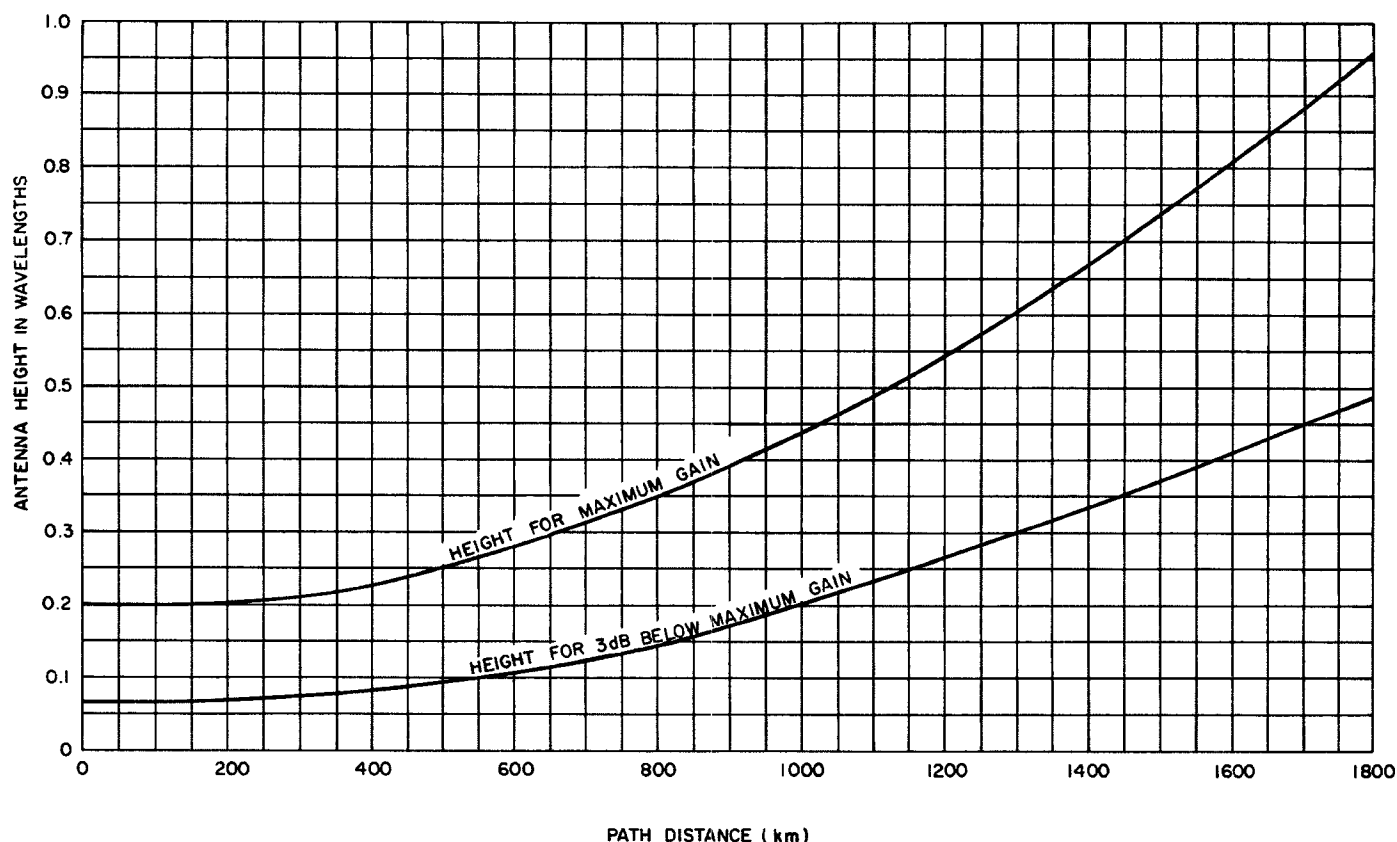


Figure 5-3. Approximate height of half-wave dipole for best F2 layer propagation.

provided the azimuth should be determined by the best available means. The accuracy required in determining the azimuth of the path is dependent upon the radiation pattern of the antenna. If the antenna beamwidth is very wide (90° angle between half-power points, for example), an error of 10° in azimuth is of little consequence. In transportable operation, the rhombic and V antennas may have such a narrow beam as to require great accuracy in azimuth determination.

(2) For paths a few hundred kilometers or less in length and in areas between 50°S and 50°N latitude, the path may be plotted on any available map and the azimuth estimated from the latitude and longitude lines on the map. For longer distances, azimuth may be determined by stretching a string on a globe between the two end points. Great care is required, however, because antennas used on long paths are apt to have narrow beams.

(3) The antenna should be erected in for the correct azimuth. Great accuracy is not required in erecting broad-beam antennas. Unless a line of known azimuth is available at the site, the direction of the path is best determined by a magnetic compass. Figure 5-4 is a map of magnetic

declination, showing the variation of the compass needle from true north. When the compass is held so that the needle points to the direction indicated for that location on the map, all directions indicated by the compass card will be true.

5-6. Antenna Grounds

a. Unless the antenna is erected over highly conducting soil considerable ground losses occur. Ground losses result in reduced reinforcement and cancellation of the direct ray by the reflected ray and in alteration of the radiation pattern. The alteration reduces the sharpness of the antenna pattern. Rocky soil in hills and mountains and dry, sandy soil have poor conductivity. Although a good ground is always desirable, it is essential for vertical quarter-wave antennas, which depend upon the reflected wave as a principle of operation. Soil characteristics must be considered in selecting an antenna site.

b. Where optimum antenna performance is essential and a site with good conductivity is not available, a ground screen must be constructed. Ground screens are described in paragraph 4-19.

c. USACEEIA performance calculations consider soil characteristics. In requesting USA-

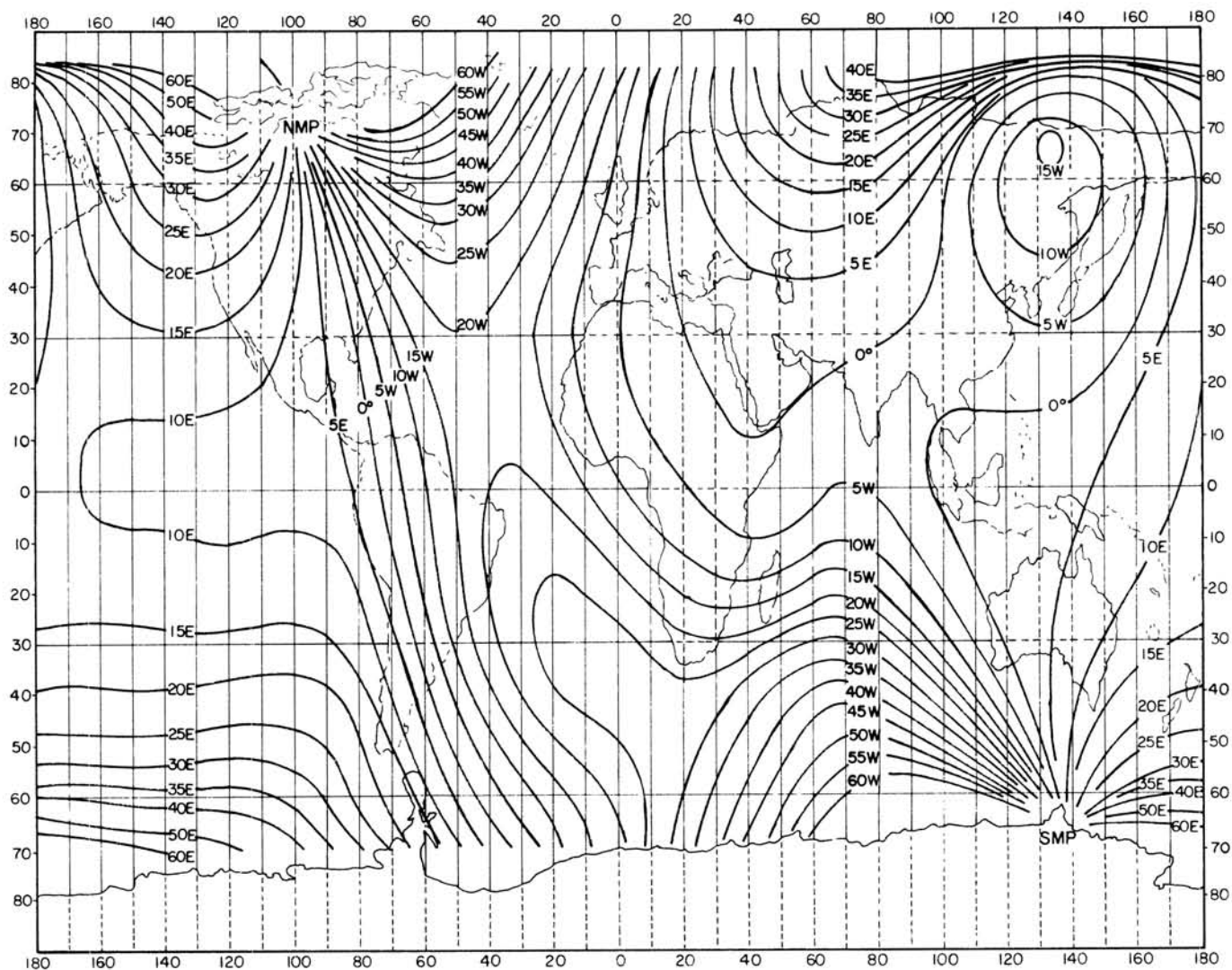


Figure 5-4. Magnetic declination over the world.

CEEIA assistance in performance predictions or in antenna selection or design, the soil characteristics should be given.

5-7. Frequency Selection for Skywave Mode

a. General. Because propagation of radio waves in the hf band is highly frequency-dependent, the selection of the proper operating frequency is of extreme importance. Under normal operating conditions, a frequency change must be made at least twice daily. When unusual conditions exist, more frequent changes are required. Whenever possible, the propagation prediction services of USACEIA should be utilized in selecting frequencies. This paragraph will discuss the use of USACEIA predictions.

b. Using Frequency Reliability Tables. One form of output of USACEIA's skywave prediction program is a frequency reliability table. A sample table is presented at figure D-1 and de-

scribed in appendix D. The use of the tables in selecting the operating frequency will be explained by examples. Refer to figure 5-5.

(1) As example 1, choose the best operating frequency for the period 0200 UT. The frequency of 7.3 MHz is the frequency giving the greatest reliability at this time interval. If interference is experienced on 7.3 MHz, the next choice would be 10.4 MHz, at a slight decrease in reliability. Note that the reliability for 9.2 MHz drops below 0.90 and multipathing increases to 0.38. At the bottom of the table the maximum usable frequency (MUF) and the frequency of optimum operation (FOT) are shown for this circuit. The FOT is 11.2 for 0200 UT. Since the reliability is computed for each of 11 frequencies, a good sampling of reliability is obtained throughout the frequency assignment.

(2) As example 2, choose the best operating frequency for 1000 UT. At this hour the highest frequency giving the greatest reliability is 12.1

GREENWICH TIME

FREQ {MHZ}	02	04	06	08	10	12	14	16	18	20	22	24	
20.2	-	-	-	.55	.72	.60	.53	.73	.83	.53	.13	.09	REL
16.2	-	-	-	.94	.93	.87	.86	.93	.96	.85	.72	.69	MP PROB
15.7	FREQUENCY CHANGE AT 0100			.72	.96	.95	.76	.90	.95	.97	.88	.78	REL
13.6				.79	.98	.95	.91	.95	.97	.99	.95	.92	REL
12.1	.79	.76	.98	.97	.96	.93	.94	.97	.99	.98	.96	.96	REL
11.5	.86	.84	.99	.97	.96	.93	.93	.97	.99	.98	.97	.97	REL
10.4	.95	.94	.99	.96	.75	.65	.90	.96	.99	.99	.99	.98	REL
9.2	.83	.88	.89	.26	FREQUENCY CHANGE AT 0500			.22	.76	.87	.83	.85	REL
7.3	.97	.98	.90	-				.61	.94	.97	.96	.96	REL
4.5	.94	.94	-	-	-	-	-	.09	.82	.94	.90	.90	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PROB
	02	04	06	08	10	12	14	16	18	20	22	24	
MUF	17.8	14.0	13.8	17.7	20.9	28.4	27.5	29.2	29.1	27.0	20.7	18.1	
LUF	13.8	11.2	11.0	14.5	17.1	22.2	21.5	23.6	23.6	20.0	15.3	14.1	

Figure 5-5. Operational frequency plan based on frequency reliability table.

MHz. Although the same reliability, 0.96, is predicted for 11.5 MHz, there is less likelihood of interfering signals at the higher frequency. Should interference be experienced at 12.1 MHz, the operator should try 11.5 MHz as a second choice. If interference is experienced on both 12.1 and 11.5, the operator should try 15.7 MHz. From the FOT table it is seen that FOT at this hour is 22.2 MHz.

(3) Usually it is desired that the stations have communications capability at all times. A plan of operational frequencies is drawn up which gives a high probability of successful communications with a minimum of frequency changes. Figure 5-5 shows such a plan based on the frequency reliability table already discussed.

5-8. Improving Communications by Signal Processing

a. General. In the discussion of the power budget (para 5-2), the importance of an adequate signal-to-noise ratio was stressed. High frequency communications are characterized by various

types of fading which result in deterioration of the signal-to-noise ratio. Methods of partially counteracting short periods of fading by signal processing are discussed in this paragraph. Some techniques improve the signal-to-noise ratio; others utilize the periods of good transmission and disregard the periods of serious fading.

b. Use of Diversity.

(1) Diversity is the reception of signals by two or more paths, followed by the choosing of the best signal or by combining all signals. Diversity is very effective in reducing errors in digital transmission. Dual diversity, using two paths, is most commonly used on hf. Quadruple diversity, using four paths, can be used on hf, but is more common on microwave links. Diversity techniques require special receiving equipment, and all except space diversity require special transmitting equipment or techniques.

(2) At a space diversity receiver site, two antennas are used and these are spaced some distances apart so that the signal from the transmitting antenna to each of the receiving antennas

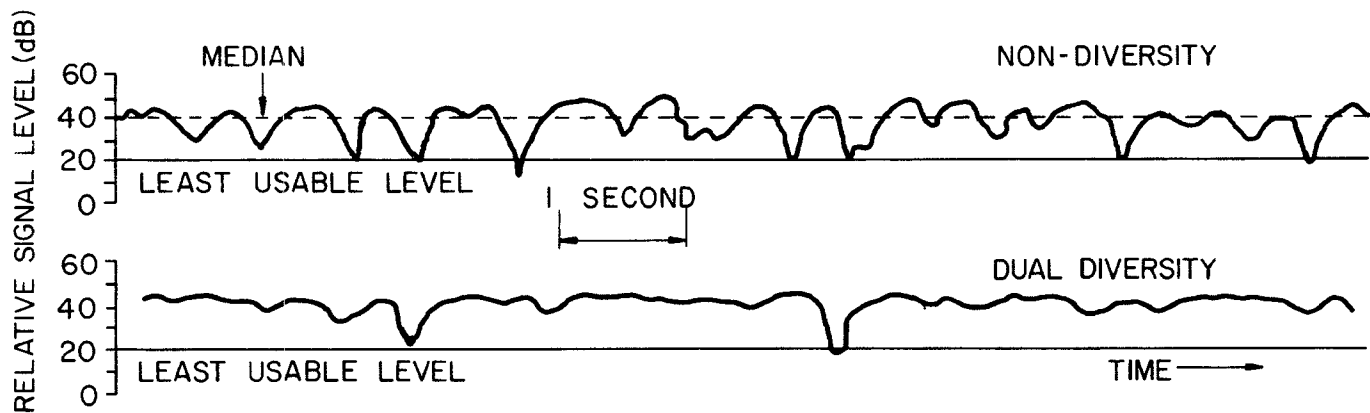


Figure 5-6. Example of diversity improvement.

travels a different path through the medium. A separate receiver is employed for each receiving antenna and some selecting or combining equipment must be used. Over a period of time, the combined diversity signal will have a higher average value than a single signal. This is shown in figure 5-6, which presents signal levels for a single signal and the combined signal from two diversity receivers. Comparison of the curves shows that the fading of the combined signal is less than that of the single signal. The effect of antenna spacing on the performance of two hf antennas and receivers in space diversity is shown in figure 5-7. It is apparent that, as the spacing is increased from zero, the improvement is very rapid initially, but beyond 7.5 wavelengths the rate of improvement is so small that no further increase would be warranted. Space diversity is the most commonly used diversity technique.

(3) Fading does not affect all frequencies equally. Frequency diversity may use two carrier frequencies or two tones on the same carrier. Both techniques are wasteful of the frequency spectrum. The use of a separate carrier means two transmitters as well as two receivers. The use of two tones on the same carrier is less effective than space diversity.

(4) Time diversity takes advantage of the fact that the error-producing factors usually occur in bursts of moderately well defined time durations. The time diversity technique places the same information two or more times in the same medium, separated by more than the normal duration of an error burst. Excellent performance improvements are possible even when the medium has a high percentage of such bursts. Its major disadvantage is that it requires at least twice the nominal throughput capacity of a nondiversity system.

(5) Where space is not available to separate

two large horizontal antennas, polarization diversity is a compromise which requires very little space yet provides almost as much advantage as is offered by space diversity, or more advantage if space separation is less than five wavelengths. The antennas can be very close to each other. In polarization diversity, one antenna is vertical and one horizontal. Each antenna is fed into a separate receiver in a manner similar to a space diversity system. Polarization diversity is effective on medium and long paths, but on long paths space diversity with adequate separation is better.

(6) One doublet antenna can be erected broadside to the distant station and one erected with the doublet ends in line with the distant station. The antennas can share a common support pole. Radio waves arriving after bouncing off the ionosphere will be randomly or elliptically polarized. The doublets arranged in a cross arrangement will function as a cross-polarized antenna to a wave arriving at a very high angle. As a result, fading will be highly uncorrelated and excellent diversity gain will result on paths up to 650 km (400 mi). Figure 5-8 shows antennas connected in the right-angle diversity configuration.

c. Coding Techniques. Both error-detecting and error-correcting codes are used to improve the bit error rate. An automatic request-repeat (ARQ) feature may be used with any full duplex circuit having an error-detecting code. The receiving terminal is configured to automatically transmit "QR" (request repeat) when an error is detected. The transmitting terminal then repeats the character or block of characters involved. ARQ works well on hf circuits.

d. Adaptive Techniques.

(1) Various techniques can be used to adapt a communications link to deteriorated conditions.

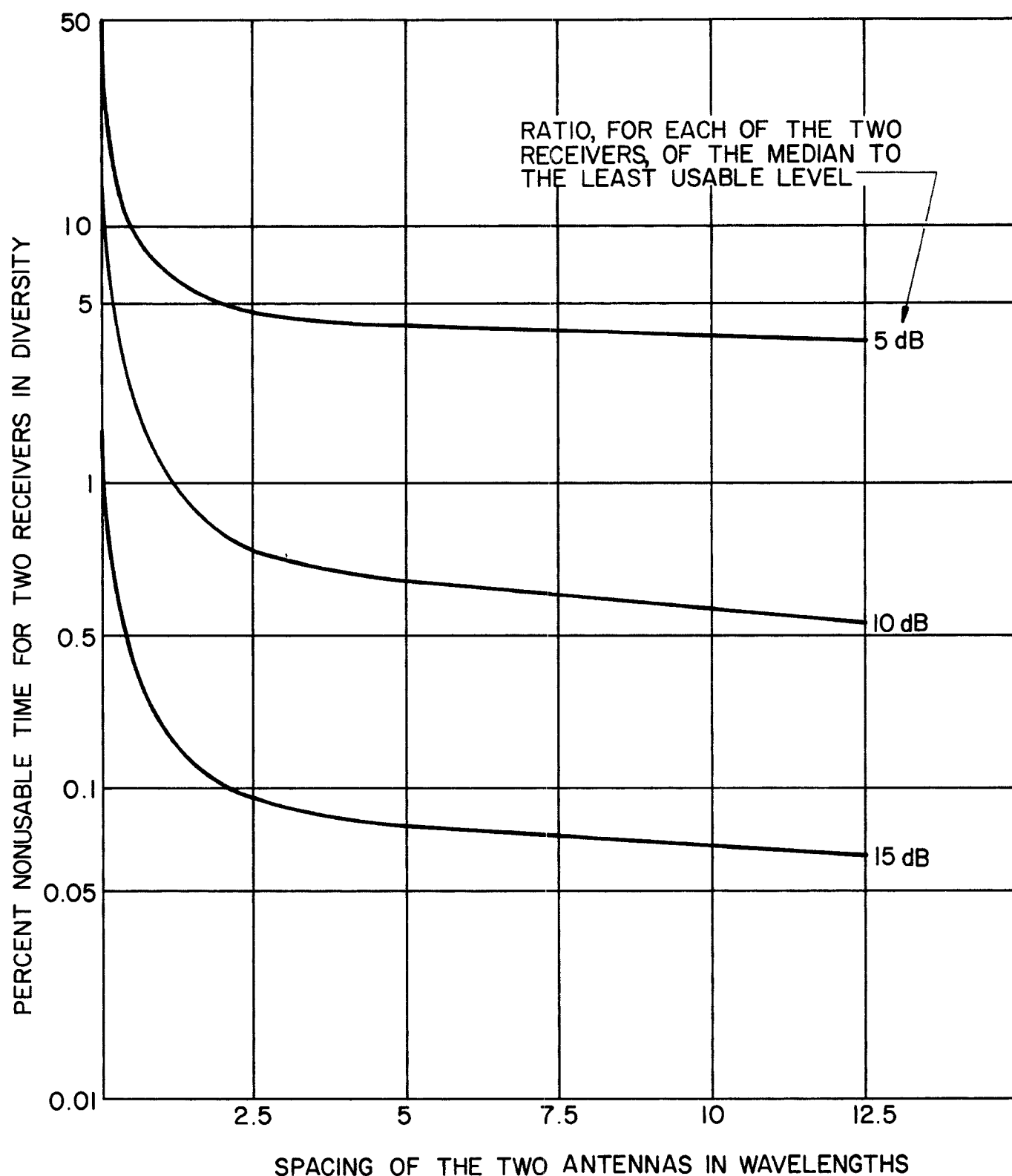


Figure 5-7. Effect of antenna spacing on diversity improvement.

Concentrating the available transmitter power over a narrower bandwidth improves the signal-to-noise ratio. Changing from two voice channels (6 kHz bandwidth) to a single voice channel (3 kHz) may allow continuation of communications

under adverse conditions. Dropping to a single teletypewriter channel will allow communications under even worse conditions.

(2) Teletypewriter service frequently uses frequency shift keying, in which one frequency is

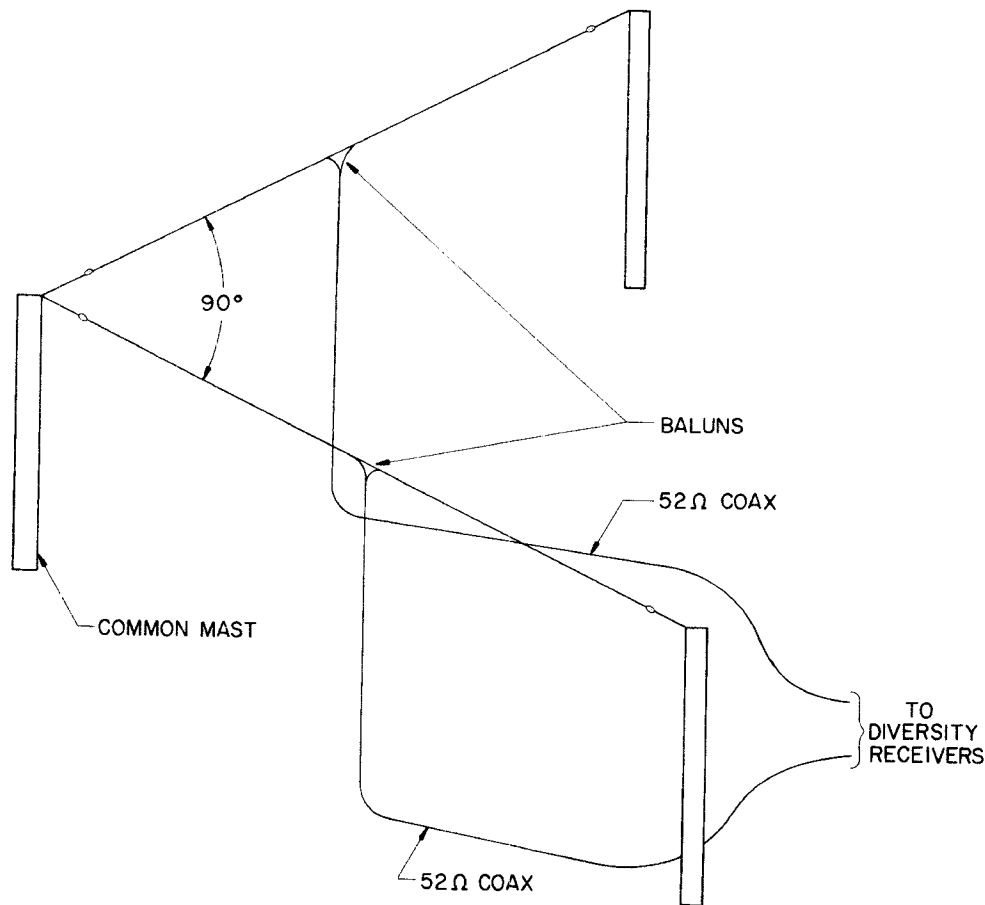


Figure 5-8. Right-angle diversity antennas.

transmitted for a "mark" and another frequency is transmitted for a "space." In general, fading is not identical on the two frequencies. Some receivers alter the method of distinguishing between the marks and spaces in response to varying propagation conditions. Significant improvement in error rates can be achieved by such techniques.

e. Noise Blanking. Noise blanking is used to combat manmade noise on voice transmissions. In one type of blanking a noise receiver, tuned somewhat above 30 MHz, is used to pick up noise. The output of the noise receiver is used to disable rf and if sections of the communications receiver during noise pulses. The result is brief blankings of the desired signal, but the absence of signal is much less disturbing to the listener than the noise bursts, and intelligibility is improved.

f. Speech Processing.

(1) Speech waveforms have much lower average power than peak power. Increasing the ratio of average power to peak power by signal processing increases the effectiveness of the signal.

(2) A simple way of improving the average to peak power ratio is by using an audio compressor circuit. Such a circuit acts as an automatic volume control.

(3) A more effective technique is clipping the peaks. Clipping introduces higher order harmonics and causes splatter. To prevent this, audio frequencies above those needed for intelligible speech are filtered out after clipping but before modulation.

(4) Rf clipping of a single sideband (ssb) signal is very effective. Improvements of 8 dB in signal-to-noise ratio can be obtained with this technique. Severe clipping (20 dB) changes the total quality of the voice signal, but intelligibility is increased. With moderate clipping (6 to 12 dB), there is little loss in naturalness of tone and intelligibility is increased considerably.

(5) *Automatic level control* can be used to maintain the rf peak of an ssb transmitter at a relatively constant level. The choice of the time constant for the circuit, which governs its reaction time, is critical. To be most effective the circuit should react to the syllables of speech.

5-9. Electromagnetic Compatibility

a. Definition. Electromagnetic compatibility (EMC) means the ability of communications-electronics equipments, subsystems, and systems, together with electro-mechanical devices (for example, motor vehicles, engine generators, and electrical tools), to operate in their intended operational environments without suffering or causing unacceptable degradation because of unwanted electromagnetic radiation or response. Electromagnetic compatibility embraces both susceptibility and vulnerability. Radio frequency interference reduction is an older term for (and is considered to be synonymous with) the newer, preferred term, electromagnetic compatibility.

(1) Design compability is electromagnetic compability achieved by incorporating engineering characteristics and features during the design stages as opposed to corrective measures adopted after equipment has been produced. EMC engineering and analysis services are available at HQ, USACEEIA (CCC-EMEO-ECD), Fort Huachuca, Arizona 85613.

(2) Operational electromagnetic compatibility is electromagnetic compatibility among C-E items of material in a real operational situation. HQ USACEEIA provides assistance to help correct operational problems caused by electromagnetic interference (EMI).

b. Sources of Interference.

(1) Some of the many sources of interference are discussed in this paragraph. A more detailed treatment of interference sources can be found in TM 11-490-5, Operational Electromagnetic Compatibility and FM 11-486-15, Telecommunications Engineering: EMC for Systems Engineers.

(2) Interference can arise from natural or from manmade causes. Although the effects of natural interference are usually geographically widespread, manmade interference usually affects only limited areas relatively close to the source.

c. Natural Interference.

(1) In the case of natural interference, the source may be as close as the earth's atmosphere or as far away as distant galaxies.

(a) Atmospheric noise includes such things as lightning discharges, which produce "static." This is well known and easily recognized by most people. Another type of atmospheric noise is "precipitation static" which is caused by rain, hail, snow, and dust storms. Most atmospheric noise consists of short, erratic bursts spread over

the rf spectrum. Such noise is usually the limiting factor in communications below 30 MHz.

(b) Solar noise is usually a long-term noise level covering the major portion of the usable frequency spectrum. It increases in level during high sunspot or solar flare activity. Some parts of the frequency spectrum may experience greater increase in noise than others. Solar noise is usually not important in the hf band.

(c) Galactic noise varies with frequency and with position in the galaxy. Galactic noise is usually not important in the hf portion of the rf spectrum.

(2) There is little we can do about natural interference except to know as much as possible about it and to use this knowledge to minimize its worst effects. For example, a knowledge of the usual effects of sunspot and solar-flare activity might enable us to anticipate which frequencies might be usable if the normal frequencies are affected by such activities.

d. Manmade Interference. As might be expected, manmade interference is most prevalent in areas where man is most prevalent, such as industrial areas and densely populated areas. It arises from intentional radiation from communications and radar equipment, and from many other equipments which emit electromagnetic energy as an unintentional but largely unavoidable byproduct of their operation. Examples of the latter are electric generators and motors and many types of medical equipment, such as X-ray and diathermy apparatus. Manmade interference falls into two categories—narrowband and broadband.

(1) Narrowband interference is interference which affects or interferes with one specific frequency or frequency band without affecting others. The various types are cochannel and adjacent channel interference. Cochannel interference is caused by one or more unwanted frequencies which fall within the pass-range of a particular system, or in one or more channels of a system, in sufficient strength to cause unacceptable interference. Adjacent channel interference is caused by one or more unwanted frequencies which, although outside the frequency range of a given receive channel, are sufficiently close in frequency and of sufficient strength to cause unacceptable interference. Usually, proper planning and the built-in facility of most receivers to reject unwanted signals are sufficient to afford protection. However, if the unwanted signals are in the same channel as the wanted signal (cochannel), the

receiver cannot tell the difference between the two. In many cases, the unwanted signals arise from authorized emissions, so the question is, why the interference? There are several possible reasons:

(a) A cochannel emission without adequate geographical separation is a typical source of interference. If the proximity is unavoidable, every effort should be made to allocate frequencies sufficiently separated to prevent the cochannel effects. This may be the case in transportable operations where it is necessary to locate receiver and transmitter adjacent to each other. If it is not possible to obtain either the required geographical or frequency separation, steps should be taken to ensure minimum power radiation and bandwidth consistent with the mission.

(b) The need for good maintenance can never be stressed too much. The effects of inadequate maintenance are magnified where potential adjacent channel interfering signals are present. Transmitter or receiver frequency drift can cause the onset of the problem. If, because of inadequate maintenance, the receiver's ability to reject frequencies outside the tuned range (its selectivity) is below standard, the likelihood of experiencing adjacent channel interference is greatly increased.

(c) Sometimes, as a result of such phenomena as sunspot activity, solar flares, or magnetic storms, unusual propagation conditions can occur. These can cause not only the temporary loss of usual operating frequencies but also the reception of signals from distant locations which normally would not be significant. The result is that the distant station can be mistaken for a local interfering station and time can be wasted in attempting to locate it. The presence of such a station, coupled with abnormal propagation conditions, should always raise the possibility that the interference is not local. Under such circumstances, a frequency change might be the only effective solution.

(d) Harmonics are multiples of a basic frequency. In a transmitter's radiated emission, there are always some harmonics and other spurious components. These are a normal outcome of the internal electronic mechanism, not only of transmitters but also of most types of electronic equipment. Usually, these components are kept at a low level compared with the fundamental radiated frequency. If, however, adequate maintenance is not carried out or circuits are maladjusted, the inevitable outcome is an increase in the level of harmonics and spurious components. These not only use up an unnecessarily high pro-

portion of available power, but also occupy space in the frequency spectrum at a level which could cause significant interference. Given a good equipment design in the first instance, this problem can be minimized only by thorough, regular maintenance of equipment, including antennas, transmission lines, switches, connectors, and grounding conductors and connections.

(e) Intermodulation occurs when two or more signals or frequencies which are simultaneously present in a circuit, such as a receiver input or a transmitter output, mix in such a way as to produce other frequencies called *intermodulation products*. Cross-modulation occurs when the intelligence on an unwanted signal is superimposed on the wanted signal. Both types of phenomena can cause frequency interference to other circuits. As in the case of harmonics, these products are usually of a low level relative to the wanted signal, and increase to an unacceptable level for the same reasons—lack of adequate maintenance or maladjustment of circuits, loose antenna leads, or connectors, corrosion or foreign matter inside equipment, and the like. Thorough maintenance will do much to keep intermodulation and crossmodulation products within tolerable limits. A pertinent point relating to detecting problems arising from all types of spurious components is that even when they are causing local interference, their levels are usually well below that of the fundamental frequency of the emitters concerned. Consequently, they can be mistaken for the output of a distant transmitter. The possibility should always be considered, unless there is a positive identification of a distant transmitter. Steps for isolation are then required. The usual start point is to arrange for each local transmitter to be removed from operation, in turn, and as briefly as possible, to see if the interference disappears. Where appropriate, approval should be obtained in advance, and whenever possible, alternative facilities should first be provided.

(2) Broadband interference is characterized by noise emission which extends over a wide frequency range, usually at an even intensity. It has many sources, but the most common cause is the periodic opening and closing of a circuit, which produces "transient" voltages and currents composed of many frequencies harmonically related to the opening/closing rate.

(a) Sudden changes of voltage are caused by surges, corona, arcing, sparking, or any interruption of an electric power circuit. The most common causes of powerline interference are: insulators covered by dust or other airborne mate-

rial, resulting in conduction of current across the material; damaged insulators; excessive voltage, which causes a breakdown in the dielectric (insulating medium) between power lines; insulation breakdown in transformers or capacitor banks; intermittent contact between metallic objects on power poles; poor ground connections between a neutral line and earth; and sharp points projecting from powerlines causing corona. Again, the most effective answer is a thorough inspection and maintenance program. Proper maintenance will minimize corona, but some corona will exist on very high voltage lines despite the best maintenance practices.

(b) Transient voltages can be generated by the normal operation of fluorescent and other arc-discharge lights or by defective starter or ballast units. The interference so caused can be conveyed to the victim receiver by direct radiation, by conduction along powerlines, or by radiation into the powerline followed by conduction along the powerline. Interference caused by defective arc-discharge lights can usually be pinpointed because of visual or aural indications from the light involved. However, as some lights cause interference while working perfectly, it may be necessary to undertake a trial and error system to locate the culprit. The main thing is not to discount a fluorescent, mercury vapor, or sodium vapor light, based on lack of visual or aural indication that it is at fault.

(c) Transient voltages are generated by arc-welding equipment and these may cause interference. Industrial quality arc-welders usually have adequate suppression incorporated. However, inadequate maintenance can result in significant radiation. The first step is to ensure that the equipment is properly installed. This includes proper shielding, bonding, bypassing, and grounding. In particular, the shortest possible grounding connection should be made. Subsequently, a thorough maintenance program will minimize problems. Inspection for loose connections, broken bonding straps, and defective filters or bypass capacitors should be carried out regularly and repairs made without delay.

(d) Interference caused by vehicular-type ignition systems is normally present over a wide frequency range (0 to 500 MHz). Although it is usually most troublesome in the frequency range from 30 to 150 MHz, it also affects the 2- to 30-MHz range. The effects can be intensified in vehicles having defective ignition harnesses. Although this interference is usually minimal at distances over 500 feet from the source, instances

have occurred of interference from greater distances. This depends largely on the sensitivity of the victim receptor. Because of the diversity of interference sources of this type (including private vehicles, lawn mowers, and the like), it is difficult to remedy. Remedial steps could include fitting noise-suppression devices to site service vehicles, denying or restricting site access to other vehicles, and careful selection of timings for lawn-cutting (periods of low communications usage).

(e) The operation of switches and relays causes transient voltages capable of producing interference. This is particularly so of electro-mechanically-operated switches or relays working at a periodic rate (such as vibrators or buzzers). Many components of this type, particularly relays, have suppression devices incorporated. Non-operation or failure of the suppressor can result in significant interference. Checking and maintenance can minimize the effects of this type of interference.

(f) In a motor or generator, a rotating commutator is usually connected to external circuitry through brushes which rest upon and slide over the commutator segments as the commutator rotates. This type of switching arrangement which will produce transient voltages capable of causing interference. The interference can be intensified by brush bounce and commutator segment surface irregularities. As the segments become pitted and dirty with carbon deposits (and this is a normal occurrence in a motor or generator), the degree of interference generated will increase. Thorough and regular maintenance (cleaning of sliding parts, replacement of brushes) will help to minimize interference.

(g) Any sort of rapidly pulsed voltage, such as radar, contains components of energy spread over a wide frequency range. Most radar systems employ some type of sweeping or rotating antenna system, which causes the intensity of any interference experienced by a receptor to vary at the antenna's sweeping rate. It is therefore fairly easy to detect and isolate. It may be possible to eliminate the interference by means of a slight change of location of the victim receptor's antenna. If this fails, a closer examination to include trial and error corrective procedures will be required. It is important in such cases to report the interference to the appropriate authority and request assistance.

(h) Industrial, scientific, commercial, and medical apparatus employ radio frequency elec-

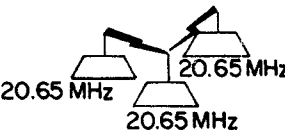
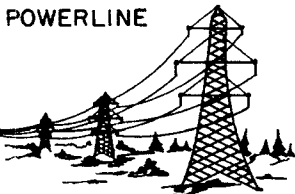
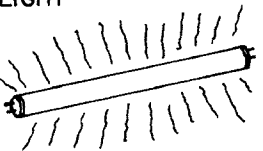
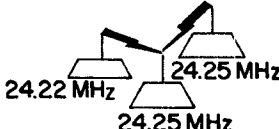
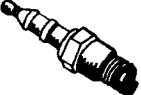
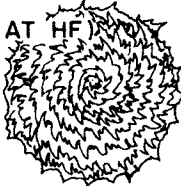
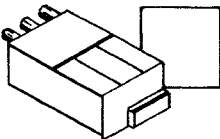
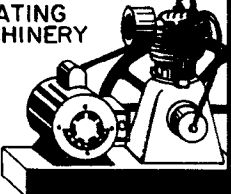
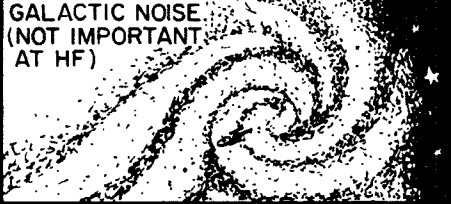
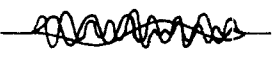
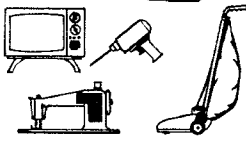
tromagnetic energy which can produce serious interference, particularly as the latter can be radiated or conducted over long distances. Such equipment includes diathermy, X-ray, microwave ovens, and induction heaters. Remedies for this type of interference might be difficult to obtain. If a problem can be traced to a local source, it may be possible to arrange for the faulty equipment to be checked out, repaired if practicable, or replaced if necessary. If the source is distant, it may not be possible to locate it because there may be no trace of the interference anywhere between the source and the victim. In such cases, "evasive" action such as a frequency change or a reassignment of channels may be the only solution. A pictorial representation of some of the sources of interference described is contained in figure 5-9.

e. Rf Interference With Audio Equipment. Rf signals are outside the passband of audio equipment and normally cause no problems. However, very strong rf signals may overload any audio

stages they are able to enter and cause severe interference. Amplitude modulated signals can be rectified in an overloaded audio stage and intelligible interference may appear at the output of the audio equipment. The cure for this problem is to shield and filter signal and power leads so that rf does not enter the audio equipment. Power leads or input cable may be acting as an antenna for rf. Sometimes it may be necessary to improve the shielding of the audio equipment which is rectifying the rf signal. Series inductances and shunt capacitors may have to be installed at the device.

f. Transfer Mechanisms.

(1) In practice, it is usually necessary to isolate and locate a source of interference before effective action can be taken to eliminate it. To determine the source, it is usually necessary to find out by what means the interfering energy is being transferred from the culprit to the victim equipment. The two basic means of transfer are conduction and radiation and, although these are

NATURAL	MANMADE		
	NARROWBAND	BROADBAND	
	COCHANNEL 	POWERLINE 	FLUORESCENT LIGHT 
ATMOSPHERIC NOISE 	ADJACENT CHANNEL 	ARC WELDING 	IGNITION NOISE 
SOLAR NOISE (NOT IMPORTANT AT HF) 	HARMONIC 	SWITCH AND RELAY 	ROTATING MACHINERY 
GALACTIC NOISE (NOT IMPORTANT AT HF) 	INTERMODULATION 	RADAR PULSE 	OTHER* 

*INCLUDES INDUSTRIAL, SCIENTIFIC, COMMERCIAL, AND MEDICAL APPARATUS

Figure 5-9. Interference sources.

quite different in nature, many instances of interference occur because of a combination of the two effects.

(a) Interference by conduction can occur when two separate circuits have an element, or part of a load, in common. Voltages from one circuit, present in the common element, affect the other circuit. One of the most frustrating aspects of the common load is that there need not be a physical connection between the two circuits. Voltages in a wire can be transferred into another wire in close proximity because of certain coupling phenomena. These effects are true of tones or other alternating currents. They are not true of steady-state direct currents but are very much a factor where direct current is being keyed or switched either periodically or aperiodically. If such effects are noted in a new installation, either the design or the installation techniques, or both, are poor. If they suddenly appear in an installation, someone has changed something. One of the most frequent instances of a physical common connection is in grounding systems. If the earth leads from various circuits are connected to a common point as close as possible to the subterranean earth system (in the ground if possible), the interference effects will usually be minimized. If, however, they are joined together to a common earth wire some distance from the subterranean earth system, the impedance of the wire between the joint and the earth system constitutes a common load and interference can be experienced in this way. Given a sound installation in the first place, subsequent conduction interference is generally caused by proximity coupling and as a result of some action taken by a member of the site staff. If the transmission power levels in a system are maladjusted, and are higher than they should be, the proximity effects between adjacent cables or pairs in a particular multipair cable are intensified. Where the transferred voltage at the correct level setting might be imperceptible, it may be sufficient to cause interference if the level is set too high. Most cables carrying different types of signals at various levels, such as TTY, vf, and control, have metal foil or other types of shields to prevent or minimize coupling of the energy to other pairs in the same cable or other cables. The shield must be unbroken and properly bonded to earth at one end. If the shield is broken or the earth disconnected for any reason, the shield becomes ineffectual and interference can occur. This usually happens when someone has been doing work close to the cable in question. The work may have been

the laying of a new cable, retermination of cables, jumpering, or some other activity not directly associated with the cable in question but close enough to it physically to cause damage. Another possibility is that in installing and connecting a new cable and carrying out associated jumpering, the incorrect type of cable or jumper may have been used (for example, nonshielded cable or jumper when shielding was needed).

(b) Many C-E and other types of electronic equipment are designed to transfer information to the desired receptor by means of radiation, or wave propagation through space. Many equipments radiate without any requirement or intention to do so. Consequently, our atmospheric environment is heavily dosed with radiated electromagnetic energy. Owing to the limitations inherent in propagation, only a small fraction of the energy radiated from an emitter usually reaches the desired receptor. Some finds its way into outer space; some experiences absorption in various media. Owing to various other phenomena, such as reflection, refraction, and diffraction, the remainder finds its way as interference into receptors for which it was not intended. Sound frequency planning, together with the bandwidth and power limitations previously discussed, usually minimizes the degree of interference which is experienced. However, it is clear that the propagation media which exist between emitter and receptor, and the conditions existing in those media at any time, will have a profound effect on how and where the energy is distributed.

(2) A knowledge of these transfer mechanisms is invaluable as a starting point in trying to locate and eliminate sources of interference. In particular, it is important to establish how interference is entering a receptor, as well as where it comes from, because the nature of the transfer mechanism often determines what remedial action can, rather than should, be taken.

(3) Methods of locating interference were discussed in paragraph 2-10d.

g. Avoidance and Minimization of EMC Problems.

(1) Although a particular installation cannot always avoid being the victim of externally generated interference, there is much that can be done within the station to prevent it from either suffering from its own interference or inflicting it on someone else.

(2) Some of the most important factors in avoiding or minimizing EMC problems are knowledge of the station or facility and the capacity to

speedily detect any abnormalities. It is essential that operating personnel have knowledge of normal meter and indicator readings, normal signal sounds, and normal visual representations of signals on instruments such as cathode ray oscilloscopes and spectrum analyzers. This should be coupled with a thorough familiarity with all aspects of station configuration, and expertise in operational and maintenance procedures. The use of accurate station logs and performance records for channels and systems will then provide invaluable data on past problems, which may throw light on current problems.

h. Electromagnetic Radiation (EMR) Hazards. Whenever electromagnetic energy is radiated

there is a potential hazard to nearby ordnance, fuel, and personnel. The hazard criteria for ordnance items depends upon the type of items and their configuration (in storage, transit, and various stages of assembly). Prior to installing and operating transmitters in the vicinity of ordnance items HQ, USACEEIA (CCC-EMEO-ECD) should be contacted to determine if electromagnetic fields will be generated which may exceed the applicable hazard criteria. For installation of transmitters near refueling operations, 5 watts/cm² is considered to be hazardous. Power density of 10 mW/cm² is considered to be hazardous to personnel. The Environmental Hygiene Agency, Aberdeen Proving Ground, should be contacted concerning hazards to personnel.

CHAPTER 6

ESTABLISHING HIGH FREQUENCY COMMUNICATIONS WHEN FREQUENCY RELIABILITY TABLES ARE NOT AVAILABLE

6-1. General

Although it has been stressed in this manual that communicators should take advantage of the engineering and advisory services of USACEEIA, situations will inevitably arise in which such services will not be available. This chapter provides guidance to be used in the absence of more specific advice. Expedient methods of choosing operating frequencies and determining great circle distances and azimuths are discussed.

6-2. Short Groundwave Paths

a. General. With the transmitter power likely to be available, 80 km (50 mi) over sea water should be taken as the upper limit for groundwave service. Path lengths that may be supported by groundwave get progressively shorter as the ground conditions become poorer. Sometimes even with relatively high power, only 2 miles or less can be covered by groundwave, even with optimum frequency selections. Controlling elements include frequency, type of antennas, and the condition of the terrain over which propagation must be accomplished. Groundwave propagation is most effective at the lower frequencies, that is, 2 to 5 MHz. Unfortunately, noise (atmospheric, man-made, etc.) has a more detrimental effect at the lower frequencies, thus requiring a careful choice of operating frequency. The use of vertical antennas is strongly recommended for groundwave propagation since the signal radiated from a vertical source is not attenuated to as great a degree as that produced by a horizontal antenna. A sloping wire antenna is also an effective radiator for reliable groundwave communications.

b. Groundwave Range Charts. "Ground Wave Propagation Charts for Selected Areas World Wide" is a book published by USACEEIA to provide guidance about groundwave propagation for selected areas of the world. Various types of earth conditions are considered. Groundwave ranges are predicted for 2, 3, 5, 7, 10, 12, 15, 20,

and 25 MHz. Appendix D contains a somewhat more complete description of the charts and provides instructions for requesting copies. Use of these charts will give the communicator an indication of the range to be expected from groundwave service and provide a basis for deciding between the use of groundwave and skywave. It must be kept in mind, however, that in hilly and mountainous terrain the groundwave range may be much less than indicated by the charts.

6-3. Short and Intermediate Skywave Paths

"Intermediate and Short Distance Skywave Propagation Charts" is a series of separate volumes published by USACEEIA. Each volume covers a specific geographic area. The charts provide propagation predictions for distances up to 2,400 km (1,500 mi). The charts are described more completely, and instructions for requesting copies are provided, in appendix D. These charts, and information provided in paragraph 6-6, can be used in selecting operating frequencies.

6-4. Air/Ground Communications

Most air/ground communications are now carried on at vhf and uhf frequencies. However, because communications at vhf and uhf frequencies are restricted to essentially a line-of-sight range, there are two important applications of hf to air/ground service. These are communications at great distances and communications with low-flying helicopters in hilly and mountainous terrain. Communications with the low-flying helicopter are little different than any other short skywave path. For communications with aircraft at distances up to 8,300 km (4,500 nmi), USACEEIA publication "Air/Ground Sky Wave Propagation Charts for Selected World Wide Stations" may be used for guidance. The short title of the publication is "A/G Book." A description of the A/G Book and instructions for requesting copies are included in appendix D.

6-5. Estimating Azimuth and Great Circle Distance

USACEEIA will provide azimuth and great circle distance computations upon request. Instructions for requesting this service are included in appendix D. It is desirable that this information be obtained from USACEEIA in all cases. It is essential that it be obtained when highly directional antennas are used on paths more than a few hundred kilometers in length. On short paths, using a half-wave dipole, any of its variations, or any other broadbeam antenna, approximations described in paragraph 5-5b may be used. The antenna should be oriented using a magnetic compass, also as described in paragraph 5-5b. With broadbeam antennas the orientation need not be precise, and any available supports which provide approximate alinement may be utilized. Distance can be scaled from any available map.

6-6. Obtaining Information by Listening

a. A considerable amount of information about propagation conditions can be obtained by listening to signals which can be received. Any signal emanating from the vicinity of the distant station may be used. If a received signal is very strong, propagation conditions are good between the transmit and receive points. Unless transmitting power and type of antenna are known, reception of a weak signal does not give much information. The signal may be weak because the transmission is beamed in another direction. Variations in signal strength do give useful information. Table 6-1 lists the international call sign prefixes. K, W, and N are basic stems for United States stations. The United States is also authorized to use call letters AAA through ALZ. During peacetime, interception of amateur radio stations is recommended. The call sign gives the country of origin. Table 6-2 lists amateur geographical prefixes. In the United States the numeral following the initial prefix usually but not always defines the geographical area (fig 6-1). Table 6-3 lists the frequencies and type of emission for the amateur bands.

b. Another source of excellent propagation data is international time signals. For example, WWV, located at Ft. Collins, Colo., and operated by the National Bureau of Standards, broadcasts time signals on 2.5, 5, 10, and 15 MHz. WWVH, located in Hawaii, broadcasts the same signals on the same frequencies with the exception of 2.5 MHz. Several countries broadcast time signals on

hf frequencies. Some of these transmissions are included in table 6-4.

c. In order to obtain propagation data by signal intercept, it is virtually essential that a tunable receiver be available so that the bands can be scanned reasonably rapidly. The antenna must have relatively constant gain and directivity characteristics across the hf band, since evaluation is based on received signal intensities. Figure 6-2 shows sample data obtained from 24-hour propagation intercept. After a couple of days of sampling, a reasonably accurate propagation chart can be constructed. In the interim, the intercept data must be used for propagation guidance. When in doubt as to which signal to use of those indicated as propagating, select the highest or next to the highest band indicated. Be very careful in the use of frequencies above 21 MHz as they may have unstable propagation characteristics.

6-7. Calling Frequencies

Calling frequencies are frequencies designated for use in making initial contact between stations. In hf operations, three or four calling frequencies may be assigned so as to have an appropriate frequency available for various ionospheric conditions.

6-8. Characteristics of Various Frequencies in the Hf Spectrum

a. All radio waves are attenuated in passing through an ionized region. The attenuation is inversely proportional to the square of the frequency. Lower frequencies are attenuated more than higher frequencies. The optimum frequency for operation is only slightly less than the maximum usable frequency. For very short distances, the best operating frequency is only slightly below the critical frequency. For greater distances, as the elevation angle of the ray decreases, the best operating frequency increases.

b. Very low frequencies in the 2 to 30 MHz band are least affected by the solar cycle and are needed, especially at night, for successful skywave transmission over short paths. For very long distances, 10 MHz at night and 20 MHz during the day are useful. The optimum frequencies increase with increasing sunspot numbers.

c. The maximum usable frequency for long paths is almost always determined by the F2 layer. At short and intermediate distances, the E layer is controlling during the daytime.

Table 6-1. International Call-Sign Prefixes

Prefix	Country	Prefix	Country
AAA-ALZ	United States	NAA-NZZ	United States
AMA-AOZ	Spain	OAA-OCZ	Peru
APA-ASZ	Pakistan	ODA-ODZ	Lebanon
ATA-AWZ	India	OEA-OEZ	Austria
AXA-AXZ	Australia	OFA-OJZ	Finland
AYA-AZZ	Argentina	OKA-OMZ	Czechoslovakia
BAA-BZZ	China	ONA-OTZ	Belgium
CAA-CEZ	Chile	OUA-OZZ	Denmark
CFA-CKZ	Canada	PAA-PIZ	Netherlands
CLA-CMZ	Cuba	PJA-PJZ	Netherlands Antilles
CNA-CNZ	Morocco	PKA-POZ	Indonesia
COA-COZ	Cuba	PPA-PYZ	Brazil
CPA-CPZ	Bolivia	PZA-PZZ	Surinam
CQA-CRZ	Portuguese Territories	QAA-QZZ	(Service abbreviations)
CSA-CUZ	Portugal	RAA-RZZ	Soviet Union
CVA-CXZ	Uruguay	SAA-SMZ	Sweden
CYA-CZZ	Canada	SNA-SRZ	Poland
DAA-DTZ	Germany	SSA-SSM	United Arab Republic
DUA-DZZ	Philippines	SSN-STZ	Sudan
EAA-EHZ	Spain	SUA-SUZ	United Arab Republic
EIA-EJZ	Ireland	SVA-SZZ	Greece
EKA-EKZ	Soviet Union	TAA-TCZ	Turkey
ELA-ELZ	Liberia	TDA-TDZ	Guatemala
EMA-EOZ	Soviet Union	TEA-TEZ	Costa Rica
EPA-EQZ	Iran	TFA-TFZ	Iceland
ERA-ERZ	Soviet Union	TGA-TGZ	Guatemala
ESA-ESZ	Estonia	THA-THZ	France and Territories
ETA-ETZ	Ethiopia	TIA-TIZ	Costa Rica
EUA-EWZ	Soviet Bielorussia	TJA-TJZ	Cameroon
EXA-EZZ	Soviet Union	TKA-TKZ	France and Territories
FAA-FZZ	France and Territories	TLA-TLZ	Central African Republic
GAA-GZZ	Great Britain	TMA-TMZ	France and Territories
HAA-HAZ	Hungary	TNA-TNZ	Congo (Brazzaville)
HBA-HBZ	Switzerland	TOA-TQZ	France and Territories
HCA-HDZ	Ecuador	TRA-TRZ	Gabon
HEA-HEZ	Switzerland	TSA-TSZ	Tunisia
HFA-HFZ	Poland	TTA-TTZ	Chad
HGA-HGZ	Hungary	TUA-TUZ	Ivory Coast
HHA-HHZ	Haiti	TVA-TXZ	France and Territories
HIA-HIZ	Dominican Republic	TYA-TYZ	Dahomey
HJA-HKZ	Colombia	TYA-TZZ	Mali
HLA-HMZ	Korea	UAA-UQZ	Soviet Union
HNA-HNZ	Iraq	URA-UTZ	Soviet Ukraine
HOA-HPZ	Panama	UUA-UZZ	Soviet Union
HQA-HRZ	Honduras	VAA-VGZ	Canada
HSA-HSZ	Thailand	VHA-VNZ	Australia
HTA-HTZ	Nicaragua	VOA-VOZ	Canada
HUA-HUZ	El Salvador	VPA-VSZ	British Territories
HVA-HVZ	Vatican City	VTa-VWZ	India
HWA-HYZ	France and Territories	VXA-VYZ	Canada
HZA-HZZ	Saudi Arabia	VZA-VZZ	Australia
IAA-IZZ	Italy and Territories	WAA-WZZ	United States
JAA-JSZ	Japan	XAA-XIZ	Mexico
JTA-JVZ	Mongolia	XJA-XOZ	Canada
JWA-JXZ	Norway	XPA-XPZ	Denmark
JYA-JYZ	Jordan	XQA-XRZ	Chile
JZA-JZZ	Irian	XSA-XSZ	China
KAA-KZZ	United States	XTA-XTZ	Upper Volta
LAA-LNZ	Norway	XUA-XUZ	Cambodia
LOA-LWZ	Argentina	XVA-XVZ	Vietnam
LXA-LXZ	Luxembourg	XWA-XWZ	Laos
LYA-LYZ	Lithuania	XXA-XXZ	Portuguese Territories
LZA-LZZ	Bulgaria	XYA-XZZ	Burma
MAA-MZZ	Great Britain	YAA-YAZ	Afghanistan

Table 6-1. International Call-Sign Prefixes—continued

Prefix	Country	Prefix	Country
YBA-YHZ	Indonesia	5VA-5VZ	Togo
YIA-YIZ	Iraq	5WA-5WZ	Western Samoa
YJA-YJZ	New Hebrides	5XA-5XZ	Uganda
YKA-YKZ	Syria	5YA-5ZZ	Kenya
YLA-YLZ	Latvia	6AA-6BZ	United Arab Republic
YMA-YMZ	Turkey	6CA-6CZ	Syria
YNA-YNZ	Nicaragua	6DA-6JZ	Mexico
YOA-YRZ	Romania	6KA-6NZ	Korea
YSA-YSZ	El Salvador	6OA-6OZ	Somali
YTA-YUZ	Yugoslavia	6PA-6SZ	Pakistan
YVA-YYZ	Venezuela	6TA-6UZ	Sudan
YZA-YZZ	Yugoslavia	6VA-6WZ	Senegal
ZAA-ZAZ	Albania	6XA-6XZ	Malagasy
ZBA-ZJZ	British Territories	6YA-6YZ	Jamaica
ZKA-ZMZ	New Zealand	6ZA-6ZZ	Liberia
ZNA-ZOZ	British Territories	7AA-7IZ	Indonesia
ZPA-ZPZ	Paraguay	7JA-7NZ	Japan
ZQA-ZQZ	British Territories	7OA-7PZ	(not allocated)
ZRA-ZUZ	South Africa	7QA-7QZ	Malawi
ZVA-ZZZ	Brazil	7RA-7RZ	Algeria
2AA-2ZZ	Great Britain	7SA-7SZ	Sweden
3AA-3AZ	Monaco	7TA-7YZ	Algeria
3BA-EFZ	Canada	7ZA-7ZZ	Saudi Arabia
3GA-3GZ	Chile	8AA-8IZ	Indonesia
3HA-3UZ	China	8JA-8NZ	Japan
3VA-3VZ	Tunisia	8OA-8RZ	(not allocated)
3WA-3WZ	Vietnam	8SA-8SZ	Sweden
3XA-3XZ	Guinea	8TA-8YZ	India
3YA-3YZ	Norway	8ZA-8ZZ	Saudi Arabia
3ZA-3ZZ	Poland	9AA-9AZ	San Marino
4AA-4CZ	Mexico	9BA-9DZ	Iran
4DA-4IZ	Philippines	9EA-9FZ	Ethiopia
4JA-4LZ	Soviet Union	9GA-9GZ	Ghana
4MA-4MZ	Venezuela	9HA-9HZ	Malta
4NA-4OZ	Yugoslavia	9IA-9JZ	Zambia
4PA-4SZ	Ceylon	9KA-9KZ	Kuwait
4TA-4TZ	Peru	9LA-9LZ	Sierra Leone
4UA-4UZ	United Nations	9MA-9MZ	Malaysia
4VA-4VZ	Haiti	9NA-9NZ	Nepal
4WA-4WZ	Yemen	9OA-9TZ	Congo (Leopoldville)
4XA-4XZ	Israel	9UA-9UZ	Burundi
4YA-4YZ	International Civil Aviation Organization	9VA-9WZ	Malaysia
4ZA-4ZZ	Israel	9XA-9XZ	Rwanda
5AA-5AZ	Libya	9YA-9ZZ	Trinidad and Tobago
5BA-5BZ	Cyprus	A2A-A2Z	<i>Recent Additions</i>
5CA-5GZ	Morocco	A3A-A3Z	Republic of Botswana
5HA-5IZ	Tanzania	A4A-A4Z	Kingdom of Tonga
5JA-5KZ	Colombia	A5A-A5Z	Oman
5LA-5MZ	Liberia	A6A-A6Z	Bhutan
5NA-5OZ	Nigeria	C2A-C2Z	United Arab Emirates
5PA-5QZ	Denmark	C3A-C3Z	Republic of Nauru
5RA-5SZ	Malagasy	L2A-L9Z	Principality of Andorra
5TA-5TZ	Mauretania	S2A-S3Z	Argentina
5UA-5UZ	Niger		Bangladesh

d. Abnormal ionospheric conditions associated with solar disturbances disrupt high frequency communications to varying degrees. In severe instances there can be an almost complete blackout.

Paths traversing the polar regions are particularly sensitive to solar influences. Appendix G discusses solar disturbances and their effects on communications.

Table 6-2. Amateur Radio Geographical Prefixes

Prefix	Country	Prefix	Country
AA-AL	USA	LZ	Bulgaria
AP	Pakistan	N	USA
BV	Taiwan	OA	Peru
BY	China	OD	Lebanon
CE	Chile	OE	Austria
CM, CO	Cuba	OH	Finland
CN	Morocco	OK	Czechoslovakia
CP	Bolivia	ON	Belgium
CR9	Macao	OX, XP	Greenland
CT	Portugal	OZ	Denmark
CT2	Azores	PA, PD, PE, PI	Netherlands
CX	Uruguay	PY	Brazil
DA, DF, DJ, DK, DL	West Germany	SK, SL, SM	Sweden
DM, DT	East Germany	SP	Poland
DU	Philippines	SU	Egypt
EA	Spain	SV	Crete, Greece
EI	Ireland	TA, TC	Turkey
EP	Iran	TF	Iceland
ET3	Ethiopia	TG	Guatemala
F	France	TI	Costa Rica
G	England	UA	USSR
GM	Scotland	VE	Canada
GW	Wales	VK	Australia
HA	Hungary	VO	Canada
HB	Switzerland	VP9	Bermuda
HC	Ecuador	VS6	Hong Kong
HH	Haiti	VU	India
HI	Dominican Rep.	W	USA
HK	Colombia	XE	Mexico
HL, HM	Korea	XU	Khmer Republic
HP	Panama	XW	Laos
HR	Honduras	XZ	Burma
HS	Thailand	YI	Iraq
HV	Vatican City	YK	Syria
HZ, 7Z	Saudi Arabia	YN	Nicaragua
I, IT	Italy	YO	Rumania
JA, JE, JG, JH, JR	Japan	YU	Yugoslavia
JY	Jordan	YV	Venezuela
K	USA	ZA	Albania
KA	US personnel in Japan	ZB	Gibraltar
KC4AA, KC4US	Antarctica	ZE	Rhodesia
KG4	Guantanamo Bay, Cuba	ZL	New Zealand
KH6	Hawaii	ZP	Paraguay
KL7	Alaska	ZS1, 2, 3, 4, 5, 6	South Africa
KP4	Puerto Rico	3V	Tunisia
KS6	US Samoa	4U	ITU Geneva
KW6	Wake Island	4X, 4Z	Israel
KZ	Canal Zone	5A	Libya
LA, LB, LF, LG, LU	Norway	9H1	Malta
LU	Argentina	9Q	Rep. of Zaire
LX	Luxembourg		

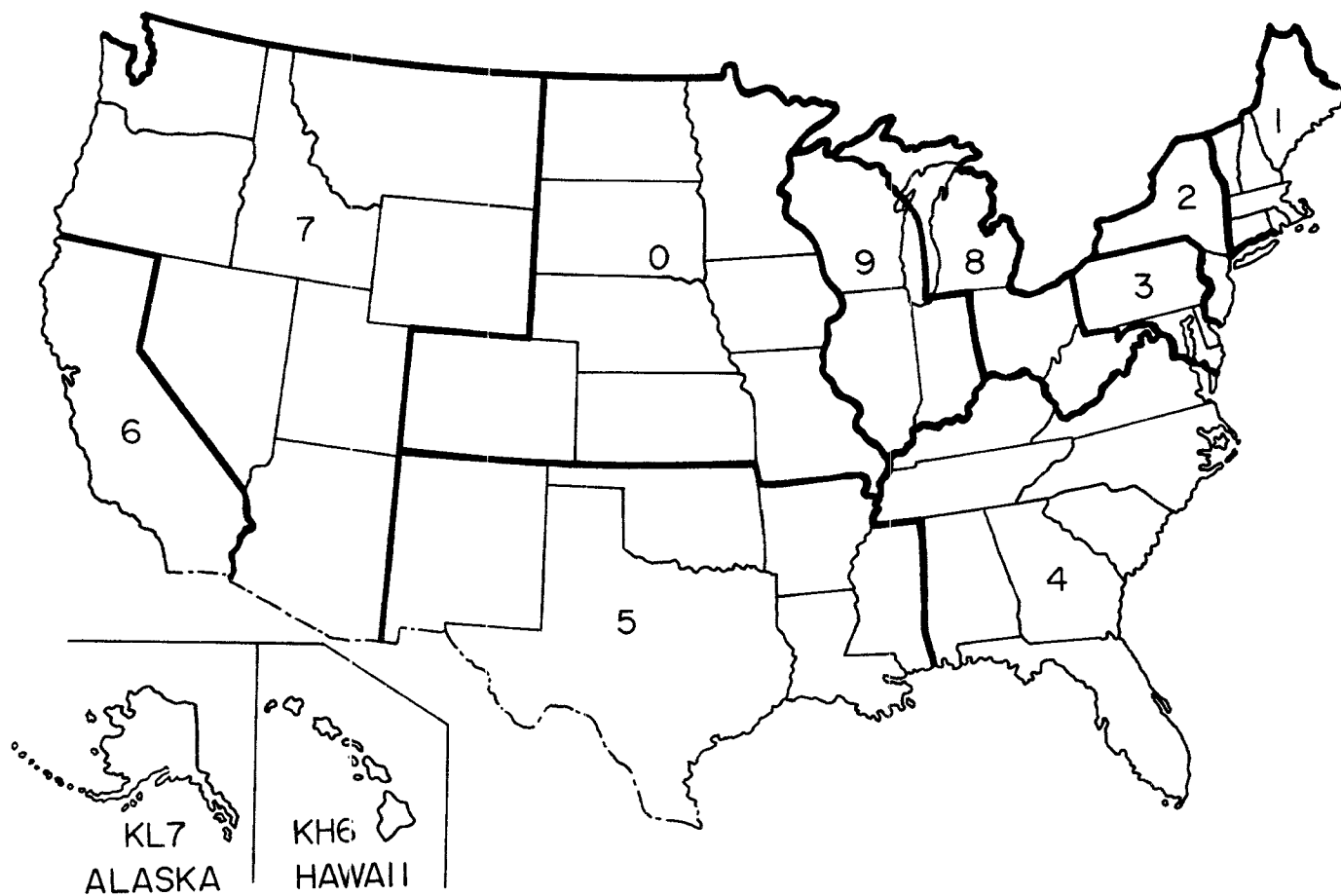


Figure 6-1. Amateur Call Area Designations.

Table 6-3. Internationally Allocated Hf Amateur Bands

Band designation	Frequency coverage	Emissions	Remarks
160 meter	1.800-2.000 (MHz)	0.1A1, 1.1F1, 3A3J, 6A3	Some power restrictions, worldwide
80 meter	3.500-3.800	0.1A1 and 1.1F1	Worldwide
75 meter	3.800-4.000	3A3J and 6A3	Worldwide
40 meter	7.000-7.100	0.1A1, 1.1F1, 3A3J, 6A3	Europe and Pacific only
	7.000-7.300	0.1A1, 1.1F1, 3A3J, 6A3	CONUS, Alaska, and South America
20 meter	14.000-14.350	0.1A1, 1.1F1, 3A3J, 6A3	Worldwide
15 meter	21.000-21.450	0.1A1, 1.1F1, 2A3J, 6A3	Worldwide
10 meter	28.000-29.700	0.1A1, 2.1F1, 3A3J, 6A3	Worldwide

Table 6-4. Standard Frequency and Time Stations

Call Sign	Location name	Latitude and longitude	Carrier frequency (MHz)
ATA	New Delhi, India	28° 34' N 77° 19' E	10
CHU	Ottawa, Canada	45° 18' N 75° 45' W	3.330 7.335 14.670
HBN	Neuchatel, Switzerland	47° 00' N 06° 57' E	5
JJY	Tokyo, Japan	35° 42' N 139° 31' E	2.5 5 10 15
LOL	Buenos Aires, Argentina	34° 37' S 58° 21' W	5 10 15
MSF	Rugby, United Kingdom	52° 22' N 01° 11' W	2.5 5 10
OMA	Prague, Czechoslovakia	50° 07' N 14° 35' E	2.5
RWM/ RES	Moscow, Soviet Union	55° 45' N 37° 18' L	5 10 15
VNG	Lyndhurst, Victoria, Australia	38° 00' S 145° 12' E	5.425 7.515 12.005
WWV	Fort Collins, Colo., USA	40° 41' N 105° 2' W	2.5 5 10 15
WWVH	Maui, Hawaii, USA	20° 46' N 156° 28' W	5 10 15
ZUO	Olifantsfontein, South Africa	25° 58' S 28° 14' E	5
ZUO	Johannesburg, South Africa	26° 11' S 28° 04' E	10

CHAPTER 7

PROBLEMS OF SHORT SKYWAVE PATHS

7-1. General

Short skywave paths of 650 km (400 mi) or less present certain difficulties. For one, groundwave propagation may interfere with the desired skywave propagation. For another, takeoff angles up to almost 90° are required.

7-2. Possibility of Interference Between Groundwave and Skywave

Sea water and ground of high conductivity cause groundwaves to propagate very efficiently. Where these conditions exist between the terminals of a short skywave path, the groundwave can cause destructive interference. Since vertically polarized waves result in better groundwave propagation, antennas intended for skywave paths where groundwave interference may be expected should be horizontally polarized to provide the maximum discrimination against the unwanted groundwave. Selection of an operating frequency as close as possible to the FOT will also tend to discriminate against groundwave interference because the higher frequencies are less effective in groundwave propagation.

7-3. High Takeoff Angles

a. Short skywave paths of course require high takeoff angles. Such angles introduce problems of antenna selection and construction. They also severely curtail the choice of operating frequencies.

b. Antenna types that will produce high takeoff angles generally have low gains.

c. The range between LUF and FOT, especially at night, is extremely small, limiting the choice of operating frequencies. Moreover, the proximity of the FOT to the lower frequencies magnifies the noise problems. The lower hf spectrum is more congested. It is also more susceptible to interference from natural causes such as lightning, precipitation and snow static, etc.

7-4. Low Operating Frequency

a. FOT has been defined as equal to approxi-

mately 85% of the MUF. The MUF in turn is a function of the critical frequency, f_c , and the angle of incidence, i , of the transmitted beam, that is, the angle between the direction of transmission and a line perpendicular to the surface of the earth. The critical frequency has been defined as the highest frequency that will be reflected back to earth when transmitted straight up (angle of incidence = 0°). The MUF is related to critical frequency and angle of incidence by

$$\text{MUF} = f_c \sec i. \quad (7-1)$$

As i increases with increasing transmission distance, $\sec i$ becomes larger and the MUF becomes higher. The corollary applies to short paths: the MUF is close to the critical frequency.

b. The lower limit on the range of frequencies that can be successfully used for skywave transmission is set by the LUF. Since the LUF does not vary in the same manner as the MUF (it is a function of the parameters which determine signal-to-noise ratio), the useful frequency range varies in accordance with the transmission distance. The shorter the distance, the smaller the range. For distances up to about 400 km (250 mi) nighttime frequencies range between 2 and 4 MHz, and daytime frequencies, between 4 and 8 MHz.

7-5. High Atmospheric Noise

a. Atmospheric noise is produced mostly by lightning and varies with location, season, time of day, and weather. In general, it is lower at the higher latitudes.

b. Atmospheric noise is higher at the lower frequencies. This provides one means for guarding against it: using the highest possible frequency, i.e., operating at or near the critical frequency. Since charted MUF are statistical medians derived from empirical data, there are times when actual MUF will be higher. Operation at higher frequencies during these times will reduce the atmospheric noise. Interception and analysis of

other signals emanating from the location of the distant terminal can provide information on usable frequencies. If available, data collected by an oblique ionospheric sounder will aid in selecting the operating frequency.

c. Other means for protecting against atmospheric noise consist of the use of directional antennas and high transmitter powers.

7-6. High Level of Interference From Other Stations

The lower hf frequency ranges are heavily congested. Large portions are allocated to aeronautical, maritime, broadcast and amateur uses. Consequently, electromagnetic interference (EMI) constitutes a severe problem at these frequencies.

7-7. High Level of Manmade Noise

Manmade noise is higher at the lower hf frequencies. It comes mostly from electrical machinery or internal-combustion-engine ignition systems within 100 m or so of the receiving antennas. It can also come from power lines in the vicinity. High-voltage transmission lines can produce interfering signals at even greater distances. The same is true for certain high-frequency medical devices such as diathermy equipment. Manmade noise levels, of course, vary widely with location, the noisiest being urban areas. Manmade noise is propagated primarily by groundwave, although it may also travel by skywave at frequencies below about 20 MHz. Manmade noise may also be transmitted over power lines. (Manmade noise is treated in chapter 2, paragraph 2-10d.)

7-8. Big Antennas

Big antennas are required to provide reasonable gain at the lower hf frequencies. The gain of big antennas is limited, however, by losses introduced by the extreme lengths of the antenna conductors. The size of the antennas is also a disadvantage from the point of view of installation in the tactical environment.

7-9. Low Gain Antennas

The antennas used for short skywave paths are usually low gain antennas. For vehicular communications, bent verticals and inverted L type antennas are used. For fixed applications, the doublet is the most commonly used antenna, although its structural requirements sometimes lead to the use of an inverted V antenna. The multiple doublet antenna (fig 7-1) may be used where it is necessary to operate well over several frequencies, although it is structurally more complex than either the single doublet or inverted V. This is a compromise antenna. If used for transmitting, it requires some "cut and try" to function within a reasonable VSWR for the frequencies involved. It consists of two or more doublets tied together and fed directly with 52-ohm coaxial cable. Very little interaction is noticeable and the antenna works well, provided the separation between wires is maintained, especially at the feed point. Using an antenna such as this will normally require use of three supporting masts, the center one to prevent sag. For broadband requirements, there are transportable log-periodic antennas such as the one shown in figure 7-2. Table 7-1 summarizes the

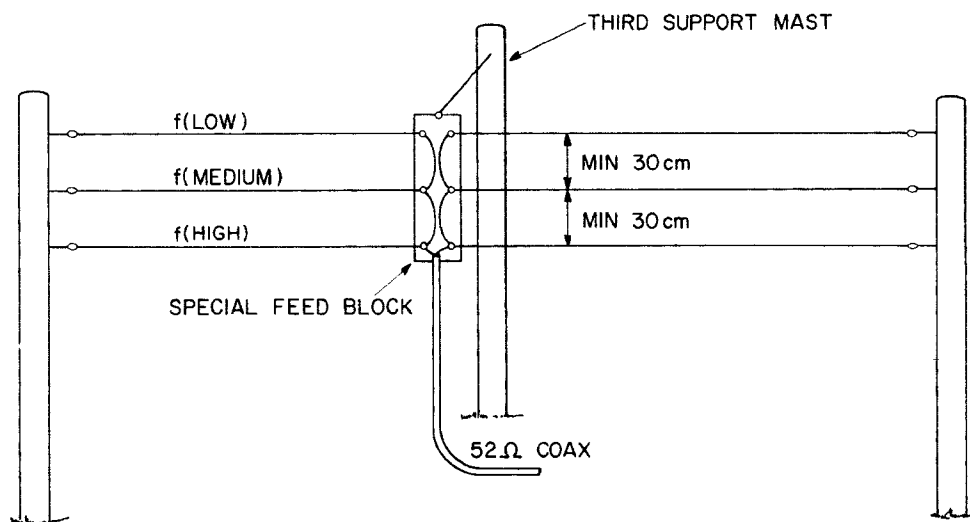


Figure 7-1. Multiple doublet antenna.

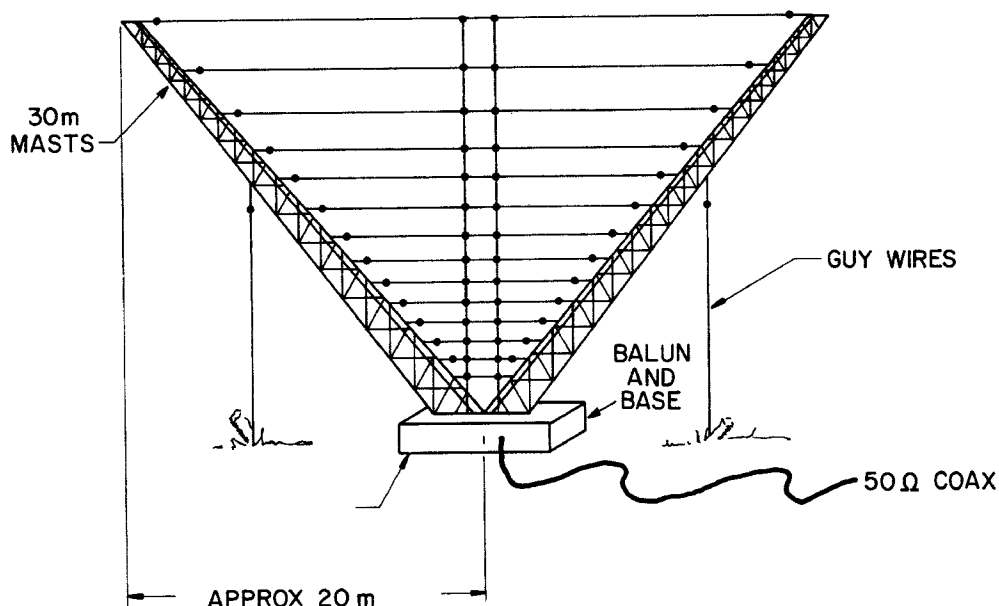


Figure 7-2. Transportable log-periodic antenna.

Table 7-1. Summary of Low Gain Antennas for Short Skywave Paths

Type	Description	Application	Construction
Doublet	$\lambda/2$ dipole constructed of wire, center fed. Can be a tape dipole.	Used for short and medium distance communication. Useful radiation normally 40° – 90° , good to about 1,300 km.	Wire, center fed normally with 52–72 Ω coax supported between masts or poles. Maximum radiation perpendicular to plane of wire. No tuning unit required.
Inverted V	$\lambda/2$ dipole constructed of wire or tape, center fed. Uses only one mast.	Used for short and medium distance communication. Slightly less efficient than normal doublet, particularly at short distances.	Wire, center fed normally with 52 Ω coax. Supported with single mast at center, ends drooping to ground, otherwise same basic characteristics as a doublet. No tuning unit required.
Multiple doublet	Compromise broadband antenna for hf.	Basically same results as doublet, but operates well on several frequencies in band.	Two or three doublets, all connected together. Fed with 52 Ω coax cable, wires separated by minimum of 30 cm.
Log-periodic	High-angle (90° – 40°) broadband antenna.	Used for very short path, 0 to 1,300 km, transmit or receive, up to 2.5 kW pep, 4–30 MHz.	Two 45° towers each approximately 30 m high supporting a log-periodic dipole curtain.

characteristics of the various low gain antennas used for short skywave paths.

7-10. The Dipole as a Field Antenna

The horizontal dipole, or doublet antenna, provides a ready solution to the problem of achieving high takeoff angles in the field. Doublets are easily installed and can even use trees of suitable height

for masts. For frequencies between 2 and 15 MHz, a height above ground of 10 m (33 ft) is recommended. A major disadvantage of the doublet antenna is its frequency sensitivity. To facilitate frequency changes, the segmented doublet (fig 4-16) can be used. It consists of an antenna with insulators at appropriate intervals such that they can be jumpered to change the electrical length of the antenna.

CHAPTER 8

HIGH FREQUENCY COMMUNICATIONS UNDER UNUSUAL CONDITIONS

8-1. General

a. Some problems in hf communications are peculiar to certain geographic areas and certain types of terrain. These problems are discussed in this chapter. The polar region is one such area. The polar ionosphere is very susceptible to the effects of solar disturbances. Hence, communication paths which traverse the arctic are disturbed more frequently than those at lower latitudes. Mountainous areas present problems both because of the rugged terrain and the poor soil conductivity. The dense vegetation in jungles absorbs groundwaves and presents problems in providing a clear path for skywaves. Special rules for antenna alignment apply near the magnetic equator. Desert areas present problems because of poor ground and atmospheric noise caused by blowing dust.

b. A high-altitude nuclear explosion creates intense ionization over a large portion of the earth. This ionization has very disruptive effects on many types of radio communications, including high frequency paths.

8-2. Special Problems in the Arctic

a. Groundwave Mode. The arctic regions have very poor groundwave propagation characteristics, poorer than those for sandy soil, forests, and even jungle. Hence, in those regions the skywave mode is the most practical means of hf communications.

b. Skywave Mode. Because of the effect of the earth's magnetic field on charged particles arriving from the sun, the ionosphere is affected more over the polar regions than over temperate or equatorial regions. These disturbances disrupt communications not only between two arctic stations but also between any two stations whose great circle connecting-path crosses a polar region.

(1) *Spread-F.* Spread-F may occur during an ionospheric storm and is particularly disruptive to communications in the arctic, where it is sometimes known as "auroral flutter." Spread-F occurs because of the breaking up or dispersion of the

F2 reflecting region. This brings about rapid fading which is disastrous to hf communications.

(2) *Shortwave fade.* A shortwave fade (or blackout) is a form of sudden ionospheric disturbance created when high energy X-rays and ultraviolet rays strike the ionosphere. This radiation causes a sudden increase in ionization of the D-layer which greatly increases the absorption of hf radio signals. Typically, a shortwave fade lasts 1 to 4 hours and occurs during the daylight hours. Generally, the low end of the high frequency band is affected first. The LUF rises rapidly and may result in a sudden radio blackout.

(3) *Polar cap absorption.* High-energy protons from the sun are deflected toward the poles by the earth's magnetic field. Here they produce increased ionization in the D-layer. Severe absorption of hf radio waves results and may last for several days (especially in the sunlit polar cap). Polar-cap absorption is generally followed by a geomagnetic storm.

8-3. Special Problems in Mountainous Terrain

a. Because of intervening terrain, it is frequently not feasible to use groundwave in mountainous areas. Hence, skywave may be used even for very short distances. For short skywave paths, nearby ridges and peaks will not be a problem because of the high angles being employed. For long paths, it may be difficult to obtain the horizon clearance required for low elevation angles.

b. Forested areas should be avoided, if possible. Signal attenuation in a thick forest will be similar to that in a jungle environment (para 8-4).

c. Poor ground will usually be found in mountainous areas. This is especially true on ridges and peaks. The use of a ground screen may be necessary.

8-4. Special Problems in a Jungle Environment

a. Groundwave attenuation in jungles and in other areas of dense vegetation is much greater than in other environments. The amount of at-

tenuation is dependent on the density of the foliage. Groundwave communications are limited to a mile or so for normal transmitter power when the transmitting and receiving antennas are located within the jungle growth. Figure 8-1 compares field intensities of various environments for a vertically polarized wave.

b. Another peculiarity of radio wave propagation in the jungle is the result of the discontinuity in the radio refractive index which exists at the jungle-air interface. This discontinuity may refract or reflect radio waves, as shown in figure 8-2. A wave may be refracted so as to travel along the interface, with a small amount of its energy being refracted or scattered to the receiving antenna. Such a wave is called a *lateral wave*.

c. Because of the jungle growth, the ground-reflected wave, the direct wave, and the wave reflected by the jungle-air interface are at least

60 dB below the lateral wave at distances greater than 1 km (0.6 mi). The lateral wave (the wave refracted by the jungle-air interface) is stronger than the skywave for very short distances. The distance at which the lateral wave and the skywave components become equal depends upon the time of day and the frequency being used. Because of attenuation of the lateral wave, the skywave will usually be stronger at long distances.

d. Raising antennas to minimize the portion of the path through the jungle foliage will improve communications. The horizontal doublet will probably be the best antenna, even for short distances. Sloping antennas should not be used.

8-5. Antenna Orientation Near the Magnetic Equator

a. In the usual cases of groundwave and skywave communications, the transmitting and receiving antennas are aimed so that their main

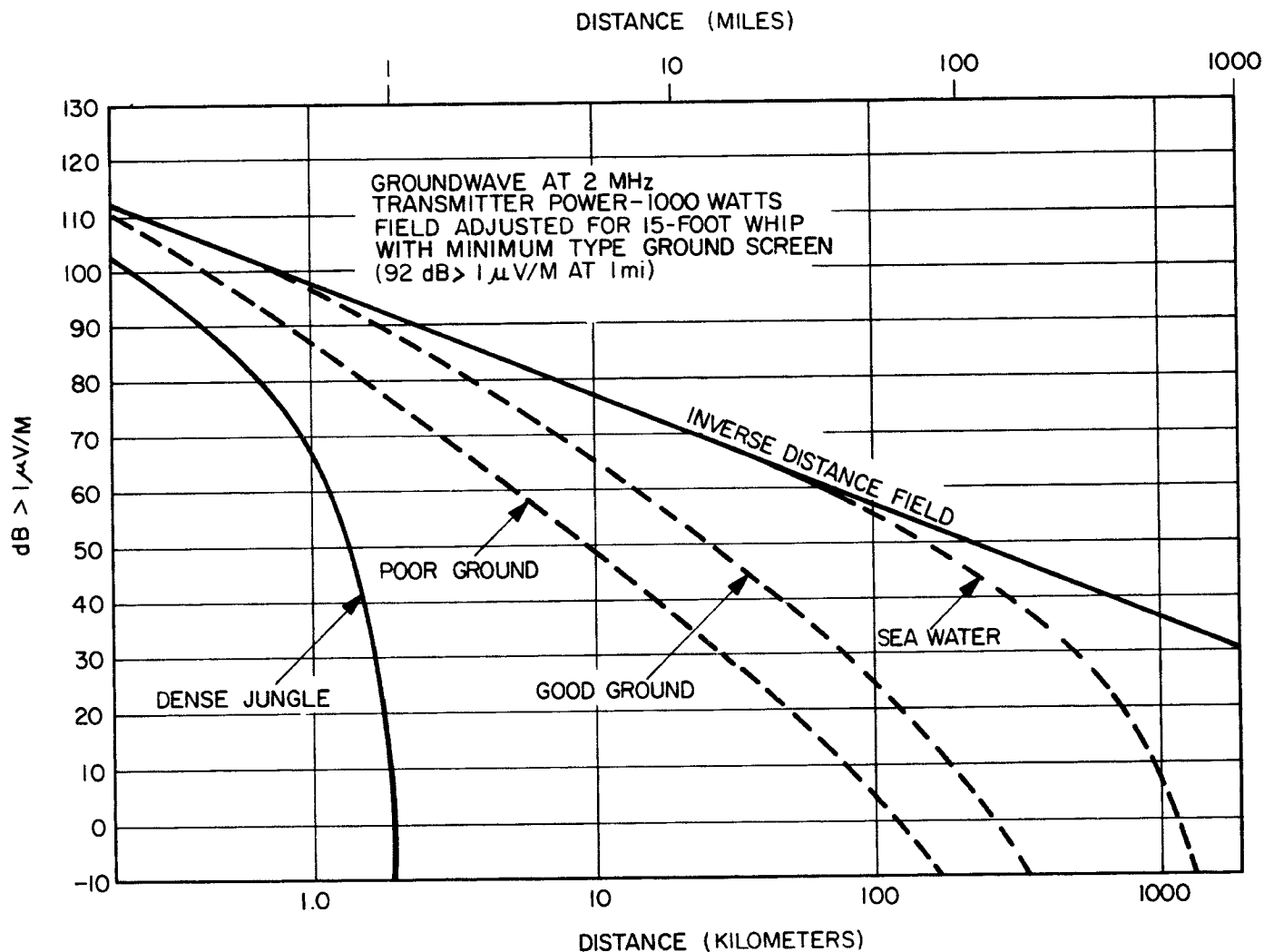
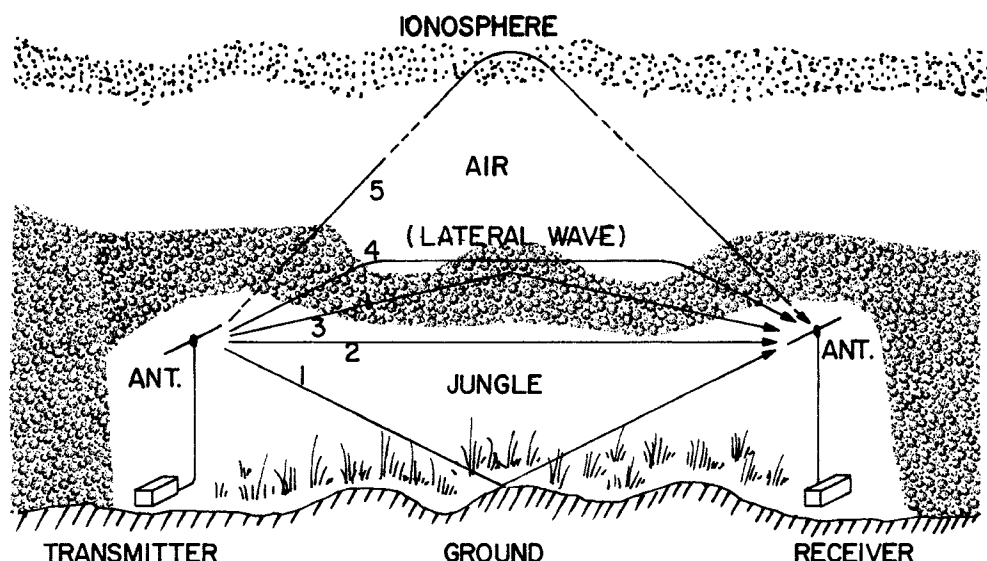


Figure 8-1. Groundwave field intensity for vertically polarized antenna over different types of terrain.



1. GROUND-REFLECTED WAVE
2. DIRECT WAVE
3. WAVE REFLECTED FROM JUNGLE-AIR INTERFACE
4. LATERAL WAVE
5. SKYWAVE

Figure 8-2. Radio paths in a jungle environment.

beams point to each other. However, theory and experiments indicate a possible variation from this rule for nearly vertical incidence skywave communications at latitudes near the magnetic equator. The ordinary wave (2-6b) is desirable because it usually has less attenuation than the extraordinary wave. The ordinary wave is best received in areas near the magnetic equator, on paths of 100 km (60 mi) or less, by alining dipole antennas in a north-south direction, regardless of the direction of propagation. Considering the effect of the magnetic field over the geomagnetic equatorial region, it has been shown that such usage of north-south dipoles for both transmitting and receiving—and hence utilization of the ordinary wave mode—would involve a relative gain over the orthodox antenna arrangement of 6 to 18 dB, depending on ionospheric conditions at the time.

b. However, during certain periods of the day, especially at near-vertical incidence (on very short-distance paths), when the transmission frequency is very near the ordinary wave critical frequency, the extraordinary wave may be more stable or less absorbed than the ordinary wave. In such cases, east-west alinement should be used.

c. The following recommendations are made for paths up to about 100 km (60 mi) near the geomagnetic equator.

(1) Horizontal half-wave dipole antennas are preferable to vertical quarter-wave whip antennas in all environments, especially in the dense forest; other horizontally polarized antennas, singly or in arrays, may be used in place of dipoles.

(2) Horizontal dipoles for both transmitting and receiving should be oriented in the magnetic north-south direction for paths shorter than 100 km (60 mi). For paths greater than 100 km (60 mi) use the classic broadside dipole alinement.

(3) Should the quality of reception become poor because the signal strength is dropping low into the background noise, either or both of the two following procedures should be tried:

(a) Shift the operating frequency to the next one allocated. Shift to the lower frequency first, then to the higher one if necessary.

(b) Try east-west dipoles if this alternative is available.

(4) If facilities for diversity reception are provided in the system, the cross north-south and east-west antennas should be used for receiving and the north-south antenna for transmitting.

8-6. Operating in a Post-Nuclear Environment

a. An atmospheric nuclear burst always produces large disturbances in the atmosphere. For bursts below about 25 km, the most significant disturbed region is a well defined fireball having dimensions ranging from fractions of a kilometer to tens of kilometers. If the burst occurs at shallow depths below the earth's surface, or at very low altitudes, i.e., a hundred meters or so above the earth's surface, large dust clouds are generated in the atmosphere in addition to the fireball.

b. For bursts at altitudes of about 80 km and higher the affected regions extend far beyond the fireball and are measured in hundreds of kilometers. For bursts between 25 and 80 km, there is a transition from the well defined fireball to the extended affected regions.

c. The increase of ionization in the D, E, and F layers by a nuclear burst may have several effects on hf propagation. First absorption can be increased, and the hf wave can be at least partially, if not totally, absorbed. The refraction characteristics of the E and F layers may also be significantly changed. Where absorption has not been total, there will be a change in propagation characteristics. Where there has been intense ionization of the D layer, hf waves may be totally absorbed in this layer. Or, only a portion of the wave may be absorbed and the remainder refracted back to the earth. In this instance, there

will be attenuation of the hf signal and changes in the hf propagation geometry.

d. Low altitude bursts could cause some interference for both the skywave and groundwave modes of propagation. This interference will not be significant unless the bursts are close enough to the transmitting and receiving antennas to affect a significant portion of the main transmission path. Noise from the fireball will also be a source of interference. Interference from dust will be negligible.

e. The interference with hf skywave communications from high altitude bursts may last many hours. The other types of interference discussed above are expected to last only a few seconds. Table 8-1 summarizes the effects of a nuclear burst on hf communications.

8-7. Special Problems in the Desert

a. Blowing dust often causes very high levels of atmospheric noise. This noise can be minimized by maintaining an antenna and surrounding area which are free of sharp points.

b. Due to the poor ground in a desert, special care should be taken to establish a good man-made ground for the vertical antenna to work against.

c. For safety reasons, metallic objects should be grounded with a good static discharge path to ground.

Table 8-1. Effects of Nuclear Burst on Hf Communications

Burst region	Mode of hf propagation	Approximate diameter of affected area	Degree of disturbance	Duration
Below 25 km	Groundwave and skywave	Km to tens of km	Blackout for very near stations, otherwise degraded performance	Few seconds
80 km and above	Skywave	Hundreds of km	Blackout to degraded performance	Minutes to several hours

Note. For bursts between 25 km and 80 km there is a transition of effects from those of a low altitude burst.

CHAPTER 9

MIXED PATH DISTANCES

9-1. Hf Nets

In the operation of hf nets it is not unusual to encounter a mix of path distances which requires special consideration. (In planning a net, early coordination with a propagation engineer is recommended.) The mix may be a combination of groundwave and skywave modes, or it may be a wide range of distances using only the skywave mode. A multipoint net can consist of fixed stations, mobile stations, and manpack stations (fig 9-1). After considering service requirements, the configuration and concept of operation of the net are developed. In some cases, it is impossible to meet the requirements with a single antenna at any one time. Important considerations are—

- a.* The service radius or area coverage required.
- b.* Antenna type and power output of radio transmitters involved.
- c.* Directions of stations from the net control station.
- d.* Requirement for all net stations to hear all other stations or only for all net stations to hear the net control station.
- e.* Special aeronautical and land mobile net entry requirements.

9-2. Sample Hf Net

a. Consider an hf net which has the net control station in San Antonio, Tex. The net control station requires hf contact with all Texas land mobile and fixed stations, and also requires contact with stations in Illinois, Missouri, Indiana, Kentucky, and other Midwest states. There is no requirement for each station to talk directly to all other stations or hear all transmissions directed to all other stations. A requirement does exist for all stations to talk directly with the net control station. Two radio positions at the net control station, should be considered, one using frequencies in the lower portion of the hf band to serve the Texas stations and one using higher frequencies to serve

the Midwest stations. Midway stations can use the frequency they receive best (fig 9-2).

b. Since the maximum distance involved is no more than 1,300 km (800 mi), the consideration of antennas for the net control station would include several doublets for the Texas stations and a log periodic, sloping to the ground, for the midwest stations. The doublets and log-periodic antennas should be at heights suitable for the required takeoff angles. The transmit and receive antennas should be separated to minimize receiver blocking or desensitization. A rotatable log-periodic antenna, in addition to the fixed antennas, would provide increased flexibility in operation.

c. Horizontal or sloping doublets would be considered for the nearby stations; and log-periodic antennas, aimed at the net control station would be considered for the distant station.

d. There may be a requirement in a similar net for all participants to hear all transmissions from the net control station. In this case, all transmissions are made on several frequencies simultaneously.

9-3. Intersite Links

Where transmitter and receiver sites are separated by several thousand feet or more, intersite links consisting of low power line-of-sight radio links are generally employed. A typical terminal is shown in figure 9-3.

9-4. Air-to-Ground

A conical monopole antenna is frequently used at the ground station for hf air-to-ground communications because of its omnidirectional, broadband coverage. It can be used for continuous monitoring and for short-range, two-way communications. For effective communications at greater distances, the conical monopole might be supplemented by a rotatable log-periodic. (The conical monopole is discussed in FM 11-486-3.)

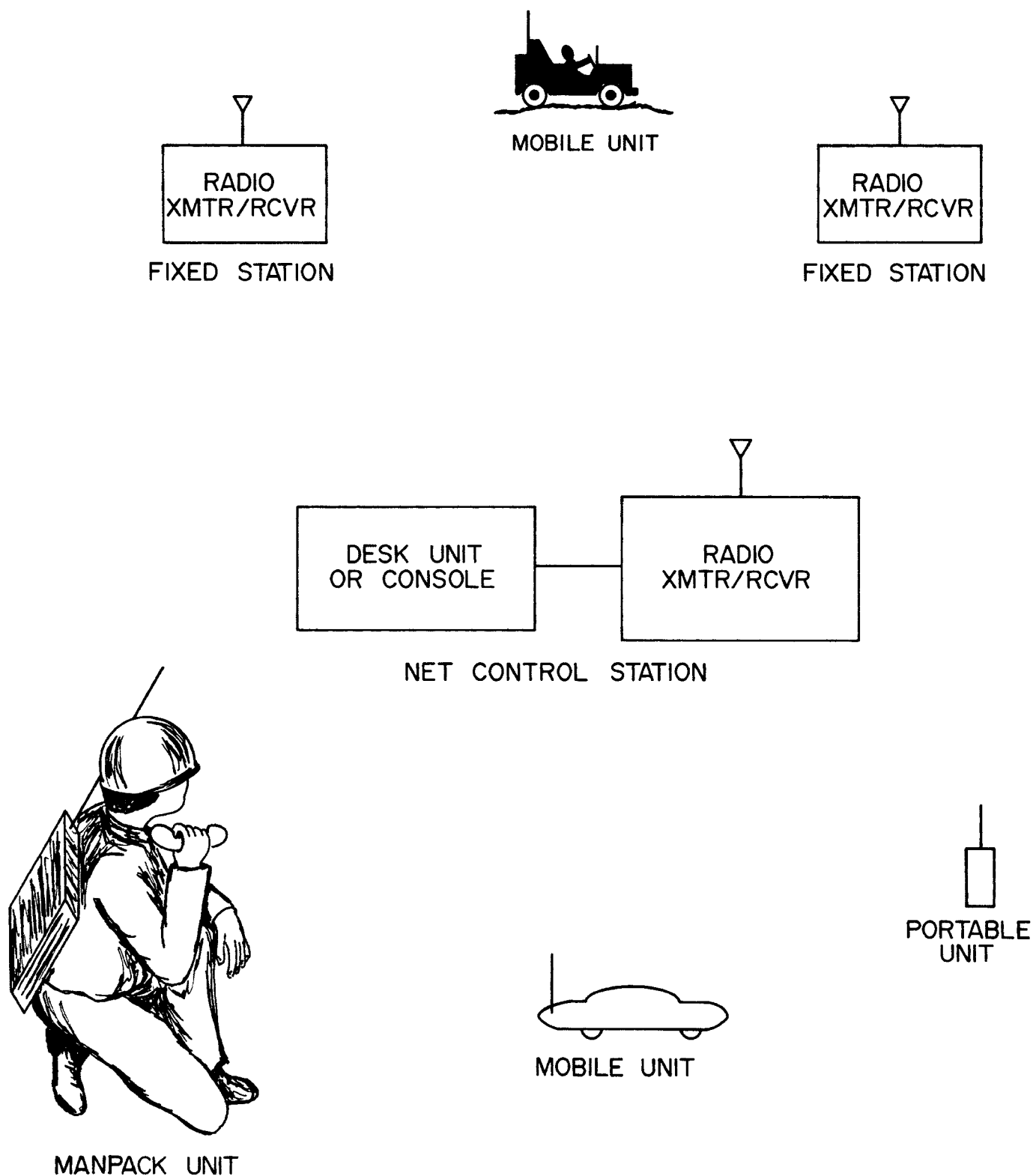
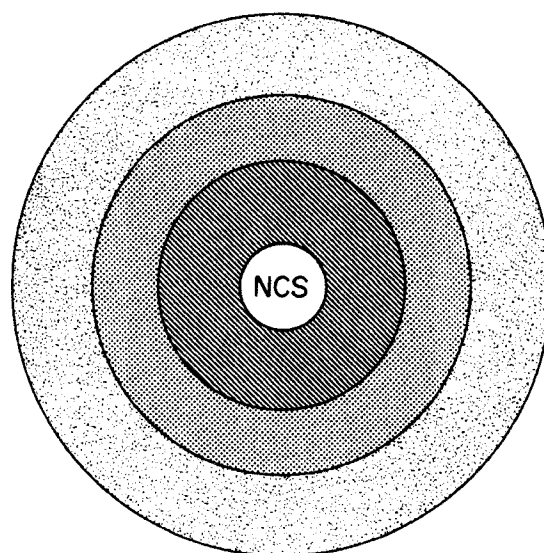


Figure 9-1. Multipoint hf radio net.

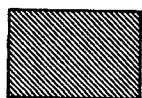
9-5. Complex Nets

In complex nets, it may be necessary for the net control to monitor and to have transmit capability on many frequencies simultaneously. Multiple receivers can operate from one broadband antenna

by the use of an rf antenna multicoupler. Outputs for 4, 8, and 12 receivers are common and more outputs can be provided. Sometimes the rf multicouplers are combined with antenna switches for greater flexibility. Antenna patch panels and



AREA USING HIGHER
FREQUENCY



AREA USING LOWER
FREQUENCY



AREA USING FREQUENCY WHICH
PROVIDES BEST PERFORMANCE

Figure 9-2. Two-frequency multipoint net.

switching matrices are used for even greater capabilities. A diagram of this arrangement is shown in figure 9-4. Figure 9-5 diagrams a net which

requires the use of many frequencies because of the large volume of traffic between net control and the individual stations.

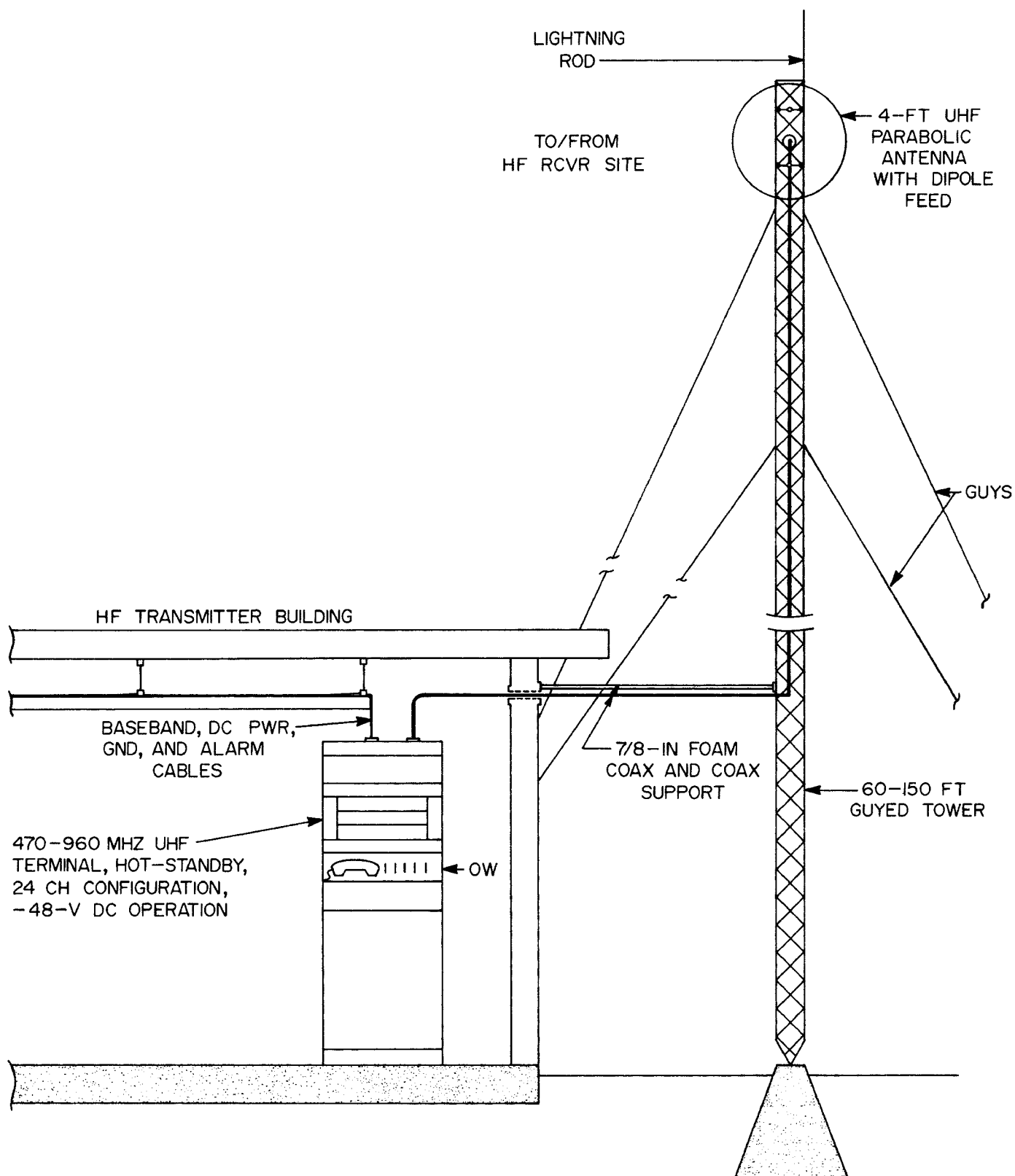


Figure 9-3. Example of uhf radio terminal configuration at hf transmitter site.

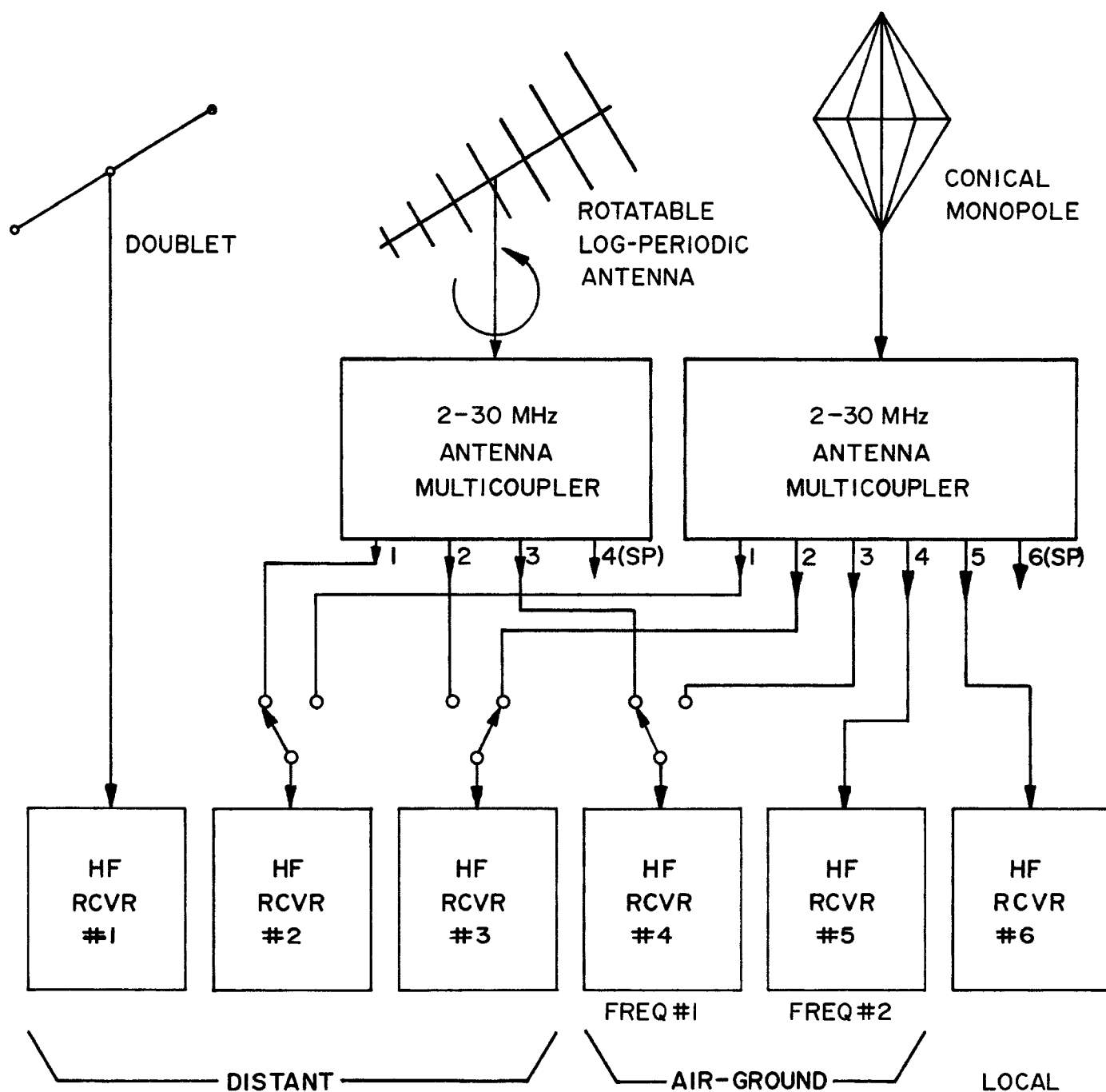
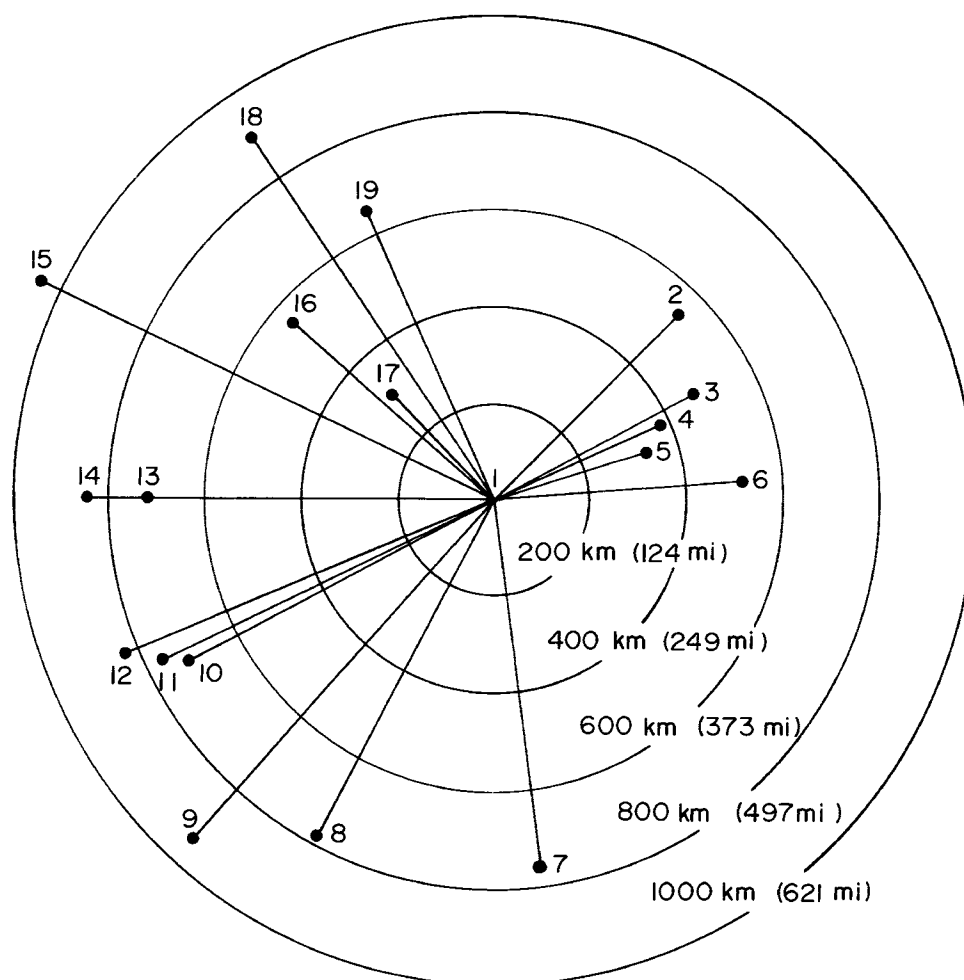


Figure 9-4. Example of receive multicoupler and antenna switching network.



STATION KEY

SITE	NAME	AZIMUTH	DISTANCE	
			km	mi
1	NET CONTROL	----	----	----
2	SITE A	25°	510	317
3	SITE B	60°	437	272
4	SITE C	62°	385	289
5	SITE D	70°	357	222
6	SITE E	84°	462	287
7	SITE F	173°	773	480
8	SITE G	196°	796	495
9	SITE H	207°	927	576
10	SITE I	241°	698	434
11	SITE J	244°	736	457
12	SITE K	246°	807	501
13	SITE L	270°	699	434
14	SITE M	270°	831	516
15	SITE N	295°	1075	668
16	SITE O	309°	579	360
17	SITE P	312°	304	189
18	SITE Q	326°	899	559
19	SITE R	333°	632	393

Figure 9-5. Hf net using many frequencies.

APPENDIX A

REFERENCES

Army Regulations (AR)

- | | |
|--------|---|
| 310-25 | Dictionary of United States Army Terms. |
| 310-50 | Authorized Abbreviations and Brevity Codes. |

Army Pamphlets (PAM)

- | | |
|-------|--|
| 310-1 | Index of Administrative Publications. |
| 310-3 | Index of Doctrinal, Training, and Organization Publications. |
| 310-4 | Index of Technical Manuals, Technical Bulletins, Supply Manuals (types 7, 8, and 9), Supply Bulletins, and Lubrication Orders. |

Field Manuals (FM)

- | | |
|-----------|---|
| 11-486-1 | Telecommunications Engineering: Planning Considerations. |
| 11-486-3 | Telecommunications Engineering: Transmission and Circuit Layout. |
| 11-486-19 | Telecommunications Engineering: Systems Engineering and Project Implementation. |

Technical Manuals (TM)

- | | |
|----------|--|
| 11-490-5 | Operational Electromagnetic Compatibility. |
| 11-666 | Antennas and Radio Propagation. |

APPENDIX B

DECIBELS

B-1. Expressing Power Ratios in Decibels

a. The decibel (dB) is a unit used to express the ratio between two powers. By definition, the ratio in dB's equals $10 \log_{10} P_1/P_2$, where P_1 and P_2 represent the powers being compared. The dB notation is applied to a wide variety of power ratios, such as the gain of an antenna over an isotropic radiator, the free space loss between antennas, or the signal-to-noise ratio at any point in a system.

b. Consider an amplifier which yields an output of 9.7 watts with an input of 0.9 watt.

$$\begin{aligned} \text{dB's} &= 10 \log_{10} 9.7/0.9 \\ &= 10 \log_{10} 10.78 \\ &= 10 \times 1.03 \\ &= 10.3 \end{aligned}$$

The amplifier has a gain of 10.3 dB.

c. Table B-1 shows the relation of dB to power ratio for a range from 1.0 to 10.0 and provides rules for extending the table. The table shows that a 3-dB change is equal to changing power by a factor of two. A 10-dB change is equal to changing power by a factor of 10. The smallest change detectable by the human ear is about 1 dB. Usually, a change of less than $\frac{1}{2}$ dB is not important. However, a series of such variations can be of extreme importance in system operation. For example, on a long coaxial transmission line having hundreds of repeaters, the gain of the repeaters must be held to a very close tolerance to keep the overall performance within acceptable limits.

d. Consider a transmission line which receives 100 watts from a transmitter and delivers 89 watts to the associated antenna.

$$\begin{aligned} \text{dB} &= 10 \log_{10} 89/100 \\ &= 10 \log_{10} 0.89 \\ &= 10 \times (-0.0506) \\ &= -0.506 \end{aligned}$$

The negative sign indicates a loss. Hence, the transmission line has a loss of $\frac{1}{2}$ dB.

B-2. Advantages in the Use of Decibels

a. Because dBs are logarithmic, performance calculations are reduced from a series of multiplications to a series of addition and subtraction steps. Consider a system with the following characteristics:

(1) The transmission line at the transmitter has a loss of 10 percent.

(2) The transmit antenna has a gain of four over an isotropic radiator.

(3) The free space loss between transmit and receive antennas is 99.999 999 999 9 percent of the transmitted signal. (Received power is 0.000 000 000 000 1 percent of transmitted signal.)

(4) The receive antenna has a gain of two over an isotropic antenna.

(5) The transmission line at the receive end has an efficiency of 85 percent.

What is the ratio of received power (P_R) to transmitted power (P_T)?

b. To solve without dB:

$$\begin{aligned} P_R/P_T &= 0.9 \times 4 \times 1 \times 10^{-15} \times 2 \times 0.85 \\ &= 6.12 \times 10^{-15} \end{aligned}$$

Note that the large value of path loss is difficult to handle.

c. To solve with dB:

(1) Transmit transmission line loss equals $10 \log 0.9 = 0.5$ dB.

(2) Transmit antenna gain equals $10 \log 4 = 6$ dB.

(3) Free space loss equals $10 \log 1 \times 10^{15} = 150$ dB.

(4) Receive antenna gain equals $10 \log 2 = 3$ dB.

(5) Receive transmission line loss equals $10 \log 0.85 = 0.7$ dB.

(6) P_R/P_T in dB = $5 + 6 - 150 + 3 - 0.7 = -142$ dB.

To verify that the two methods give consistent results: $10 \log 6.12 \times 10^{-15} = -142$.

d. Because the use of dB is simpler, it is customary to express gains and losses in dB, although amplifier and antenna gains are sometimes also expressed as ratios.

Table B-1. Relation of Decibel to Power Ratio

Decibel values are along top and left edge; power ratios are in body of table.

dB	0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0	1.00	1.02	1.05	1.07	1.10	1.12	1.15	1.17	1.20	1.23
	1.00	.977	.955	.933	.912	.891	.871	.851	.832	.813
	1.26	1.29	1.32	1.35	1.38	1.41	1.45	1.48	1.51	1.55
1	.794	.776	.759	.741	.724	.708	.692	.676	.661	.646
	1.58	1.62	1.66	1.70	1.74	1.78	1.82	1.86	1.91	1.95
2	.631	.617	.603	.589	.575	.562	.550	.537	.525	.513
	2.00	2.04	2.09	2.14	2.19	2.24	2.29	2.34	2.40	2.46
3	.501	.490	.479	.468	.457	.447	.437	.427	.417	.407
	2.51	2.57	2.63	2.69	2.75	2.82	2.88	2.95	3.02	3.09
4	.398	.389	.380	.372	.363	.355	.347	.339	.331	.324
	3.16	3.24	3.31	3.39	3.47	3.55	3.63	3.72	3.80	3.89
5	.316	.309	.302	.295	.288	.282	.275	.269	.263	.257
	3.98	4.07	4.17	4.27	4.37	4.47	4.57	4.68	4.79	4.90
6	.251	.245	.240	.234	.229	.224	.219	.214	.209	.204
	5.01	5.13	5.25	5.37	5.50	5.62	5.75	5.89	6.03	6.17
7	.200	.195	.191	.186	.182	.178	.174	.170	.166	.162
	6.31	6.46	6.61	6.76	6.92	7.08	7.24	7.41	7.59	7.76
8	.158	.155	.151	.148	.145	.141	.138	.135	.132	.129
	7.94	8.13	8.32	8.51	8.71	8.91	9.12	9.33	9.55	9.77
9	.126	.123	.120	.117	.115	.112	.110	.107	.105	.102
	10.0									
10	.100									

The following rules are used to extend the table:

Add to dB	Multiply ratio by	Subtract from dB	Divide ratio by
10	10	10	10
20	100	20	100
30	1,000	30	1,000
10n	10n	10n	10n

Sign of dB is positive for ratios greater than one, and negative for ratios less than one.

B-3. Use of Decibels with Voltage and Current Ratios

a. If the impedance of a circuit remains constant, the power is proportional to the square of the voltage. The ratio of powers in such circuits is the same as the ratios of the squares of the voltages.

$$\begin{aligned}
 \text{dB} &= 10 \log V_1^2/V_2^2 \\
 &= 10 \log (V_1/V_2)^2 \\
 &= 20 \log V_1/V_2
 \end{aligned}$$

To use this relation, it is essential that the impedances associated with the voltages be the same.

b. Although not commonly used, the dB notation can also be used with current ratios. The formula is the same as for voltage ratios.

$$\text{db} = 20 \log I_1/I_2$$

As with voltages, the impedances must be equal. Table B-2 shows the relation of dB to voltage and current ratios.

B-4. Use of Decibels to Express Power Levels

By designating a reference power, dB can be used to express absolute power values. When 1 mW is used as the reference, the unit is designated *dBm*. A power of 2 mW is 3 dB above 1 mW and is designated 3 dBm. Figure B-1 shows the relation of dBm to milliwatts. When 1 W is used as the reference, the unit is designated *dBW*. A 1 kW transmitter has a power of 30 dBW.

B-5. Use of Decibels to Express Sound Intensity

By defining a reference level, dB nomenclature can be used to express sound intensity. The generally accepted reference level is 1×10^{-16} watts per square centimeter. Sound level meters are calibrated to read the level in dB above this reference. Properly, such readings are designated as *dBap* (dB above reference acoustic pressure). In popular usage, the designation dB alone is frequently used. Table B-3 gives some typical sound levels.

Table B-2. Relation of Decibel to Current or Voltage Ratio (Equal Impedances)

Decibel values are along top and left edge; voltage and current ratios are in body of table.

	0	.1	.2	.3	.4	.5	.6	.7	.8	.9
	1.00	1.01	1.02	1.04	1.05	1.06	1.07	1.08	1.10	1.11
0	1.00	.989	.977	.966	.955	.944	.933	.923	.912	.902
	1.12	1.14	1.15	1.16	1.18	1.19	1.20	1.22	1.23	1.25
1	.891	.881	.871	.861	.851	.841	.832	.822	.813	.804
	1.26	1.27	1.29	1.30	1.32	1.33	1.35	1.37	1.38	1.40
2	.794	.785	.776	.767	.759	.750	.741	.733	.724	.716
	1.41	1.43	1.45	1.46	1.48	1.50	1.51	1.53	1.55	1.57
3	.708	.700	.692	.684	.676	.668	.661	.653	.646	.638
	1.58	1.60	1.62	1.64	1.66	1.68	1.70	1.72	1.74	1.76
4	.631	.624	.617	.610	.603	.596	.589	.582	.575	.569
	1.78	1.80	1.82	1.84	1.86	1.88	1.91	1.93	1.95	1.97
5	.562	.556	.550	.543	.537	.531	.525	.519	.513	.507
	2.00	2.02	2.04	2.07	2.09	2.11	2.14	2.16	2.19	2.21
6	.501	.495	.490	.484	.479	.473	.468	.462	.457	.452
	2.24	2.26	2.29	2.32	2.34	2.37	2.40	2.43	2.46	2.48
7	.447	.442	.437	.432	.427	.422	.417	.412	.407	.403
	2.51	2.54	2.57	2.60	2.63	2.66	2.69	2.72	2.75	2.79
8	.398	.394	.389	.385	.380	.376	.372	.367	.363	.359
	2.82	2.85	2.88	2.92	2.95	2.99	3.02	3.05	3.09	3.13
9	.355	.351	.347	.343	.339	.335	.331	.327	.324	.320
	3.16	3.20	3.24	3.27	3.31	3.35	3.39	3.43	3.47	3.51
10	.316	.313	.309	.305	.302	.298	.295	.292	.288	.285
	3.55	3.59	3.63	3.67	3.72	3.76	3.80	3.85	3.89	3.94
11	.282	.279	.275	.272	.269	.266	.263	.260	.257	.254
	3.98	4.03	4.07	4.12	4.17	4.22	4.27	4.32	4.37	4.42
12	.251	.248	.245	.243	.240	.237	.234	.232	.229	.226
	4.47	4.52	4.57	4.62	4.68	4.73	4.79	4.84	4.90	4.95
13	.224	.221	.219	.216	.214	.211	.209	.207	.204	.202
	5.01	5.07	5.13	5.19	5.25	5.31	5.37	5.43	5.50	5.56
14	.200	.197	.195	.193	.191	.188	.186	.184	.182	.180
	5.62	5.69	5.75	5.82	5.89	5.96	6.03	6.10	6.17	6.24
15	.178	.176	.174	.172	.170	.168	.166	.164	.162	.160
	6.31	6.38	6.46	6.53	6.61	6.68	6.76	6.84	6.92	7.00
16	.158	.157	.155	.153	.151	.150	.148	.146	.145	.143
	7.08	7.16	7.24	7.33	7.41	7.50	7.59	7.67	7.76	7.85
17	.141	.140	.138	.136	.135	.133	.132	.130	.129	.127
	7.94	8.04	8.13	8.22	8.32	8.41	8.51	8.61	8.71	8.81
18	.126	.124	.123	.122	.120	.119	.117	.116	.115	.114
	8.91	9.02	9.12	9.23	9.33	9.44	9.55	9.66	9.77	9.89
19	.112	.111	.110	.108	.107	.106	.105	.104	.102	.101
	10.00									
20	.100									

1. The sign of dB is positive for ratios greater than one, and negative for ratios less than one.

2. To extend the table, add 20n to dB, which is ratio multiplied by 10n.

B-6. Other Uses of Decibel Nomenclature

In addition to dBm, dBW, and dBap, many specialized dB designations have been devised. Those most commonly used in communications are discussed below.

a. The unit dBm0 expresses signal or noise power measurements made at, or referred to, a *zero level transmission point*. The unit is used especially with test tones. If the normal level at the zero level transmission point is 0 dBm, then 9 dBm0 is 9 dBm for that point in that system. At a point in that system normally 5 dB below the zero level transmission point, 0 dBm0 for that

point is -5 dBm. If a level of -10 dBm is normal for the zero level transmission point, 0 dBm0 for that point is -10 dBm. For a new transmission level point in the same system which is 5 dB below the zero level transmission point, 0 dBm0 is -15 dBm.

b. The relative signal level (dBr) is the ratio of power, expressed in dB, at the measurement point to the power at the zero transmission level point (0 TLP). A test point which should read 16 dB below the zero level reference point is a -16 dBr point. A signal at a -16 dBr point which is 1 dB too high reads -15 dBr.

c. The term dBm is used to denote *dB above*

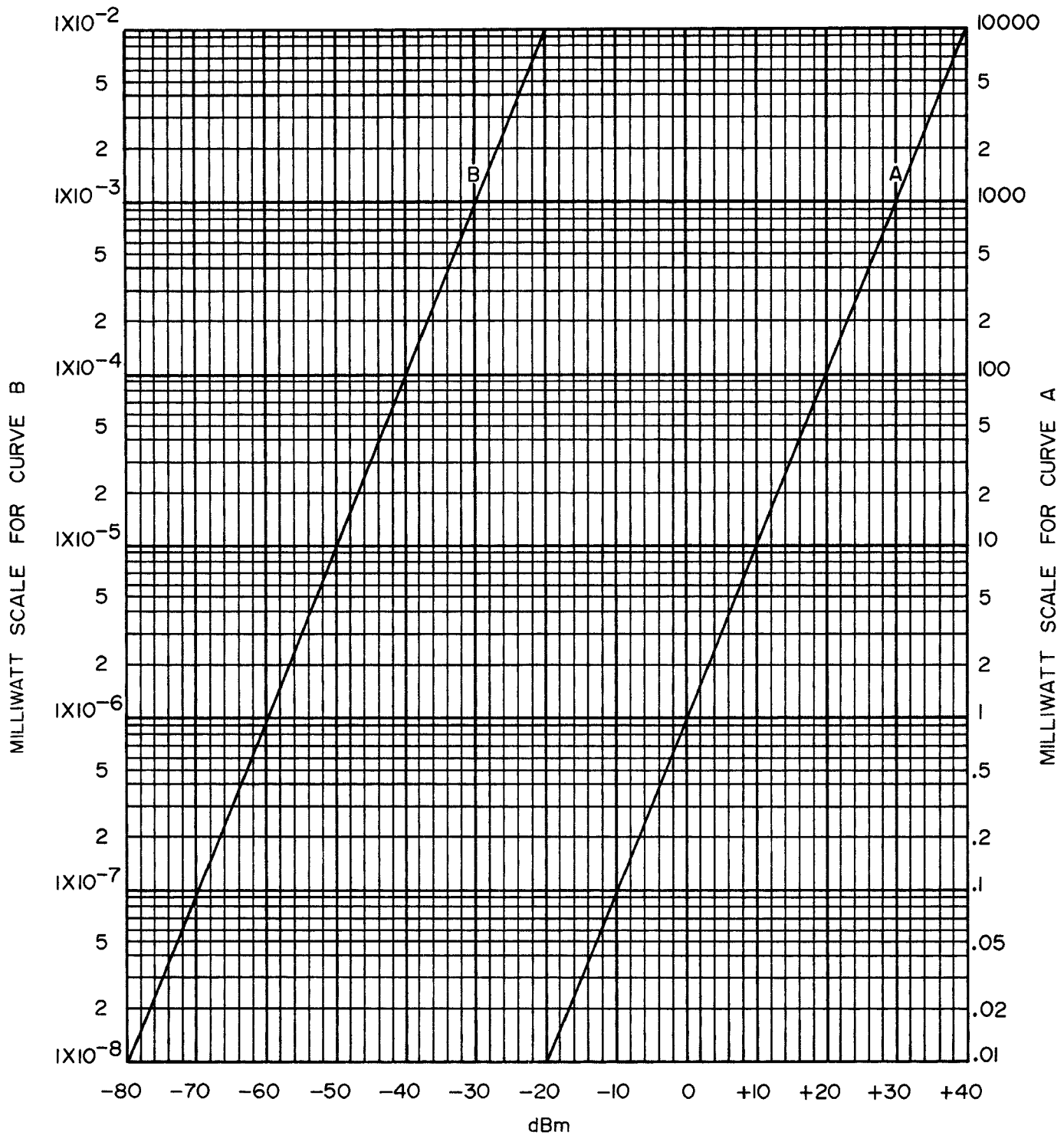


Figure B-1. Relation of dBm to milliwatts.

reference noise. The reference is one picowatt (10^{-12} watts; -90 dBm) of a 1,000-Hz tone measured through the weighting network. The term dBrnc indicates that the measurement is made through the C-message weighting network. The C-message network matches the 500-type telephone. The term dBrnc0 indicates that the C-message weighting network is used and that the

reference level is the noise power at the 0 TLP. The use of dBrnc0 and dBrnc0 allows noise measurement made at different points to be readily compared. They indicate a measurement made at or referred to the 0 TLP. For example, a noise reading of 43 dBrnc made at a +7 transmission point level is 36 dBrnc0 ($42 \text{ dBrnc} - [+7 \text{ dBr}] = 36 \text{ dBrnc0}$).

Table B-3. Typical Sound Levels

Source of sound	Sound level; dBrp*
Jet plane at 100 feet; pain	130
Propeller plane takeoff at 20 feet; thunder	120
Riveting shop	110
Superhighway; New York subway	90
Noisy street	85
Very noisy office; telephone use difficult	60
Average room	50
Quiet room	30
Broadcast studio	25
Soundproof vault; threshold of audibility	10
Average speech at lips	110
Average speech at microphone diaphragm	100
Average speech at 4 feet	70
Average whisper at 4 feet	25

*Decibels above 1×10^{-16} W/cm²

B-7. Relations Between Decibel Units

a. Figure B-2 shows the relation between the various units. The figure provides a number of equations for converting among the units. Some obvious relations are—

- (1) To convert from dBrn to dBrnc, subtract 2.
- (2) to convert from dBrnc, add 2.
- (3) To convert from dBrn to dBm, subtract 90.
- (4) To convert from dBm to dBrm, add 90.
- (5) To convert from dBrnc to dBm, subtract 88.
- (6) To convert from dBm to dBrnc, add 88.

Comparable rules apply to measurements referred to the zero transmission level point.

b. To convert from a unit in the upper half of the figure to a unit in the lower half, or vice versa, it is necessary to enter a dB value by the "TLP" block. This value is the normal level of the TLP above the 0 TLP. If the TLP being considered is at the same level as the 0 TLP, the numerical values in the two halves of the figure are the same.

c. Consider an example in which the normal level at the TLP is 7 dB above the 0 TLP and noise at that point measures 17 dBrn. Subtracting 7 from 17 gives 10 Brn0. Subtracting 3 from 10 gives dBrnc to dBrnc0. Subtracting 90 from 17 gives -73 dBm. Subtracting 7 from -73 gives -80 dBm0. Adding 88 to -80 gives 8 dBrnc0. Consistent values are obtained regardless of route followed.

B-8. Addition of Powers Expressed in Decibels

a. When two powers are expressed in dB with

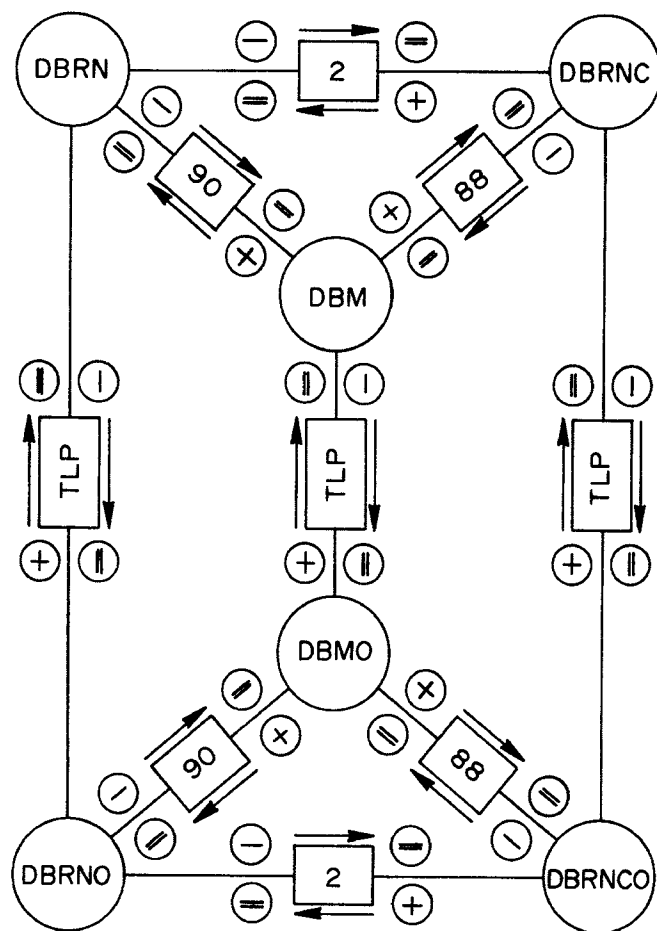


Figure B-2. Relations among decibel units.

respect to some reference level (for example, noise power in dBrn0 or signal power in dBm), their sum or difference is not obtained by adding or subtracting the individual dB values. The powers themselves must be added or subtracted and then the sum or difference must be converted to dBrn0, dBm, or dB with respect to the reference used. Figure B-3 is a graph for determining directly the dB value corresponding to the sum or difference of two powers, the values of which are known in terms of dB with respect to some reference.

b. For example, given the two powers 10 dBm and 6.5 dBm. First, find the arithmetical difference between them (3.5 dB). Spot this 3.5 dB value on the abscissa (horizontal scale) of the graph. A projection of this 3.5 dB point upward intersects the subtraction curve at 2.6 on the left vertical scale. To obtain the difference in *power* between the original value given, subtract 2.6 from the larger of the original values ($10 - 2.6 = 7.4$ dBm). To obtain the sum of the two powers, project the 3.5 dB point upward until it intersects the *addition* curve (at 1.6 on the right vertical

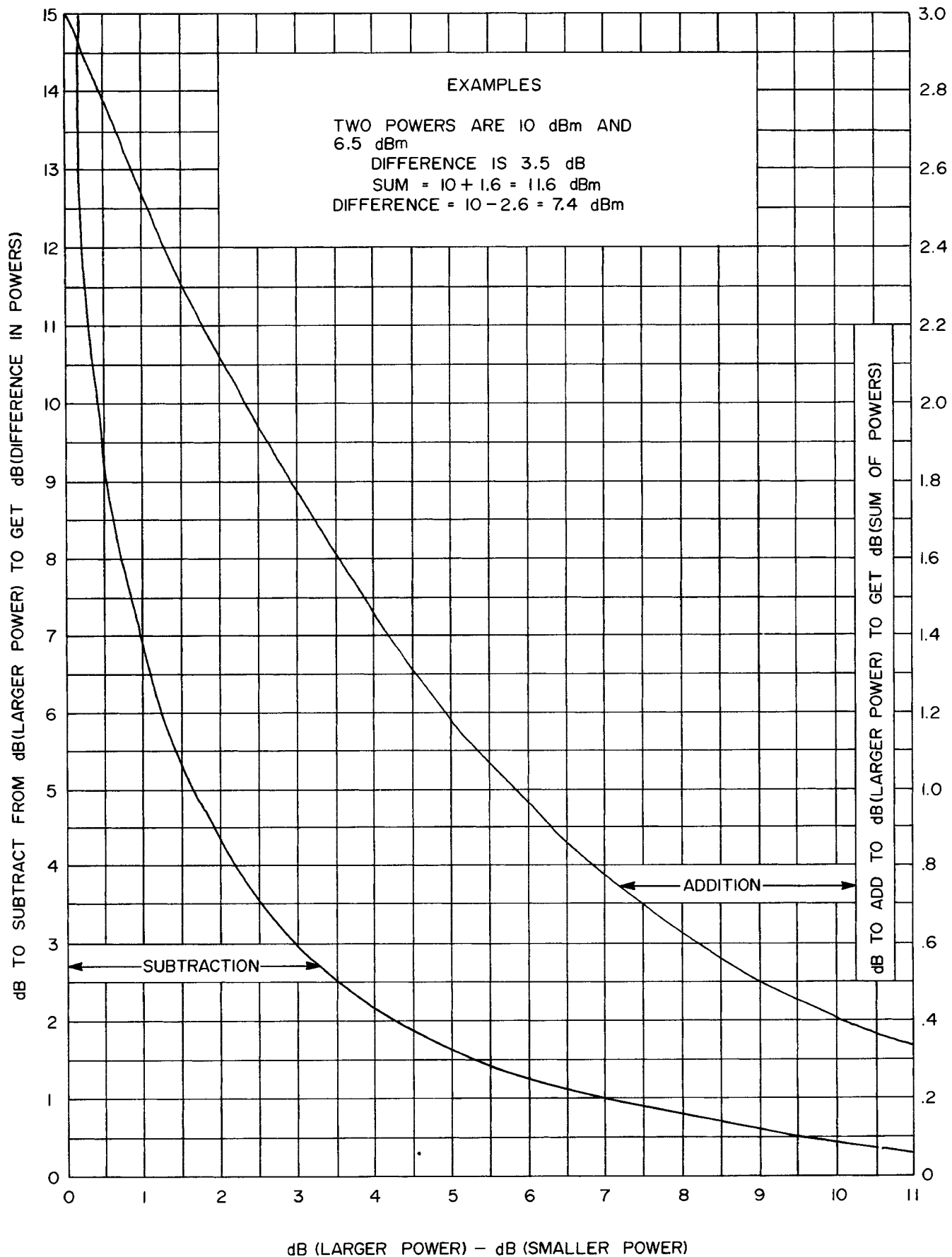


Figure B-3. DB corresponding to sum or difference of two powers.

scale). Then add 1.6 to the larger original value ($10 + 1.6 = 11.6$ dBm).

c. For a short-cut method of adding two powers expressed in dB above a common reference point, take the difference between the two dB values and find in the following chart the number corresponding to that difference; then add this number to the larger dB value. Most of these tabulated values are taken from the addition curve in figure B-3.

Difference in dB	Add to larger value	Difference in dB	Add to larger value
0	3.0 dB	7	0.8 dB
0.5	2.8	8	0.6
1.0	2.5	9	0.5
1.5	2.3	10	0.4
2.0	2.1	11	0.3
3.0	1.8	13	0.2
4.0	1.5	16	0.1
5.0	1.2	20	0.0
6.0	1.0		

d. When there are more than two powers to be added, the resultant may be obtained as shown in the following example. Assume that it is necessary to find the number of dBm that corresponds to

the total of the powers represented by these values of dBm: 10, 9.7, 6.5, 6.5, and 3. One method of solution by pairing is as follows:

Pairs of values	Difference between the 2 values	DB to add (from addition curve of figure B-3)	Resultant obtained after value from curve is added to larger of pair
10, 6.5	3.5	1.6	$10 + 1.6 = 11.6$
6.5, 3	3.5	1.6	$6.5 + 1.6 = 8.1$

Use the two resultants obtained above (11.6 and 8.1) as a pair:

11.6, 8.1	3.5	1.6	$11.6 + 1.6 = 13.2$
-----------	-----	-----	---------------------

e. To add a number of equal powers expressed in dBm (or dBrn0 or dBrnc), the sum may be found by adding the number obtained from the following formula to the value of one power:

Number of dB to be added $= 10 \log$ (number of equal powers).

For example, assuming four powers are each 30 dBm:

Number of dB to be added $= 10 \log 4 = 6$.
Adding 6 dB to the power of 30 dBm gives a total power of 36 dBm.

APPENDIX C

REQUIRED SIGNAL-TO-NOISE RATIOS FOR VARIOUS GRADES OF SERVICE

C-1. General

The primary factor in determining the circuit reliability is the signal-to-noise ratio, which is directly associated with a grade of service. The grade of service defines the type of communication desired. The grade of service may state the percentage of error-free messages in teletype transmissions, the intelligibility of voice transmissions, or the percentage of satisfied observers of facsimile transmissions. A minimum required signal-to-noise ratio is associated with the desired grade of service. This ratio is a function of the class of emissions, modulation index, modulation rate, and signaling codes, and includes the effects of fading, error-correcting schemes, noise reducers, optimum modulation and detection techniques, and diversity schemes. The user of the system may specify the grade of service or the signal-to-noise ratio. Representative signal-to-noise ratios for various grades of service are given in table C-1.

C-2. Bandwidth Considerations

a. Note that this table lists the required S/N ratio for the signal in the occupied bandwidth relative to noise in a 1-Hz bandwidth. To convert the values to the ratio S/N (signal to noise in the occupied bandwidth), subtract the dB equivalent of the actual signal bandwidth from the tabulated values.

b. For example, the tabulated required S/N for 6A3 emission, double sideband am, dual diversity, fading, operator-to-operator, is 48 dB. The equivalent required S/N for both signal and noise in a 3-kHz bandwidth

$$\begin{aligned}
 &= 48 \text{ dB} - 10 \log \text{ signal bandwidth} \\
 &= 48 \text{ dB} - 10 \log 3000 \text{ dB} \\
 &= 48 \text{ dB} - 10 \times 3.5 \text{ dB} \\
 &= 48 \text{ dB} - 35 \text{ dB} \\
 &= 13 \text{ dB}
 \end{aligned}$$

Table C-1. Required Signal-to-Noise Ratio in Occupied Bandwidth Relative to Noise in a 1-Hz Bandwidth

Radio telephone	Signal-to-noise ratio (dB)					
	Operator-to-operator ¹			Good commercial quality		
	Stable condition	Fading condition		Stable condition	Fading condition	
		No diversity	Dual diversity		No diversity	Dual diversity
6A3 Double sideband am	50	51	48	67	75	70
3A3 Single sideband am						
3A3A (reduced carrier)	48	49	46	65	73	68
3A3J (suppressed carrier)	47	48	45	64	72	67
6A3 Independent sideband am						
6A3 (2 voice channels)	49	50	47	66	74	69
9A3B (3 voice channels)	49	50	47	66	74	69
12A3B (4 voice channels)	50	51	48	67	75	70

¹ See note below table.

Table C-1. Required Signal-to-Noise Ratio in Occupied Bandwidth Relative to Noise
in a 1-Hz Bandwidth—continued

Radio teletypewriter	Signal-to-noise ratio (dB)					
	Character error rate ²					
	10 ⁻²		10 ⁻³		10 ⁻⁴	
	No diversity	Dual diversity	No diversity	Dual diversity	No diversity	Dual diversity
1.1F1						
FSK, 60 wpm, 1500-Hz filter						
Start-stop	56	51	63	59	69	64
Synchronous	51	47	59	55	66	61
3A7J						
SSB, suppressed carrier, 16 teletypewriter subchannels, each subchannel ± 42.5 Hz						
FSK, 110-Hz filter, 100 wpm						
Start-stop	62	59	69	65	75	70
Synchronous	57	53	65	61	72	67
6A9B						
ISB, 1 voice channel and 16 teletypewriter subchannels, each subchannel ± 42.5 Hz,						
FSK, 110-Hz filter, 100 wpm						
Start-stop	65	60	72	66	78	71
Synchronous	60	56	68	64	75	70
12A9B						
ISB, 2 voice channels and 32 teletypewriter subchannels, each subchannel ± 42.5 Hz,						
FSK, 110-Hz filter, 100 wpm						
Start-stop	66	62	73	68	79	73
Synchronous	61	57	69	65	76	71

¹ For 90 percent intelligibility of related words.² 5-unit code, no error control schemes.

APPENDIX D

HIGH FREQUENCY RADIO WAVE PROPAGATION ENGINEERING SERVICES AVAILABLE FROM USACEEIA

D-1. Summary of Services Available

High frequency radio wave propagation services available from the US Army Communications-Electronics Engineering Installation Agency (USACEEIA) are listed below and are discussed more completely in subsequent paragraphs.

a. Rf system performance analyses, including propagation predictions.

b. Antenna selection and antenna design.

c. Great-circle distance and azimuth calculations.

d. Advice on radio-wave propagation engineering matters.

e. Publications related to radio-wave propagation.

(1) Groundwave propagation charts for selected areas worldwide.

(2) Intermediate and short-distance skywave propagation charts.

(3) Air/ground skywave propagation charts for selected worldwide stations.

(4) Weekly Ionospheric Activity Report.

D-2. Requesting Service

a. *Availability of Services.* The services described in this appendix are available to all Department of Defense organizations activities. Requests may be made by mail or by telegraph. Telephonic consultation service is also available.

b. *Mail Requests.* Mail requests should be addressed to—

Commander
USACEEIA
ATTN: CCC-EMEO-PED
Fort Huachuca, Arizona 85613

c. *Telegraphic Requests.* Telegraphic requests should be addressed—

CDRUSACEEIA FT HUACHUCA AZ
//CCC-EMEO-PED//

d. *Telephonic Consultation.* Telephonic consultation service is available during duty hours, 1430 Z to 2315 Z. Phone numbers are AUTOVON 879-6779 or commercial (602) 538-6779.

D-3. Rf System Performance Analyses

a. *Performance Analysis Summary.* For a new system or radio net exercise, the "performance analysis summary" is offered. These summaries usually are in technical report format and describe propagation conditions for the paths in question, provide antenna design or selection recommendations, and give an accurate estimate of the system performance in terms of circuit reliability for a given type and grade of service. In some cases, the requester is interested in an analysis which provides advice as to antenna design and frequency selection for development of an operations plan for a full-scale communications exercise. At other times the requester needs to evaluate a number of alternative system configurations. A recently innovated figure of merit for making comparison analyses in accordance with military standards is being found to be useful for system designers and program managers.

b. *High Frequency Skywave Propagation Prediction.*

(1) *Description.* The USACEEIA Sky Wave Prediction Program is used to describe the ionospheric conditions expected to occur during the lifetime of the hf radio system. The program is capable of predicting path conditions for the frequency range of 2 to 30 MHz. These point-to-point propagation predictions are available on a one-time basis or a recurrent monthly or quarterly basis. Ionospheric conditions which are predicted for a given path from the worldwide data base are—

(a) Probability of the seven most predominant modes.

(b) Takeoff angle in degrees above horizontal.

- (c) Virtual height of the reflection layer.
- (d) Time delay of the signal.
- (e) Free space path loss.
- (f) Absorption loss.
- (g) Ground reflection loss.
- (h) Fraction of days that a particular mode is expected to exist at that hour during the specific month.

Many options of output form are available. One that is very useful to operators is the frequency reliability table. A sample table is reproduced as figure D-1.

(2) Frequency reliability tables.

(a) The heading gives information as to geographic location of transmitter and receiver, antenna dimensions, transmitter power, type of service, etc. Element length and height above ground may be given in meters for fixed physical lengths. When the electrical length or height remains fixed (the physical length varies with frequency) the dimension will be shown. For example, half-wave elements will be shown as "L-0.5." An antenna that is $\frac{1}{4}$ wavelength above ground would be shown as "H-0.25."

(b) No attempt is made to predict the probability of interference from signals originating from other transmitters. However, as a general rule, more interference can be expected on the lower frequencies. For this reason, when two or more frequencies show the same reliability, the highest frequency should be tried first.

(c) The frequency reliability table shown in figure D-1 is for Fort Huachuca, Arizona transmitting to Fort Sill, Oklahoma in April 1978. In the upper right hand corner of the table, "DRXB178R32" is an internal control number for USACEIA.

(d) In the upper left hand block, "100 PROJECT" is the project number for USACEIA. "DCA NO." precedes the DCA circuit number if there is such a number for the circuit. "ANT.TBL- 1 78082" is an internal control number for USACEIA.

(e) In the upper center block, "USACEIA FREQUENCY RELIABILITY TABLE" is the title. "MP POWER = 10DB DELAY = 2. OMS" refers to the probability that signals from the transmitter may propagate over different paths and thus arrive at the receiver at slightly different times and at different values of signal strength. For teletypewriter emissions, when two or more signals are within 10 dB of each other in strength and the delay equals or exceeds 2.0 milliseconds, distortion may result and errors may

appear in the message. "725.1 STAT. MILES 1166.9 KM" is the terrestrial length of the circuit.

(f) In the upper right hand block, "APRIL 1978" is the month for which the predictions apply. "SSN72.0" is the predicted sunspot number for that month. "TOS 1.1F1" is the type of service or emission. "REQD S/N = 57 DB" is the ratio of the signal in the occupied bandwidth to the noise in a 1 Hz bandwidth required to ensure a character error rate of not more than one error in one thousand characters for 1.1F1 emission with dual diversity. (2dB is allowed in the 57dB for transmitter coupling and transmission line losses.)

(g) In the "TX" block "FT HUACHUCA, ARIZONA" is the location of the transmitter. "69.56N—110.40W" is the set of geographical coordinates for that location. "69.56 TRUE, 57.41 MAGN." is the bearing of the propagation path from the transmitter. "FREQ 2-30" is the frequency range of the transmitting antenna in MHz. "ANTENNA DIPOLE" is the type of transmitting antenna. "H 10.0" is the height of the antenna in meters. "L—.5" is the length of the antenna in wavelengths. "A 0.0" indicates that no angle is associated with the antenna. "POWER = .400KW" is the output power of the transmitter.

(h) In the RX block, "FT SILL, OKLAHOMA" is the location of the receiver. "34.68N—98.42W" is the set of geographic coordinates for that location. "256.12 TRUE, 247.76 MAGN." is the bearing of the propagation path from the receiver. "FREQ 2-30" is the frequency range of the receiving antenna in MHz. "ANTENNA DIPOLE" is the type of receiving antenna. "H 10.0" is the height of the antenna in meters. "A 0.0" indicates that no angle is associated with the antenna. "3MHz MAN-MADE NOISE = -164 DBW" is the manmade noise in a 1 Hz bandwidth in a quiet rural location.

(i) In the table itself, the column labeled "FREQ" shows a maximum of 11 frequencies for which reliabilities are shown. When the trunk is a part of the DCA system, the frequencies may be taken from the DCA Frequency Directory. Where more than 11 frequencies are assigned, computations are made for frequencies selected to be representative of other nearby frequencies. In some cases the user may wish to designate a frequency complement. When the frequency complement is not known, eleven frequencies in the 2-30 MHz band are arbitrarily assigned for the purpose of computation. In each case, the computer rounds the frequencies to the nearest 0.1 MHz before they are printed, although the cal-

DRX8178R32

100 1001 ***** U S A C E E I A ***** APRIL 1978
 PROJECT EME * FREQUENCY RELIABILITY TABLE * SSN 72.0
 DCA NO. * MP POWER = 10DB DELAY = 2.0MS * TOS 1.1F1
 ANT.TBL- 1 78082 * 725.1 STAT.MILES 1166.9 KM * REQD S/N =57 DB

 TX FT.HUACHUCA, ARIZONA * RX FT.SILL, OKLAHOMA
 31.57N - 110.40W * 34.68N - 98.42W
 69.56 TRUE, 57.41 MAGN. * 256.12 TRUE, 247.76 MAGN.
 FREQ ANTENNA H L A * FREQ ANTENNA H L A
 2-30 DIPOLE 10.0 -.5 0.0 * 2-30 DIPOLE 10.0 -.5 0.0
 POWER = .400KW * 3MHZ MAN-MADE NOISE = -164 DBW

GREENWICH TIME

FREQ	02	04	06	08	10	12	14	16	18	20	22	24	
25.0	-	-	-	-	-	-	-	-	-	-	-	-	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
20.0	-	-	-	-	-	-	-	-	-	-	.03	.03	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
18.0	.05	-	-	-	-	-	-	-	-	-	.15	.16	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
16.0	.19	-	-	-	-	-	-	.09	.02	.06	.39	.40	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
14.0	.43	-	-	-	-	-	.12	.41	.90	.96	.93	.95	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
12.0	.90	.21	-	-	-	-	.51	.97	.99	.99	.98	.99	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
10.0	.98	.55	.29	.22	.15	.25	.99	.99	.99	.99	.99	.99	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
8.0	.97	.96	.90	.86	.74	.89	.99	.99	.50	.41	.93	.96	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
6.0	.88	.90	.89	.87	.91	.92	.94	.42	-	-	.23	.84	REL
	-	-	-	-	-	-	-	.40	-	-	.23	-	MP PB
4.0	.55	.54	.49	.46	.64	.67	.43	-	-	-	-	.30	REL
	-	-	-	-	-	-	-	-	-	-	-	-	MP PB
2.0	.02	.07	.08	.08	.20	.07	-	-	-	-	-	-	REL
	-	.04	.06	.05	.12	.07	-	-	-	-	-	-	MP PB
	02	04	06	08	10	12	14	16	18	20	22	24	
MUF	13.3	9.8	8.7	8.5	8.1	8.5	11.8	13.1	14.9	15.3	15.3	15.4	
FOT	9.6	7.1	6.7	6.5	6.2	6.5	9.7	10.8	13.0	13.3	11.9	12.0	

DASHES SIGNIFY THAT RELIABILITIES ARE ZERO

VALUES GIVEN IN MP PB LINES INDICATE THE FRACTION OF TIME MULTIPATH CAN BE EXPECTED TO CAUSE SYSTEM DEGRADATION

PROVIDE CIRCUIT UPDATE INFORMATION TO:
 U.S.ARMY COMMUNICATIONS-ELECTRONICS ENGINEERING INSTALLATION AGENCY
 ATTN: CCC-EMEO-PED
 FORT HUACHUCA ARIZONA 85613
 AUTOVON 879-6779

Figure D-1. Sample frequency reliability table.

culations are carried to three decimal places. The top row of the table shows Greenwich time at two-hour intervals, the times for which reliabilities are computed. Reliability ("REL") is defined as the percentage of days of the month that successful communication may be expected for a given two-hour period, month, and sunspot number. In the example shown, successful communication would be described as a character error rate not to exceed one character per thousand. If a lower character error rate is desired, the required signal-to-noise ratio must be increased and the reliabilities will change. The "MP PB" lines in the table show the probability of multipath at a given frequency and time.

(j) In order to provide the most accurate service, the parameters upon which the reliabilities are based must be correct. If the information is incorrect as to the type of antennas or their dimensions, transmitter output power, type of service, or geographical coordinates, the reliabilities may not be accurate. The user of the tables is urged to check the information in the heading and provide necessary corrections to the address at the bottom of the table.

(3) *Using a frequency reliability table.* The operator will normally choose the frequency having the greatest reliability at that time. If interference is experienced, the next highest reliability is tried. A frequency change plan will be developed which will provide a high degree of communications reliability without excessive changes of frequency.

(4) *Color propagation charts.*

(a) Upon request, USACEEIA can make available color propagation charts, similar to figure D-2. Colors are used to show regions of time and frequency in which a given quality of hf communications is predicted to be possible for 90 percent or more of the undisturbed days in the month. For example, a chart might have green for 12A9B service, yellow for 6A9B service, and red for order-wire quality 3A3J. Such a chart tells an operator not only where the best frequency is located at any hour, but also when to anticipate a need to reduce channel loading in order to keep a circuit on the air. This is extremely important when operating such radio terminals as the AN/TSC-25, AN/TSC-26, and AN/TSC-38B.

(b) The sample chart is for the hf link from MacDill AFB to Fort Huachuca, for the month of September 1975. Shown are the hours of the day in universal time and a frequency scale ranging from 2 to 30 MHz. Contour lines show predicted regions of best operation. For

example, at 0200 hours the best operating frequencies for 12A9B service fall between 6 and 10 MHz. After 0300 hours, 12A9B service is not predicted to be supported at any frequency. However, a reduced type of service consisting of one voice channel and one teletypewriter channel is shown to be usable on frequencies from 4.5 to 8 MHz. At 0700, 12A9B service can be resumed at about 8 MHz.

(c) Color propagation charts are used in a manner similar to the frequency reliability tables. However, use of the color charts eliminates the need for the operator to use several frequency reliability tables simultaneously, one for each type of service.

(5) *Information required from user.*

(a) Transmitter location, including latitude and longitude.

(b) Receiver location, including latitude and longitude.

(c) Power in watts delivered to transmitting antenna, identified as peak envelope power or carrier power.

(d) A complete description of the physical characteristics (fig D-3) of the transmitting antenna, including manufacturer's type number (if applicable). Information should be verified by actual physical inspection as far as practical, since changes may have been made that are not recorded on the station records.

(e) A complete description of the physical characteristics (fig D-3) of the receiving antenna, including spacing between diversity antennas and manufacturer's type number (if applicable). Information should be verified by actual physical inspection as far as practical, since changes may have been made that are not recorded on the station records.

(f) Angle subtended by any horizon obstruction in direction of transmission or reception. If no angle is indicated, zero will be assumed.

(g) Type of ground at transmitting and receiving sites. Specify dielectric constant and conductivity, if available, or classify ground as poor or good. If no ground information is provided, poor ground (dielectric constant of 4 and conductivity of .001 mhos/m) will be assumed.

(h) Descriptions of the terrain roughness and slopes in the reflection zones of the antennas. If no descriptions are furnished, smooth earth will be assumed. The earth can be considered to be smooth if the surface of the reflection zone has no abrupt changes greater than 0.1 wavelength of the highest operating frequency and has

MACDILL AFB TO FORT HUACHUCA

EMISSION DESIGNATION	NO DIVERSITY S/N (dB)	GRADE OF SERVICE
12A9B (GREEN)	73	10^{-3} BER
6A9B (YELLOW)	72	10^{-3} BER
3A3J (RED)	48	90% VOICE INTELLIGIBILITY

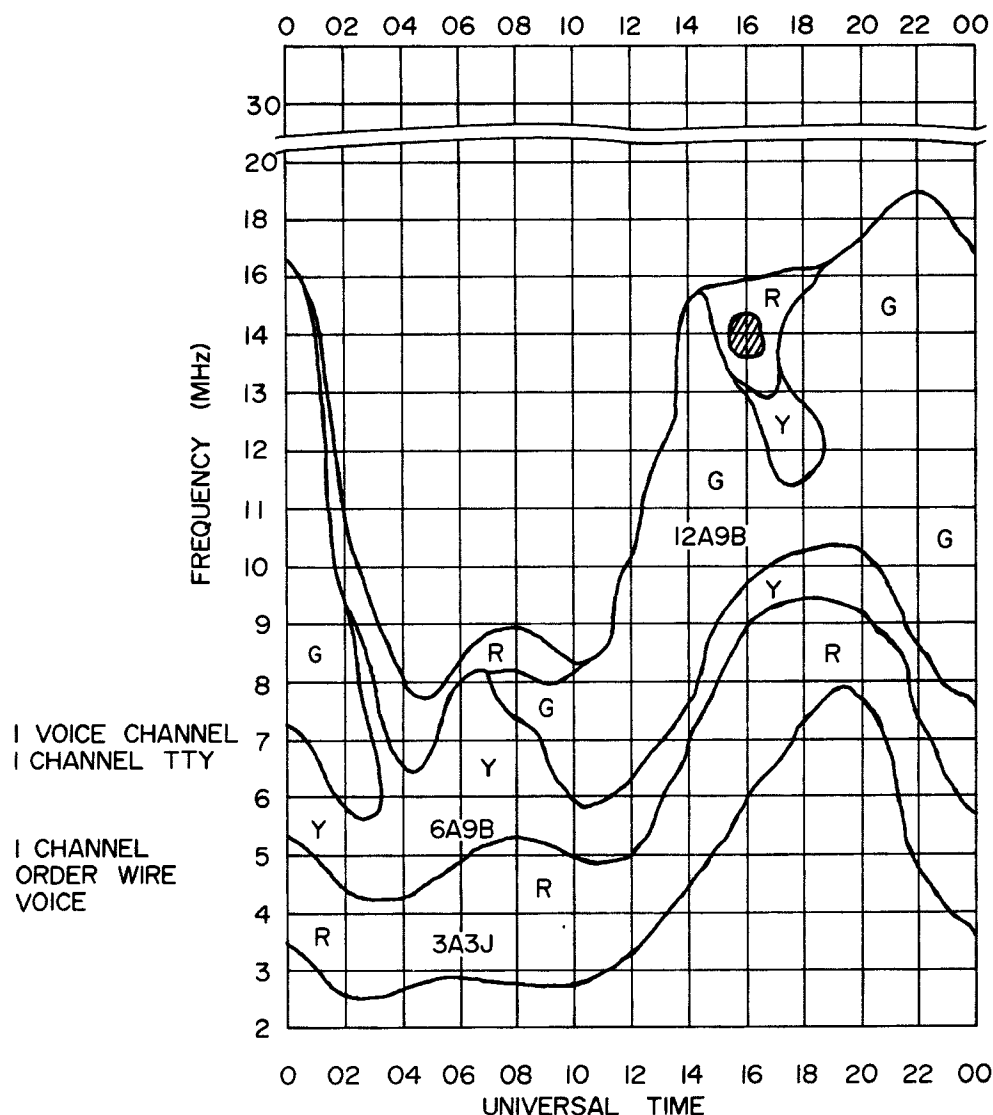


Figure D-2. Color propagation chart.

no slope greater than 10 percent in the direction of transmission or reception and five percent normal thereto.

(i) Median level of manmade radio noise at the receiving area as recorded with a short vertical lossless antenna, expressed in dB referenced to 1 watt in a 1-Hz bandwidth at 3 MHz, or

classification of the receiving area as urban (-136.5 dBw), suburban (-148.6 dBw), or quiet rural (-165.6 dBw). An urban location is defined as one within the industrial or business area of a large city, suburban as one near a large city or within a small town, and rural as one well removed from all populated areas and chosen to

1. RHOMBIC

height _____
 leg length _____
 tilt angle _____
 true bearing _____
 frequency range _____

2. VERTICAL

length _____
 length of radials _____
 radius of radials _____
 number of radials _____
 frequency range _____

3. HORIZ. $\lambda/2$ DIPOLE

height _____
 true bearing _____
 frequency range _____

4. HORIZONTAL YAGI

feed height _____
 driven element length _____
 reflector length _____
 true bearing _____
 director length _____
 no. of elements _____
 director spacing _____
 reflector spacing _____
 frequency range _____

5. VERTICAL DIPOLE

feed height _____
 length _____
 frequency range _____

6. SLOPING VEE

feed height _____
 leg length _____
 half-apex angle _____
 true bearing _____
 terminated height _____
 frequency range _____

7. INVERTED 'L'

height _____
 length _____
 true bearing _____
 frequency range _____

8. SLOPING RHOMBIC

feed height _____
 leg length _____
 half-apex angle _____
 true bearing _____
 terminal height _____
 frequency range _____

9. SLOPING WIRE

feed height _____
 length _____
 slope from horiz. _____
 true bearing _____
 frequency range _____

10. CONSTANT GAIN

gain _____
 frequency range _____

11. HORIZONTAL LPA

feed height _____
 rear element length _____
 slope from vertical _____
 true bearing _____
 unloaded transmission line
 impedance _____
 angle between array axis and
 element tips _____
 TAU _____
 no. of elements _____
 frequency range _____

12. TILTED DIPOLE

feed height _____
 length _____
 angle of element from
 horizontal _____
 true bearing _____
 frequency range _____

13. HALF RHOMBIC

leg length _____
 angle from ground _____
 true bearing _____
 frequency range _____

Figure D-3. Antenna description form.

be as free as possible of manmade radio noise. If nothing is specified, a quiet rural location will be assumed.

(j) Type of service including emission designation and complete description (for example—12A9B, three voice channels, 16-channel frequency-division multiplexing, noncoherent frequency-shift keying, 100 w/m/ch; start-stop teletypewriter; limiter-discriminator demodulator; 100-Hz bandpass filter in each channel).

(k) Maximum allowable bit error rate for data systems, maximum allowable character error rate for teletypewriter systems, and minimum grade of service (just usable or good commercial quality) for voice systems.

(l) Type of diversity, if any.

(m) If multipathing calculations are required, minimum difference in delayed signal power in dB and maximum difference in delay time in milliseconds. If multipathing figures are not provided, 10 dB and 2 milliseconds will be assumed.

(n) Operating frequencies. If operating frequencies are not provided or there are more than 11 of them, 11 representative frequencies will be used.

(o) The DCA trunk designator, if any.

(p) A statement as to whether the predictions should be for one time, monthly recurrent, or quarterly recurrent.

(q) Number of copies of each required.

(r) Organization mailing address.

(s) Name and telephone number of person who may be contacted for informal coordination of details of request.

(t) If predictions are required for both directions of a path, the information outlined in (a) through (s), above, should be provided for each direction.

c. Groundwave Propagation Predictions.

(1) *Description.* Computerized, theoretical techniques are available for predicting groundwave coverage for systems using vertically polarized antennas. Predictions are for a given signal-to-noise ratio at the output of the receiving antenna. These predictions are used in a manner similar to the skywave predictions. Groundwave propagation is much more stable than skywave propagation, and fewer frequency changes are required.

(2) *Information required from user.* The following information is required for groundwave system performance analysis, in addition to those

items listed under Skywave Propagation Predictions.

(a) Description of propagation path in terms of dielectric constant and conductivity of ground or water. If the path is inhomogeneous, specify the distances from the reference point to the junctions. If the ground constants are not known, classify ground in terms of poor or good, and water in terms of fresh or salt.

(b) For regions having heavy forestation or jungle cover, describe distances from the reference point that this area includes.

(c) For regions which are considered rough (very hilly terrain or mountainous), a terrain path profile should be included.

D-4. Antenna Selection and Antenna Design

USACEEIA has the capability to give advice regarding antenna selection and, if necessary, to design antennas to meet special needs. Requests for such services should include the information required for a high frequency skywave or for a groundwave rf system performance analysis, along with any antenna size, height, or other limitations. Preferably, a site plot plan of existing antenna fields should be provided. Include the name and telephone number of a person who may be contacted for informal coordination.

D-5. Great-Circle Distance and Azimuth Calculations

Great-circle computations are very laborious when performed by hand, using logarithmic and trigonometric tables. However, USACEEIA has computer programs for making these calculations and will furnish this information on request. Whenever the combination of path length and antenna directivity makes the azimuth critical, this service should be requested. Information required of the user is latitude and longitude of transmitter and receiver, to the nearest minute.

D-6. Publications

a. Groundwave Propagation Charts for Selected Areas Worldwide.

(1) This document provides predicted distance ranges for groundwave propagation at 2, 3, 5, 7, 10, 12, 15, 20, 25, and 30 MHz. Currently 24 areas, covering the continental United States, Central America, Europe, Africa and parts of Asia, South America, and Greenland, are defined. Predictions of this document cover all seasons. They are applicable to any year and are issued on a one-time basis until such time as revisions war-

rant a reissue. Requests for these charts need state only the number of copies desired.

(2) The charts are for 15-foot whips, 32-foot whips, 50-foot sloping wire, and one-quarter wavelength antennas. These antennas are representative of those used with tactical equipment and are not the best to use for maximum groundwave range. Ten categories of transmitter powers cover from 5 to 4,000 watts. Various services ranging from Morse code CW to two 3-kHz independent sidebands are included. Three types of earth condition are considered: poor ground, sea water, and, when applicable, rice paddies. The charts are for vertically polarized signals.

b. Intermediate and Short Distance (ISD) Sky-wave Propagation Charts.

(1) "Intermediate and Short Distance Sky-

wave Propagation Charts" is a series of separate volumes providing propagation predictions. The area covered by each volume is shown in figure D-4. Within each volume the predictions are given for six seasonal intervals: January-February, March-April, May-June, July-August, September-October, and November-December. The charts show the upper and lower limiting frequencies for communications in the 2- to 30-MHz band for distances up to 2,400 km. Predicted values of frequencies are shown for the solar activity extremes for a normal solar cycle (solar activity indexes of 10 and 110). Values of frequency predictions may be obtained for any time during the normal solar cycle by means of interpolation.

(2) Within each seasonal interval the predictions are given in a series of 15 charts. The first

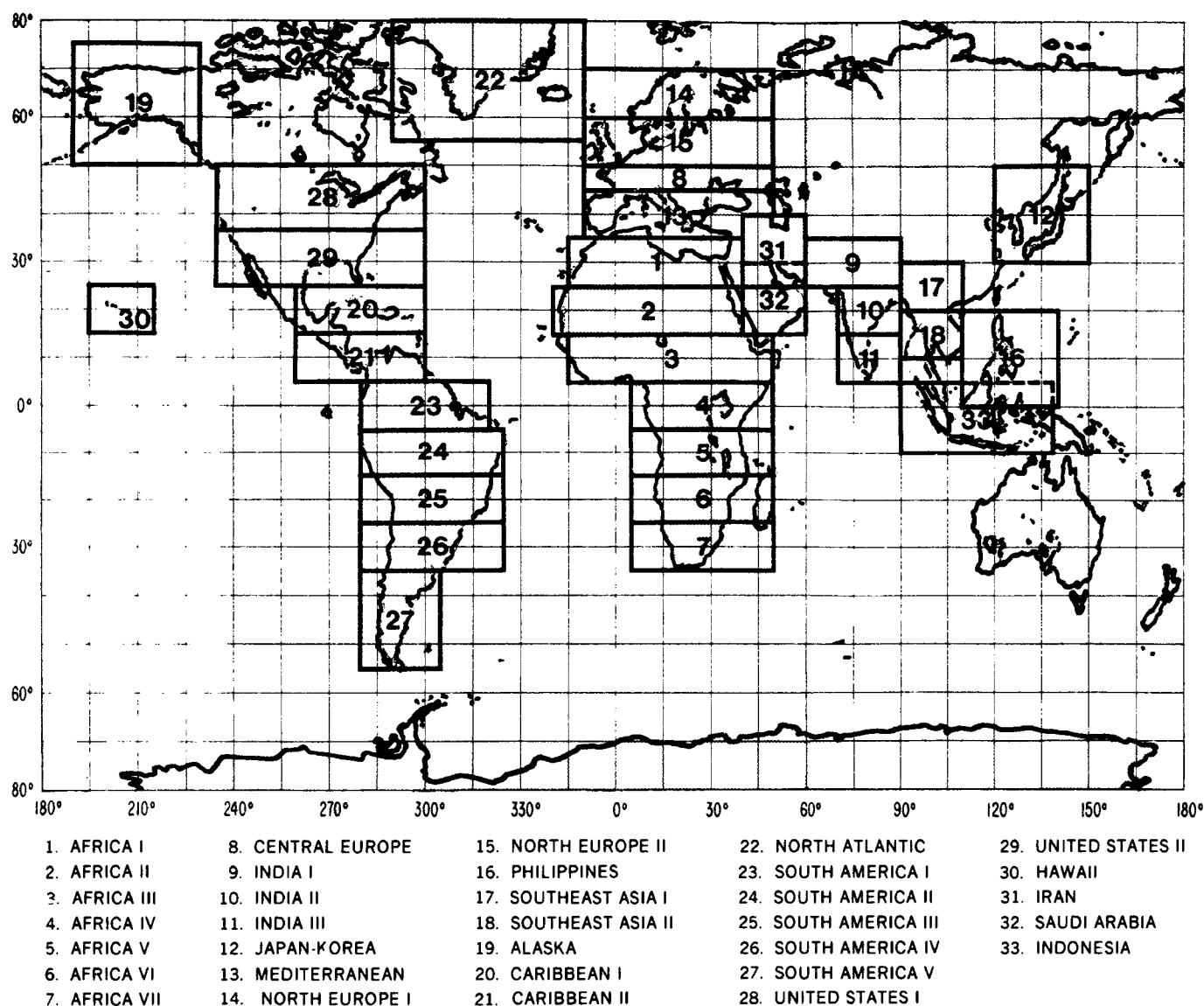


Figure D-4. Key to areas covered by short and intermediate distance skywave chart.

chart provides MUF and FOT values for each hour. The other charts provide LUF values at even 2-hour intervals. The MUF and FOT values are tabulated at six distance intervals—160 km, 400 km, 800 km, 1,200 km, 1,600 km, and 2,400 km. At each distance interval the MUF and FOT are given at each hour of local time for the solar activity extremes of a normal solar cycle.

(3) The LUF charts are prepared for the following antenna types:

- (a) Horizontal $\lambda/2$ dipole, 33 feet (10 m) high.
- (b) Horizontal $\lambda/2$ dipole, 66 feet (20 m) high.
- (c) Vertical 15-foot whip.
- (d) Vertical 32-foot whip.
- (e) Inverted L.
- (f) Sloping long wire.
- (g) Sloping V.

c. Air/Ground Skywave Propagation Charts for Selected Worldwide Stations. These charts (two volumes) provide predicted MUF for various distances between 41 selected ground stations and aircraft flying within a 8,300 km (4,500 nmi) radius. The selected stations include both CONUS and overseas locations. Predictions are for various sunspot numbers, times of day, and seasons. Requests for these charts should specify the number of copies desired.

d. Using the Groundwave, ISD, and Air/Ground Charts. Because of their general nature, the user must locate the charts applying to his particular parameters. Although not as conveni-

ent to use and not as accurate as point-to-point predictions, the operator can determine the best frequencies and best antennas to use for his particular situation. In some cases he may find that with the equipment available continuous communication is unlikely, but that certain times are quite favorable.

e. Weekly Ionospheric Activity Report. The actual day-by-day ionospheric activity over a 7-day period is published in the weekly Ionospheric Activity Report. These reports are extracted from the National Oceanographic and Atmospheric Administration messages and alerts from the Global Weather Central at Offutt AFB. A request to be placed upon the list to receive these reports should state the duration of the need for the reports.

D-7. Consulting Service

The services of USACEEIA are available for problems involving radio wave propagation and antenna design. In addition to the expertise of the personnel, many computer programs are available for handling propagation and antenna design problems. The initial request should describe the problem and provide a name and phone number for informal coordination. If it is necessary to establish a long hf path, the point-to-point propagation services and the engineering services of USACEEIA should always be utilized. Successful long hf paths require high power, as well as proper selection of frequencies and antennas and good siting.

APPENDIX E

IONOSPHERIC SOUNDINGS

E-1. Ionospheric Measurements

Measurements of ionospheric conditions are useful in adding to the general understanding of the ionosphere and its role in radio propagation. Measurements can also assist in selecting operating frequencies. Two types of sounding devices are in use, the vertical and the oblique.

E-2. Measurement of Layer Heights and Critical Frequency

The simplest method of measuring heights of ionospheric layers is by means of a vertical incidence sounding. A vertical-incidence sounding station is a combination of a transmitter and a receiver placed side-by-side, often using the same antenna. The transmitter sends out pulses which

are swept through the frequency range from about 0.75 to 25 MHz. The receiver is synchronized with the frequency of the transmitter. The output of the receiver is coupled to a display (usually, a cathode-ray tube) which displays the time delay between the transmitted and returned pulse at each frequency. A permanent record can be obtained by photographing the display. Assuming that the waves travel to and back from the reflection point at the velocity of light in free space, the time delay can be used to determine the virtual height of the ionized layers. This display of height versus frequency is called an *ionogram*. A simplified ionogram is shown in figure E-1. The figure shows that at the lowest frequency of the sweep, the signal is reflected from the E layer. As frequency is increased, the height of reflection in-

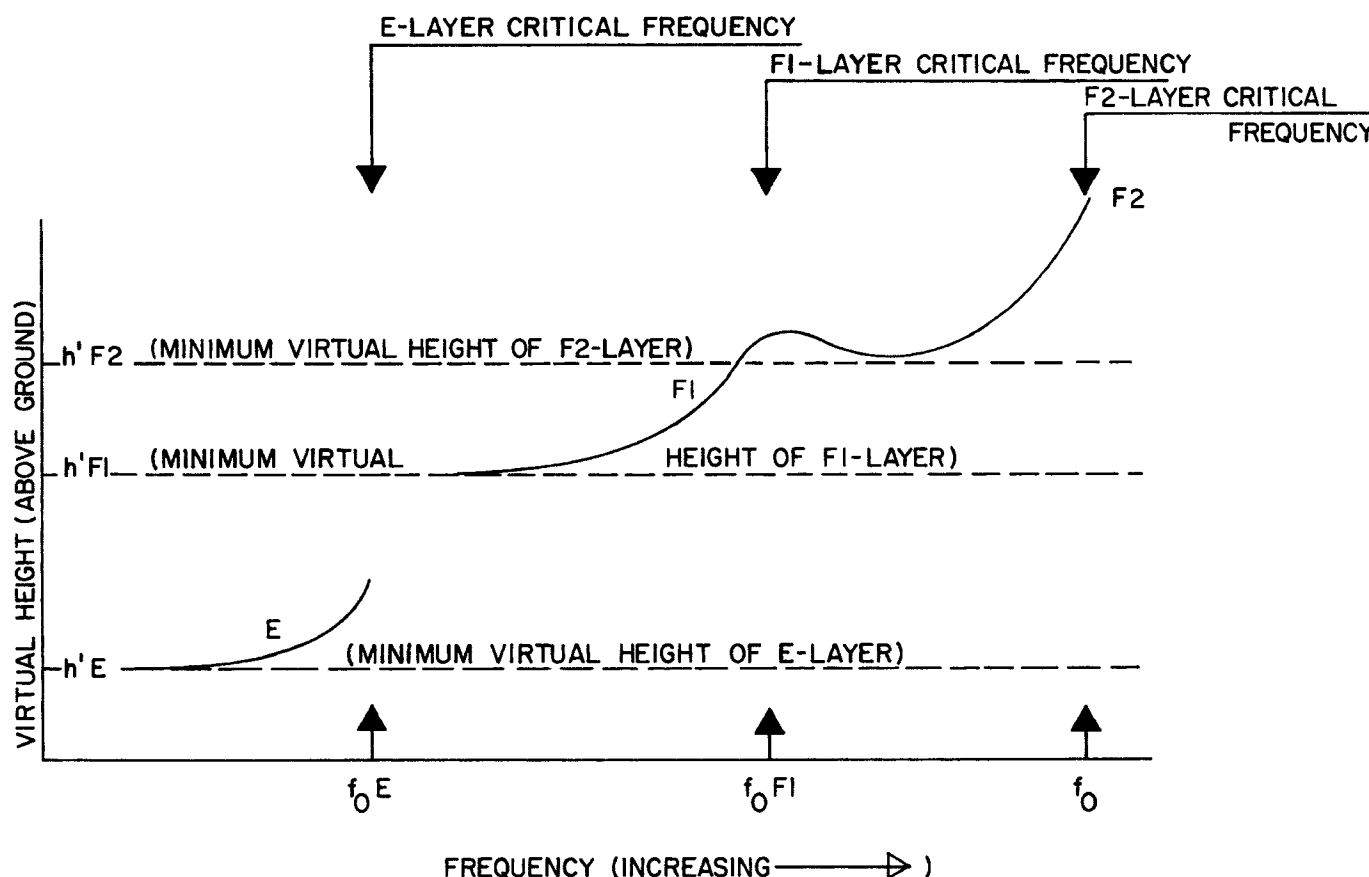


Figure E-1. Simplified vertical incidence ionogram.

creases, until, at the frequency designated " $f_o E$ " on the figure, the E layer is penetrated ($f_o E$ can be thought of as the critical frequency for the E layer). Similarly, the F1 layer is picked up and then penetrated. Above the critical frequency, no return signal is received. An actual ionogram may also show a spurious E layer and may give returns which can be interpreted as magnetoionic splitting. These ionograms are useful in estimating propagation path frequencies and angles of arrival.

E-3. Oblique Ionospheric Sounders

By separating the transmitter and receiver of the ionospheric sounding set, information about the actual communications path can be obtained. Figure E-2 is a simplified drawing prepared from an

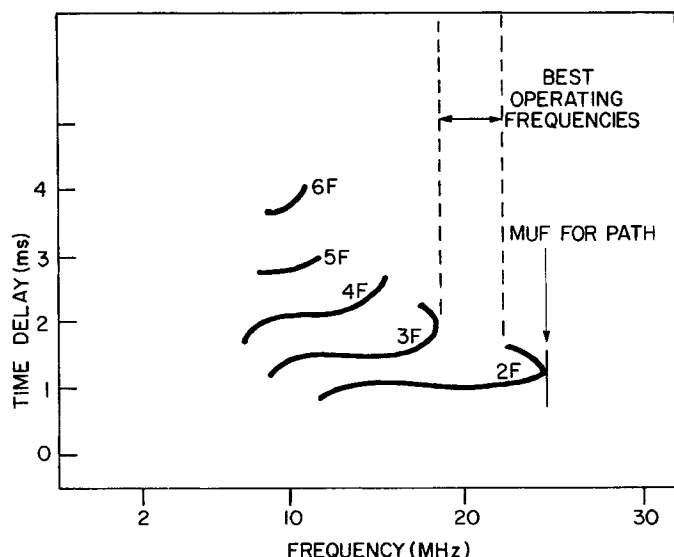


Figure E-2. Simplified oblique incidence ionogram.

oblique ionogram. Unlike the vertical incidence ionogram, the vertical scale on an oblique ionogram cannot be directly interpreted as height. The data for the figures were obtained over a 4,000-km path during daytime in winter. The ionogram has been interpreted by the analyst as showing two, three, four, five, and six hops involving the F layer. The ionogram indicates the best operating frequency to be in the 20- to 22-MHz range. Other frequencies with apparently good received signal strength involve multipath reception and its attendant interference and distortion.

E-4. The AN/TRQ-35 Tactical Frequency Management System (Chirpsounder)

The AN/TRQ-35 is a transportable oblique incidence ionospheric sounding system. A block diagram of the system is shown in figure E-3. Transmitter power can be adjusted from 0.2 watt to 10,000 watts as required. Transmitter power is usually less than 100 watts. When used with communications transmitters of 2.5 kW or less, the AN/TRQ-35 signal can be diplexed through the existing transmit antenna. When used with higher power communications transmitters, a separate transmit antenna is required for the AN/TRQ-35. Selectable frequencies on the AN/TRQ-35 transmitter can be blanked to avoid interference with operating circuits. The use of the AN/TRQ-35 allows operators to select frequencies to suit actual operating conditions. Tests have shown that outages are greatly reduced when real-time soundings are available. The AN/TRQ-35 does not assist in the initial contact, since manual synchronization of transmitter and receiver is required. The AN/TRQ-35 yields an ionogram similar to that shown in figure E-2.

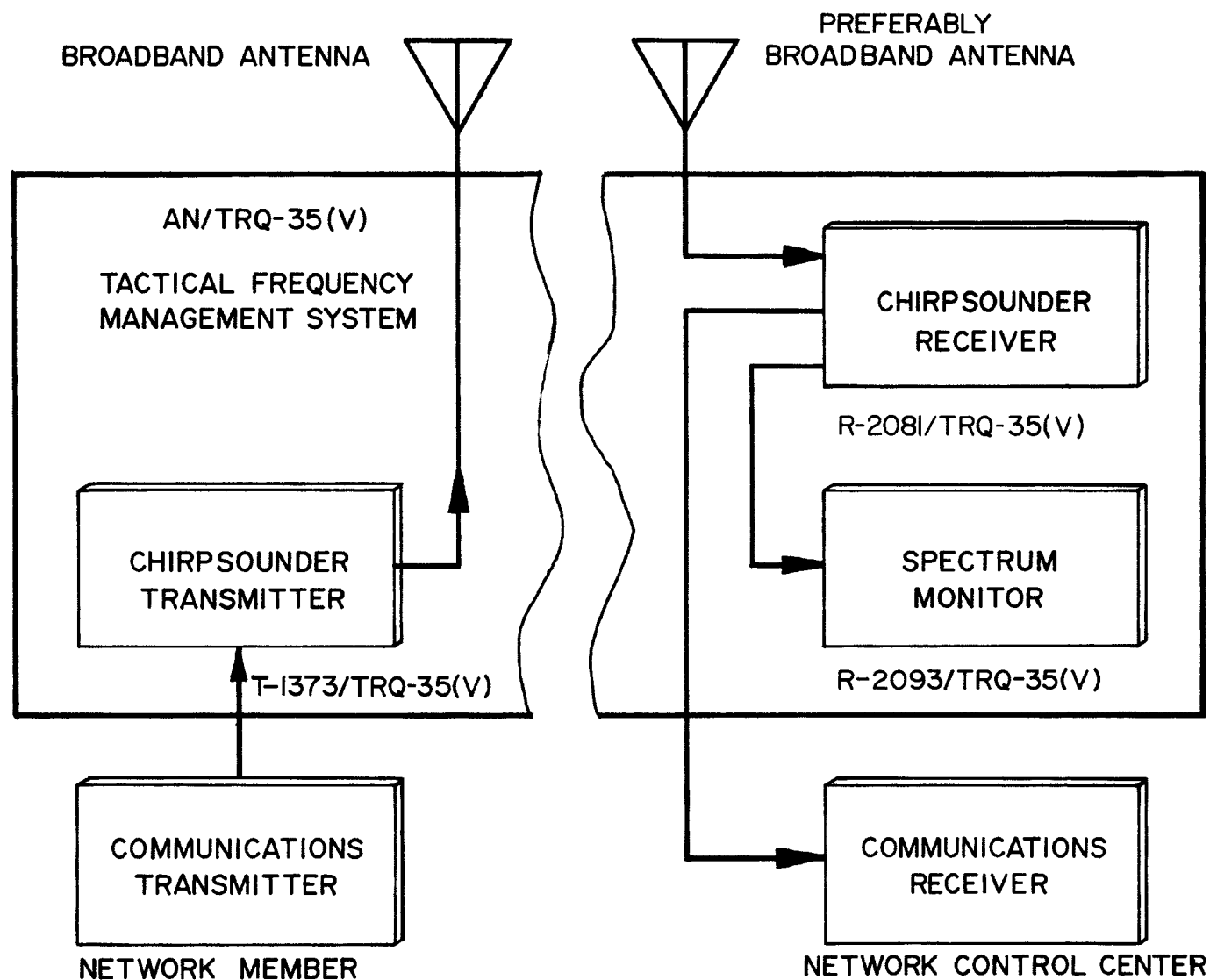


Figure E-3. Block diagram of the AN/TRQ-35 Chirpsounder.

APPENDIX F

FRESNEL ZONE CONSIDERATIONS

F-1. Ground-Reflected Rays

Ground-reflected rays are of importance because, depending upon the phase relationship, they may either reinforce or weaken the direct ray. Following the guidance given in the body of the text of this manual regarding antenna siting and antenna height will ensure reinforcement rather than cancellation. However, to obtain optimum propagation, very careful site preparation is required and the antenna must be placed at the proper height. Such attention to detail is usually not possible when using transportable equipment, but it is warranted at permanent sites.

F-2. Fresnel Zone Description

- a. The concept of Fresnel zones as described in

this manual applies to horizontal antennas and to resonant vertical antennas that are an even number of half-wavelengths long. Because of the length this entails in the hf band, such vertical antennas are very rare. The first Fresnel zone is defined as the first area in front of the antenna from which the reflected rays will reinforce the direct ray at the desired elevation angle. The geometry of the first Fresnel zone is shown in figure F-1. The reference ray is the reflected ray which is parallel to the direct ray and has a path length one-half wavelength longer than the direct ray. There is a phase shift of 180° in a ray upon reflection. After reflection, the reference ray is in phase with the direct ray. The ellipse outlined by the dashed line represents the area from which

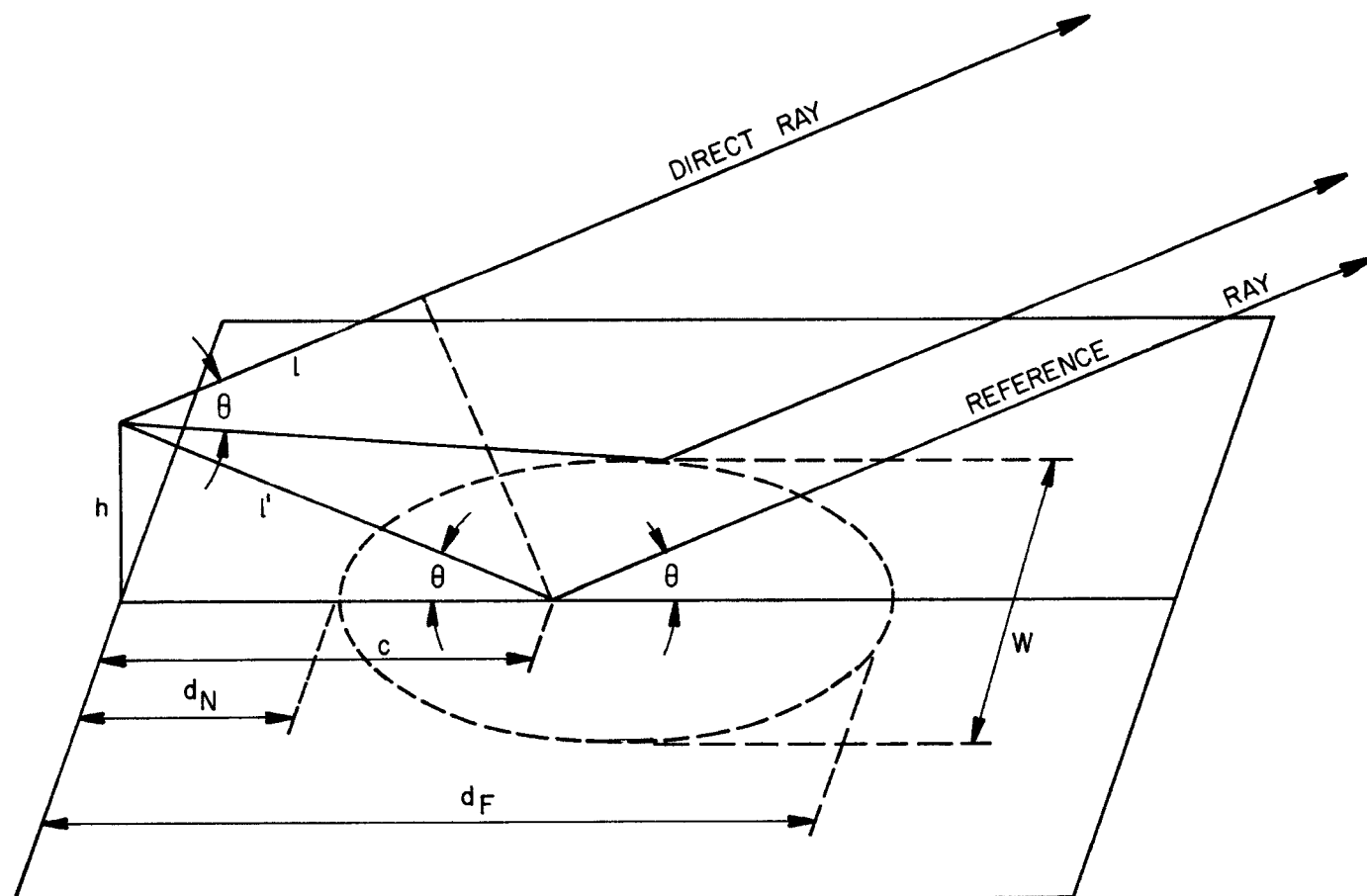


Figure F-1. Geometry of the first Fresnel zone.

the reflected rays will be less than 180° from the phase of the reference ray. All reflections from the elliptical area will reinforce the direct ray. The geometry of the first Fresnel zone can be determined if the wavelength (frequency) and the elevation angle (θ in the diagram) are known. The area within the first Fresnel zone should be smooth and free of obstacles.

b. Second, third, and higher order Fresnel zones surround the first Fresnel zone. They are of less importance because at the greater distance from the transmitter the reflected rays are weaker. In the odd-numbered Fresnel zones the reflected ray strengthens the direct ray; in the even-numbered zones, it weakens the direct ray.

F-3. Determining Elevation Angle

a. The elevation angle is a function of total path length between transmitter and receiver, the number of hops (reflections), and the height of the ionospheric reflecting layer. The height of the E layer is typically taken to be 110 km (70 mi). The height of the F1 layer can be taken as 200 km (125 mi), and the F2 layer, as 280 to 360 km (175 to 225 mi), depending upon season, latitude, and time of day. Determine the elevation angle by using paragraph b or c below.

b. For E-layer propagation, divide the path length by 2,000 km (1,250 mi); round to the lower integer but not to less than one (to obtain lowest elevation angle). Divide path length by this integer to obtain hop distance. Using this hop distance and the estimated E-layer height of 110 km (70 mi), obtain the elevation angle from figure F-2. If angle is negative, read as zero degrees.

c. For F-layer propagation, divide the path distance by 4,000 km (2,500 mi); round to the lower integer but not to less than one. Divide the path length by this integer to obtain the hop distance. Using this hop distance and the assumed height of the F layer, obtain the elevation angle from figure F-2. If angle is negative, read as zero degrees.

F-4. Determining Antenna Height

Determine antenna height by using elevation angle, frequency, and figure F-3. As an alternative, antenna height can be calculated from the equation

$$h = \frac{\lambda}{4 \sin \theta}$$

If the elevation angle is less than about 5°, a reasonable antenna height can be obtained only on sloping terrain, as discussed in paragraph F-7. The distance between the base of the antenna and the point of reflection of the reference ray can be determined with the use of figure F-4. The near and far edges of the first Fresnel zone are obtained from figures F-5 and F-6, respectively. The width of the zone is obtained from figure F-7.

F-5. Allowable Roughness

For reflections from the Fresnel zone to reinforce the direct ray, the Fresnel zone must be free of obstruction and without appreciable roughness. Table F-1 lists the acceptable upper limits of irregularities in the Fresnel zone. The symbols correspond to those used on figure F-1. There should be no buildings, trees, or other obstacles in the Fresnel zone. In addition, no reflected ray from any portion of the first Fresnel zone should be obstructed by buildings, trees, or terrain. The size of the first Fresnel zone and the unobstructed areas are greatest for low elevation angles and least for high elevation angles.

Table F-1. Fresnel Zone Roughness Criteria

Reflection zone section	Maximum departure from ideal smoothness
From d_n to 0.2 d_F	0.2h
From 0.2 d_F to 0.6 d_F	0.5h
From 0.6 d_F to d_F	h

F-6. Compromise in First Fresnel Zone Criteria

By accepting a degradation of about 2 dB, the compromise shown in table F-2 below may be made, rather than adhering strictly to the first Fresnel zone geometry.

Table F-2. Compromise Fresnel Zone

Fresnel zone geometry	Compromise
d_F	0.6 d_F
d_N	1.6 d_N
W	0.7 W

F-7. Sloping Sites

For very low elevation angles, reasonable antenna heights can be obtained only on sites which slope downward in the direction of the radio path. The effect of the downward sloping terrain is to give an apparent increase in the height of the antenna, as shown in figure F-8. For the use of figures F-3 through F-7 on sloping ground, the angle of

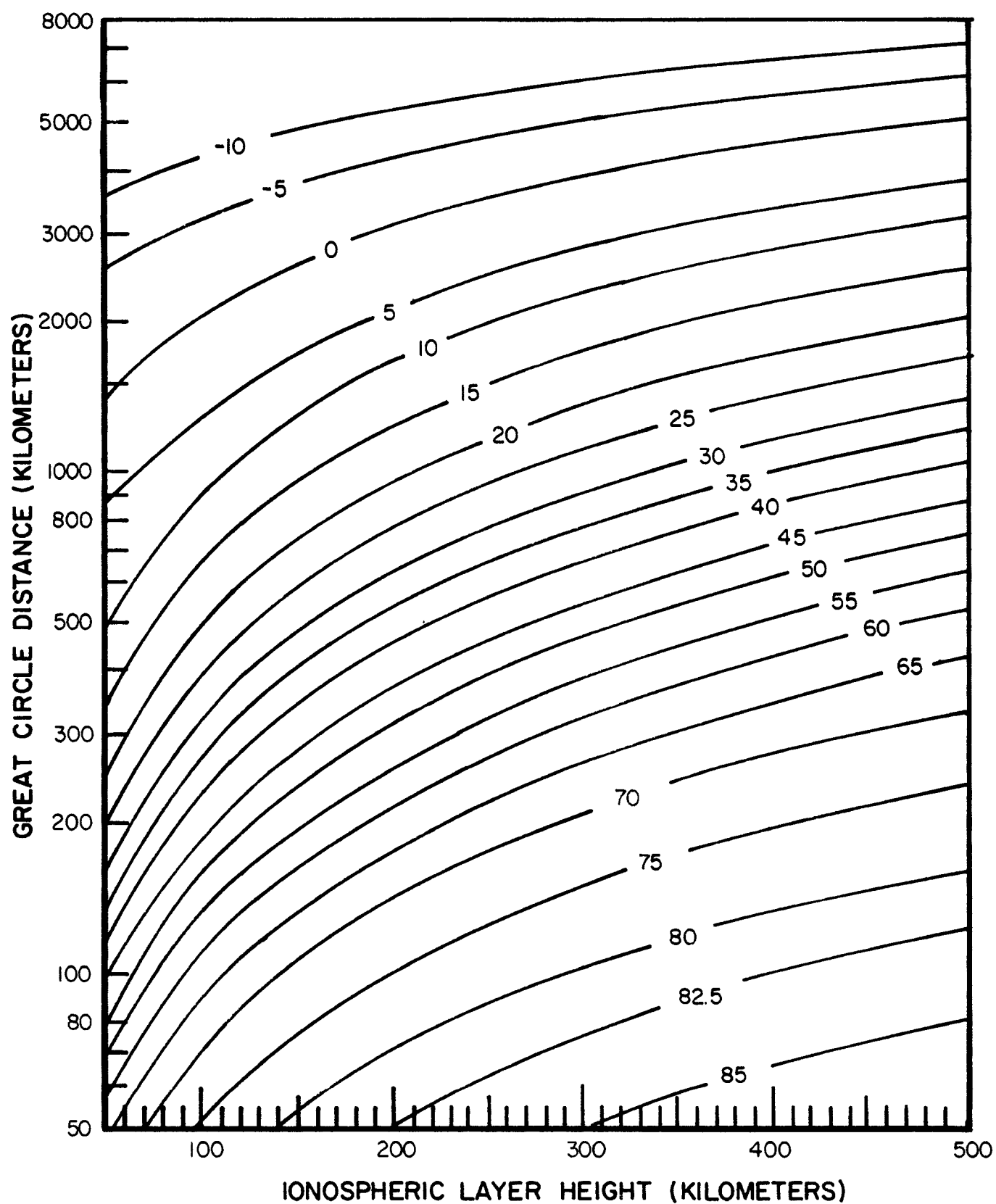


Figure F-2. Elevation angle as a function of distance and ionospheric layer height.

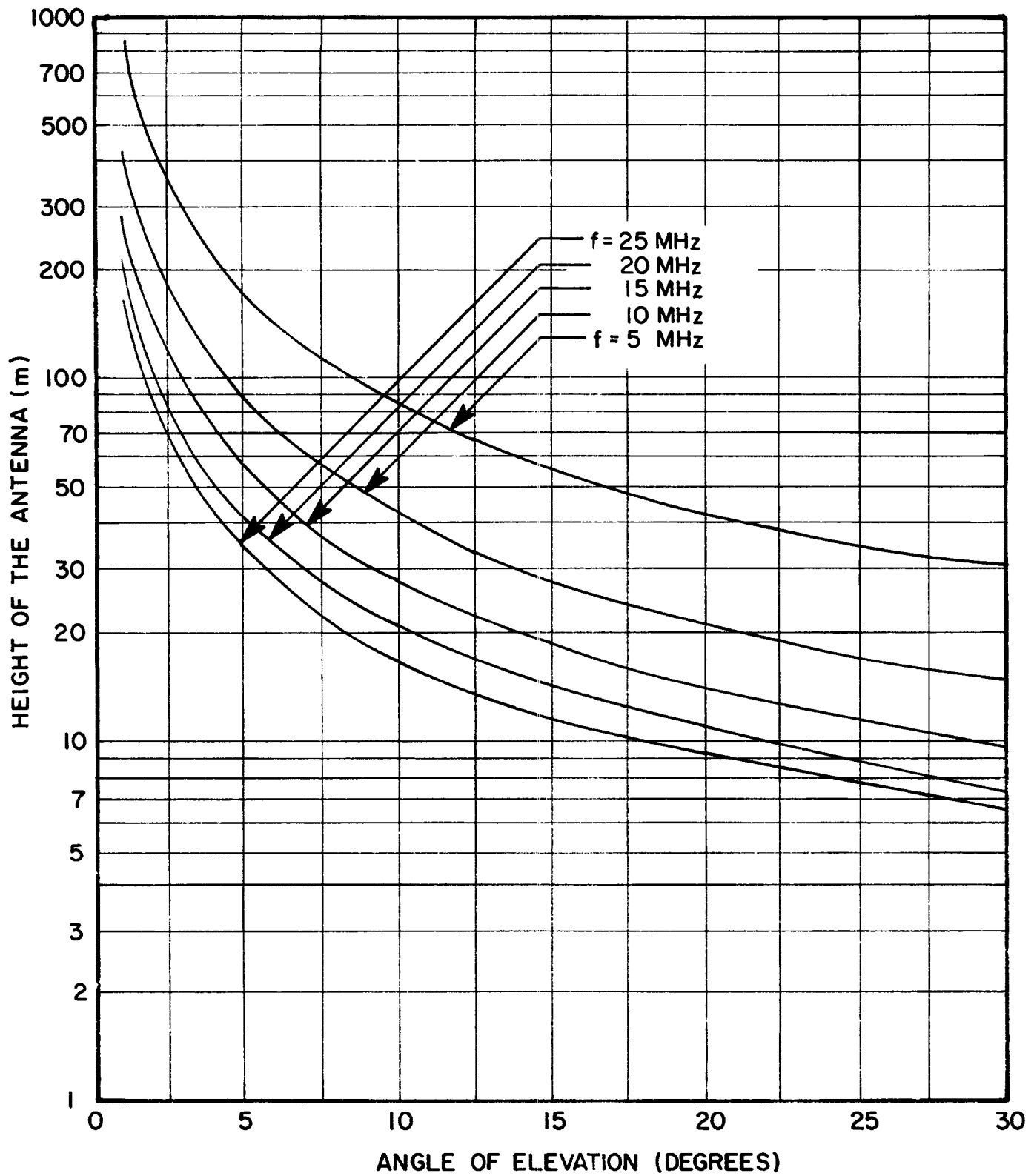


Figure F-3. Antenna height as a function of elevation angle and frequency.

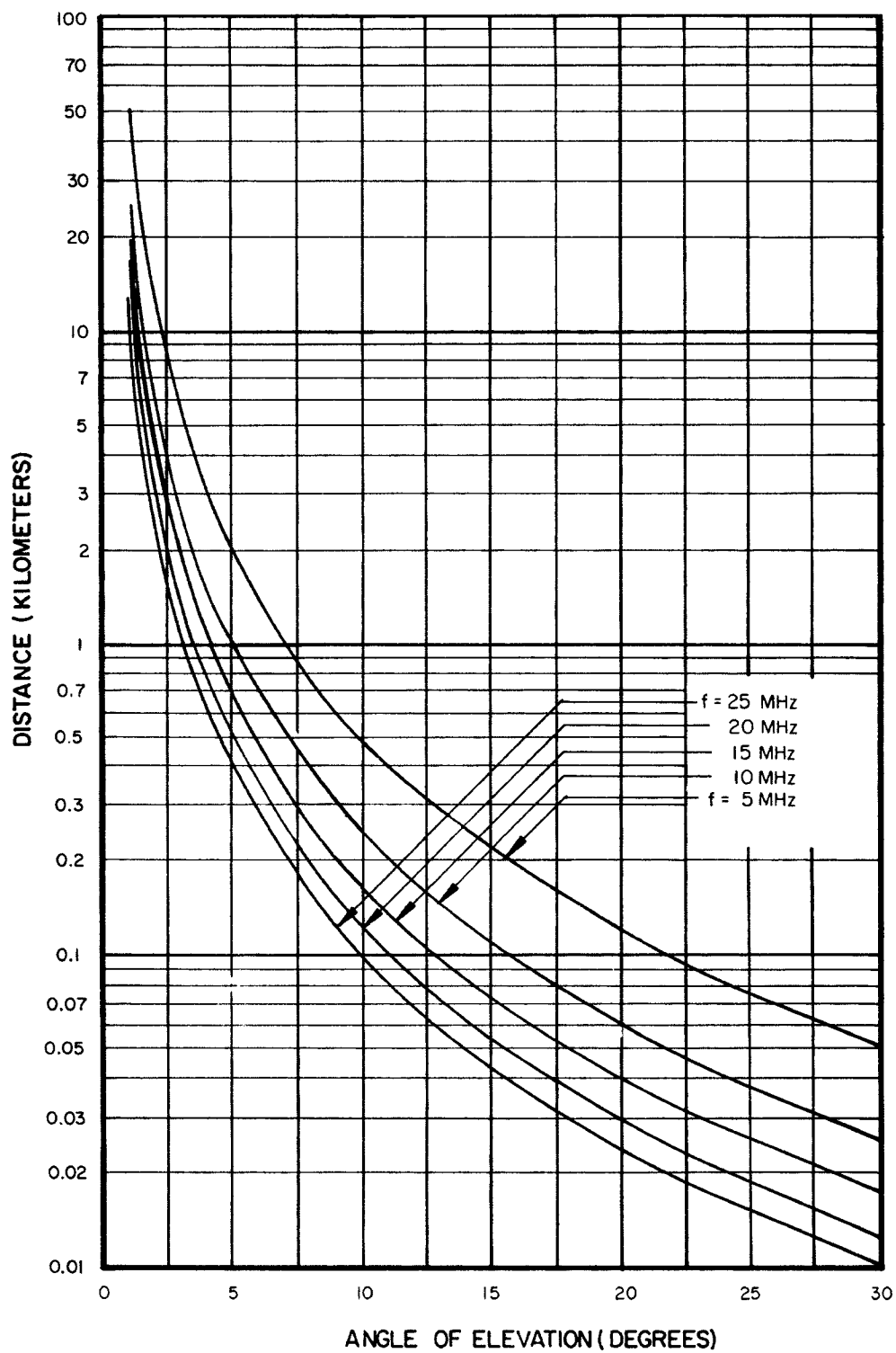


Figure F-4. Horizontal distance from antenna to point of reflection of reference ray.

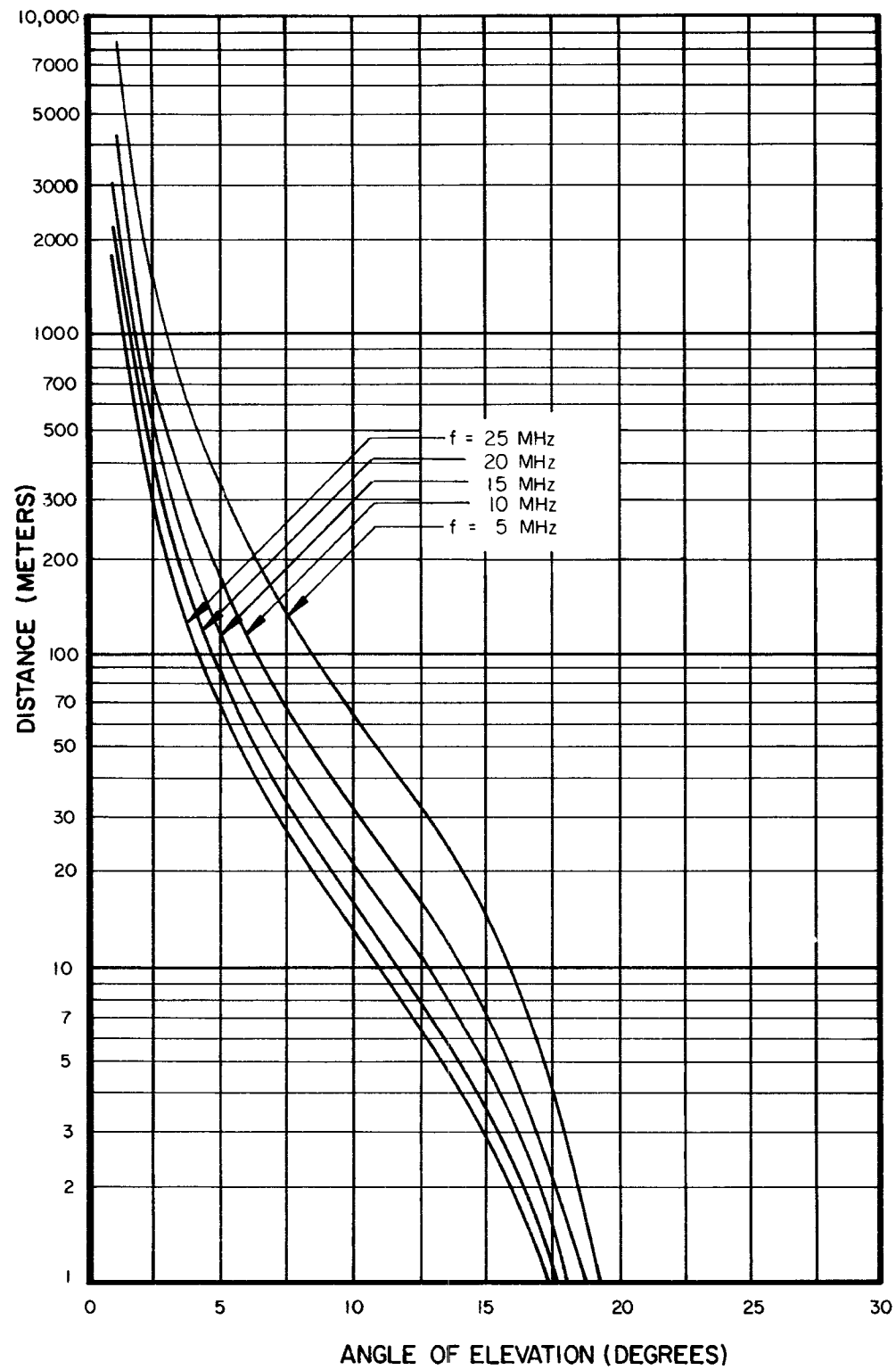


Figure F-5. Horizontal distance from antenna to near edge of first Fresnel zone.

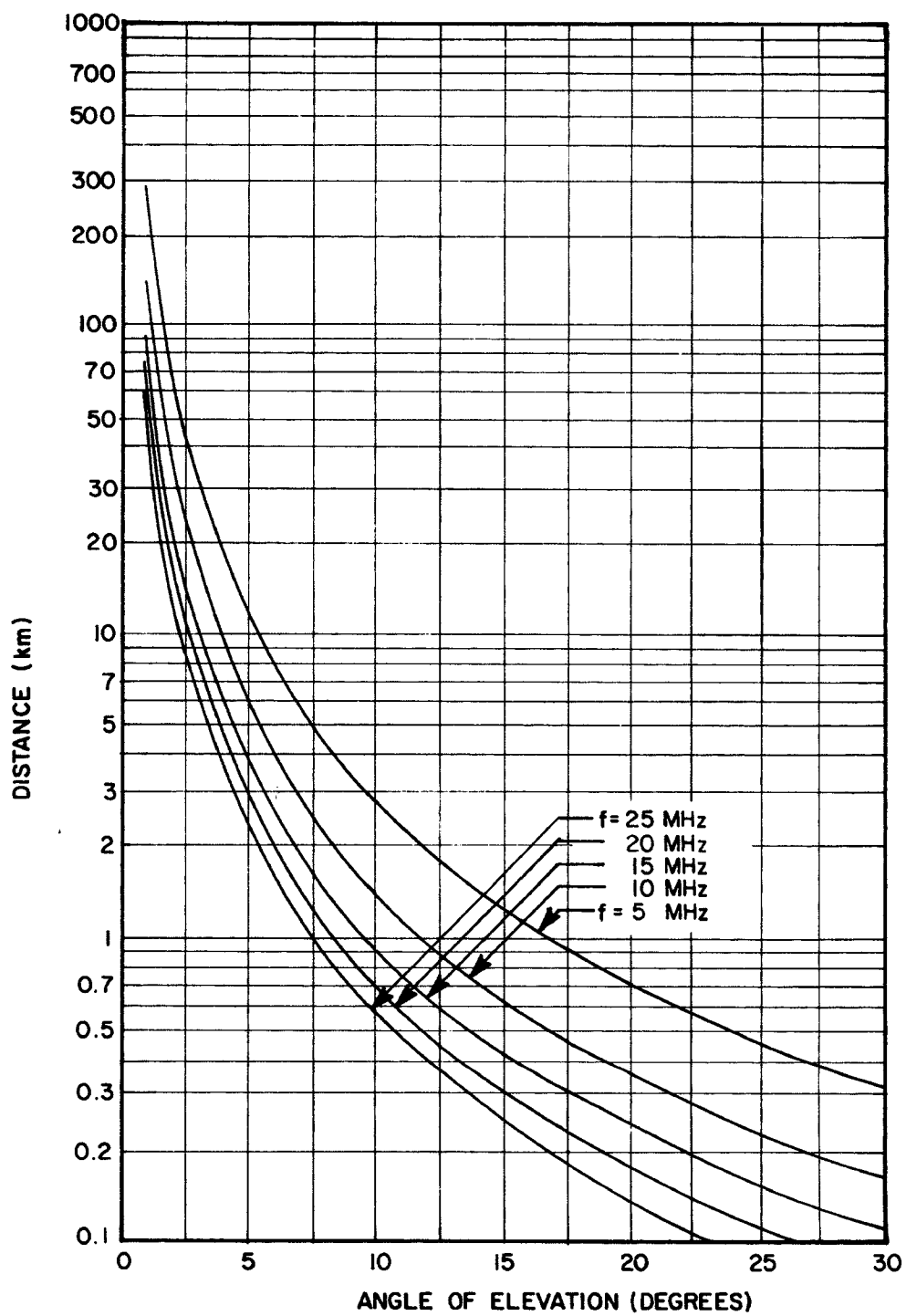


Figure F-6. Horizontal distance from antenna to far edge of first Fresnel zone.

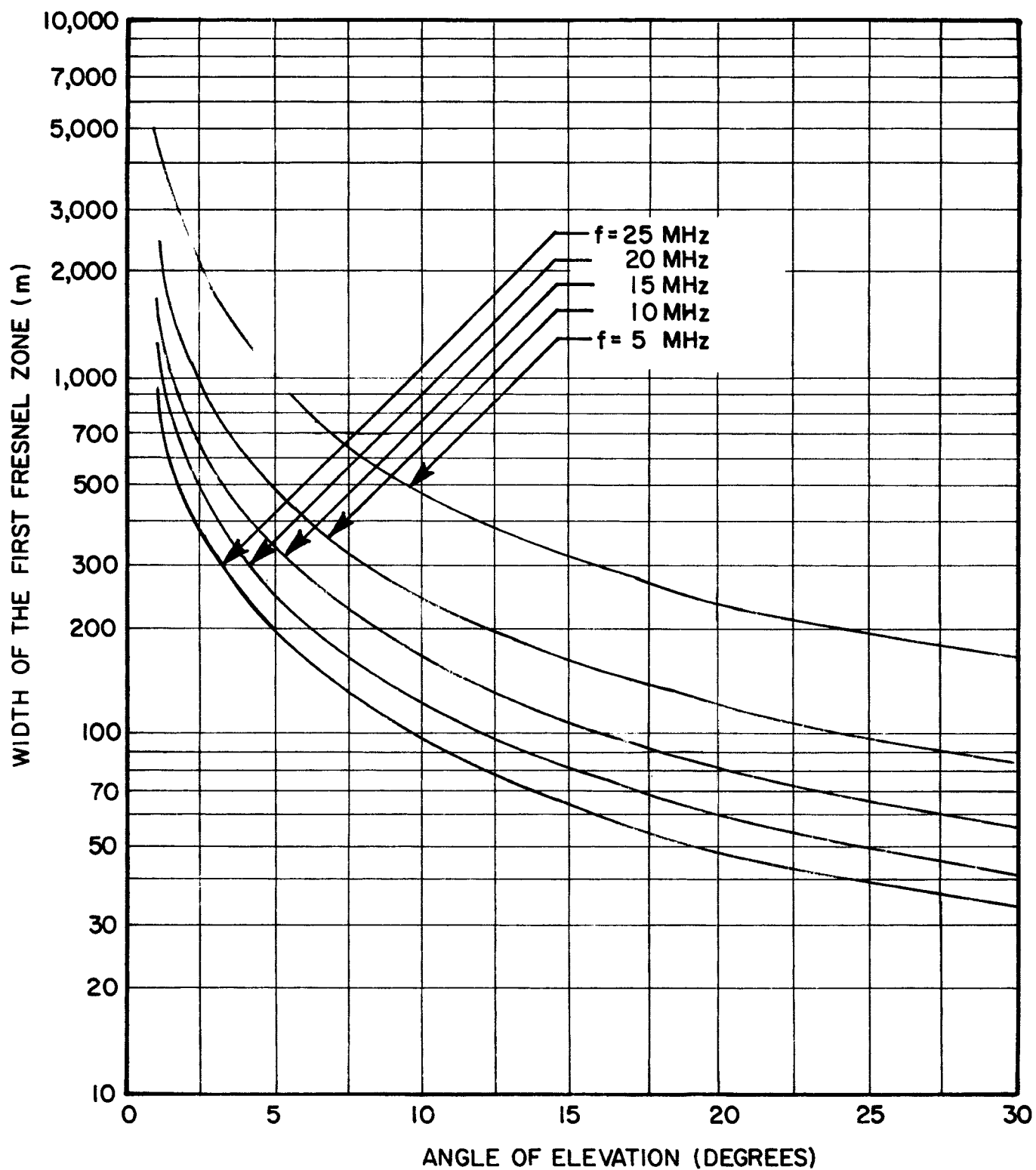


Figure F-7. Width of first Fresnel zone.

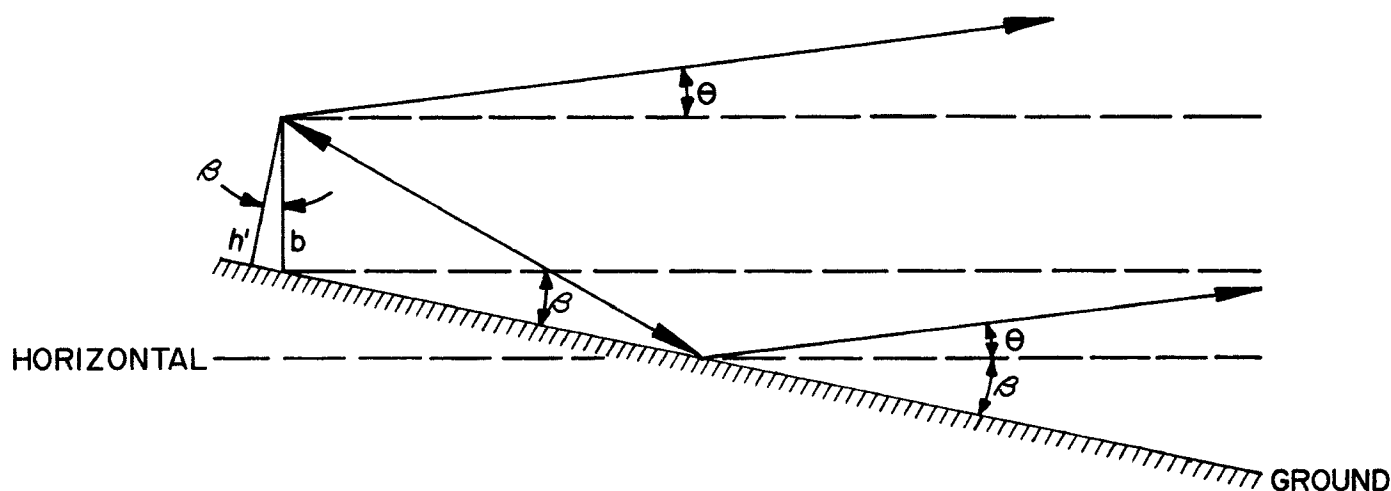


Figure F-8. The geometry of reflection from a sloping site.

slope can be added to the elevation angle. Instead of using figure F-3, one can determine the antenna height by the equation

$$h = \frac{\lambda}{4 \sin (\theta + \beta)}$$

F-8. Elevated Sites

Tower heights may also be reduced by locating the antenna on elevated terrain, like a cliff. An elevated site is shown in figure F-9. The symbols are the same as those used in figure F-1. Note that the reflection from the far end of the first Fresnel zone clears the cliff on the right.

F-9. Other Terrain Considerations

An area with good ground (moist black soil, swampland, or lakes) should be selected if possible. Dry, rocky land should be avoided.

F-10. Siting Vertical Antennas

a. In the hf band, vertical antennas will usually be a quarter-wavelength electrically and will depend upon reflections from the ground to provide the image antenna. Good ground is even more essential for the quarter-wave vertical antenna than for horizontal antennas. Unless the ground

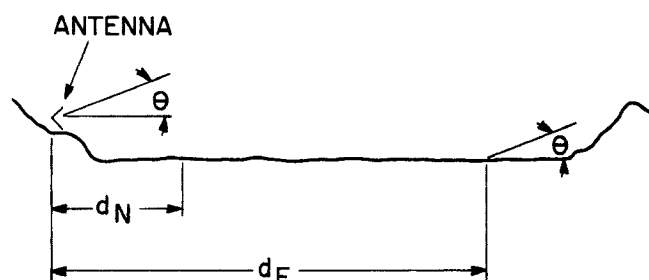


Figure F-9. An elevated site.

has good conductivity, a ground screen must be provided for optimum radiation.

b. When vertical antennas are used in ground-wave service, any obstruction will cast a radio shadow. Hence the site around the antenna should be as free from obstructions as possible. When vertical antennas are used in long-distance sky-wave service, no obstructions should be higher than about one-half the elevation angle of the radio ray.

c. The requirements for good ground and freedom from obstructions tend to be contradictory. The best ground is apt to be in the low-lying land. The solution may be to choose an elevated location and use a ground screen.

APPENDIX G

EFFECTS OF SOLAR ACTIVITY ON HF COMMUNICATIONS

G-1. Introduction

This appendix discusses solar activities and solar disturbances and their effects on the ionosphere. Also covered are the effects on radio wave propagation, and the use of solar and ionospheric predictive services. Corrective techniques for attempting to maintain uninterrupted radio communications during disturbances are suggested. This appendix supplements the discussion of ionospheric variables and their relation to communications in paragraph 2-9.

G-2. Solar Activities

a. Quiet Solar Structure.

(1) The radiating surface of the sun has a temperature of approximately 6,000 K (10,000° F). Most of the sun's energy is in the visible and infrared portions of the spectrum, and a relatively small amount is in the ultraviolet and even less in the X-ray region. Nevertheless, the ultraviolet and soft X-ray radiations are the important ionizing agents. Figure G-1 shows the relation of the sun's spectrum to radio waves.

(2) The surface of the sun which we can observe is the *photosphere*. Sunspots show on the photosphere as dark areas. The photosphere, as seen from the earth, appears to rotate about once in 27 days at the sun's midlatitudes. Directly above the photosphere is the *chromosphere*.

(3) Above the chromosphere is the *corona*, which is the outer atmosphere of the sun. The corona, in the form of the solar wind, extends past the earth. The corona is composed of fully ionized plasma. Because of its high temperature, over a million degrees absolute, the corona radiates in the X-ray and short ultraviolet wavelengths. As solar activity increases, shock waves from lower in the sun's atmosphere propagate outward through the corona, sometimes accompanied by gigantic bubbles of hot plasma. These dynamic features are thought to contribute to the solar wind. Individual particles in the solar wind travel from the sun to the earth in about 3 to 5 days.

(4) The stream of particles in the solar wind is deflected by the earth's magnetic field, which in turn is distorted by the stream of ionized particles. Figure G-2 illustrates the interaction. The representation of the earth on the figure is not to scale.

b. The Active Sun.

(1) Much solar activity is generated in vertical columns in the solar atmosphere overlying sunspots. Figure G-3 is a schematic representation of an active region. As shown in the figure, the magnetic field extends great distances. It is the magnetic field that creates changes from the quiet sun conditions which are important to communicators.

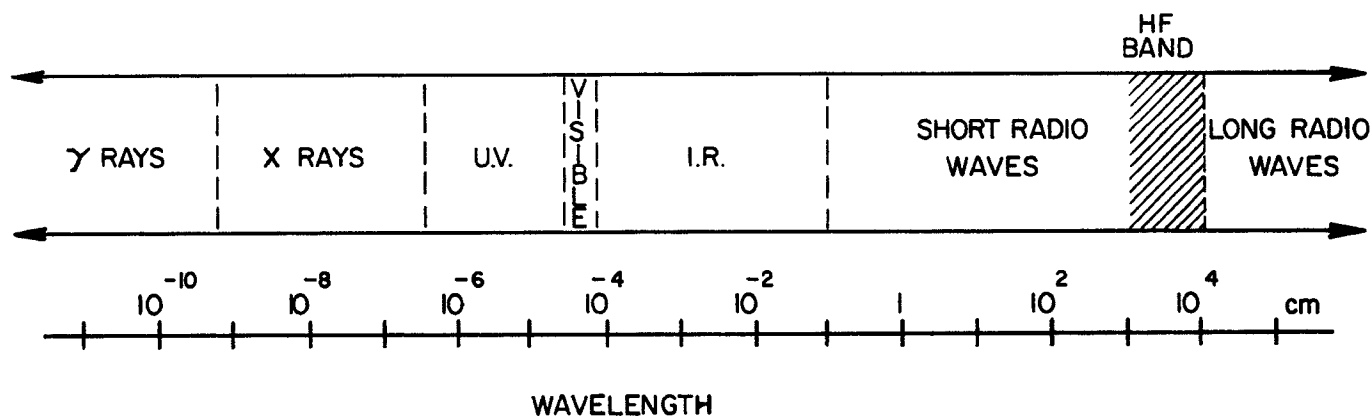


Figure G-1. Wavelengths of solar and radio spectrum.

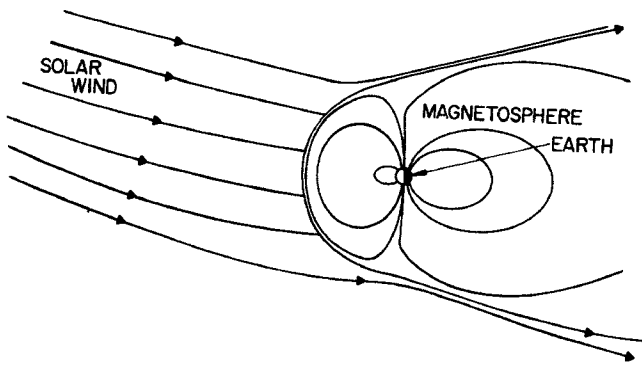
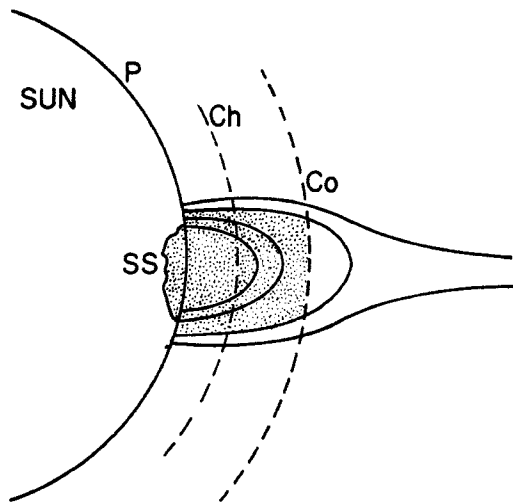


Figure G-2. Interaction of solar wind with earth's magnetic field.



LEGEND:

P = PHOTOSPHERE
 Ch = CHROMOSPHERE
 SS = SUNSPOTS
 Co = CORONA
 LOOPS ARE MAGNETIC FIELD LINES.
 DOTTED AREA IS THE ACTIVE REGION.

Figure G-3. Schematic representation of an active region on the sun.

(2) Active chromosphere regions show plages, associated with magnetic fields, and arch filament systems. Plages are large, luminous patches of calcium and hydrogen clouds that appear near disturbed areas of the sun. The arch filament systems are likely precursors of nearby activity. Filaments are narrow lines, either dark or bright, and are called arch filaments if they curve back to the sun.

(3) Active coronal regions are characterized by hot structures called coronal plages. They emit strong X-rays and often remain a month or more

after the underlying photospheric and chromospheric phenomena have disappeared.

(4) Other phenomena of an active sun are solar flares, eruptive prominences, surges, bursts of radio emission, proton emissions, and shock waves.

(5) Many of the indications of an active sun are accompanied by variations in the solar wind. The locations of high speed streams in the solar wind are correlated with the locations of coronal holes on the solar surface. These coronal holes are regions of very low emissivity in the corona and are thought to be located where the magnetic field lines are open to interplanetary space. It is also known that recurrent geomagnetic disturbances are related to high speed streams in the solar wind and, hence, to coronal holes. Coronal hole monitors have become important tools in the prediction of geomagnetic indices. Generally, a maximum of geomagnetic activity occurs about 3 to 4 days after a coronal hole passes the central portion of the sun's disk. This is about the time required for disturbances in the solar wind to reach the earth's magnetic field.

c. *Geomagnetic Activity.* Variations in the solar wind result in changes in the earth's magnetic field. This reaction explains the reasonably good correlation between solar activity and the earth's magnetic storms. A commonly used measure of geomagnetic activity, the K index, uses a roughly logarithmic scale of 0 through 9. It represents magnetogram variations over a 3-hour period. The point of measurement is sometimes indicated by a subscript: K_B means Boulder, Colo; K_{FR} means Fredericksburg, Va. The A_K index is an average of the eight 3-hour indexes for a day. Sometimes, A is used to mean A_K . When indexes from many stations are combined to give a planetary average, the index is called K_P or A_P .

G-3. Propagation Effects Associated With Solar Disturbances

a. Vertical and oblique incidence ionograms show directly the effect of solar disturbances on the ionosphere. Figure G-4 shows analyzed ionograms taken during quiet and disturbed conditions. The ionograms show a compression of usable frequencies which is characteristic of disturbed conditions.

b. Figure G-5, based on analyzed vertical incidence ionograms, shows the effects of an *ionospheric storm*. During an ionospheric storm the ionosphere becomes turbulent and loses its normal

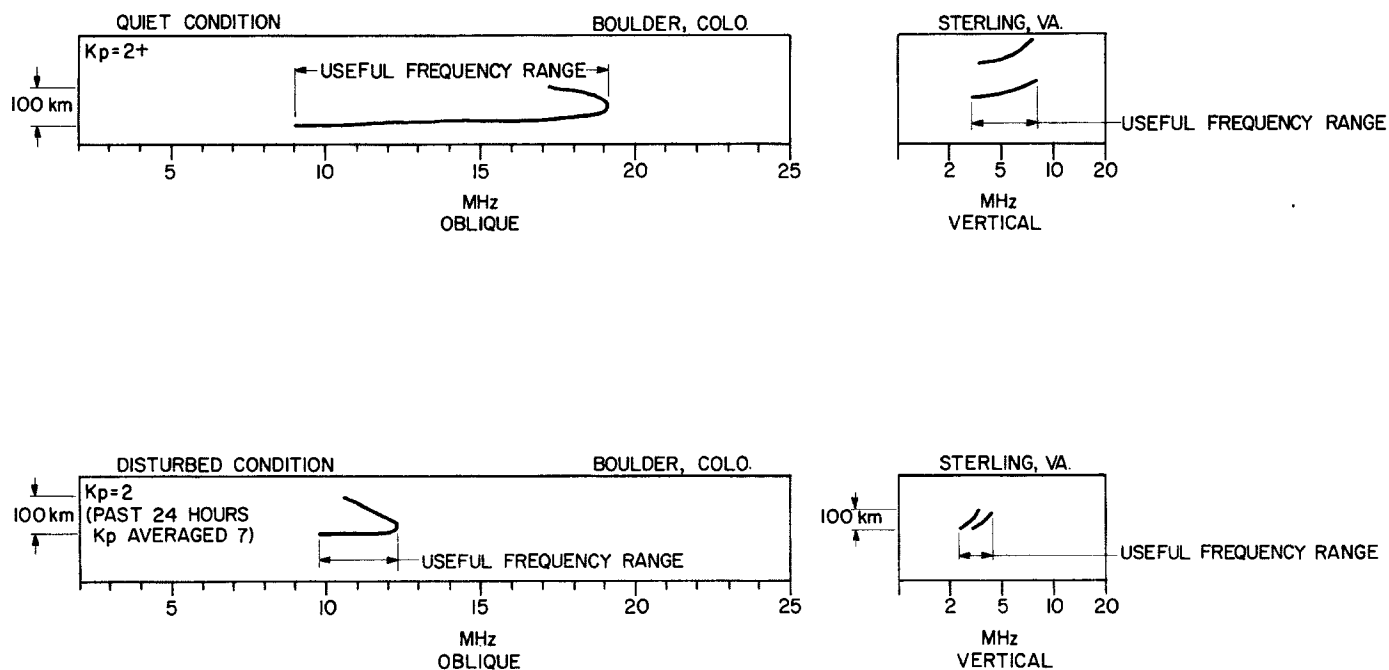


Figure G-4. Sterling, Va., to Boulder, Colo., ionograms under quiet and disturbed conditions.

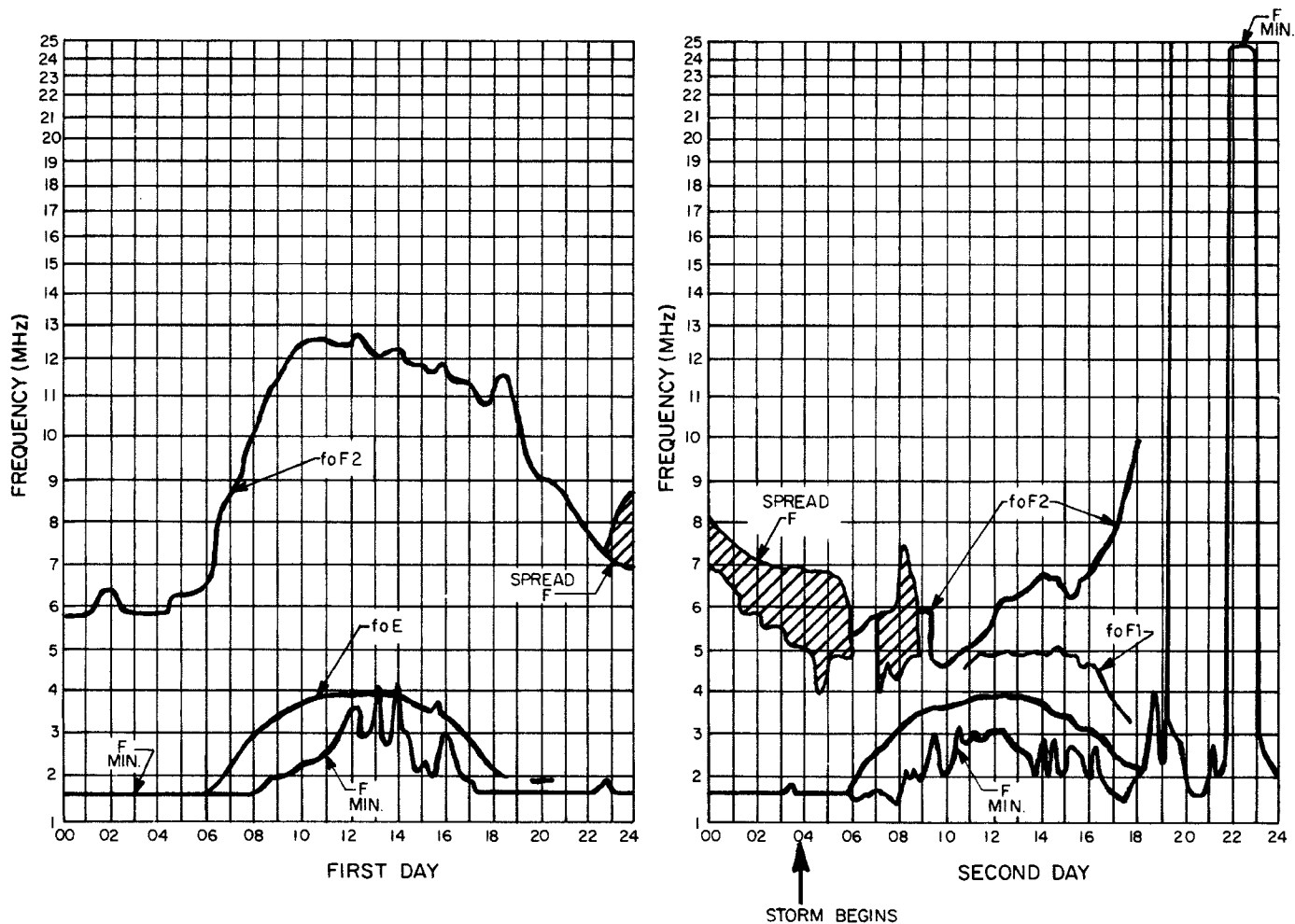


Figure G-5. Effect of ionospheric storm on critical frequency.

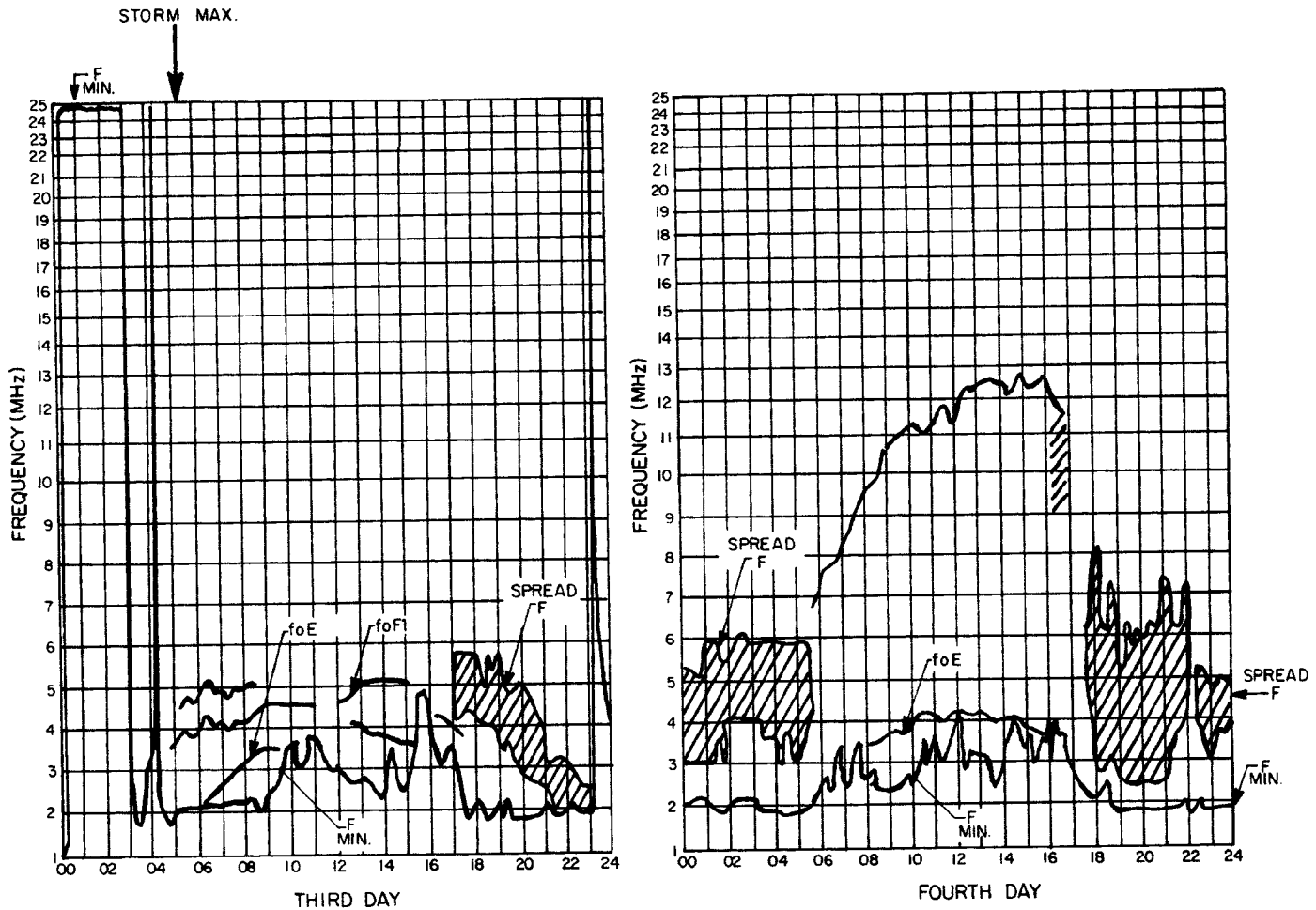


Figure G-5.—continued

stratification. The result is that radio wave propagation becomes very erratic, and it is often necessary to lower the working frequency in order to maintain communication. Figure G-5 also shows the phenomenon known as *spread F*, in which the distinct F layers disappear. Sometimes the duration of a pulse reflected from the F region may be as much as ten times greater than the duration of the incident pulse, causing the ionogram to have a "smeared" appearance. In middle and higher latitudes the occurrence is essentially a storm phenomenon. In equatorial regions it occurs on magnetically quiet days and disappears with the onset of a magnetic storm. The cause of the ionospheric storm is thought to be the emission of a burst of electrified particles from the sun. The fact that the storms sometimes tend to occur at 27-day intervals, the period of rotation of the sun, indicates that there are active areas on the sun which produce the phenomenon. The effects of ionospheric storms may last for several days. As solar activity increases, the intensity and rate

of magnetic storms and ionospheric disturbances also increase.

c. Ionospheric effects of solar disturbances are usually greater at high geomagnetic latitudes. Geomagnetic coordinates are obtained by shifting the normal geographic latitude and longitude grid so that the poles correspond to the magnetic poles. Figure G-6 is a world map of geomagnetic latitudes.

d. A dramatic irregular variation in the ionosphere is a *sudden ionospheric disturbance* (SID), which produces a radio blackout lasting from a few minutes to an hour or more. The phenomenon is caused by sudden bright eruptions on the sun that produce a large increase in the electromagnetic ionizing radiations that reach the D layer. The increase of ionization density in this layer can result in a complete absorption of all skywave signals having a frequency greater than about 1 MHz. An SID is often accompanied by disturbances in the geomagnetic field. Figure G-7 shows

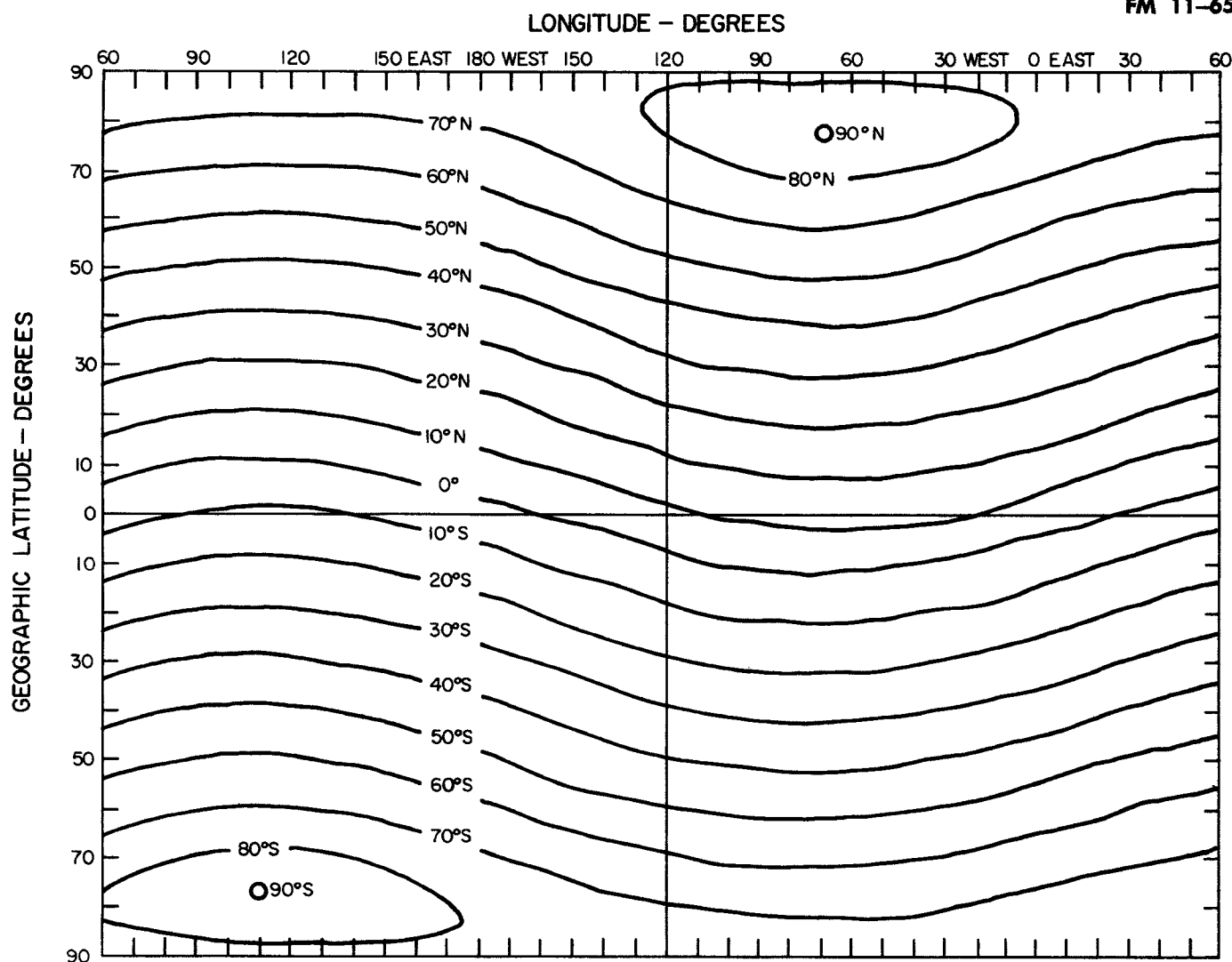


Figure G-6 Geomagnetic latitudes.

how an SID affected a radio path. It also shows the concurrent magnetic disturbance. The effect never occurs at night because of the earth's shielding of the effective solar electromagnetic radiation.

e. If the ionization of a reflecting layer changes very rapidly, the radio path may be shortened or lengthened abruptly, resulting in a change in the received frequency. This phenomenon is called the Doppler effect. Figure G-8 is an example of a sudden frequency deviation which resulted from a solar flare. Another transmission peculiarity associated with solar flares is *polar cap absorption* (PCA). PCA may cause a complete blackout of paths crossing the affected area. Although there is a lack of data for the south polar cap, it is believed that PCA events occur simultaneously in both hemispheres. PCA is believed to be due primarily to the ionizing effects of high energy solar protons penetrating down to altitudes of the order of 50 km. Observations of PCA are almost always preceded by a major solar flare on

the visible hemisphere of the sun. The time interval between the appearance of the flare and the start of the PCA ranges from a fraction of an hour to several hours. The duration is usually about 3 days but may be as short as 1 day; the longest may be about 10 days.

f. Geomagnetic and ionospheric storms also influence circuit quality, as shown in figure G-9. This figure shows that the higher geomagnetic latitude paths are more degraded, as indicated by the larger amplitudes of the circuit disturbance ratings.

g. Table G-1 shows the changes in the ionosphere associated with solar and geomagnetic events. Table G-2 shows communication anomalies associated with solar and geomagnetic events. Solar disturbances have a wide range in type and intensity. Also, the reactions of solar events with the ionosphere and the earth's magnetic field are very complex. Therefore, tables G-1 and G-2 summarize only the usually associated events. The

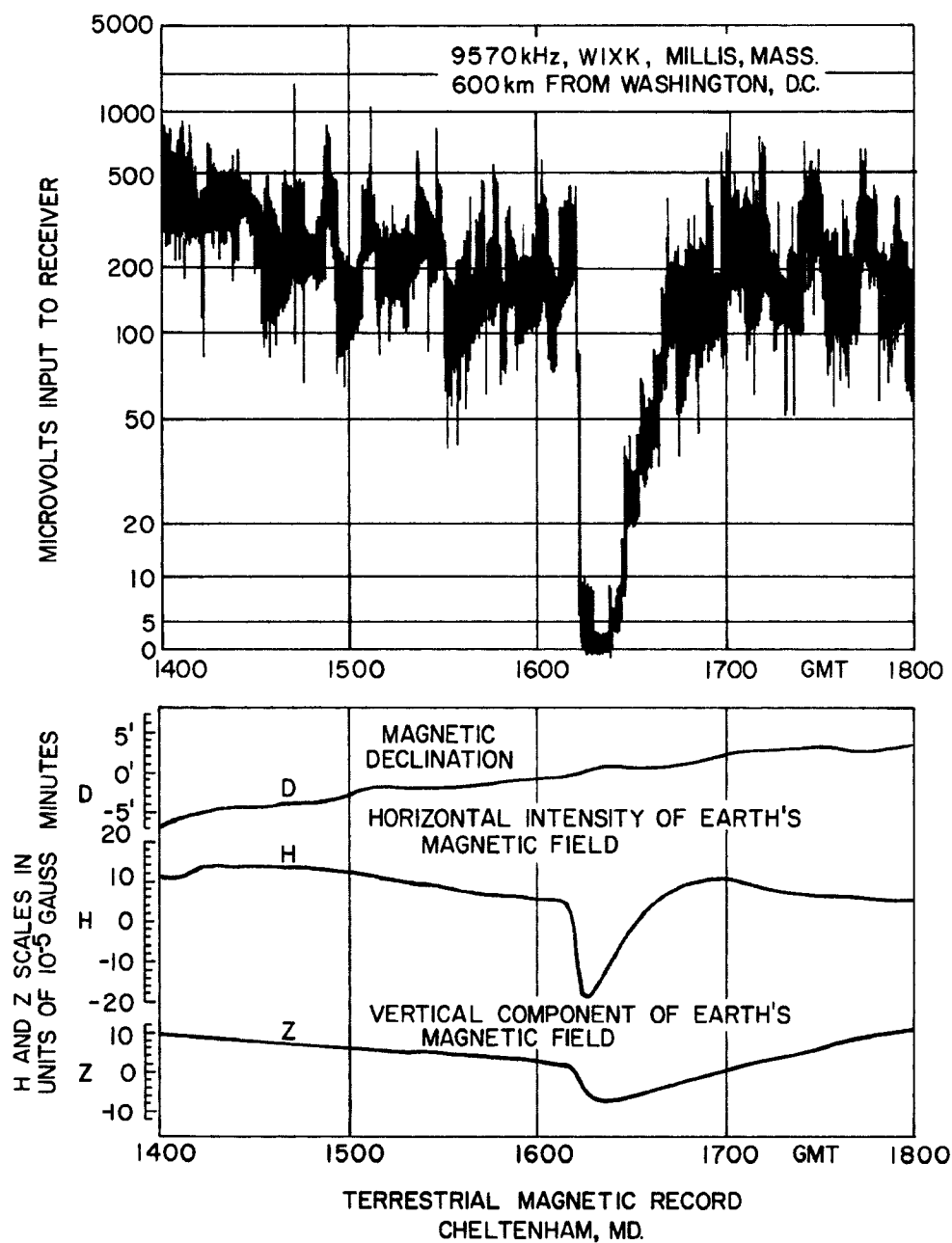


Figure G-7. Example of shortwave fadeout and magnetic disturbance.

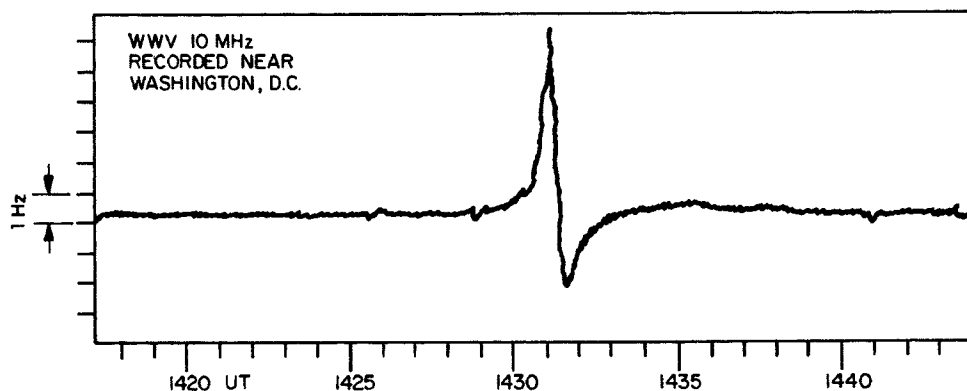


Figure G-8. Effect of solar flare on received frequency.

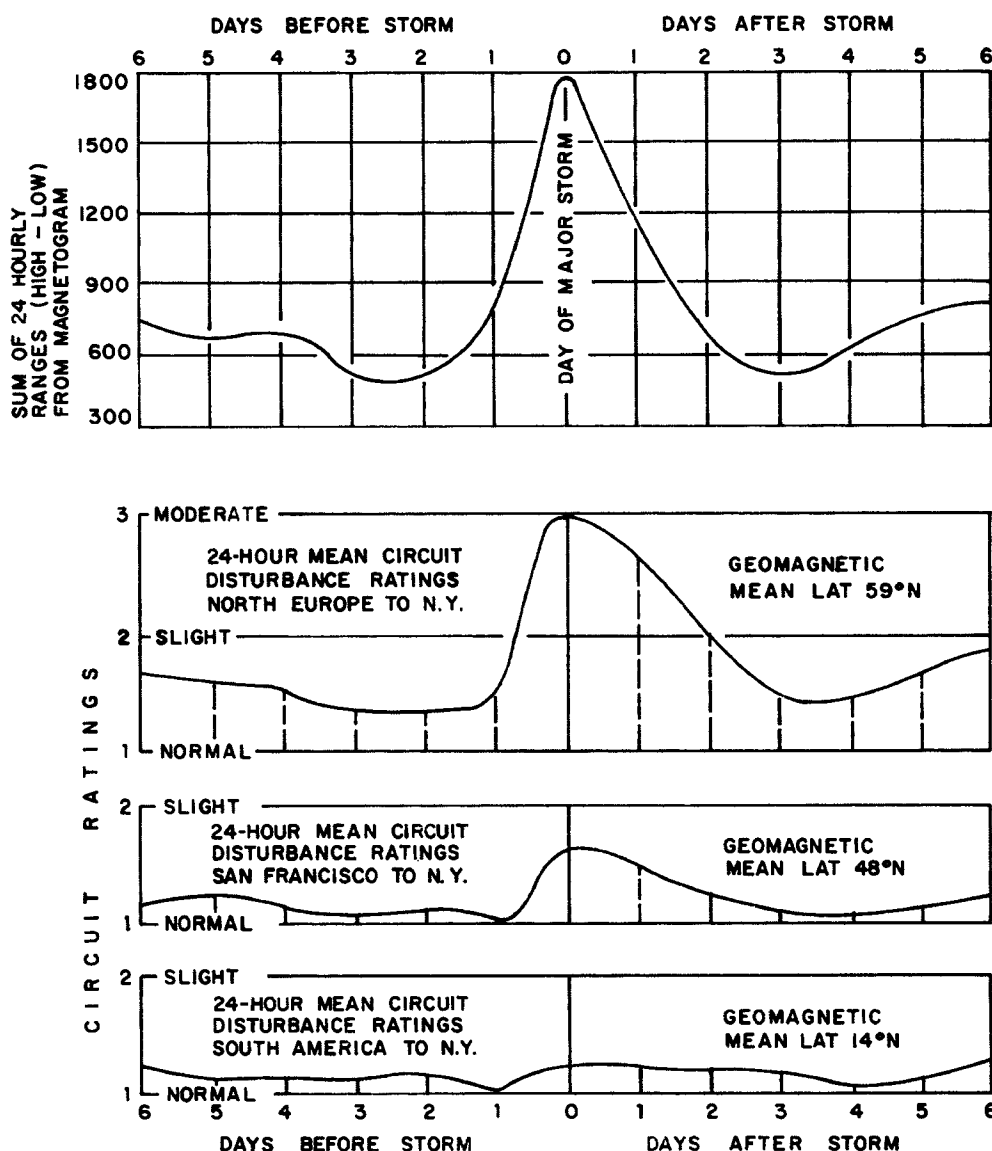


Figure G-9. Latitude effect on traffic disturbances.

tables are simplified presentations of some complex phenomena. In individual cases, many exceptions to the tables will be found.

G-4. The Use of Solar and Ionospheric Predictive Services

a. Air Force Global Weather Central.

(1) The Air Force Global Weather Central at Offutt AFB, Neb., issues periodic bulletins on solar and geomagnetic activity and radio propagation conditions. Figure G-10 is a sample of the primary hf radio propagation report issued daily at 0600Z. Secondary reports containing only Part I are issued daily at 0000Z, 1200Z, and 1800Z. Figure G-11 is a description of the primary and secondary hf radio propagation reports. Figure G-12 is a sample of GWC's seven-day outlook

issued every Friday at 0800Z. This report includes an outlook for expected hf radio propagation conditions for the seven-day period. Figure G-13 is a sample of the extended period forecast issued every Monday at 0800Z. This report gives a general forecast of solar and geomagnetic activity for several weeks in advance, and also contains an outlook for expected hf radio propagation conditions for the following seven days.

(2) In addition to issuing the regular bulletins, the Global Weather Central has a duty forecaster available at all times to respond to questions regarding hf propagation. The duty forecaster can be reached at AUTOVON numbers 271-5871 or 271-2709. Figure G-14 summarizes the services of Global Weather Central pertaining to hf propagation.

Table G-1. Ionospheric Changes Associated With Solar and Geomagnetic Events

Solar or geomagnetic event	D layer	E layer	F1 layer	F2 layer	Sporadic-E (Es) layer	Spread F
Geomagnetic activity	Increases electron concentration. Effect increases at higher magnetic latitude	Critical frequency not affected significantly. But the ionization at the lower ledge between the D and E regions is increased by about two times		MUF correlation can be positive or negative depending on magnetic latitude. Strong negative correlation with MUF at high magnetic latitudes. Weak positive correlation at low magnetic latitudes	Electron density can increase by factor of ten	Occurrence increases from higher to lower latitudes. Exception in southern hemisphere, when its occurrence increases poleward. Type depends on magnetic latitude
Magnetic storm associated with blackout	Absorption increases and lowest useful frequency increases with an increase in K_p		F region splits into F1 and F2 layers if not already split. Latitude dependent	Daytime critical frequency decreases at higher latitudes. At sunspot maximum the effect is shorter but more severe. At equatorial latitudes the critical frequency increases		
Solar flare	<i>Day only:</i> Immediate increase of electron density due to UV and X-ray. Result absorption <i>after-effects</i> . Cosmic rays delayed to several hours, resulting in possible PCA	Increase of electron density due to EUV and X-ray. Most transient Doppler due to this effect in E region	Decrease in ionization; delayed by day or more	Decrease in ionization; delayed by day or more		
X-ray radiation of chromospheric flares	SID. Greatest increase in ionization at 60-70 km mostly due to soft X-rays ($\lambda \approx 1-5\text{\AA}$) in flare					
10.7 cm flux		No close correlation with daily values of E-layer parameters. Good correlation with mean monthly values				Equatorial type increases with solar activity. High decreases with solar activity

Table G-1. Ionospheric Changes Associated With Solar and Geomagnetic Events—continued

Solar or geomagnetic event	D layer	E layer	F1 layer	F2 layer	Sporadic-E (Es) layer	Spread F
Ionospheric storm	Increased electron concentration		F1 may be created	Critical frequency depressed	Frequency occurs	Occurrence increases from higher to lower latitudes
PCA with major solar flare	Absorption due to ionization by high energy solar protons at about 50 to 70 km					Enhanced over both polar regions
SID	Increased absorption within minutes to hours	Not much change in critical frequency, but increase in ionization in lower region.				

Note. No entry in a block indicates that there is insufficient evidence of a clear relationship.

Table G-2. Communication Anomalies Associated With Solar and Geomagnetic Events

Solar or geo-magnetic event	Possible associated events											Associated circuit quality			Effect on MUF		
	SID/SWF	Ionospheric storm	Magnetic storm	SWF	Noise	Signal attenuation	Multipath	Sporadic fading	Doppler	PCA	Blackout	Poor	Fair	Normal	Decrease	Normal	Increase
Magnetic activity Severity of related event is roughly proportional to magnitude of magnetic disturbance					X				X	X	X	X			Middle and high geomagnetic latitudes, for F2 MUFs only	F2 MUF very weak dependency on average K values below 30° geomagnetic latitude	Due to sporadic E. (In auroral regions, E, occurrence more frequent as K increases)
Solar flare	X	X	X	X	X	X	X	X	X	X	X	X			In low latitudes 2 or 3 days past optical flare the F2 MUF is depressed		
High value of solar 10.7 cm flux																	X
SID											X	X					
Ionospheric storm		X	X		X		X		X		X	X			Temperate and high latitude FT MUF depressed. Spread F occurrence may require shift to lower frequency to minimize signal degradation	From equator to 30° geomagnetic latitude	In equatorial latitudes, when spread F occurs, operate nearer to MUF to minimize spread F signal degradation
Noise												X	X				
X-ray burst	X											X					
Polar cap absorption (PCA) (solar flare associated)					X	X					X				F2 MUF only during later stages of magnetic storm		
Sporadic E								X				X	X				X
Spread F							X					X			In temperate and high latitudes spread F occurrence may require shift to lower frequency to minimize signal degradation		In equatorial latitudes when spread F occurs, shift to higher than normal frequency to minimize spread F signal degradation

SUBJ: HF RADIO PROPAGATION REPORT
 SPACE ENVIRONMENTAL SUPPORT BRANCH
 AIR FORCE GLOBAL WEATHER CENTRAL
 PRIMARY HF RADIO PROPAGATION MESSAGE ISSUED AT 230600Z MAR 77
 PART I. SUMMARY 230000 TO 230600 MAR 77/
 FORECAST 230600 TO 231200Z MAR 77.

		QUADRANT			
		I	II	III	IV
		Ø TO 9Ø W	9Ø W TO 18Ø	18Ø TO 9ØE	9ØE TO Ø
REGION	POLAR	N6/+4Ø	N6	N7/+3Ø	N7
	AURORAL	U5/+4Ø	N5	N6/+25	U6
	MIDDLE	N7/+2Ø	N7	N7/+35	N7
	LOW	N7	N7	N7	N7
	EQUATORIAL	N7	N7	N7	N7

PART II. GENERAL DESCRIPTION OF HF RADIO PROPAGATION CONDITIONS
 OBSERVED DURING THE 24 HOUR PERIOD ENDING 222400Z MAR 77 AND
 FORECAST CONDITIONS FOR THE NEXT 24 HOURS.

THE GEOMAGNETIC FIELD REACHED MINOR STORM LEVEL BRIEFLY AROUND
 1Ø3ØZ AND SUBSIDED SHORTLY THEREAFTER. THIS CREATED SHORT PERIODS
 OF SIGNAL FADE AND ABSORPTION ON AURORAL AND POLAR PATHS.

ENHANCED MID LATITUDE SPORADIC E HAS BEEN OBSERVED DURING THE
 POST MIDNIGHT HOURS AND IS EXPECTED TO CONTINUE TO OCCUR DURING
 THE FORECAST PERIOD. THE GEOMAGNETIC FIELD IS EXPECTED TO REMAIN
 UNSETTLED DURING THE NEXT 24 HOURS.

PART III. SUMMARY OF SOLAR FLARE INDUCED IONOSPHERIC DISTURBANCES
 FOR THE 24 HOUR PERIOD ENDING 222400Z MAR 77: NONE

PROBABILITY FOR THE NEXT 24 HOURS: NIL

PART IV. OBSERVED/FORECAST F 1Ø/K

THE OBSERVED 1Ø.7 CM FLUX FOR 22 MAR 77 WAS Ø75. THE PREDICTED
 1Ø.7 CM FLUX FOR 23, 24, AND 25 MAR 77 IS Ø75, Ø76, AND Ø77. THE
 OBSERVED K VALUE FOR 22 MAR 77 WAS Ø3. THE FORECAST K VALUES FOR
 23, 24, AND 25 MAR 77 ARE Ø3, Ø3, AND Ø3.

Figure G-10. Sample Global Weather Central primary propagation report.

(1). Part I contains a grid of observed and six hour forecast hf propagation conditions. The grid is made up of four longitude quadrants and five latitude zones. Quadrants are defined as:

Quadrant I	0° to 90° W.
Quadrant II	90° W to 180°.
Quadrant III	180° to 90° E.
Quadrant IV	90° E to 0°.

Zones will be:

Polar	Greater than 70° from the equator.
Auroral	55° to 70° from the equator.
Midlatitude	From 30° to 55° from the equator.
Low Latitude	From 10° to 30° from the equator.
Equatorial	Within 10° of the equator.

All coordinates are geographic. A simple literal numerical scheme will be used to define the observed and forecast propagation conditions within each area described by the grid, 20 areas in all. The literal part will be used to describe the propagation conditions within the area during the previous six hours, using the following convention:

W - Poor	U - Fair	N - Normal.
----------	----------	-------------

The numerical indicator will describe the forecast conditions expected during the next six hours, using the following convention:

1 Useless	2 Very poor
3 Poor	4 Poor to fair
5 Fair	6 Fair to good
7 Good	8 Very good
9 Excellent.	

In addition to the observed/forecast propagation quality within each area, maximum usable frequency (MUF) deviation of more than 20 percent from the seasonal means will be so indicated.

(2). Part II (optional on secondary report). This part will include a commentary on unusual conditions within each area, such as, geomagnetic substorm activity, enhanced sporadic E or spread F conditions, polar cap absorption, etc. General terms will be used to make this portion of the message useful to a variety of communicators and frequency managers.

(3). Part III. This part will include disturbances reported during the previous radio day due to solar flare activity. This section will include start and end times, whether the event was confirmed on an hf system, and the frequency range affected. It will also give a probability of occurrence of solar flare induced short wave fades using the following convention:

Nil	Less than 1 percent
Slight	1 to 20 percent
Moderate	20 to 50 percent
Strong	Greater than 50 percent

(4). Part IV. This part will include the observed/forecast daily F10 (10 cm solar radio flux) and K value (measure of geomagnetic activity on a scale from 0 to 9). The forecast will be for the next three days.

Note: The primary bulletins (HFUS1), conforming to the above format, will be issued at 0600Z daily.

The secondary bulletins (HFUS2), which may consist only of Part I of the format, will be issued at 0000Z, 1200Z, and 1800Z.

Figure G-11. Explanation of Global Weather Central propagation reports.

SUBJ: SEVEN DAY OUTLOOK ISSUED 180800Z MARCH 1977.

SOLAR ACTIVITY: SOLAR ACTIVITY IS EXPECTED TO BE LOW TO VERY LOW DURING THE FORECAST PERIOD. OCCASIONAL CLASS C X-RAY EVENTS MAY OCCUR WITH NEWLY DEVELOPING REGIONS.

GEOMAGNETIC ACTIVITY: THE GEOMAGNETIC FIELD SHOULD BE UNSETTLED THE FIRST TWO AND LAST TWO DAYS AND QUIET THE REMAINDER OF THE PERIOD.

HF PROPAGATION: HF RADIO PROPAGATION CONDITIONS WILL BE GENERALLY GOOD AT LOW AND MIDDLE LATITUDES. SOME PERIODS OF FAIR TO POOR CONDITIONS WILL ACCOMPANY SUNRISE AT LOW LATITUDES. HIGH LATITUDE CONDITIONS WILL BE FAIR TO POOR ON NIGHTTIME TRANSAURORAL CIRCUITS DUE TO NOISE, SIGNAL ABSORPTION, AND SPORADIC FADING. WORST CONDITIONS WILL BE DURING THE FIRST TWO AND LAST TWO DAYS. OTHERWISE, HIGH LATITUDE CONDITIONS WILL BE FAIR TO GOOD.

Note: This outlook is issued each Friday at 0800Z.

Figure G-12. Sample Global Weather Central seven-day outlook.

SUBJ: AFGWC EXTENDED PERIOD FORECAST ISSUED 210800Z MAR 1977.

FORECAST VALID FOR THE PERIOD 21 MAR-16 APR 1977.

SOLAR ACTIVITY: SOLAR ACTIVITY SHOULD CONTINUE AT VERY LOW LEVELS DURING THE FORECAST PERIOD.

GEOMAGNETIC ACTIVITY: UNSETTLED TO ACTIVE CONDITIONS CAN BE EXPECTED 21 TO 24 MAR, AND AGAIN FROM 4 TO 11 APRIL. MINOR STORM CONDITIONS ARE LIKELY DURING THE LATTER PERIOD. GENERALLY QUIET TO UNSETTLED CONDITIONS SHOULD PREVAIL DURING THE REMAINDER OF THE FORECAST PERIOD.

HF PROPAGATION OUTLOOK VALID 21 MAR - 27 MAR 1977.

HF RADIO PROPAGATION CONDITIONS WILL BE GENERALLY GOOD ON LOW AND MIDLATITUDE CIRCUITS THROUGHOUT THE FORECAST PERIOD. HIGH LATITUDE CONDITIONS WILL BE GENERALLY FAIR TO GOOD DURING THE DAYLIGHT HOURS AND FAIR TO OCCASIONALLY POOR DURING THE NIGHTTIME HOURS. DEGRADED CONDITIONS WILL BE DUE TO ABSORPTION, FADING, AND NOISE.

Note: This forecast is issued each Monday at 0800Z.

Figure G-13. Sample Global Weather Central extended period forecast.

b. *Environmental Research Laboratories.* The Environmental Research Laboratories of the National Oceanic and Atmospheric Administration provide observations and forecasts of solar and geomagnetic activity from their Space Environmental Services Center. Their services do not,

however, include propagation observations and forecasts. Information is available from WWV broadcasts and by telephone, as shown in figure G-15. Figure G-16 provides an explanation of the format and content of the WWV service. Figure G-17 is a copy of the form used in preparing the

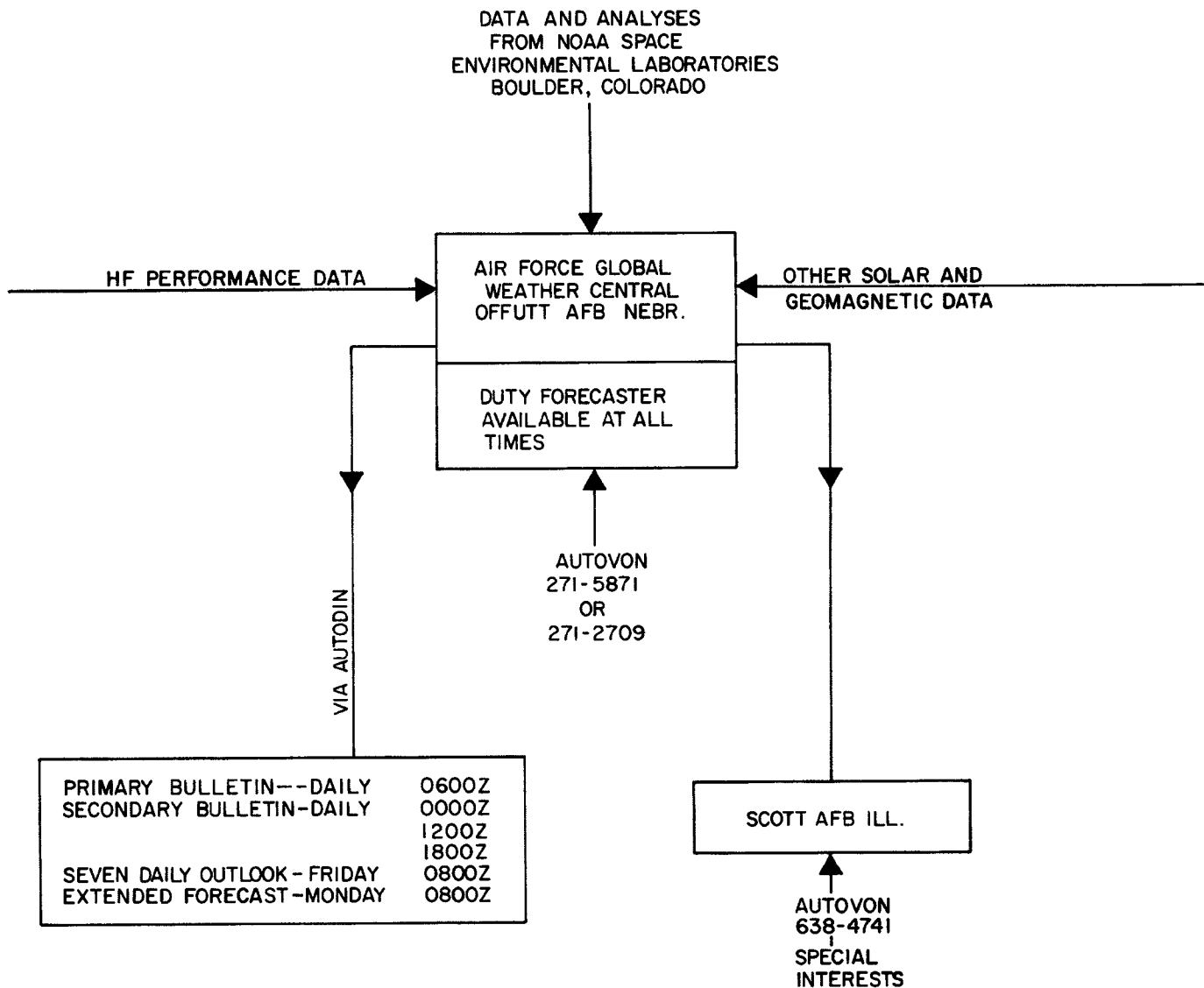


Figure G-14. Global Weather Central hf propagation services.

telephone recording, which shows the format of the recording. Copies of figure G-17 can also be used to facilitate copying the recording.

G-5. Maintaining Hf Radio Communications During Disturbed Conditions

a. General. The forecasting of high frequency propagation conditions is based upon detailed knowledge of current solar, geomagnetic, ionospheric, and radio wave propagation conditions together with known ionospheric responses to specific types of solar activity. The tendency for ionospheric disturbances to recur at about 27-day intervals is also considered. Keep well informed about solar activity and ionospheric conditions in order to better interpret forecasts and better react to deteriorating propagation conditions. Lis-

ten to the WWV broadcasts and, if possible, obtain the Global Weather Central reports. Listen to commercial and amateur stations to learn what frequencies are being propagated well. When propagation conditions deteriorate, attempt to determine the type of disturbance before taking adaptive action.

b. Communicating During an SID.

(1) At the onset of an SID, signals fade away entirely within minutes and may not return for 2 hours or more. Only the sunlit hemisphere is affected. Low latitudes, where the sun's rays are perpendicular or nearly so, are affected more than high latitudes. An SID occurs at the same time a bright solar flare is observed.

(2) Everything appears to go dead at once. Even background noise disappears. During an

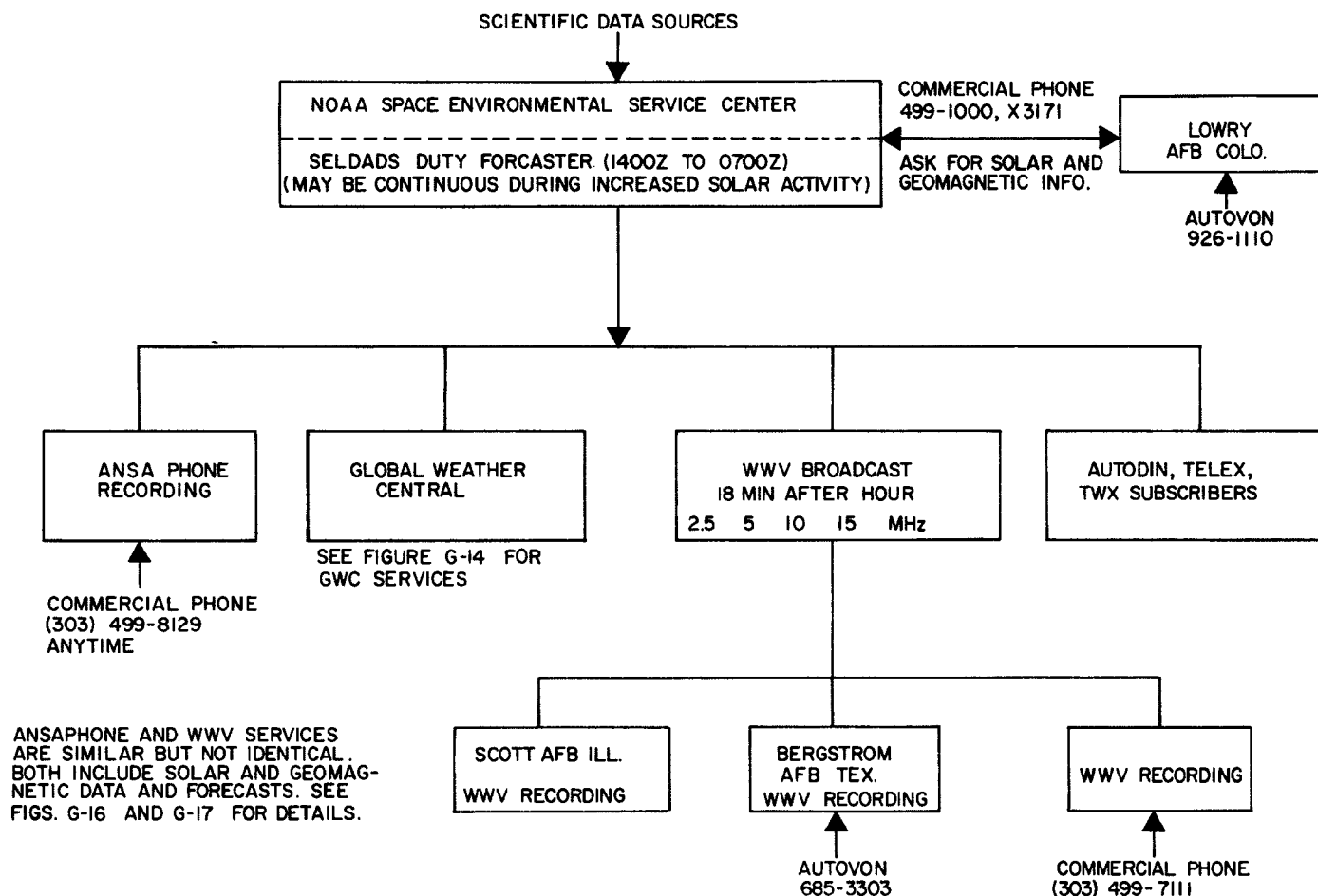


Figure G-15. Services of Environmental Research Laboratories.

SID, use the highest frequency possible. High frequencies are the last to go out and the first to come back. If even the high portion of the band becomes unusable, wait until the fadeout ends. Resist the urge to troubleshoot apparently dead receiving equipment.

c. Communicating During an Ionospheric Storm.

(1) Very large solar flares tend to be followed in 1 or 2 days by geomagnetic and ionospheric storms. During an ionospheric storm, signals on frequencies normally received well drop to very low levels, become erratic, and often dis-

appear entirely. The high portion of the hf band is most affected, and it is affected to the same extent in both sunlit and dark hemispheres. High latitude paths are affected more than low, and auroral paths are most severely disturbed.

(2) To maintain communications, reduce frequency, increase power or drop channels, and route around auroral zones. Such measures may allow communications to continue during the storm. In the most severe storms, even the low-altitude paths become unusable. The storm may last for several hours or several days.

Geophysical alerts are broadcast on WWV at 18 minutes after the hour.

The messages are changed each day at 0000 GMT. Part A of the message which gives the Boulder K index is updated every six hours.

Part B which gives the solar-terrestrial conditions for the previous day is 1) The 1700Z solar flux from Ottawa at 2800 MHz. The A index is the estimated A value for Fredericksburg. 2) Solar activity is classified as:
 Very low = usually only quiet regions on the solar disk and no more than five of these; fewer than ten class-C subflares without centimetric radio burst or SID observed or expected.

Low = usually more than five but less than ten quiet regions on the solar disk; only Class-C subflares without centimetric radio bursts or SID observed or expected.

Moderate = eruptive regions on the solar disk; fewer than five class-M X-ray events with centimetric radio bursts and SID observed or expected.

High = active regions on solar disk; several class-M X-ray events with centimetric radio bursts and strong SID; and/or one to two importance-2 chromospheric flares or class-X X-ray events observed or expected.

Very high = region capable of producing protons on the sun; one or more chromospheric flares of importance-2 or greater; with outstanding centimetric radio bursts (500 flux units or greater), class-X X-ray bursts, and major SID observed or expected.

3) The geomagnetic field is classified as:
 Quiet = $A \leq 7$, usually no K-indexes > 2 .

Unsettled = $7 < A < 15$, usually no K-indexes > 3 .

Active = $15 < A < 30$, a few K-indexes of 4.

4) The geomagnetic storms are classified as:

Minor = $30 \leq A < 50$, K-indexes mostly 4 and 5.

Major = $A \geq 50$, some K-indexes 6 or greater. Sudden commencements indicated by beginning time given to nearest minute.
 Gradual commencements indicated by beginning time to nearest hour.

Figure G-16. WWV geophysical alert format.

5) the rest of the report is as follows:

Solar Flares

Class C = any solar X-ray burst with a peak flux at 1-8 Å of less than 10^{-2} erg cm⁻² sec⁻¹.

Class M = a solar X-ray burst with peak flux at 1-8 Å greater than or equal to 10^{-2} but less than 10^{-1} erg cm⁻² sec⁻¹.

Class X = a solar X-ray burst with peak flux at 1-8 Å greater than or equal to 10^{-1} erg cm⁻² sec⁻¹.

Major Solar Flare = a flare of optical importance ≥ 2 B with a centimetric radio outburst of 500 flux units or more; or an X-ray event of class-X intensity of duration ≥ 180 min regardless of optical flare importance.

Proton Flare = protons by satellite detectors (or polar cap absorption by riometer) have been observed in time association with the H α flare.

Satellite-Level Proton Event = a proton enhancement detected by Earth-orbiting satellites with particle flux ≥ 10 MeV at least ten times background.

Polar Cap Absorption = proton-induced absorption ≥ 2 dB as measured by a 30 MHz riometer located within the polar cap.

Stratwarm = reports of stratospheric warmings in the high latitude regions of the winter hemisphere of the Earth, associated with gross distortions of the normal circulation associated with the winter season.

Part C. This is a forecast for the next day which is based on the forecaster's data and knowledge of solar activity.

Notes:

WWV frequencies are 2.5, 5, 10, and 15 MHz. Information identical to that broadcast can be obtained by phoning (303) 499-7111 at 18 minutes after the hour. The same information can be obtained by calling AUTOVON 685-3303 (Bergstrom AFB, Austin, Tex.) at 18 minutes past the hour. Geophysical alerts are not transmitted by WWVH.

Figure G-16.—continued

Information obtained by calling (303) 499 8129

DATE _____

ANSAPHONE RECORDING

A. BOULDER K-INDEX -- GIVE DATE ALSO

- () THE BOULDER K-INDEX AT 0000 UT WAS _____
- () THE BOULDER K-INDEX AT 0300 UT WAS _____
- () THE BOULDER K-INDEX AT 0600 UT WAS _____
- () THE BOULDER K-INDEX AT 0900 UT WAS _____
- () THE BOULDER K-INDEX AT 1200 UT WAS _____
- () THE BOULDER K-INDEX AT 1500 UT WAS _____
- () THE BOULDER K-INDEX AT 1800 UT WAS _____
- () THE BOULDER K-INDEX AT 2100 UT WAS _____

B. SOLAR TERRESTRIAL CONDITIONS FOR _____ FOLLOW:
 ALWAYS INCLUDE THE FIRST THREE, ADD OTHERS AS APPROPRIATE.

- () SOLAR FLUX _____ AND A-INDEX _____
- () SOLAR ACTIVITY WAS _____
- () GEOMAGNETIC ACTIVITY WAS _____
- () MAJOR SOLAR FLARE AT _____ UT _____
- () PROTON FLARE AT _____ UT _____
- () PROTON EVENT ON SATELLITE _____
- () PCA EVENT

C. THE FORECAST FOR _____ FOLLOWS:
 ALWAYS INCLUDE THE FIRST TWO, ADD OTHERS AS APPROPRIATE.

- () SOLAR ACTIVITY WILL BE _____
- () GEOMAGNETIC FIELD WILL BE _____
- () GEOMAGNETIC STORM EXPECTED _____
- () PROTON FLARE EXPECTED _____

Figure G-17. Space Environment Services Center telephone service format.

APPENDIX H

HF EXERCISE

OPERATIONAL CHECKLIST

H-1. Before the Exercise

- a.* Request propagation charts.
- b.* Request frequencies.
- c.* Compare complement of assigned frequencies to predictions. How many hours of operation are predicted? Is an antenna analysis needed to improve predicted time availability?
- d.* Set up initial time-versus-frequency matrix.
- e.* Check antenna kits for ground rods, extra copper wire, bug nuts, etc.
- f.* Make sure the antenna shown on the propagation predictions is one that can be built.
- g.* Set up a tentative schedule of frequency-change times based on the frequency-usage versus time-of-day forecast given in the propagation predictions. Schedule shift or crew changes at times when the frequency change probability is low.
- h.* Arrange to have a spare, continuously tuneable, monitor receiver at the site for listening to other stations (short-wave broadcast, time, MARS, amateur, etc) to check on current propagation conditions and to troubleshoot the HF system being operated in the exercise. (Weak signals in that system may be equipment related rather than propagation related.)
- i.* Is duplicate equipment necessary to assist in minimizing time required in determining useable frequencies and making frequency changes?
- j.* Write the Communications Operating Instructions (COI) based on previous items and in the best manner to achieve the mission goals.

H-2. Selecting a Site

- a.* Are there any significant horizon obstructions?
- b.* Is there enough room for the antennas? Avoid putting antenna in the foreground of another antenna.

c. Is it likely that there will be a co-site problem with the antennas?

d. Is the site far enough removed from industrial areas, hospitals, power lines, and highways so that manmade noises will be at the lowest practical level?

H-3. Preparing the Site

- a.* With regard to transmission lines:
 - (1) Keep them (especially transmit lines) as short as possible.
 - (2) Avoid sharp bends.
 - (3) Keep receiving lines separated from transmitting lines.
 - (4) Make sure all connectors are OK.
 - (5) Clearly mark the lines to avoid damage or accidents.
- b.* Make sure antenna dimensions and headings are correct.
- c.* Make sure all wire connections in the antenna system are tightly clamped.
- d.* Ground (or suitably protect) all metallic items that may become a hazard to personnel. Large items, such as fences, should be grounded every eight feet.
- e.* Avoid long ground leads for equipment. Use multiple grounds if necessary.
- f.* Use good grounds on antennas.
- g.* Provide an environment for the operators that is as quiet as possible.
- h.* Consider safety of the people involved.

H-4. During the Exercise

- a.* Compile data on—
 - (1) Propagation outages on an hourly basis.
 - (2) Daily circuit reliability versus predicted reliability.
 - (3) Comparison of actual and predicted frequency changes.

(4) Adjustment of frequency-versus-time matrix as determined by experience.

b. Maintain close contact with the USAF Global Weather Central to follow ionospheric conditions.

c. Consider reducing the number of channels on the circuit when propagation is poor.

d. Make a daily inspection of the antenna field, radio equipment, transmission lines, power system, and grounding.

e. If outages occur, troubleshooting should begin by looking for the symptoms, making a determination of the problem, and effecting a solution in

one or more of the following four basic problem areas.

- (1) Propagation outage
- (2) Improper frequency usage
- (3) Antenna design/electrical failure
- (4) Transmitter/receiver fault

H-5. After the Exercise

Prepare an after-action report, which should:

- a.* Compare predicted and actual system performance.
- b.* Assess achievement of mission goals.
- c.* Document troubleshooting actions.

GLOSSARY

A

active region of sun—A localized, transient region of the solar atmosphere in which sunspots, faculae, plages, flares, etc., may be observed.

amplitude modulation (am)—A form of modulation in which the amplitude of a carrier wave is varied in accordance with the instantaneous value of the modulating signal.

angle of incidence—The acute angle between a ray and the normal to the surface on which the ray is incident.

angstrom (A)—A unit of length = 1×10^{-8} cm.

antenna—Any structure or device used to collect or radiate electromagnetic waves.

antenna matching—The process of adjusting impedance so that the input impedance of an antenna equals the characteristic impedance of its transmission line.

antenna tuner—Also called antenna loading unit. A device used to electrically lengthen or shorten an antenna by using lumped inductance and capacitance. In addition, it matches the antenna to the feedline impedance.

aperiodic antenna—An antenna designed to have an approximately constant input impedance over a wide range of frequencies, e.g., terminated rhombic antennas and wave antennas. Synonym: nonresonant antenna.

arch filament system—A system of small, arched filaments which are often a sign of rapid or continued growth in an active solar region.

ARQ system—See error-detecting and feedback system.

AUTODIN—A world-wide data communications network of the Defense Communications System; acronym for AUTOMATIC Digital Network.

B

balanced—Electrically symmetrical with respect to ground.

balun—A device used for matching an unbalanced impedance to a balanced impedance.

bandwidth—The difference between the limiting frequencies of a continuous frequency band. The bandwidth of a device is the difference between the limiting frequencies within which performance in respect to some characteristic falls within specified limits.

baud—The unit of modulation rate. One baud corresponds to a rate of one unit interval per second.

beam width—The angle between the half-power points (3-dB points) of the main lobe of the antenna pattern when referenced to the peak power point of the antenna pattern. It is measured in degrees and normally taken off the horizontal radiation pattern.

binary digit (bit)—

1. In pure binary notation, either of the characters, 0 or 1.

2. A unit of information equal to one binary decision or the designation of one of two possible and equally likely states of anything used to store or convey information.

bit error rate (BER)—The number of erroneous bits divided by the total number of bits over some stipulated period of time.

C

carrier—

1. A wave suitable for modulation by an intelligence-bearing signal to be transmitted over a communication system.

2. An unmodulated emission. The carrier is usually a sinusoidal wave or a recurring series of pulses.

chromosphere—The layer of the solar atmosphere between the photosphere and corona.

corona—The outermost layer of the solar atmosphere.

corona discharge—Luminous discharge due to ionization of the air surrounding a conductor around which exists a voltage gradient exceeding a certain critical value.

cosmic noise—The broad spectrum of radio noise arriving at Earth from sources outside the solar system.

circuit reliability—The percentage of time a circuit was available to the user during a specified period of scheduled availability. As used in hf circuit analysis, the fraction of days that successful communications may be expected at a given hour within the month. Synonym: time availability.

code—A set of unambiguous rules specifying the manner in which data may be represented.

critical frequency—The limiting frequency below which a wave component is reflected by, and above which it penetrates through, an ionospheric layer at vertical incidence.

D

decibel (dB)—The standard unit for expressing transmission gain or loss and relative power ratios. They are expressed in terms of the logarithm to the base 10 of a power ratio.

$$\text{dB} = 10 \log_{10} \frac{P_1}{P_2}$$

$$\text{db} = 20 \log_{10} \frac{E_1}{E_2} = 20 \log_{10} \frac{I_1}{I_2}$$

dBm—dB referred to one milliwatt; employed in communications work as a measure of absolute power values. Zero dBm equals one milliwatt.

dBW—dB referred to one watt. Zero dBW equals one watt.

demodulation—A process wherein a wave resulting from previous modulation is processed to derive a wave having substantially the characteristics of the original modulating wave.

diffraction—The bending of radio, sound, or light waves around an object, barrier, or aperture.

dipole antenna—A radiator, usually fed at the center, producing a maximum of radiation in the plane normal to its axis. The length specified is the overall length.

direct wave—Wave that travels directly between the transmitter and receiver without reflections from any object.

diversity reception—That method of radio reception whereby, in order to minimize the effects of fading, a resultant signal is obtained by combination or selection, or both, of two or more inde-

pendent sources of received-signal energy which carry the same modulation or intelligence, but which may vary in their fading characteristics at any given instant.

Doppler effect—The phenomenon evidenced by the change in the observed frequency of a sound or radio wave caused by a time rate of change in the effective length of the path of travel between the source and the point of observation.

double sideband transmission—That method of sideband transmission in which both sidebands are transmitted.

dual diversity—The simultaneous combining of, or selection from, two independently fading signals through the use of space, frequency, angle, time, or polarization characteristics.

E

effective radiated power (ERP)—The power supplied to the antenna multiplied by the power gain of the antenna in a given direction.

electrical length—The length expressed in wavelengths, radians, or degrees of an electromagnetic wave.

electromagnetic compatibility (EMC)—The ability of communications-electronics equipments, subsystems, and systems, together with electromechanical devices (for example, vehicles, engine generators, and electric tools), to operate in their intended operational environments without suffering or causing unacceptable degradation because of unwanted electromagnetic radiation or response.

electromagnetic interference (EMI)—The phenomenon resulting when electromagnetic energy causes unacceptable or undesirable responses, malfunction, degradation, or interruption of the intended operation of an electronic equipment, subsystem, or system. Synonym: radio frequency interference

emission—The classification of the modulated or unmodulated output wave of a radio transmitter. The designator is developed using the bandwidth of the signal, the type of modulation, and characteristics of the transmission. An example is 6A3, which is a signal 6 kHz wide, amplitude modulated with voice.

end effect—Effect of capacitance at the ends of an antenna. The actual length of a half-wave antenna is about 5 percent less than half a wave-

length, with this percentage increasing to about 6 percent at frequencies above 50 MHz.

error-correcting code—A code in which each telegraph or data signal conforms to specific rules of construction so that departures from this construction in the received signals can be automatically detected. This permits the automatic correction, at the receiving terminal, of some or all of the errors. Such codes require more signal elements than are necessary to convey the basic information.

error-detecting code—A code in which each telegraph or data signal conforms to specific rules of construction, so that departures from this construction in the received signals can be automatically detected. Such codes require more signal elements than are necessary to convey the basic information.

error-detecting and feedback system—A system employing an error-detecting code and so arranged that detection of an error automatically initiates a request for retransmission. Synonyms: ARQ system; request-repeat system.

F

facula—A bright region of the solar photosphere seen in white light; white light plage.

fading—The variation, with time, of the intensity or relative phase, or both, of any or all frequency components of a received radio signal due to changes in the characteristics of the propagation path with time.

field strength—The intensity of an electric, magnetic, or electromagnetic field at a given point, normally used to refer to the rms value of the electric field, expressed in volts per meter.

flux unit—A measure of the intensity of solar activity equal to 10^{-22} W Hz⁻¹ m⁻².

free space—Empty space with no free electrons or ions present. The term also implies remoteness from material objects that could cause reflection of radio waves.

frequency modulation (fm)—The form of modulation in which the instantaneous frequency of a sine wave carrier is caused to depart from the carrier frequency by an amount proportional to the instantaneous value of the modulating signal.

frequency shift keying—A form of frequency modulation in which the modulating signal shifts the output frequency between predetermined val-

ues. Commonly, the instantaneous frequency is shifted between two discrete values termed the mark and space frequencies.

Fresnel zone—A cigar-shaped shell of circular cross section surrounding the direct path between a transmitter and a receiver. For the first Fresnel zone, the distance from the transmitter to any point on this shell and on the receiver is one half-wavelength longer than the direct path; for the second Fresnel zone, two half-wavelengths, etc.

G

galactic radio noise—Random noise originating outside the earth's atmosphere.

gamma—A unit of magnetic field intensity equal to 1×10^{-5} gauss.

geolert—A message broadcast daily by the Space Environment Services Center, Boulder, Colo., which summarizes solar-geophysical activity for the past 2 hours and forecasts activity for the next 24 hours.

geomagnetic storm—A worldwide disturbance of the earth's magnetic field, distinct from regular diurnal variations.

grade of service—The quality specification of a communication channel or system. It may be stated in terms of signal-to-noise ratio, bit-error-rate, message throughput rate, call blocking probability, etc.

ground constants—The electrical constants of the earth, such as conductivity and dielectric constants. These values vary with the chemical composition of the earth, moisture content, and frequency.

groundwave—A radio wave that is propagated over the earth and is affected by the presence of the ground. Ground waves do not depend upon the ionosphere for propagation.

ground screen—A network of wires underneath an antenna, usually buried a few inches, used to improve grounding.

H

hertz (Hz)—The international unit denoting one cycle per second.

I

impedance—The total passive opposition offered to the flow of an alternating current. It consists

of a combination of resistance, inductive reactance, and capacitive reactance.

ionosphere—The region of the atmosphere, extending from roughly 50 km to 400 km altitude, in which there is appreciable ionization.

ionospheric disturbance—A sudden increase in the ionization of the D-region of the ionosphere, caused by solar flares, which results in greatly increased radio wave absorption.

ionospheric scatter—The propagation of radio waves by scattering due to irregularities in the ionosphere.

isotropic antenna—A hypothetical antenna that radiates or receives equally in all directions.

K

K-index—A measure of geomagnetic activity.

L

log-periodic antenna—A broadband, multielement, unidirectional, narrow-beam antenna.

lowest useful frequency (LUF)—The lowest frequency in the hf band at which the received field intensity is sufficient to provide the required signal-to-noise ratio on 90 percent (80 percent for some northern areas in USACEEIA's Intermediate and Short Distance Skywave Book) of the undisturbed days of the month.

M

mark—In binary communications, one of the two significant conditions of modulation.

maximum usable frequency (MUF)—The upper limit of the frequencies that can be used at a specified time for radio transmission between two points and involving propagation by reflection from the regular ionized layers of the ionosphere. MUF is a median frequency applicable to 50 percent of the days of a month, as opposed to 90 percent cited for the lowest useful frequency (LUF) and the optimum traffic frequency (FOT).

modem—(Acronym for MOdulator-DEModulator). A device that modulates and demodulates signals. Modems are primarily used for converting digital signals into quasi-analog signals for transmission and for reconverting the quasi-analog signals into digital signals.

modulation—The process, or result of the process, of varying certain characteristics of a signal,

called a carrier, in accordance with a message signal.

monopole antenna—An antenna, usually vertical, either a quarter-wavelength or tuned to a quarter-wavelength, which depends upon reflections from the ground to complete its radiation pattern.

multicoupler—A device which simultaneously couples several receivers to a single antenna with little or no signal loss over a wide bandwidth (assuming an amplification device in multicoupler). Also called a branching amplifier. A transmitter multicoupler couples more than one transmitter to an antenna with no interaction.

multipath—The propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths. Multipath effects range from constructive reinforcement to destructive cancellation of the signal.

N

noise—Unwanted or disturbing energy introduced into a communications system from manmade and natural sources.

O

optimum traffic frequency (FOT)—The highest frequency that is predicted to be available for skywave transmission over a particular path at a particular hour for 90 percent of the days of the month.

P

parity—In binary-coded systems, a condition obtained with a self-checking code such that in any permissible code expression the total number of 1's or 0's is always even or always odd.

path loss—The decrease in power in transmission from one point to another. It is usually expressed in decibels, and in radio systems is taken to be the loss between transmitting and receiving antennas.

peak envelope power (PEP)—The average power supplied to the antenna transmission line by a radio transmitter during one radio frequency cycle at the highest crest of the modulation envelope, taken under conditions of normal operation.

phase modulation (pm)—Modulation in which the phase of a sine-wave carrier deviates from the

no-signal phase by an amount proportional to the modulating wave.

photosphere—The lowest layer of the solar atmosphere which we can see.

polar cap absorption—A large increase in the ionospheric absorption at high magnetic latitudes, due to increases in the D-region electron density, caused by solar rays.

polarization—That property of a radiated electromagnetic wave describing the time-varying direction and amplitude of the electric field vector.

power gain of an antenna—The ratio of the power required at the input of a reference antenna to the power supplied to the input of the given antenna to produce, in a given direction, the same field at the same distance. When not specified otherwise, the figure expressing the gain of an antenna refers to the gain in the direction of the radiation main lobe.

pulse modulation—The use of a series of pulses to convey information.

Q

quadruple diversity—The term applied to the simultaneous combining of, or selection from, four independently fading signals and their detection through the use of space, frequency, angle, time, or polarization characteristics or combinations thereof.

R

radiation—In radio communication, the emission of energy in the form of electromagnetic waves.

radiation pattern—The variation of the field intensity of an antenna as a function of direction.

refraction—The bending of a sound, radio, or light wave as it passes obliquely from one medium to another medium in which its speed is different.

reliability—The probability that an item will perform its intended function for a specified interval under stated conditions.

rf bandwidth—The difference between the highest and the lowest emission frequencies in the region of the carrier or principal carrier frequency.

rhombic antenna—An antenna composed of long-wire radiators comprising the sides of a rhombus. It is usually terminated and unidirectional; when unterminated, it is bidirectional.

S

signal-to-noise ratio—The ratio of the amplitude of the desired signal to the amplitude of noise signals at a given point in time, usually expressed in dB.

single-sideband transmission—That method of sideband transmission in which only one sideband is transmitted. The carrier may be suppressed.

skip distance—The minimum distance between the transmitting station and the point of return to the earth of the transmitted wave reflected from the ionosphere.

skip zone—A region within the transmission range wherein signals from a transmitter are not received. It may be between the farthest points reached by the ground wave and nearest points at which the refracted skywaves come back to earth or between areas served by skywaves.

skywave—A radio wave that travels upward in space from the antenna. It may or may not be returned to earth from the ionosphere.

solar flares—Solar phenomena which have the effect of producing significant but short-term (3–10 days) increases in the level of ionization of the ionosphere.

solar wind—Outward flux of solar material from the corona which passes the earth with a velocity of 250 to 400 km/s and with proton and electron densities of $7/\text{cm}^3$ to $20/\text{cm}^3$.

sporadic E propagation—Radio wave propagation by means of returns from irregular ionization that appears at heights of about 90 km to 120 km.

standing wave—Distribution of current and voltage on a transmission line formed by two sets of waves traveling opposite directions, and characterized by the presence of a number of points of successive maxima and minima in the distribution curves. This also occurs on resonant antennas. In antenna feed lines it indicates power reflected from the antenna and is not desirable.

standing wave ratio (SWR)—The ratio of the amplitude of a standing wave at a point of maximum voltage to the amplitude at a point of minimum voltage.

sudden ionospheric disturbance (SID)—Abnormally high ionization densities in the D region caused by an occasional sudden outburst of ultraviolet light on the sun (solar flare). This results in a

sudden increase in radio-wave absorption, which is the most severe in the upper mf and lower hf frequencies.

sunspots—A solar phenomenon which affects the degree of ionization of the ionosphere. A long-term sun spot cycle exists, having about 11 years between successive maxima or successive minima.

T

transmission loss—In communications, the decrease in power during transmission from one point to another, usually expressed in decibels.

tropospheric scatter—The propagation of radio waves by scattering as a result of irregulari-

ties or discontinuities in the physical properties of the troposphere.

U

ultraviolet (uv)—That part of the electromagnetic spectrum between 50 and 4000 Å.

V

V antenna—A directional antenna consisting of two straight conductors set at an angle to each other and fed at the junction.

W

whip antenna—A flexible vertical monopole used with transportable and mobile equipment.

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